RecoTrackerEntranceCompare: entrance seeding diagnostic

A: original saved trajectory. B: truth position only. C: full truth boundary.
D: truth momentum only, when enabled. No truth constraint upstream of entrance.
Control: fresh backward propagation from all-reconstructed boundary quantities.
All distances: mm. Momenta: MeV/c. Times: ns. Phase: degrees. Pitch: mrad.
Signed width s68 = (q84.1345-q15.8655)/2. Standard deviation includes tails.
Nonnegative errors use 68% and 95% containment, not signed Gaussian widths.
All statistics use unbinned finite values, including histogram under/overflow.
NaN values and unavailable states are counted, not converted to zero.
dt is corrected for the common proton-bunch clock; dt_beam includes that clock.
Fixed-z comparisons do not measure an independent longitudinal resolution.
Target branch is unique, has initial production direction, and precedes entrance.
Principal target/foil results require completed transport; partial roots are counted.
Residual plots use available valid states; always compare their sample sizes.
Event efficiencies keep the identical generated-primary event denominator.
Full-truth residuals can include stochastic material effects and transport error.
Linear angular moments depend on wrapping; consult the circular statistics.
Input and propagated covariance-invalid counts are separate; legacy status bits overlap.
ROOT histograms remain in the input file. JSON/CSV retain full-precision tables.

A: tracker entrance / baseline_direct

```
ent_dx         N=  17258 NaN=    0
  mean=1.1692 med=1.2785 std=29.949 s68=6.1155
  |err| q68=6.2328 q95=24.568 q99=119.62 tail |err|>40: 450 (2.607%)
ent_dy         N=  17258 NaN=    0
  mean=0.47158 med=0.45225 std=27.789 s68=6.0735
  |err| q68=6.0309 q95=23.78 q99=112.49 tail |err|>40: 453 (2.625%)
ent_dr         N=  17258 NaN=    0
  mean=13.917 med=6.5126 std=38.433 s68=6.6935
  |err| q68=9.7803 q95=34.979 q99=184.82 tail |err|>40: 737 (4.270%)
ent_dp         N=  17258 NaN=    0
  mean=-0.25776 med=-0.055892 std=2.2189 s68=0.21193
  |err| q68=0.21035 q95=1.2061 q99=9.821 tail |err|>2: 566 (3.280%)
ent_dpz        N=  17258 NaN=    0
  mean=-0.17734 med=-0.047307 std=1.5455 s68=0.37288
  |err| q68=0.36896 q95=1.2024 q99=6.2981 tail |err|>2: 476 (2.758%)
ent_pitch      N=  17258 NaN=    0
  mean=0.0022997 med=-0.068759 std=22.94 s68=9.1999
  |err| q68=9.1173 q95=28.081 q99=82.523 tail |err|>20: 1581 (9.161%)
ent_dt         N=  17258 NaN=    0
  mean=0.32054 med=0.38234 std=0.79585 s68=0.34704
  |err| q68=0.58312 q95=1.4569 q99=3.5881 tail |err|>3: 252 (1.460%)
ent_dt_beam    N=  17258 NaN=    0
  mean=0.32173 med=0.36157 std=0.94823 s68=0.66449
  |err| q68=0.79607 q95=1.7692 q99=3.7109 tail |err|>3: 278 (1.611%)
```

A: tracker entrance / baseline_reflected

```
ent_dx         N=  13583 NaN=    0
  mean=1.1535 med=1.2231 std=29.327 s68=6.4758
  |err| q68=6.5813 q95=26.117 q99=104.83 tail |err|>40: 390 (2.871%)
ent_dy         N=  13583 NaN=    0
  mean=0.27156 med=0.4209 std=28.896 s68=6.4613
  |err| q68=6.392 q95=24.76 q99=109.25 tail |err|>40: 375 (2.761%)
ent_dr         N=  13583 NaN=    0
  mean=14.465 med=6.8785 std=38.565 s68=7.1668
  |err| q68=10.419 q95=37.929 q99=191.22 tail |err|>40: 626 (4.609%)
ent_dp         N=  13583 NaN=    0
  mean=-0.25107 med=-0.060942 std=2.5412 s68=0.21271
  |err| q68=0.21174 q95=1.1705 q99=8.0588 tail |err|>2: 437 (3.217%)
ent_dpz        N=  13583 NaN=    0
  mean=-0.14782 med=-0.043111 std=2.1436 s68=0.38456
  |err| q68=0.3791 q95=1.2329 q99=5.0182 tail |err|>2: 342 (2.518%)
ent_pitch      N=  13583 NaN=    0
  mean=-0.23299 med=-0.24187 std=21.681 s68=9.2848
  |err| q68=9.2412 q95=26.548 q99=66.072 tail |err|>20: 1191 (8.768%)
ent_dt         N=  13583 NaN=    0
  mean=0.32065 med=0.37022 std=0.73158 s68=0.35294
  |err| q68=0.58284 q95=1.3831 q99=3.2884 tail |err|>3: 159 (1.171%)
ent_dt_beam    N=  13583 NaN=    0
  mean=0.32079 med=0.34493 std=0.88535 s68=0.67214
  |err| q68=0.78956 q95=1.7221 q99=3.2891 tail |err|>3: 163 (1.200%)
```

A: tracker entrance / control_direct

```
ent_dx         N=  17258 NaN=    0
  mean=1.1692 med=1.2785 std=29.949 s68=6.1155
  |err| q68=6.2328 q95=24.568 q99=119.62 tail |err|>40: 450 (2.607%)
ent_dy         N=  17258 NaN=    0
  mean=0.47158 med=0.45225 std=27.789 s68=6.0735
  |err| q68=6.0309 q95=23.78 q99=112.49 tail |err|>40: 453 (2.625%)
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  |err| q68=9.7803 q95=34.979 q99=184.82 tail |err|>40: 737 (4.270%)
ent_dp         N=  17258 NaN=    0
  mean=-0.25776 med=-0.055892 std=2.2189 s68=0.21193
  |err| q68=0.21035 q95=1.2061 q99=9.821 tail |err|>2: 566 (3.280%)
ent_dpz        N=  17258 NaN=    0
  mean=-0.17734 med=-0.047307 std=1.5455 s68=0.37288
  |err| q68=0.36896 q95=1.2024 q99=6.2981 tail |err|>2: 476 (2.758%)
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  |err| q68=9.1173 q95=28.081 q99=82.523 tail |err|>20: 1581 (9.161%)
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  |err| q68=0.58312 q95=1.4569 q99=3.5881 tail |err|>3: 252 (1.460%)
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  mean=0.32173 med=0.36157 std=0.94823 s68=0.66449
  |err| q68=0.79607 q95=1.7692 q99=3.7109 tail |err|>3: 278 (1.611%)
```

A: tracker entrance / control_reflected

```
ent_dx         N=  13583 NaN=    0
  mean=1.1535 med=1.2231 std=29.327 s68=6.4758
  |err| q68=6.5813 q95=26.117 q99=104.83 tail |err|>40: 390 (2.871%)
ent_dy         N=  13583 NaN=    0
  mean=0.27156 med=0.4209 std=28.896 s68=6.4613
  |err| q68=6.392 q95=24.76 q99=109.25 tail |err|>40: 375 (2.761%)
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  |err| q68=10.419 q95=37.929 q99=191.22 tail |err|>40: 626 (4.609%)
ent_dp         N=  13583 NaN=    0
  mean=-0.25107 med=-0.060942 std=2.5412 s68=0.21271
  |err| q68=0.21174 q95=1.1705 q99=8.0588 tail |err|>2: 437 (3.217%)
ent_dpz        N=  13583 NaN=    0
  mean=-0.14782 med=-0.043111 std=2.1436 s68=0.38456
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  mean=0.32065 med=0.37022 std=0.73158 s68=0.35294
  |err| q68=0.58284 q95=1.3831 q99=3.2884 tail |err|>3: 159 (1.171%)
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```

A: tracker entrance / momentum_direct

```
ent_dx         N=  17258 NaN=    0
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  |err| q68=6.2328 q95=24.568 q99=119.62 tail |err|>40: 450 (2.607%)
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  mean=-0.25776 med=-0.055892 std=2.2189 s68=0.21193
  |err| q68=0.21035 q95=1.2061 q99=9.821 tail |err|>2: 566 (3.280%)
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  |err| q68=0.36896 q95=1.2024 q99=6.2981 tail |err|>2: 476 (2.758%)
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  |err| q68=0.79607 q95=1.7692 q99=3.7109 tail |err|>3: 278 (1.611%)
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A: tracker entrance / momentum_reflected

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  |err| q68=6.5813 q95=26.117 q99=104.83 tail |err|>40: 390 (2.871%)
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  |err| q68=10.419 q95=37.929 q99=191.22 tail |err|>40: 626 (4.609%)
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  mean=0.32079 med=0.34493 std=0.88535 s68=0.67214
  |err| q68=0.78956 q95=1.7221 q99=3.2891 tail |err|>3: 163 (1.200%)
```

C: target production plane / baseline_direct

```
tar_dx         N=  14769 NaN=    0
  mean=-1.0719 med=0.30359 std=145.95 s68=107.84
  |err| q68=106.81 q95=345.28 q99=423.27 tail |err|>400: 300 (2.031%)
tar_dy         N=  14769 NaN=    0
  mean=-2.1972 med=0.077335 std=141.68 s68=103.84
  |err| q68=102.83 q95=338.88 q99=418.08 tail |err|>400: 254 (1.720%)
tar_dr         N=  14769 NaN=    0
  mean=151.42 med=101.8 std=135.84 s68=150.65
  |err| q68=189.21 q95=425.98 q99=465.88 tail |err|>400: 1250 (8.464%)
tar_phase      N=  14769 NaN=    0
  mean=-2.6884 med=-2.6977 std=64.491 s68=51.316
  |err| q68=51.176 q95=150.04 q99=174.01 tail |err|>90: 2546 (17.239%)
tar_center     N=  14769 NaN=    0
  mean=8.4273 med=5.9291 std=11.22 s68=4.6306
  |err| q68=8.2352 q95=21.309 q99=55.287 tail |err|>40: 249 (1.686%)
tar_radius     N=  14769 NaN=    0
  mean=-2.8798 med=-1.2705 std=10.793 s68=4.1126
  |err| q68=3.9999 q95=16.095 q99=50.666 tail |err|>20: 553 (3.744%)
tar_dp         N=  14769 NaN=    0
  mean=-0.57348 med=-0.15931 std=2.3126 s68=0.51723
  |err| q68=0.4459 q95=2.8407 q99=10.881 tail |err|>2: 1046 (7.082%)
tar_dpz        N=  14769 NaN=    0
  mean=0.43868 med=0.14441 std=2.979 s68=1.6122
  |err| q68=1.5564 q95=6.6806 q99=10.949 tail |err|>2: 3622 (24.524%)
tar_pitch      N=  14769 NaN=    0
  mean=-12.322 med=-5.1693 std=60.363 s68=35.444
  |err| q68=33.686 q95=133.86 q99=222.12 tail |err|>20: 7441 (50.383%)
tar_dt         N=  14769 NaN=    0
  mean=0.65043 med=0.47115 std=1.7013 s68=1.0678
```

C: target production plane / baseline_direct

```
  |err| q68=1.1338 q95=3.7978 q99=6.8748 tail |err|>10: 35 (0.237%)
tar_dt_beam    N=  14769 NaN=    0
  mean=0.65281 med=0.48503 std=1.7756 s68=1.2315
  |err| q68=1.308 q95=3.8534 q99=6.8846 tail |err|>10: 37 (0.251%)
```

C: target production plane / baseline_reflected

```
tar_dx         N=  11609 NaN=    0
  mean=-2.6572 med=-1.3826 std=217.65 s68=243.82
  |err| q68=242.06 q95=401.56 q99=448.01 tail |err|>400: 606 (5.220%)
tar_dy         N=  11609 NaN=    0
  mean=5.7487 med=6.0683 std=216.95 s68=241.43
  |err| q68=239.52 q95=402.55 q99=445.34 tail |err|>400: 614 (5.289%)
tar_dr         N=  11609 NaN=    0
  mean=276.09 med=302.41 std=135.1 s68=154.66
  |err| q68=374.82 q95=451.4 q99=478.6 tail |err|>400: 2679 (23.077%)
tar_phase      N=  11609 NaN=    0
  mean=6.9226 med=13.289 std=102.6 s68=120.49
  |err| q68=120.33 q95=170.46 q99=177.8 tail |err|>90: 5680 (48.928%)
tar_center     N=  11609 NaN=    0
  mean=11.395 med=8.6893 std=11.669 s68=6.625
  |err| q68=11.964 q95=28.535 q99=54.43 tail |err|>40: 269 (2.317%)
tar_radius     N=  11609 NaN=    0
  mean=-3.828 med=-1.962 std=11.263 s68=5.6114
  |err| q68=5.7086 q95=20.512 q99=50.889 tail |err|>20: 606 (5.220%)
tar_dp         N=  11609 NaN=    0
  mean=-0.77015 med=-0.24427 std=2.4604 s68=0.85315
  |err| q68=0.77364 q95=4.2401 q99=11.1 tail |err|>2: 1413 (12.172%)
tar_dpz        N=  11609 NaN=    0
  mean=-0.57406 med=-0.29548 std=3.3514 s68=2.8702
  |err| q68=2.9018 q95=7.1163 q99=10.591 tail |err|>2: 5285 (45.525%)
tar_pitch      N=  11609 NaN=    0
  mean=15.384 med=8.226 std=65.915 s68=57.399
  |err| q68=59.067 q95=140.39 q99=210.85 tail |err|>20: 7565 (65.165%)
tar_dt         N=  11609 NaN=    0
  mean=-0.34246 med=-0.12579 std=3.4761 s68=2.8597
```

C: target production plane / baseline_reflected

```
  |err| q68=2.8518 q95=7.2394 q99=11.263 tail |err|>10: 191 (1.645%)
tar_dt_beam    N=  11609 NaN=    0
  mean=-0.34254 med=-0.11499 std=3.5061 s68=2.9217
  |err| q68=2.9136 q95=7.314 q99=11.302 tail |err|>10: 189 (1.628%)
```

C: target production plane / control_direct

```
tar_dx         N=  14093 NaN=    0
  mean=-0.90531 med=0.17447 std=148.57 s68=112.54
  |err| q68=111.4 q95=348.63 q99=423.69 tail |err|>400: 299 (2.122%)
tar_dy         N=  14093 NaN=    0
  mean=-2.3372 med=-0.039221 std=144.28 s68=108.24
  |err| q68=106.9 q95=342.8 q99=419.03 tail |err|>400: 254 (1.802%)
tar_dr         N=  14093 NaN=    0
  mean=155.55 med=106.83 std=136.75 s68=153.16
  |err| q68=197.08 q95=427.46 q99=466.92 tail |err|>400: 1247 (8.848%)
tar_phase      N=  14093 NaN=    0
  mean=-2.6917 med=-2.8823 std=65.597 s68=53.656
  |err| q68=53.282 q95=150.78 q99=174.21 tail |err|>90: 2521 (17.888%)
tar_center     N=  14093 NaN=    0
  mean=8.3505 med=5.9075 std=10.919 s68=4.6541
  |err| q68=8.2053 q95=20.994 q99=53.155 tail |err|>40: 227 (1.611%)
tar_radius     N=  14093 NaN=    0
  mean=-2.8677 med=-1.2524 std=10.679 s68=3.9843
  |err| q68=3.8849 q95=15.735 q99=50.598 tail |err|>20: 507 (3.598%)
tar_dp         N=  14093 NaN=    0
  mean=-0.57704 med=-0.16008 std=2.3094 s68=0.52039
  |err| q68=0.44544 q95=2.8463 q99=10.829 tail |err|>2: 997 (7.074%)
tar_dpz        N=  14093 NaN=    0
  mean=0.45589 med=0.15148 std=2.988 s68=1.6428
  |err| q68=1.5738 q95=6.7539 q99=10.982 tail |err|>2: 3514 (24.934%)
tar_pitch      N=  14093 NaN=    0
  mean=-12.401 med=-5.1124 std=60.181 s68=35.319
  |err| q68=33.573 q95=135.1 q99=222.03 tail |err|>20: 7033 (49.904%)
tar_dt         N=  14093 NaN=    0
  mean=0.6711 med=0.47755 std=1.7033 s68=1.0902
```

C: target production plane / control_direct

```
  |err| q68=1.1453 q95=3.8166 q99=6.8956 tail |err|>10: 34 (0.241%)
tar_dt_beam    N=  14093 NaN=    0
  mean=0.67297 med=0.49489 std=1.7791 s68=1.2457
  |err| q68=1.3193 q95=3.8802 q99=6.9079 tail |err|>10: 37 (0.263%)
```

C: target production plane / control_reflected

```
tar_dx         N=  11789 NaN=    0
  mean=-2.7322 med=-0.73847 std=216.05 s68=240.56
  |err| q68=238.81 q95=401.2 q99=447.55 tail |err|>400: 606 (5.140%)
tar_dy         N=  11789 NaN=    0
  mean=5.4969 med=4.8037 std=215.23 s68=238.61
  |err| q68=236.53 q95=401.43 q99=445.4 tail |err|>400: 612 (5.191%)
tar_dr         N=  11789 NaN=    0
  mean=272.42 med=298.2 std=137.21 s68=158.47
  |err| q68=373.44 q95=450.43 q99=478.44 tail |err|>400: 2691 (22.826%)
tar_phase      N=  11789 NaN=    0
  mean=7.0797 med=12.07 std=101.79 s68=119.39
  |err| q68=119.38 q95=170.15 q99=177.86 tail |err|>90: 5659 (48.002%)
tar_center     N=  11789 NaN=    0
  mean=11.382 med=8.6891 std=11.609 s68=6.606
  |err| q68=11.971 q95=28.497 q99=54.47 tail |err|>40: 273 (2.316%)
tar_radius     N=  11789 NaN=    0
  mean=-3.8003 med=-1.9449 std=11.217 s68=5.6097
  |err| q68=5.7014 q95=20.459 q99=50.82 tail |err|>20: 613 (5.200%)
tar_dp         N=  11789 NaN=    0
  mean=-0.76397 med=-0.24251 std=2.4549 s68=0.85126
  |err| q68=0.77414 q95=4.2497 q99=11.08 tail |err|>2: 1435 (12.172%)
tar_dpz        N=  11789 NaN=    0
  mean=-0.5745 med=-0.28851 std=3.3479 s68=2.8601
  |err| q68=2.8927 q95=7.1015 q99=10.6 tail |err|>2: 5342 (45.313%)
tar_pitch      N=  11789 NaN=    0
  mean=15.351 med=8.0527 std=65.729 s68=57.25
  |err| q68=58.938 q95=140.09 q99=210.85 tail |err|>20: 7652 (64.908%)
tar_dt         N=  11789 NaN=    0
  mean=-0.34217 med=-0.10605 std=3.468 s68=2.8528
```

C: target production plane / control_reflected

```
  |err| q68=2.8497 q95=7.2433 q99=11.211 tail |err|>10: 191 (1.620%)
tar_dt_beam    N=  11789 NaN=    0
  mean=-0.34204 med=-0.10917 std=3.4982 s68=2.9213
  |err| q68=2.9091 q95=7.2921 q99=11.271 tail |err|>10: 190 (1.612%)
```

C: target production plane / momentum_direct

```
tar_dx         N=  14392 NaN=    0
  mean=1.2609 med=1.1007 std=131.37 s68=82.592
  |err| q68=81.544 q95=324.12 q99=420 tail |err|>400: 232 (1.612%)
tar_dy         N=  14392 NaN=    0
  mean=-1.2632 med=0.30938 std=131.11 s68=84.295
  |err| q68=82.997 q95=324.21 q99=418.2 tail |err|>400: 235 (1.633%)
tar_dr         N=  14392 NaN=    0
  mean=130 med=74.917 std=132.47 s68=134.47
  |err| q68=149.46 q95=417.23 q99=470.5 tail |err|>400: 981 (6.816%)
tar_phase      N=  14392 NaN=    0
  mean=0.93139 med=0.34116 std=57.141 s68=39.368
  |err| q68=38.871 q95=140.22 q99=172.07 tail |err|>90: 1889 (13.125%)
tar_center     N=  14392 NaN=    0
  mean=13.846 med=8.2717 std=28.429 s68=7.136
  |err| q68=11.712 q95=34.393 q99=128.83 tail |err|>40: 580 (4.030%)
tar_radius     N=  14392 NaN=    0
  mean=-1.6469 med=-0.7284 std=5.421 s68=2.8394
  |err| q68=2.7101 q95=9.5984 q99=24.082 tail |err|>20: 209 (1.452%)
tar_dp         N=  14392 NaN=    0
  mean=-0.32539 med=-0.059316 std=1.0373 s68=0.33771
  |err| q68=0.24594 q95=1.5366 q99=5.0332 tail |err|>2: 535 (3.717%)
tar_dpz        N=  14392 NaN=    0
  mean=0.40642 med=0.11787 std=2.3373 s68=1.3327
  |err| q68=1.2782 q95=5.43 q99=9.2902 tail |err|>2: 2791 (19.393%)
tar_pitch      N=  14392 NaN=    0
  mean=-9.29 med=-3.4963 std=46.69 s68=27.735
  |err| q68=26.511 q95=108.39 q99=185.63 tail |err|>20: 6083 (42.267%)
tar_dt         N=  14392 NaN=    0
  mean=0.60096 med=0.4561 std=1.4407 s68=0.81999
```

C: target production plane / momentum_direct

```
  |err| q68=0.94285 q95=3.286 q99=5.9747 tail |err|>10: 15 (0.104%)
tar_dt_beam    N=  14392 NaN=    0
  mean=0.60139 med=0.47475 std=1.5279 s68=1.0325
  |err| q68=1.1368 q95=3.3639 q99=6.0753 tail |err|>10: 16 (0.111%)
```

C: target production plane / momentum_reflected

```
tar_dx         N=  12145 NaN=    0
  mean=-0.59744 med=-0.32594 std=217.43 s68=239.98
  |err| q68=238.48 q95=405.51 q99=450.7 tail |err|>400: 686 (5.648%)
tar_dy         N=  12145 NaN=    0
  mean=5.8839 med=4.0471 std=215.47 s68=235.61
  |err| q68=234.13 q95=404.26 q99=446.87 tail |err|>400: 673 (5.541%)
tar_dr         N=  12145 NaN=    0
  mean=272.27 med=296.58 std=140.03 s68=162.07
  |err| q68=374.89 q95=454.03 q99=486.41 tail |err|>400: 2900 (23.878%)
tar_phase      N=  12145 NaN=    0
  mean=5.8463 med=9.9438 std=101.18 s68=118.68
  |err| q68=118.5 q95=170.9 q99=178.32 tail |err|>90: 5765 (47.468%)
tar_center     N=  12145 NaN=    0
  mean=16.631 med=10.998 std=27.839 s68=8.8646
  |err| q68=15.37 q95=40.929 q99=131.44 tail |err|>40: 627 (5.163%)
tar_radius     N=  12145 NaN=    0
  mean=-2.6995 med=-1.2554 std=8.2566 s68=4.5627
  |err| q68=4.6521 q95=16.168 q99=34.898 tail |err|>20: 420 (3.458%)
tar_dp         N=  12145 NaN=    0
  mean=-0.52068 med=-0.14228 std=1.7581 s68=0.66077
  |err| q68=0.58213 q95=3.2482 q99=7.9504 tail |err|>2: 1113 (9.164%)
tar_dpz        N=  12145 NaN=    0
  mean=-0.59527 med=-0.27007 std=2.9728 s68=2.467
  |err| q68=2.5068 q95=6.3036 q99=9.6986 tail |err|>2: 4791 (39.448%)
tar_pitch      N=  12145 NaN=    0
  mean=14.339 med=6.701 std=58.289 s68=48.463
  |err| q68=51.118 q95=124.53 q99=192.63 tail |err|>20: 7239 (59.605%)
tar_dt         N=  12145 NaN=    0
  mean=-0.32281 med=-0.050817 std=3.2022 s68=2.5907
```

C: target production plane / momentum_reflected

```
  |err| q68=2.5492 q95=6.5883 q99=10.358 tail |err|>10: 141 (1.161%)
tar_dt_beam    N=  12145 NaN=    0
  mean=-0.32317 med=-0.087005 std=3.2386 s68=2.6535
  |err| q68=2.6147 q95=6.6894 q99=10.423 tail |err|>10: 144 (1.186%)
```

C: target production plane / position_direct

```
tar_dx         N=  13975 NaN=    0
  mean=-2.6182 med=-1.0248 std=149.79 s68=115.15
  |err| q68=113.68 q95=348.95 q99=427.99 tail |err|>400: 313 (2.240%)
tar_dy         N=  13975 NaN=    0
  mean=-2.3483 med=-0.17718 std=145.35 s68=108.92
  |err| q68=107.42 q95=342.01 q99=421.22 tail |err|>400: 259 (1.853%)
tar_dr         N=  13975 NaN=    0
  mean=157.1 med=107.97 std=137.46 s68=153.48
  |err| q68=198.82 q95=427.43 q99=471.07 tail |err|>400: 1243 (8.894%)
tar_phase      N=  13975 NaN=    0
  mean=-3.6256 med=-3.382 std=66.084 s68=54.361
  |err| q68=53.723 q95=151 q99=173.92 tail |err|>90: 2553 (18.268%)
tar_center     N=  13975 NaN=    0
  mean=12.374 med=7.7945 std=23.359 s68=6.6033
  |err| q68=11.061 q95=30.495 q99=97.663 tail |err|>40: 447 (3.199%)
tar_radius     N=  13975 NaN=    0
  mean=-2.7651 med=-1.2048 std=9.9591 s68=4.0162
  |err| q68=3.9028 q95=15.433 q99=46.498 tail |err|>20: 486 (3.478%)
tar_dp         N=  13975 NaN=    0
  mean=-0.56379 med=-0.15549 std=2.1049 s68=0.51412
  |err| q68=0.44387 q95=2.7581 q99=10.018 tail |err|>2: 966 (6.912%)
tar_dpz        N=  13975 NaN=    0
  mean=0.42181 med=0.11665 std=3.011 s68=1.6581
  |err| q68=1.6031 q95=6.9129 q99=11.001 tail |err|>2: 3543 (25.352%)
tar_pitch      N=  13975 NaN=    0
  mean=-11.317 med=-4.4366 std=60.598 s68=35.548
  |err| q68=33.904 q95=137.52 q99=222.87 tail |err|>20: 7044 (50.404%)
tar_dt         N=  13975 NaN=    0
  mean=0.64586 med=0.46063 std=1.722 s68=1.1007
```

C: target production plane / position_direct

```
  |err| q68=1.1465 q95=3.8174 q99=6.9554 tail |err|>10: 34 (0.243%)
tar_dt_beam    N=  13975 NaN=    0
  mean=0.64938 med=0.47195 std=1.7959 s68=1.2625
  |err| q68=1.3238 q95=3.8991 q99=6.9655 tail |err|>10: 37 (0.265%)
```

C: target production plane / position_reflected

```
tar_dx         N=  11688 NaN=    0
  mean=-4.3313 med=-1.936 std=216.21 s68=238.13
  |err| q68=236.73 q95=401.94 q99=448.39 tail |err|>400: 614 (5.253%)
tar_dy         N=  11688 NaN=    0
  mean=4.4439 med=4.1612 std=214.32 s68=236.29
  |err| q68=234.3 q95=400.5 q99=448.18 tail |err|>400: 596 (5.099%)
tar_dr         N=  11688 NaN=    0
  mean=271.47 med=296.26 std=137.91 s68=160.06
  |err| q68=373.28 q95=450.32 q99=480.44 tail |err|>400: 2638 (22.570%)
tar_phase      N=  11688 NaN=    0
  mean=6.4453 med=10.406 std=101.8 s68=119.56
  |err| q68=119.71 q95=170.24 q99=178.31 tail |err|>90: 5650 (48.340%)
tar_center     N=  11688 NaN=    0
  mean=15.008 med=10.503 std=22.693 s68=8.2342
  |err| q68=14.461 q95=37.291 q99=91.677 tail |err|>40: 498 (4.261%)
tar_radius     N=  11688 NaN=    0
  mean=-3.7664 med=-1.8894 std=11.008 s68=5.6687
  |err| q68=5.8175 q95=20.524 q99=49.946 tail |err|>20: 612 (5.236%)
tar_dp         N=  11688 NaN=    0
  mean=-0.77096 med=-0.23885 std=2.3755 s68=0.84811
  |err| q68=0.78578 q95=4.2684 q99=10.957 tail |err|>2: 1425 (12.192%)
tar_dpz        N=  11688 NaN=    0
  mean=-0.50384 med=-0.25059 std=3.4016 s68=2.9422
  |err| q68=2.9502 q95=7.1858 q99=10.794 tail |err|>2: 5376 (45.996%)
tar_pitch      N=  11688 NaN=    0
  mean=14.074 med=7.2745 std=67.139 s68=58.653
  |err| q68=60.167 q95=141.98 q99=213.33 tail |err|>20: 7624 (65.229%)
tar_dt         N=  11688 NaN=    0
  mean=-0.27605 med=-0.0616 std=3.5413 s68=2.9123
```

C: target production plane / position_reflected

```
  |err| q68=2.9116 q95=7.366 q99=11.46 tail |err|>10: 201 (1.720%)
tar_dt_beam    N=  11688 NaN=    0
  mean=-0.27655 med=-0.058861 std=3.5697 s68=2.9821
  |err| q68=2.9506 q95=7.4568 q99=11.417 tail |err|>10: 199 (1.703%)
```

C: target production plane / truth_direct

```
tar_dx         N=  14337 NaN=    0
  mean=-0.39233 med=-0.10987 std=126.94 s68=77.24
  |err| q68=76.436 q95=315.54 q99=412.83 tail |err|>400: 201 (1.402%)
tar_dy         N=  14337 NaN=    0
  mean=-1.0244 med=-0.24496 std=126.53 s68=79.71
  |err| q68=78.523 q95=313.4 q99=415.52 tail |err|>400: 207 (1.444%)
tar_dr         N=  14337 NaN=    0
  mean=124.31 med=70.418 std=129.12 s68=129.17
  |err| q68=142.01 q95=414.31 q99=458.57 tail |err|>400: 904 (6.305%)
tar_phase      N=  14337 NaN=    0
  mean=-0.61455 med=-0.34413 std=56.248 s68=37.98
  |err| q68=37.478 q95=139.35 q99=171.96 tail |err|>90: 1843 (12.855%)
tar_center     N=  14337 NaN=    0
  mean=6.1819 med=4.6365 std=6.157 s68=3.7138
  |err| q68=6.4748 q95=15.798 q99=31.76 tail |err|>40: 85 (0.593%)
tar_radius     N=  14337 NaN=    0
  mean=-1.532 med=-0.63214 std=5.3319 s68=2.7776
  |err| q68=2.6107 q95=9.121 q99=23.24 tail |err|>20: 199 (1.388%)
tar_dp         N=  14337 NaN=    0
  mean=-0.31134 med=-0.049648 std=1.0446 s68=0.32774
  |err| q68=0.23627 q95=1.5135 q99=5.061 tail |err|>2: 524 (3.655%)
tar_dpz        N=  14337 NaN=    0
  mean=0.32639 med=0.088223 std=2.1998 s68=1.2891
  |err| q68=1.2429 q95=5.1031 q99=8.7082 tail |err|>2: 2665 (18.588%)
tar_pitch      N=  14337 NaN=    0
  mean=-7.6918 med=-2.8386 std=43.886 s68=26.549
  |err| q68=25.643 q95=101.34 q99=172.03 tail |err|>20: 5907 (41.201%)
tar_dt         N=  14337 NaN=    0
  mean=0.21665 med=0.028671 std=1.1827 s68=0.57631
```

C: target production plane / truth_direct

```
  |err| q68=0.5436 q95=2.4881 q99=4.7377 tail |err|>10: 10 (0.070%)
tar_dt_beam    N=  14337 NaN=    0
  mean=0.21714 med=0.070501 std=1.2867 s68=0.84543
  |err| q68=0.82837 q95=2.6108 q99=4.7668 tail |err|>10: 10 (0.070%)
```

C: target production plane / truth_reflected

```
tar_dx         N=  12091 NaN=    0
  mean=-1.0901 med=-0.7453 std=215.81 s68=237.12
  |err| q68=235.56 q95=403.54 q99=446.95 tail |err|>400: 663 (5.483%)
tar_dy         N=  12091 NaN=    0
  mean=5.5779 med=3.8626 std=213.07 s68=234.48
  |err| q68=233.26 q95=401.65 q99=443.94 tail |err|>400: 629 (5.202%)
tar_dr         N=  12091 NaN=    0
  mean=269.71 med=293.87 std=138.79 s68=161.38
  |err| q68=373.43 q95=449.95 q99=477.62 tail |err|>400: 2833 (23.431%)
tar_phase      N=  12091 NaN=    0
  mean=5.2667 med=10.241 std=101.2 s68=119.26
  |err| q68=118.89 q95=170.31 q99=178.12 tail |err|>90: 5685 (47.018%)
tar_center     N=  12091 NaN=    0
  mean=9.7717 med=7.5955 std=9.2306 s68=5.8182
  |err| q68=10.487 q95=24.602 q99=43.463 tail |err|>40: 154 (1.274%)
tar_radius     N=  12091 NaN=    0
  mean=-2.5606 med=-1.1284 std=8.0111 s68=4.4476
  |err| q68=4.4957 q95=15.524 q99=34.948 tail |err|>20: 399 (3.300%)
tar_dp         N=  12091 NaN=    0
  mean=-0.51371 med=-0.12604 std=1.721 s68=0.6501
  |err| q68=0.57265 q95=3.1736 q99=7.9285 tail |err|>2: 1079 (8.924%)
tar_dpz        N=  12091 NaN=    0
  mean=-0.46749 med=-0.21464 std=2.8297 s68=2.4043
  |err| q68=2.4342 q95=5.9788 q99=9.0268 tail |err|>2: 4614 (38.161%)
tar_pitch      N=  12091 NaN=    0
  mean=11.757 med=5.605 std=55.41 s68=47.445
  |err| q68=49.229 q95=118.12 q99=178.69 tail |err|>20: 7087 (58.614%)
tar_dt         N=  12091 NaN=    0
  mean=-0.50411 med=-0.24206 std=2.9629 s68=2.4147
```

C: target production plane / truth_reflected

```
  |err| q68=2.4158 q95=6.2764 q99=9.5935 tail |err|>10: 101 (0.835%)
tar_dt_beam    N=  12091 NaN=    0
  mean=-0.50436 med=-0.2697 std=3.0031 s68=2.4795
  |err| q68=2.4674 q95=6.3378 q99=9.7245 tail |err|>10: 105 (0.868%)
```

C: phase statistics, wrapping-aware

```
baseline_direct        N=14769
  circular mean=-3.3655 deg, R=0.57856, circular std=59.94 deg
baseline_reflected     N=11609
  circular mean=73.142 deg, R=0.057893, circular std=136.77 deg
control_direct         N=14093
  circular mean=-3.5121 deg, R=0.56465, circular std=61.258 deg
control_reflected      N=11789
  circular mean=59.822 deg, R=0.063434, circular std=134.56 deg
momentum_direct        N=14392
  circular mean=0.79999 deg, R=0.6655, circular std=51.707 deg
momentum_reflected     N=12145
  circular mean=47.999 deg, R=0.065877, circular std=133.63 deg
position_direct        N=13975
  circular mean=-4.2811 deg, R=0.55793, circular std=61.896 deg
position_reflected     N=11688
  circular mean=57.355 deg, R=0.059598, circular std=136.07 deg
truth_direct           N=14337
  circular mean=-0.66816 deg, R=0.67506, circular std=50.793 deg
truth_reflected        N=12091
  circular mean=50.944 deg, R=0.069839, circular std=132.19 deg
```

C: common-cohort configuration comparison

```
Same tracks valid in every enabled configuration; N=23486
This sensitivity cohort supplements, and never replaces, failure-inclusive efficiencies.
baseline_direct
  tar_dr N=12406 med=115.25 s68=151.53 |err|q95=426.09
  tar_phase N=12406 med=-2.8344 s68=55.833 |err|q95=150
  tar_dt N=12406 med=0.4853 s68=1.0983 |err|q95=3.7258
baseline_reflected
  tar_dr N=11080 med=301.38 s68=154.31 |err|q95=449.38
  tar_phase N=11080 med=11.813 s68=119.82 |err|q95=170.41
  tar_dt N=11080 med=-0.15394 s68=2.8249 |err|q95=7.1859
control_direct
  tar_dr N=12406 med=115.07 s68=151.74 |err|q95=426.13
  tar_phase N=12406 med=-2.7568 s68=55.852 |err|q95=150.01
  tar_dt N=12406 med=0.48651 s68=1.0981 |err|q95=3.7327
control_reflected
  tar_dr N=11080 med=301.41 s68=154.35 |err|q95=449.22
  tar_phase N=11080 med=12.038 s68=119.77 |err|q95=170.18
  tar_dt N=11080 med=-0.15346 s68=2.8228 |err|q95=7.1759
momentum_direct
  tar_dr N=12406 med=77.781 s68=129.55 |err|q95=413.16
  tar_phase N=12406 med=0.74221 s68=39.906 |err|q95=138.83
  tar_dt N=12406 med=0.46253 s68=0.80817 |err|q95=3.157
momentum_reflected
  tar_dr N=11080 med=296.09 s68=161.64 |err|q95=451.22
  tar_phase N=11080 med=9.0485 s68=118.97 |err|q95=170.9
  tar_dt N=11080 med=-0.077285 s68=2.5234 |err|q95=6.372
position_direct
  tar_dr N=12406 med=119.01 s68=154.25 |err|q95=427.57
  tar_phase N=12406 med=-3.2987 s68=57.462 |err|q95=151.93
```

C: common-cohort configuration comparison

```
  tar_dt N=12406 med=0.46321 s68=1.1198 |err|q95=3.7569
position_reflected
  tar_dr N=11080 med=300.49 s68=156.84 |err|q95=449.83
  tar_phase N=11080 med=10.319 s68=120.08 |err|q95=170.44
  tar_dt N=11080 med=-0.079037 s68=2.9 |err|q95=7.3416
truth_direct
  tar_dr N=12406 med=74.004 s68=127.06 |err|q95=412.37
  tar_phase N=12406 med=-0.086864 s68=38.899 |err|q95=138.02
  tar_dt N=12406 med=0.038843 s68=0.59227 |err|q95=2.4518
truth_reflected
  tar_dr N=11080 med=294.2 s68=160.84 |err|q95=449.01
  tar_phase N=11080 med=9.6147 s68=119.32 |err|q95=170.36
  tar_dt N=11080 med=-0.29256 s68=2.3913 |err|q95=6.2672
```

B: event crossing efficiencies

```
baseline: read=20000, common denominator=20000
  any_plane               17911 /    20000  fraction=0.89555
  any_annulus              7240 /    20000  fraction=0.362
  both_plane               9870 /    20000  fraction=0.4935
  both_annulus              544 /    20000  fraction=0.0272
  both_compatible             4 /    20000  fraction=0.0002
control: read=20000, common denominator=20000
  any_plane               16298 /    20000  fraction=0.8149
  any_annulus              6241 /    20000  fraction=0.31205
  both_plane               9679 /    20000  fraction=0.48395
  both_annulus              532 /    20000  fraction=0.0266
  both_compatible             4 /    20000  fraction=0.0002
momentum: read=20000, common denominator=20000
  any_plane               16456 /    20000  fraction=0.8228
  any_annulus              7406 /    20000  fraction=0.3703
  both_plane               10126 /    20000  fraction=0.5063
  both_annulus              664 /    20000  fraction=0.0332
  both_compatible             5 /    20000  fraction=0.00025
position: read=20000, common denominator=20000
  any_plane               16221 /    20000  fraction=0.81105
  any_annulus              6166 /    20000  fraction=0.3083
  both_plane               9526 /    20000  fraction=0.4763
  both_annulus              554 /    20000  fraction=0.0277
  both_compatible             4 /    20000  fraction=0.0002
truth: read=20000, common denominator=20000
  any_plane               16407 /    20000  fraction=0.82035
  any_annulus              7607 /    20000  fraction=0.38035
  both_plane               10042 /    20000  fraction=0.5021
  both_annulus              746 /    20000  fraction=0.0373
```

B: event crossing efficiencies

```
both_compatible             3 /    20000  fraction=0.00015
```

E: event availability and failure flags, including zero-track events

```
baseline
  missing_truth_events            0
  missing_input_events            0
  not_two_primary_events          0
control
  missing_truth_events            0
  missing_input_events            0
  not_two_primary_events          0
momentum
  missing_truth_events            0
  missing_input_events            0
  not_two_primary_events          0
position
  missing_truth_events            0
  missing_input_events            0
  not_two_primary_events          0
truth
  missing_truth_events            0
  missing_input_events            0
  not_two_primary_events          0
```

E: status, missing states and covariance failures

```
baseline_direct
  AmbiguousAssociation     0
  AmbiguousProductionBranch 1
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        3394
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  2614
  MissingRecoEntrance      1
  MissingTruth             0
  MultipleMCEntrances      124
  MultipleRecoEntrances    1
  NotElectron              0
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       0
  UnsupportedTrajectory    0
  all                      17384
  branch_missing           2488
  covariance_invalid       3394
  ent_valid                17258
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
```

E: status, missing states and covariance failures

```
  propagated_covariance_invalid 3374
  propagation_complete     17258
  selected                 17258
  status_0                 11831
  status_16384             2065
  status_16640             93
  status_20480             423
  status_20736             31
  status_20992             1
  status_21504             1
  status_32768             1
  status_4096              2938
  tar_valid                14769
baseline_reflected
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        3102
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  2088
  MissingRecoEntrance      0
  MissingTruth             0
  MultipleMCEntrances      113
```

E: status, missing states and covariance failures

```
  MultipleRecoEntrances    1
  NotElectron              0
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       0
  UnsupportedTrajectory    0
  all                      13697
  branch_missing           1974
  covariance_invalid       3102
  ent_valid                13583
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
  propagated_covariance_invalid 3086
  propagation_complete     13583
  selected                 13583
  status_0                 8840
  status_16384             1665
  status_16640             90
  status_20480             309
  status_20736             23
  status_21504             1
  status_4096              2769
  tar_valid                11609
baseline_unclassified
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
```

E: status, missing states and covariance failures

```
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        0
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  0
  MissingRecoEntrance      0
  MissingTruth             0
  MultipleMCEntrances      0
  MultipleRecoEntrances    0
  NotElectron              20
  NotPrimary               4
  OutsideTarget            0
  PropagationFailure       0
  UnsupportedTrajectory    0
  all                      24
  branch_missing           0
  covariance_invalid       0
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
  propagated_covariance_invalid 0
  propagation_complete     0
  status_16                4
  status_2                 20
control_direct
  AmbiguousAssociation     0
```

E: status, missing states and covariance failures

```
AmbiguousProductionBranch 1
BoundaryMismatch          0
ConfigurationDisabled     0
CrossingSearchFailure     0
Exception                 0
InvalidCovariance         5377
MissingAssociation        0
MissingClock              0
MissingInput              0
MissingMCEntrance         0
MissingProductionBranch   1472
MissingRecoEntrance       1
MissingTruth              0
MultipleMCEntrances       124
MultipleRecoEntrances     1
NotElectron               0
NotPrimary                0
OutsideTarget             0
PropagationFailure        1692
UnsupportedTrajectory     0
all                       17384
branch_missing            1472
covariance_invalid        5377
ent_valid                 17258
input_covariance_invalid  0
partial_target_available  1692
product_missing           0
propagated_covariance_invalid 5315
propagation_complete      15566
```

E: status, missing states and covariance failures

```
  selected                17258
  status_0                9154
  status_1024             1
  status_16384            1035
  status_20480            437
  status_256              124
  status_36864            1
  status_4096             4939
  status_512              1
  status_8192             1692
  tar_valid               14093
control_reflected
  AmbiguousAssociation    0
  AmbiguousProductionBranch 0
  BoundaryMismatch        0
  ConfigurationDisabled   0
  CrossingSearchFailure   0
  Exception               0
  InvalidCovariance       5057
  MissingAssociation      0
  MissingClock            0
  MissingInput            0
  MissingMCEntrance       0
  MissingProductionBranch 1794
  MissingRecoEntrance     0
  MissingTruth            0
  MultipleMCEntrances     113
  MultipleRecoEntrances   1
  NotElectron             0
```

E: status, missing states and covariance failures

```
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       119
  UnsupportedTrajectory    0
  all                      13697
  branch_missing           1794
  covariance_invalid       5057
  ent_valid                13583
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
  propagated_covariance_invalid 5002
  propagation_complete     13464
  selected                 13583
  status_0                 7160
  status_1024              1
  status_16384             1247
  status_20480             428
  status_24576             119
  status_256               113
  status_4096              4629
  tar_valid                11789
control_unclassified
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
```

E: status, missing states and covariance failures

```
  InvalidCovariance        0
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  0
  MissingRecoEntrance      0
  MissingTruth             0
  MultipleMCEntrances      0
  MultipleRecoEntrances    0
  NotElectron              20
  NotPrimary               4
  OutsideTarget            0
  PropagationFailure       0
  UnsupportedTrajectory    0
  all                      24
  branch_missing           0
  covariance_invalid       0
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
  propagated_covariance_invalid 0
  propagation_complete     0
  status_16                4
  status_2                 20
momentum_direct
  AmbiguousAssociation     0
  AmbiguousProductionBranch 1
  BoundaryMismatch         0
```

E: status, missing states and covariance failures

```
ConfigurationDisabled    0
CrossingSearchFailure    0
Exception                0
InvalidCovariance        5432
MissingAssociation       0
MissingClock             0
MissingInput             0
MissingMCEntrance        0
MissingProductionBranch  1183
MissingRecoEntrance      1
MissingTruth             0
MultipleMCEntrances      124
MultipleRecoEntrances    1
NotElectron              0
NotPrimary               0
OutsideTarget            0
PropagationFailure       1694
UnsupportedTrajectory    0
all                      17384
branch_missing           1183
covariance_invalid       5432
ent_valid                17258
input_covariance_invalid 0
partial_target_available 1682
product_missing          0
propagated_covariance_invalid 5374
propagation_complete     15564
selected                 17258
status_0                 9332
```

E: status, missing states and covariance failures

```
  status_1024              1
  status_12288             2
  status_16384             802
  status_20480             369
  status_24576             12
  status_256               124
  status_36864             1
  status_4096              5060
  status_512               1
  status_8192              1680
  tar_valid                14392
momentum_reflected
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        5126
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  1438
  MissingRecoEntrance      0
  MissingTruth             0
  MultipleMCEntrances      113
  MultipleRecoEntrances    1
  NotElectron              0
```

E: status, missing states and covariance failures

```
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       110
  UnsupportedTrajectory    0
  all                      13697
  branch_missing           1438
  covariance_invalid       5126
  ent_valid                13583
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
  propagated_covariance_invalid 5054
  propagation_complete     13473
  selected                 13583
  status_0                 7391
  status_1024              1
  status_16384             958
  status_20480             370
  status_24576             108
  status_256               113
  status_28672             2
  status_4096              4754
  tar_valid                12145
momentum_unclassified
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
```

E: status, missing states and covariance failures

```
  Exception                0
  InvalidCovariance        0
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  0
  MissingRecoEntrance      0
  MissingTruth             0
  MultipleMCEntrances      0
  MultipleRecoEntrances    0
  NotElectron              20
  NotPrimary               4
  OutsideTarget            0
  PropagationFailure       0
  UnsupportedTrajectory    0
  all                      24
  branch_missing           0
  covariance_invalid       0
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
  propagated_covariance_invalid 0
  propagation_complete     0
  status_16                4
  status_2                 20
position_direct
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
```

E: status, missing states and covariance failures

```
BoundaryMismatch         0
ConfigurationDisabled    0
CrossingSearchFailure    0
Exception                0
InvalidCovariance        5374
MissingAssociation       0
MissingClock             0
MissingInput             0
MissingMCEntrance        0
MissingProductionBranch  1573
MissingRecoEntrance      1
MissingTruth             0
MultipleMCEntrances      124
MultipleRecoEntrances    1
NotElectron              0
NotPrimary               0
OutsideTarget            0
PropagationFailure       1720
UnsupportedTrajectory    0
all                      17384
branch_missing           1573
covariance_invalid       5374
ent_valid                17258
input_covariance_invalid 0
partial_target_available 1710
product_missing          0
propagated_covariance_invalid 5310
propagation_complete     15538
selected                 17258
```

E: status, missing states and covariance failures

```
  status_0                9086
  status_1024             1
  status_12288            1
  status_16384            1079
  status_20480            484
  status_24576            10
  status_256              124
  status_4096             4889
  status_512              1
  status_8192             1709
  tar_valid               13975
position_reflected
  AmbiguousAssociation    0
  AmbiguousProductionBranch 0
  BoundaryMismatch        0
  ConfigurationDisabled   0
  CrossingSearchFailure   0
  Exception               0
  InvalidCovariance       5165
  MissingAssociation      0
  MissingClock            0
  MissingInput            0
  MissingMCEntrance       0
  MissingProductionBranch 1895
  MissingRecoEntrance     0
  MissingTruth            0
  MultipleMCEntrances     113
  MultipleRecoEntrances   1
  NotElectron             0
```

E: status, missing states and covariance failures

```
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       135
  UnsupportedTrajectory    0
  all                      13697
  branch_missing           1895
  covariance_invalid       5165
  ent_valid                13583
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
  propagated_covariance_invalid 5098
  propagation_complete     13448
  selected                 13583
  status_0                 7021
  status_1024              1
  status_16384             1262
  status_20480             498
  status_24576             135
  status_256               113
  status_4096              4667
  tar_valid                11688
position_unclassified
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
```

E: status, missing states and covariance failures

```
  InvalidCovariance        0
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  0
  MissingRecoEntrance      0
  MissingTruth             0
  MultipleMCEntrances      0
  MultipleRecoEntrances    0
  NotElectron              20
  NotPrimary               4
  OutsideTarget            0
  PropagationFailure       0
  UnsupportedTrajectory    0
  all                      24
  branch_missing           0
  covariance_invalid       0
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
  propagated_covariance_invalid 0
  propagation_complete     0
  status_16                4
  status_2                 20
truth_direct
  AmbiguousAssociation     0
  AmbiguousProductionBranch 1
  BoundaryMismatch         0
```

E: status, missing states and covariance failures

```
ConfigurationDisabled    0
CrossingSearchFailure    0
Exception                0
InvalidCovariance        5423
MissingAssociation       0
MissingClock             0
MissingInput             0
MissingMCEntrance        0
MissingProductionBranch  1233
MissingRecoEntrance      1
MissingTruth             0
MultipleMCEntrances      124
MultipleRecoEntrances    1
NotElectron              0
NotPrimary               0
OutsideTarget            0
PropagationFailure       1688
UnsupportedTrajectory    0
all                      17384
branch_missing           1233
covariance_invalid       5423
ent_valid                17258
input_covariance_invalid 0
partial_target_available 1687
product_missing          0
propagated_covariance_invalid 5363
propagation_complete     15570
selected                 17258
status_0                 9292
```

E: status, missing states and covariance failures

```
  status_1024              1
  status_16384             854
  status_20480             378
  status_24576             1
  status_256               124
  status_32768             1
  status_4096              5045
  status_512               1
  status_8192              1687
  tar_valid                14337
truth_reflected
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        5137
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  1492
  MissingRecoEntrance      0
  MissingTruth             0
  MultipleMCEntrances      113
  MultipleRecoEntrances    1
  NotElectron              0
  NotPrimary               0
```

E: status, missing states and covariance failures

```
  OutsideTarget            0
  PropagationFailure       86
  UnsupportedTrajectory    0
  all                      13697
  branch_missing           1492
  covariance_invalid       5137
  ent_valid                13583
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
  propagated_covariance_invalid 5069
  propagation_complete     13497
  selected                 13583
  status_0                 7367
  status_1024              1
  status_16384             993
  status_20480             413
  status_24576             86
  status_256               113
  status_4096              4724
  tar_valid                12091
truth_unclassified
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        0
```

E: status, missing states and covariance failures

```
MissingAssociation       0
MissingClock             0
MissingInput             0
MissingMCEntrance        0
MissingProductionBranch  0
MissingRecoEntrance      0
MissingTruth             0
MultipleMCEntrances      0
MultipleRecoEntrances    0
NotElectron              20
NotPrimary               4
OutsideTarget            0
PropagationFailure       0
UnsupportedTrajectory    0
all                      24
branch_missing           0
covariance_invalid       0
input_covariance_invalid 0
partial_target_available 0
product_missing          0
propagated_covariance_invalid 0
propagation_complete     0
status_16                4
status_2                 20
```

D: two-electron timing and vertex / baseline

```
pair rows=20000 denominator=12127 valid=9059
time_difference: N=9059, NaN=0
  mean=0.84967 med=0.50201 std=3.9924 s68=3.3047
  |err| q68=3.3078 q95=8.6726; tail >10: 3.080%
time_difference_error: N=9059, NaN=0
  mean=0.84967 med=0.50201 std=3.9924 s68=3.3047
  |err| q68=3.3078 q95=8.6726; tail >10: 3.080%
separation: N=9059, NaN=0
  mean=388.75 med=392.7 std=198.62 s68=205.22
  |err| q68=461.73 q95=763.23; tail >400: 48.129%
opening_error_deg: N=9059, NaN=0
  mean=80.96 med=83.003 std=45.474 s68=53.053
  |err| q68=111.24 q95=149.57; tail >30: 81.930%
max_vertex_dr: N=9059, NaN=0
  mean=311.67 med=342.33 std=118.43 s68=129.63
  |err| q68=395.04 q95=457.44; tail >400: 29.562%
max_vertex_dt: N=9059, NaN=0
  mean=2.767 med=2.0872 std=2.296 s68=1.9221
  |err| q68=3.1145 q95=7.3168; tail >10: 1.601%
```

D: two-electron timing and vertex / control

```
pair rows=20000 denominator=12127 valid=9561
time_difference: N=9561, NaN=0
  mean=0.81091 med=0.48225 std=3.9495 s68=3.2786
  |err| q68=3.2721 q95=8.571; tail >10: 2.981%
time_difference_error: N=9561, NaN=0
  mean=0.81091 med=0.48225 std=3.9495 s68=3.2786
  |err| q68=3.2721 q95=8.571; tail >10: 2.981%
separation: N=9561, NaN=0
  mean=380.72 med=385.88 std=198.95 s68=203.58
  |err| q68=455.25 q95=759.03; tail >400: 46.700%
opening_error_deg: N=9561, NaN=0
  mean=80.275 med=82.176 std=45.634 s68=53.478
  |err| q68=110.48 q95=149.39; tail >30: 81.236%
max_vertex_dr: N=9561, NaN=0
  mean=307.7 med=338.98 std=120.98 s68=133.47
  |err| q68=394.21 q95=456.92; tail >400: 29.150%
max_vertex_dt: N=9561, NaN=0
  mean=2.7455 med=2.0723 std=2.2843 s68=1.9103
  |err| q68=3.0938 q95=7.2759; tail >10: 1.558%
```

D: two-electron timing and vertex / momentum

```
pair rows=20000 denominator=12127 valid=10038
time_difference: N=10038, NaN=0
  mean=0.76133 med=0.44752 std=3.5673 s68=2.8889
  |err| q68=2.8596 q95=7.5903; tail >10: 2.042%
time_difference_error: N=10038, NaN=0
  mean=0.76133 med=0.44752 std=3.5673 s68=2.8889
  |err| q68=2.8596 q95=7.5903; tail >10: 2.042%
separation: N=10038, NaN=0
  mean=357.1 med=366.33 std=192.36 s68=193.41
  |err| q68=436.2 q95=714.15; tail >400: 42.210%
opening_error_deg: N=10038, NaN=0
  mean=79.592 med=81.246 std=45.874 s68=53.204
  |err| q68=110.25 q95=149.87; tail >30: 80.783%
max_vertex_dr: N=10038, NaN=0
  mean=300.39 med=329.41 std=128.07 s68=142.46
  |err| q68=390.97 q95=459.77; tail >400: 28.472%
max_vertex_dt: N=10038, NaN=0
  mean=2.4597 med=1.838 std=2.1652 s68=1.7081
  |err| q68=2.7093 q95=6.5117; tail >10: 1.086%
```

D: two-electron timing and vertex / position

```
pair rows=20000 denominator=12127 valid=9419
time_difference: N=9419, NaN=0
  mean=0.71715 med=0.37845 std=4.0182 s68=3.3498
  |err| q68=3.3142 q95=8.6362; tail >10: 3.100%
time_difference_error: N=9419, NaN=0
  mean=0.71715 med=0.37845 std=4.0182 s68=3.3498
  |err| q68=3.3142 q95=8.6362; tail >10: 3.100%
separation: N=9419, NaN=0
  mean=380.07 med=384.17 std=200.2 s68=207.83
  |err| q68=456.79 q95=751.26; tail >400: 46.534%
opening_error_deg: N=9419, NaN=0
  mean=79.851 med=81.459 std=45.118 s68=52.506
  |err| q68=109.35 q95=148.14; tail >30: 81.293%
max_vertex_dr: N=9419, NaN=0
  mean=307.39 med=337.84 std=121.46 s68=133.42
  |err| q68=393.48 q95=457.87; tail >400: 28.846%
max_vertex_dt: N=9419, NaN=0
  mean=2.7882 med=2.0887 std=2.3289 s68=1.9563
  |err| q68=3.1556 q95=7.3931; tail >10: 1.624%
```

D: two-electron timing and vertex / truth

```
pair rows=20000 denominator=12127 valid=9955
time_difference: N=9955, NaN=0
  mean=0.51771 med=0.1962 std=3.2851 s68=2.5985
  |err| q68=2.57 q95=7.0134; tail >10: 1.527%
time_difference_error: N=9955, NaN=0
  mean=0.51771 med=0.1962 std=3.2851 s68=2.5985
  |err| q68=2.57 q95=7.0134; tail >10: 1.527%
separation: N=9955, NaN=0
  mean=350.66 med=361.45 std=188.18 s68=189.54
  |err| q68=431.25 q95=695.85; tail >400: 41.236%
opening_error_deg: N=9955, NaN=0
  mean=79.763 med=81.519 std=46 s68=53.967
  |err| q68=110.01 q95=149.86; tail >30: 80.763%
max_vertex_dr: N=9955, NaN=0
  mean=296.47 med=325.55 std=126.51 s68=142.87
  |err| q68=389.61 q95=454.39; tail >400: 27.825%
max_vertex_dt: N=9955, NaN=0
  mean=2.2128 med=1.6409 std=2.046 s68=1.6672
  |err| q68=2.4832 q95=6.1793; tail >10: 0.814%
```

E: entrance-to-target correlations

```
Circular rho is unsigned association with sin/cos phase; not a slope.
Correlations use pairwise finite values; no truncation of tails.
baseline_direct:ent_dpz:tar_phase
  n=14769, circular_linear_rho=0.15573
baseline_direct:ent_dr:tar_dr
  n=14769, pearson_r=0.16597
baseline_direct:ent_dt:tar_dt
  n=14769, pearson_r=0.50968
baseline_direct:ent_pitch:tar_phase
  n=14769, circular_linear_rho=0.23647
baseline_direct:tar_phase:tar_dr
  n=14769, circular_linear_rho=0.96147
baseline_reflected:ent_dpz:tar_phase
  n=11609, circular_linear_rho=0.0051646
baseline_reflected:ent_dr:tar_dr
  n=11609, pearson_r=0.016561
baseline_reflected:ent_dt:tar_dt
  n=11609, pearson_r=0.14756
baseline_reflected:ent_pitch:tar_phase
  n=11609, circular_linear_rho=0.013541
baseline_reflected:tar_phase:tar_dr
  n=11609, circular_linear_rho=0.9646
control_direct:ent_dpz:tar_phase
  n=14093, circular_linear_rho=0.15068
control_direct:ent_dr:tar_dr
  n=14093, pearson_r=0.16018
control_direct:ent_dt:tar_dt
  n=14093, pearson_r=0.48367
control_direct:ent_pitch:tar_phase
```

E: entrance-to-target correlations

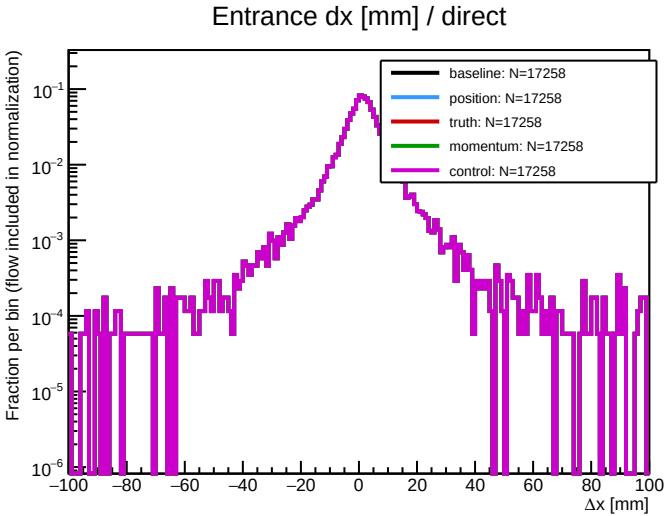
```
Circular rho is unsigned association with sin/cos phase; not a slope.
Correlations use pairwise finite values; no truncation of tails.
  n=14093, circular_linear_rho=0.23431
control_direct:tar_phase:tar_dr
  n=14093, circular_linear_rho=0.96177
control_reflected:ent_dpz:tar_phase
  n=11789, circular_linear_rho=0.0033358
control_reflected:ent_dr:tar_dr
  n=11789, pearson_r=0.018239
control_reflected:ent_dt:tar_dt
  n=11789, pearson_r=0.14886
control_reflected:ent_pitch:tar_phase
  n=11789, circular_linear_rho=0.01543
control_reflected:tar_phase:tar_dr
  n=11789, circular_linear_rho=0.96445
momentum_direct:ent_dpz:tar_phase
  n=14392, circular_linear_rho=0.041885
momentum_direct:ent_dr:tar_dr
  n=14392, pearson_r=0.21823
momentum_direct:ent_dt:tar_dt
  n=14392, pearson_r=0.51411
momentum_direct:ent_pitch:tar_phase
  n=14392, circular_linear_rho=0.052969
momentum_direct:tar_phase:tar_dr
  n=14392, circular_linear_rho=0.94588
momentum_reflected:ent_dpz:tar_phase
  n=12145, circular_linear_rho=0.014419
momentum_reflected:ent_dr:tar_dr
  n=12145, pearson_r=0.10573
```

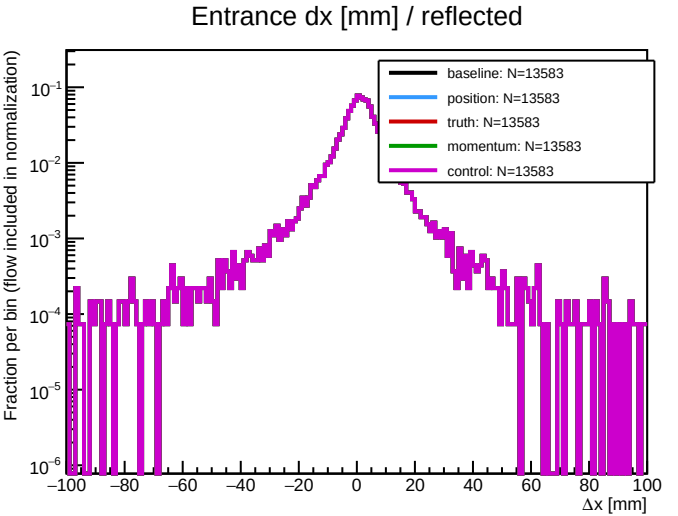
E: entrance-to-target correlations

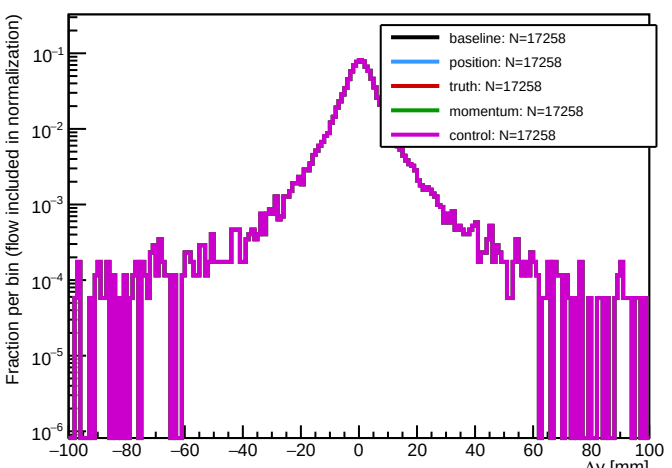
```
Circular rho is unsigned association with sin/cos phase; not a slope.
Correlations use pairwise finite values; no truncation of tails.
momentum_reflected:ent_dt:tar_dt
  n=12145, pearson_r=0.22074
momentum_reflected:ent_pitch:tar_phase
  n=12145, circular_linear_rho=0.01561
momentum_reflected:tar_phase:tar_dr
  n=12145, circular_linear_rho=0.95479
position_direct:ent_dpz:tar_phase
  n=13975, circular_linear_rho=0.14445
position_direct:ent_dr:tar_dr
  n=13975, pearson_r=0.16597
position_direct:ent_dt:tar_dt
  n=13975, pearson_r=0.47299
position_direct:ent_pitch:tar_phase
  n=13975, circular_linear_rho=0.26791
position_direct:tar_phase:tar_dr
  n=13975, circular_linear_rho=0.95512
position_reflected:ent_dpz:tar_phase
  n=11688, circular_linear_rho=0.014433
position_reflected:ent_dr:tar_dr
  n=11688, pearson_r=0.00034694
position_reflected:ent_dt:tar_dt
  n=11688, pearson_r=0.14377
position_reflected:ent_pitch:tar_phase
  n=11688, circular_linear_rho=0.014591
position_reflected:tar_phase:tar_dr
  n=11688, circular_linear_rho=0.95784
truth_direct:ent_dpz:tar_phase
```

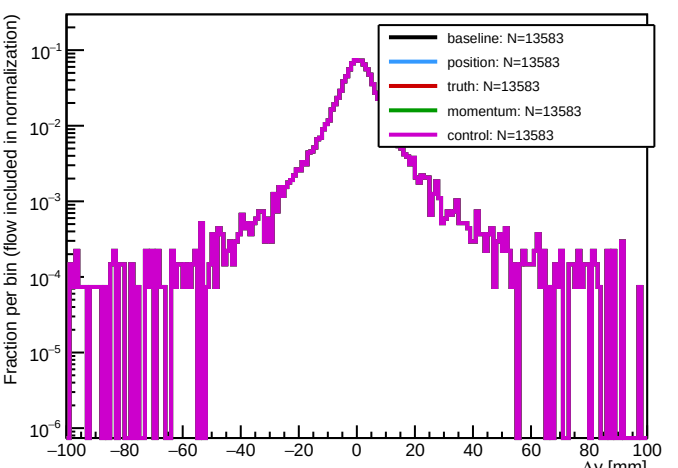
E: entrance-to-target correlations

```
Circular rho is unsigned association with sin/cos phase; not a slope.
Correlations use pairwise finite values; no truncation of tails.
  n=14337, circular_linear_rho=0.012366
truth_direct:ent_dr:tar_dr
  n=14337, pearson_r=0.028659
truth_direct:ent_dt:tar_dt
  n=14337, pearson_r=-0.001647
truth_direct:ent_pitch:tar_phase
  n=14337, circular_linear_rho=0.014258
truth_direct:tar_phase:tar_dr
  n=14337, circular_linear_rho=0.95803
truth_reflected:ent_dpz:tar_phase
  n=12091, circular_linear_rho=0.0022699
truth_reflected:ent_dr:tar_dr
  n=12091, pearson_r=0.020991
truth_reflected:ent_dt:tar_dt
  n=12091, pearson_r=0.0011754
truth_reflected:ent_pitch:tar_phase
  n=12091, circular_linear_rho=0.012785
truth_reflected:tar_phase:tar_dr
  n=12091, circular_linear_rho=0.96552
```

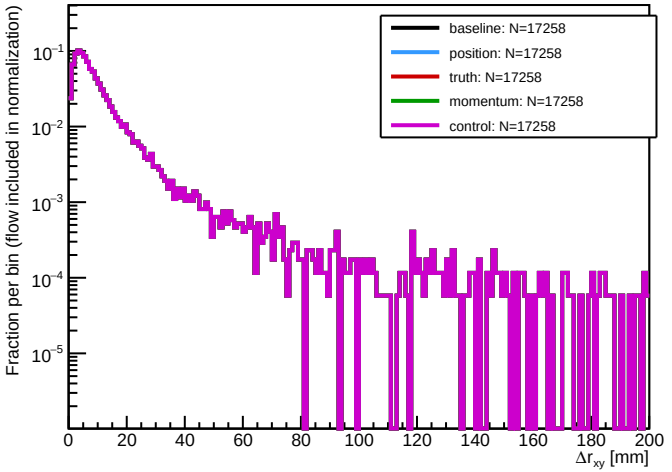

Entrance dx [mm] / direct
baseline: N=17258
position: N=17258
truth: N=17258
momentum: N=17258
control: N=17258
Fraction per bin (flow included in normalization)
Δx [mm]


Entrance dx [mm] / reflected
baseline: N=13583
position: N=13583
truth: N=13583
momentum: N=13583
control: N=13583
Fraction per bin (flow included in normalization)
Δx [mm]

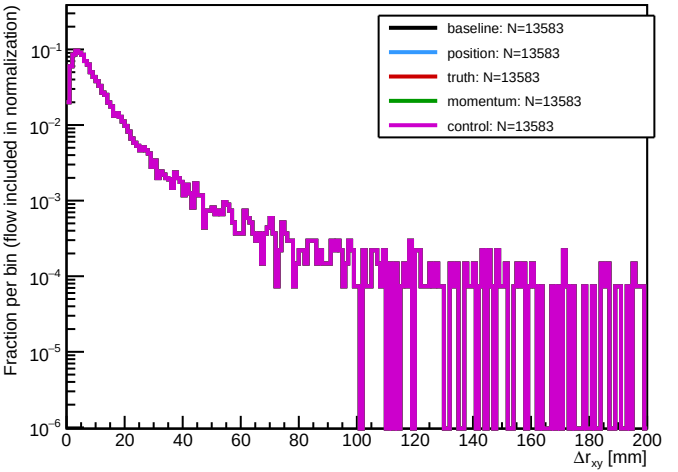

Entrance dy [mm] / direct
baseline: N=17258
position: N=17258
truth: N=17258
momentum: N=17258
control: N=17258
Fraction per bin (flow included in normalization)
Δy [mm]


Entrance dy [mm] / reflected
baseline: N=13583
position: N=13583
truth: N=13583
momentum: N=13583
control: N=13583
Fraction per bin (flow included in normalization)
Δy [mm]

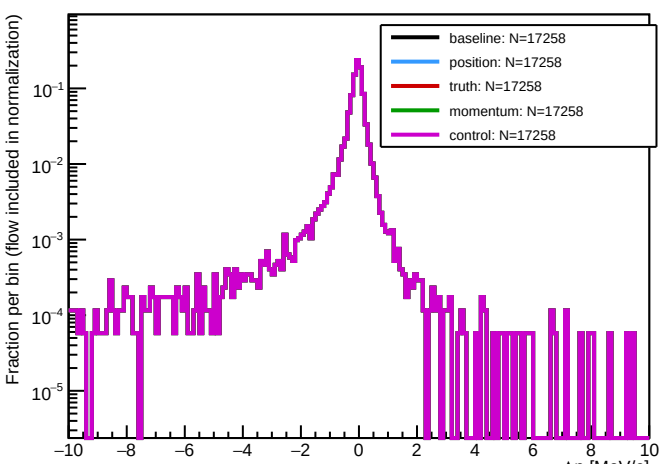
Entrance transverse error [mm] / direct



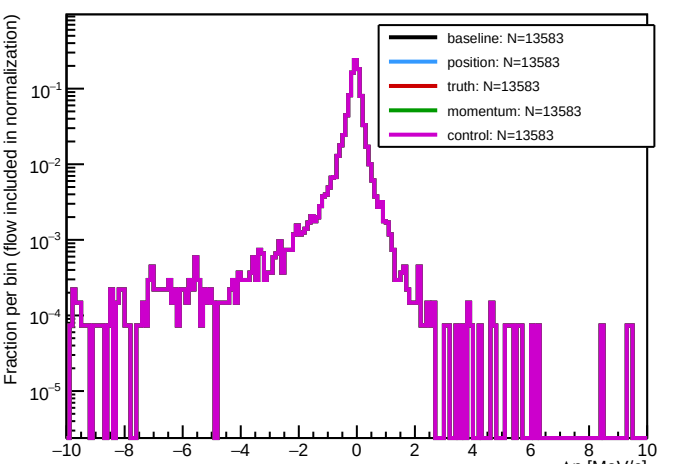
Entrance transverse error [mm] / reflected



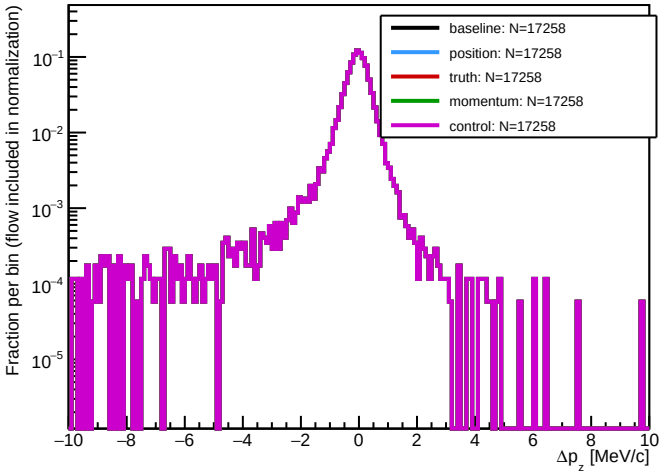
Entrance momentum error [MeV/c] / direct



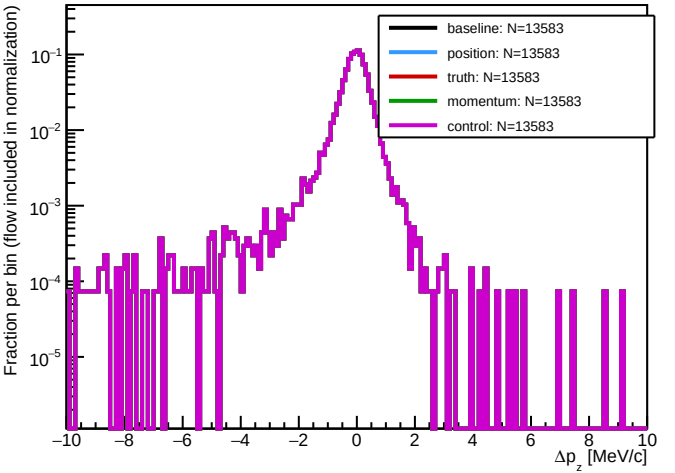
Entrance momentum error [MeV/c] / reflected



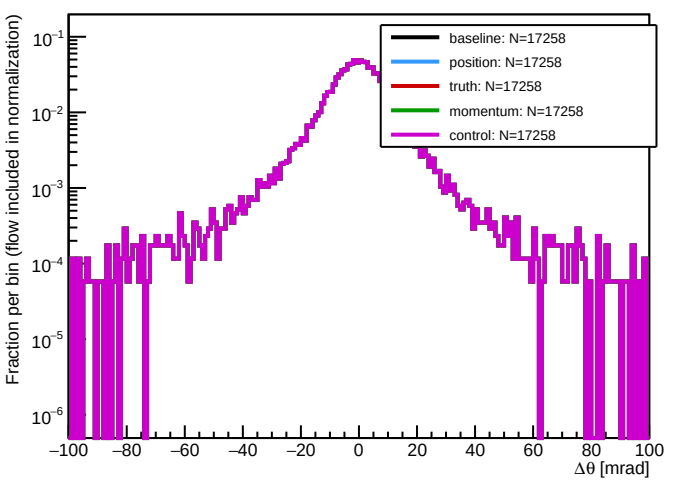
Entrance pz error [MeV/c] / direct



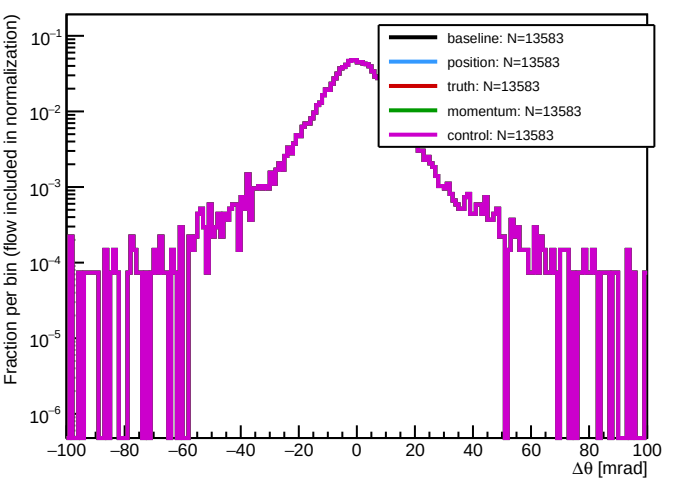
Entrance pz error [MeV/c] / reflected



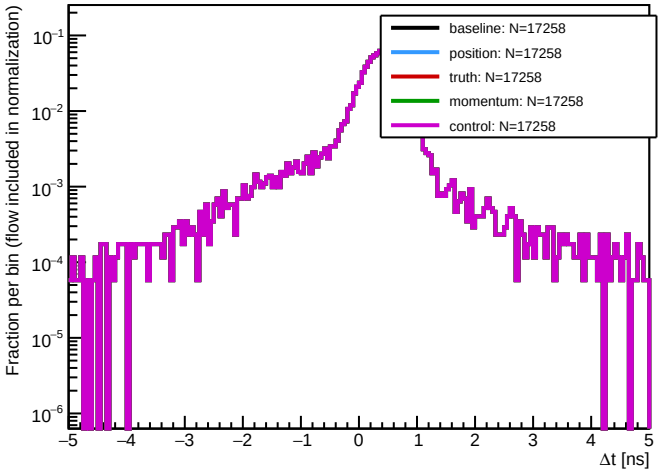
Entrance pitch error [mrad] / direct



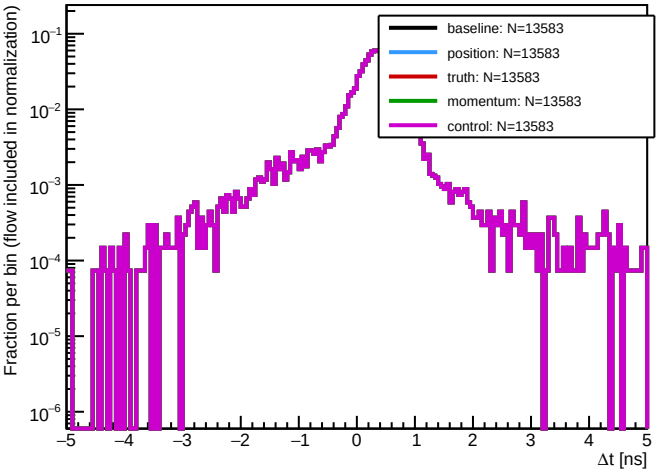
Entrance pitch error [mrad] / reflected



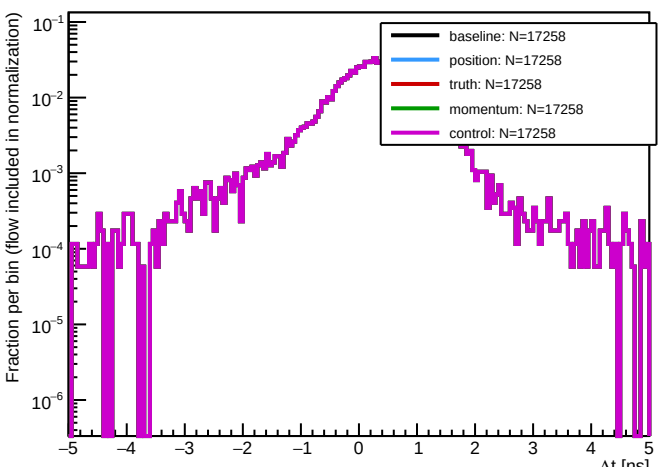
Entrance time error, clock corrected [ns] / direct



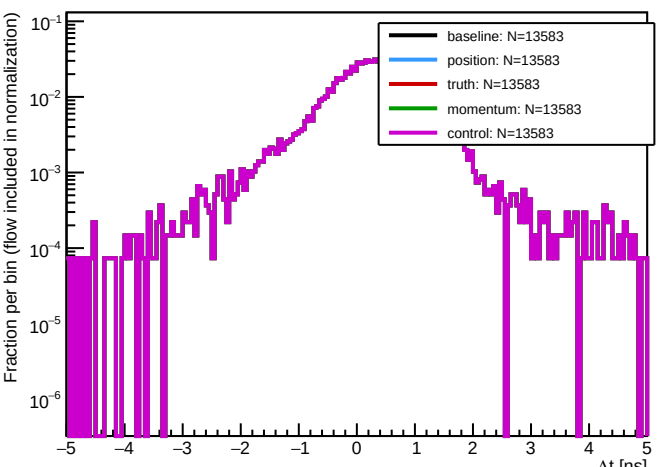
Entrance time error, clock corrected [ns] / reflected

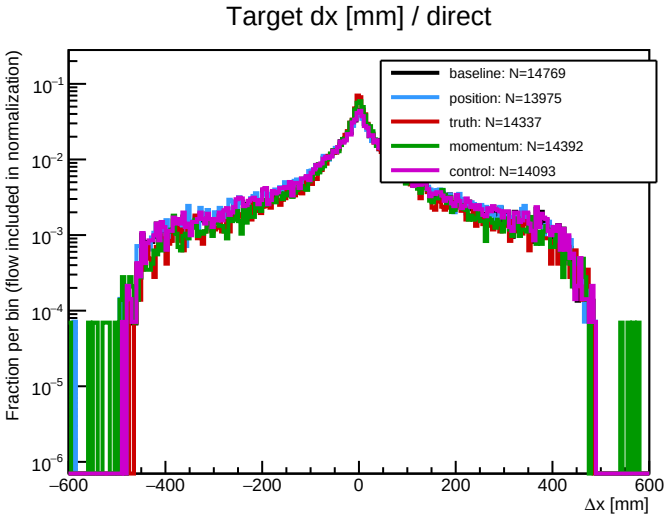


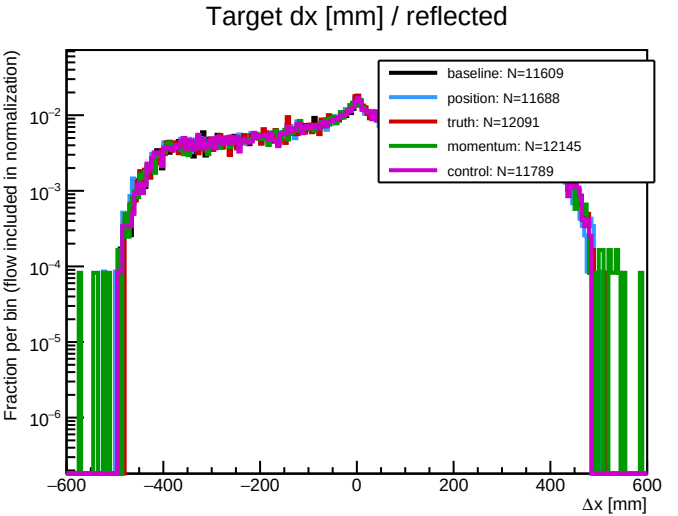
Entrance time error, bunch clock included [ns] / direct

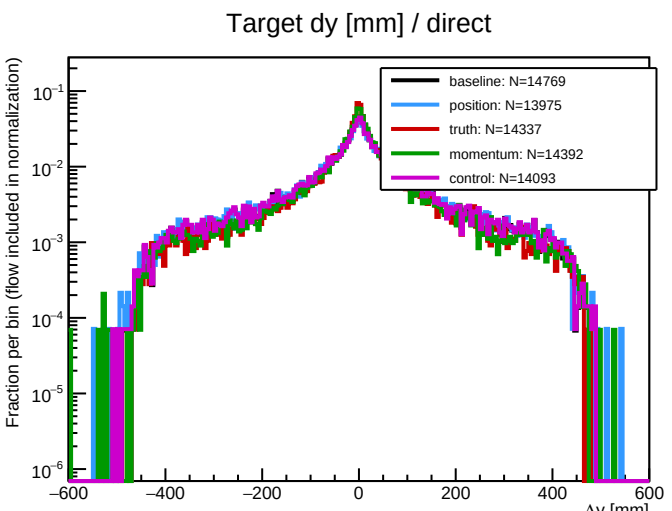


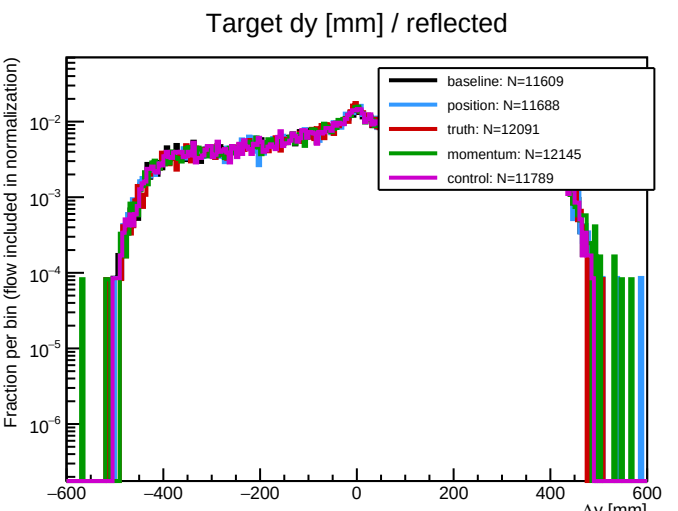
Entrance time error, bunch clock included [ns] / reflected



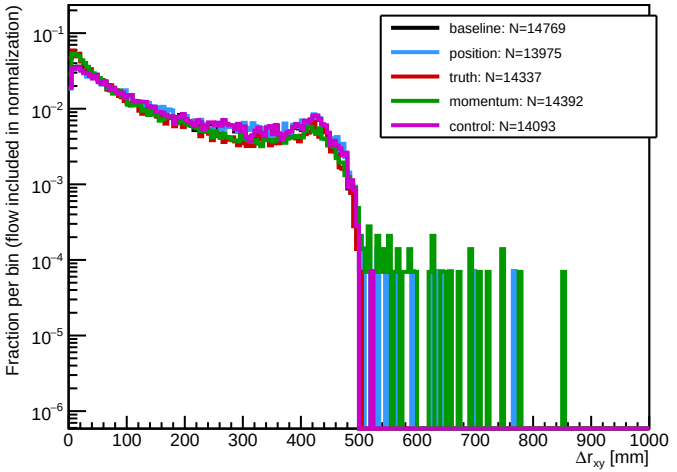

Target dx [mm] / direct
baseline: N=14769
position: N=13975
truth: N=14337
momentum: N=14392
control: N=14093
Fraction per bin (flow included in normalization)
Δx [mm]


Target dx [mm] / reflected
baseline: N=11609
position: N=11688
truth: N=12091
momentum: N=12145
control: N=11789
Fraction per bin (flow included in normalization)
Δx [mm]

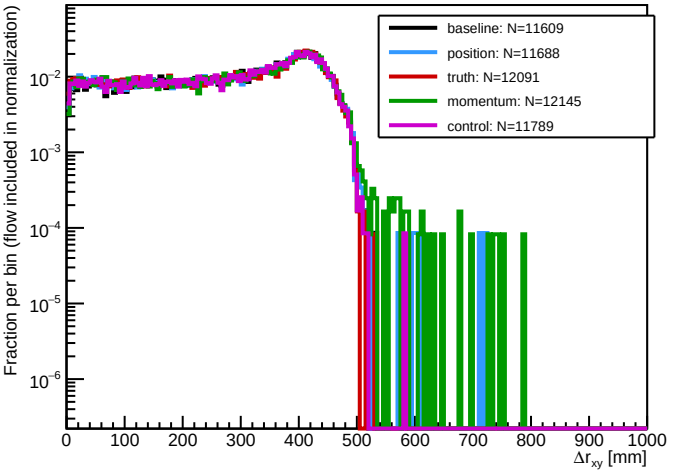

Target dy [mm] / direct
baseline: N=14769
position: N=13975
truth: N=14337
momentum: N=14392
control: N=14093
Fraction per bin (flow included in normalization)
Δy [mm]


Target dy [mm] / reflected
baseline: N=11609
position: N=11688
truth: N=12091
momentum: N=12145
control: N=11789
Fraction per bin (flow included in normalization)
Δy [mm]

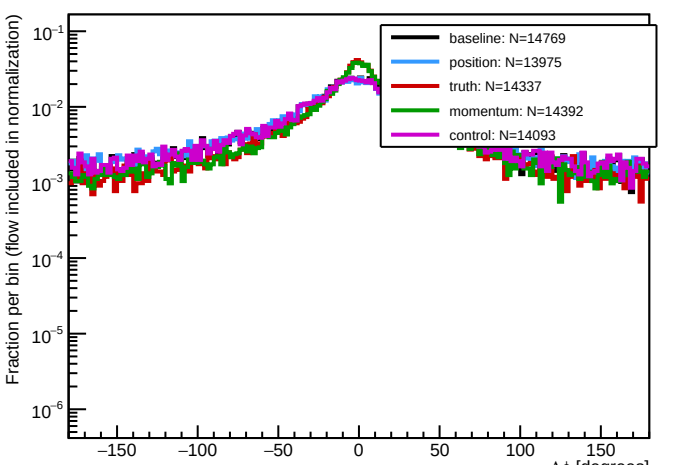
Target transverse error [mm] / direct



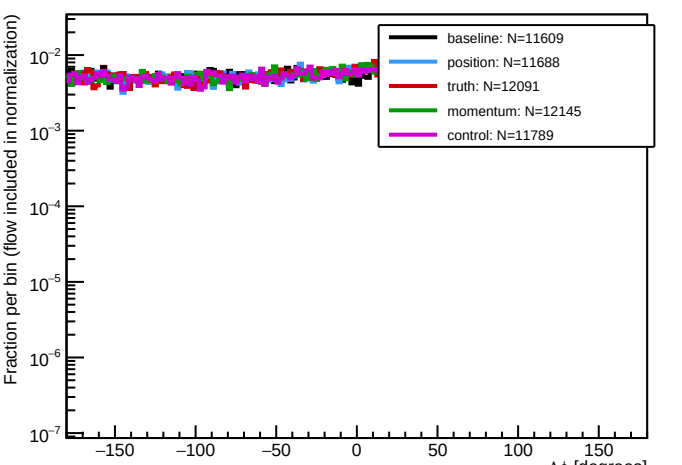
Target transverse error [mm] / reflected



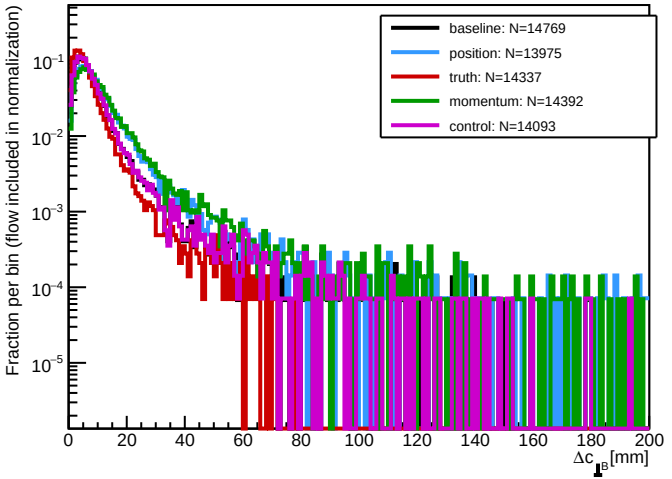
Target wrapped phase error [degrees] / direct



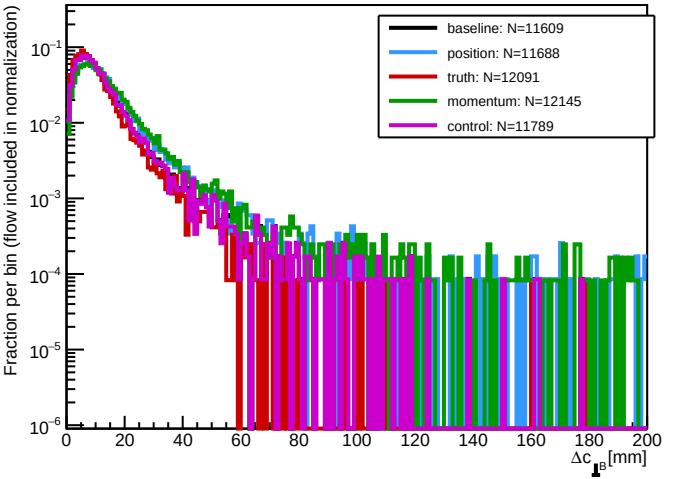
Target wrapped phase error [degrees] / reflected



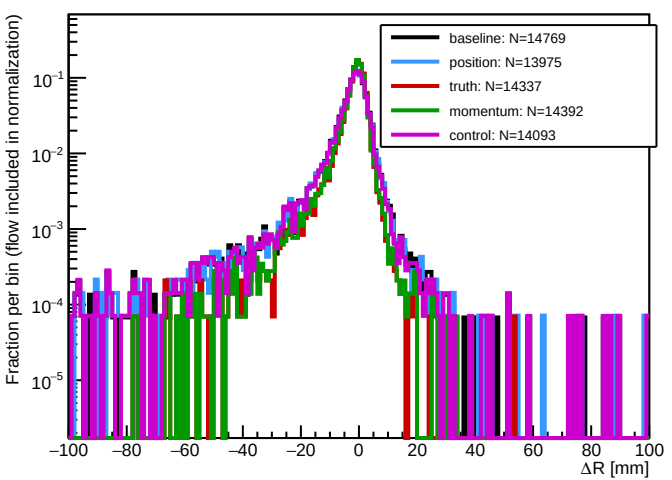
Target local orbit-center distance [mm] / direct



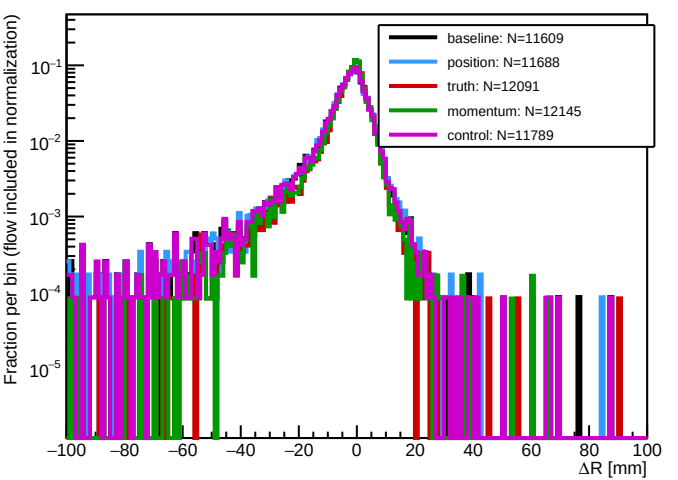
Target local orbit-center distance [mm] / reflected

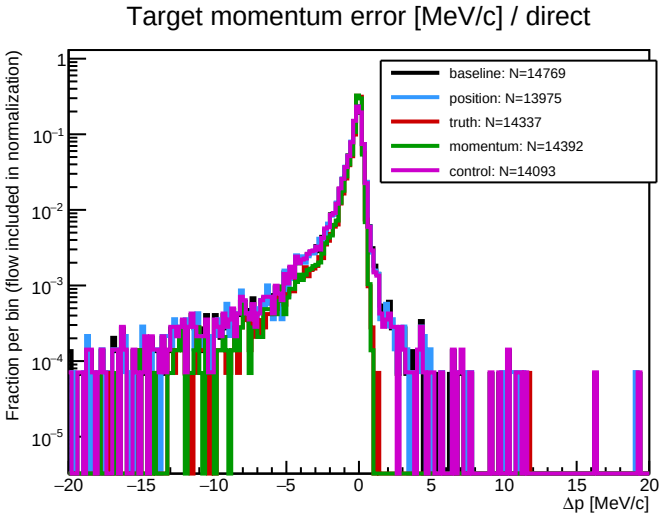


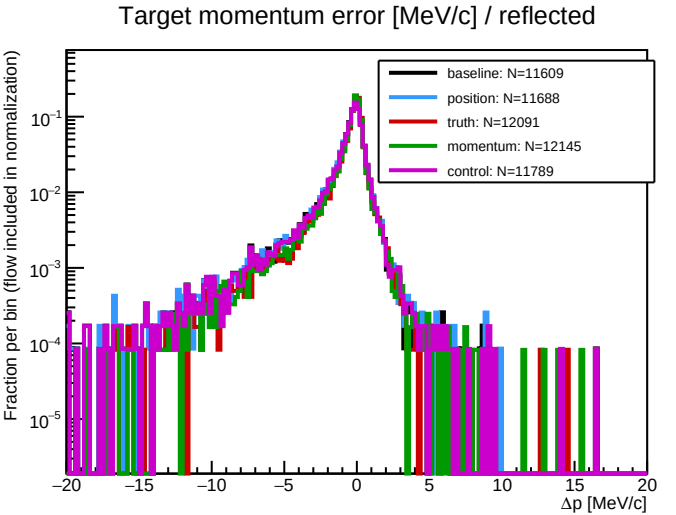
Target local orbit-radius error [mm] / direct

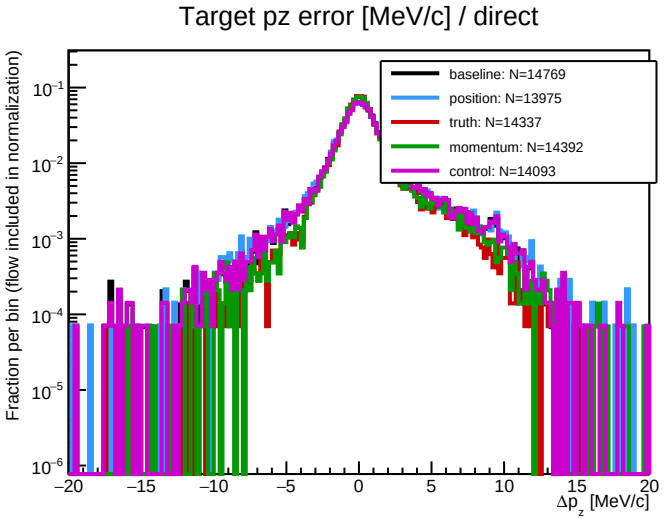


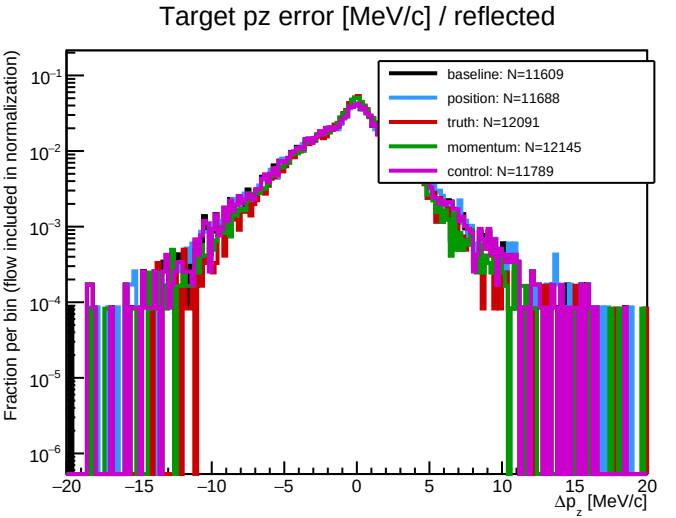
Target local orbit-radius error [mm] / reflected



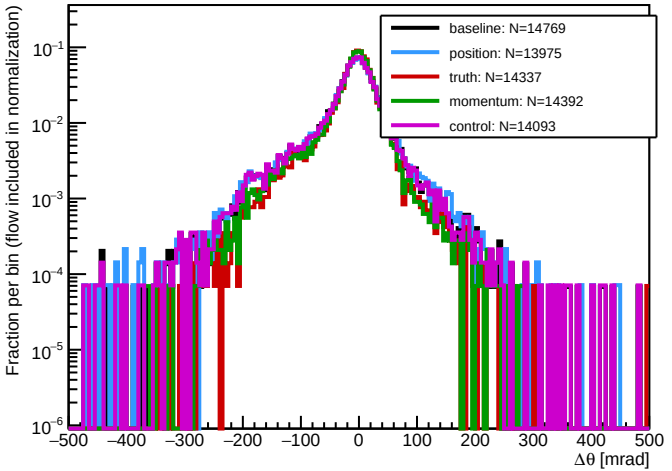

Target momentum error [MeV/c] / direct
baseline: N=14769
position: N=13975
truth: N=14337
momentum: N=14392
control: N=14093
Fraction per bin (flow included in normalization)
Δp [MeV/c]


Target momentum error [MeV/c] / reflected
baseline: N=11609
position: N=11688
truth: N=12091
momentum: N=12145
control: N=11789
Fraction per bin (flow included in normalization)
Δp [MeV/c]

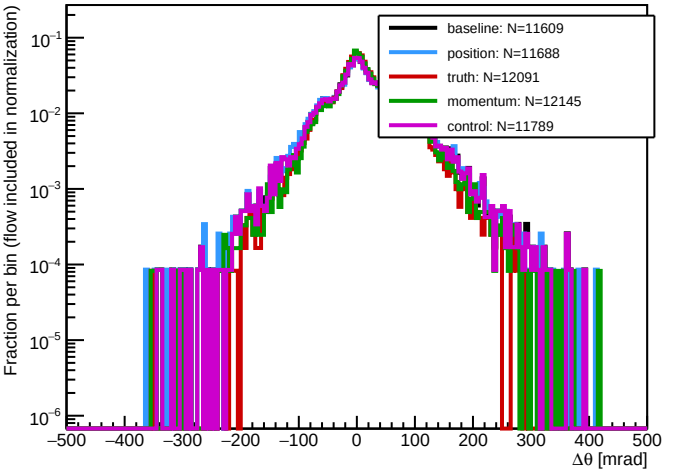

Target pz error [MeV/c] / direct
baseline: N=14769
position: N=13975
truth: N=14337
momentum: N=14392
control: N=14093
Fraction per bin (flow included in normalization)
Δp_z [MeV/c]


Target pz error [MeV/c] / reflected
baseline: N=11609
position: N=11688
truth: N=12091
momentum: N=12145
control: N=11789
Fraction per bin (flow included in normalization)
Δp_z [MeV/c]

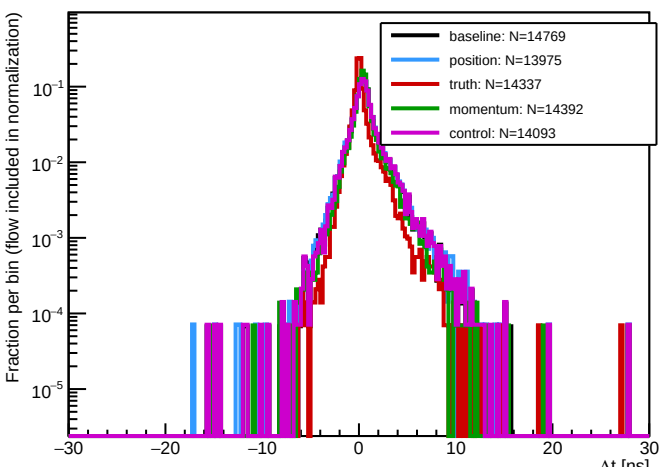
Target pitch error [mrad] / direct



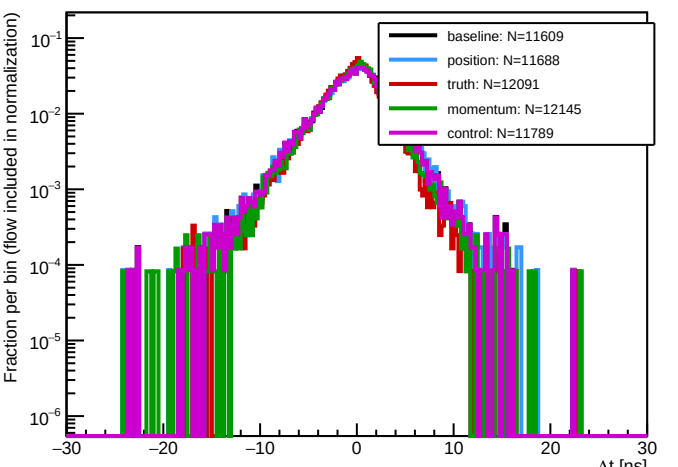
Target pitch error [mrad] / reflected

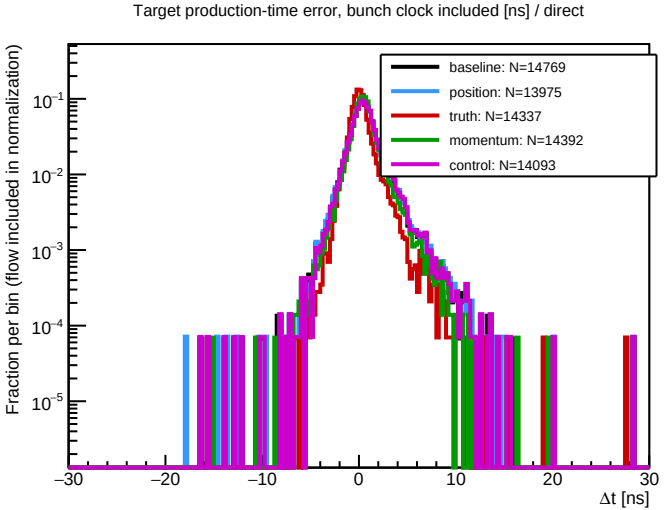


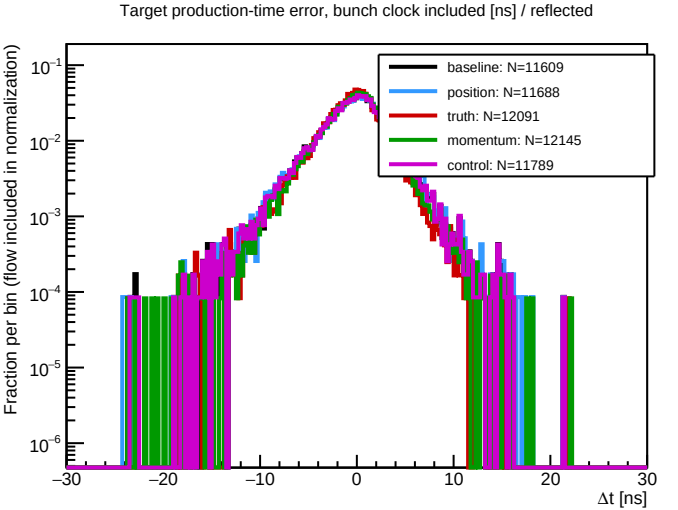
Target production-time error, clock corrected [ns] / direct

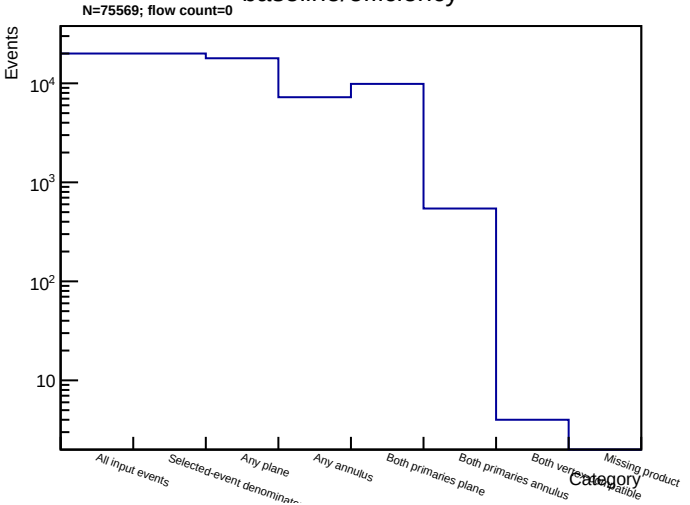


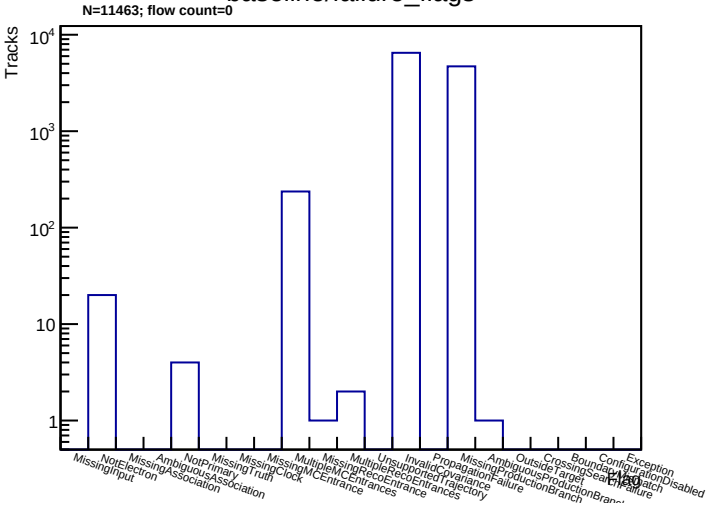
Target production-time error, clock corrected [ns] / reflected

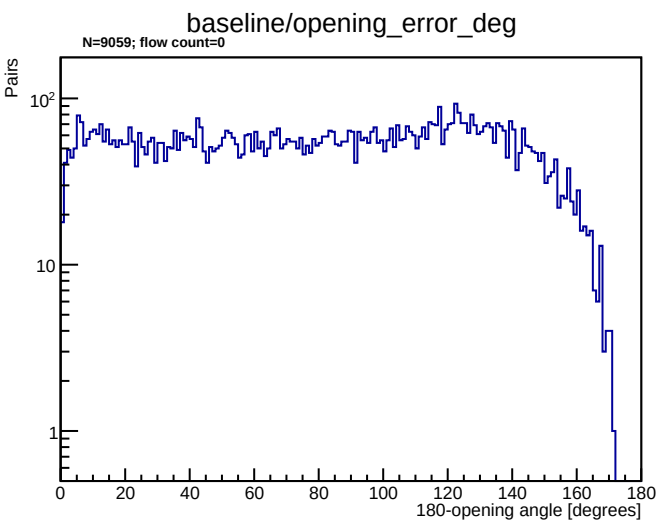


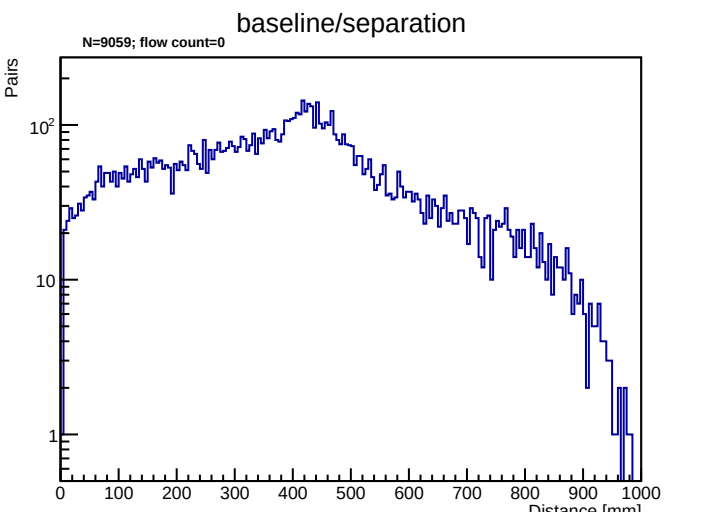

Target production-time error, bunch clock included [ns] / direct
Fraction per bin (flow included in normalization)
Δt [ns]
baseline: N=14769
position: N=13975
truth: N=14337
momentum: N=14392
control: N=14093

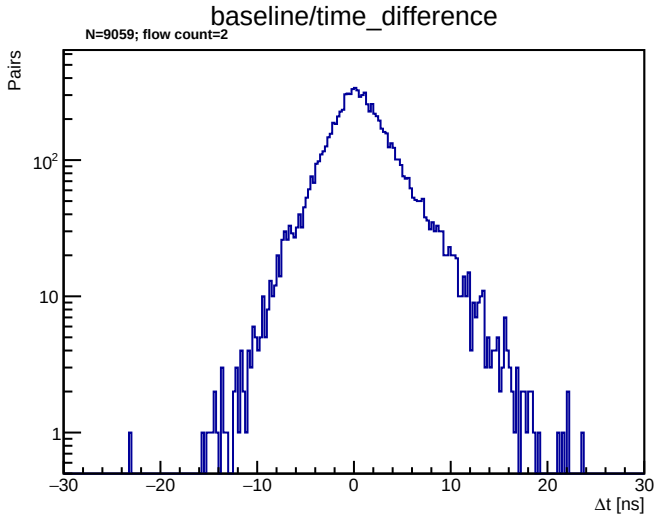

Target production-time error, bunch clock included [ns] / reflected
Fraction per bin (flow included in normalization)
Δt [ns]
baseline: N=11609
position: N=11688
truth: N=12091
momentum: N=12145
control: N=11789

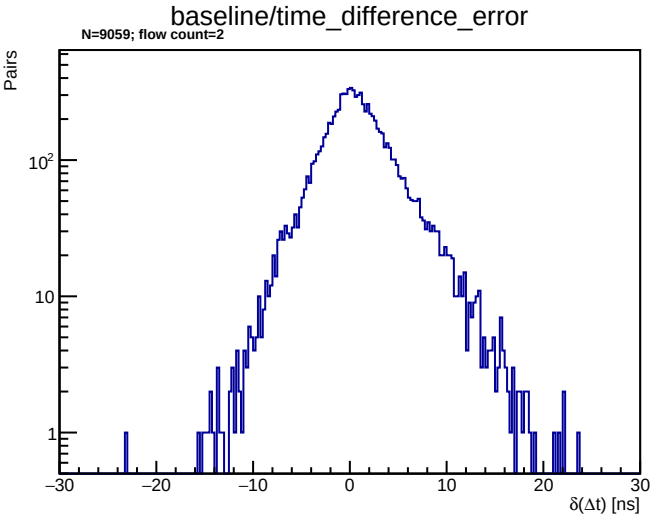

baseline/efficiency
N=75569; flow count=0
Events
Category
All input events
Selected-event denomina
Any plane
Any annulus
Both primaries plane
Both primaries annulus
Both ver
Missing product
ailable

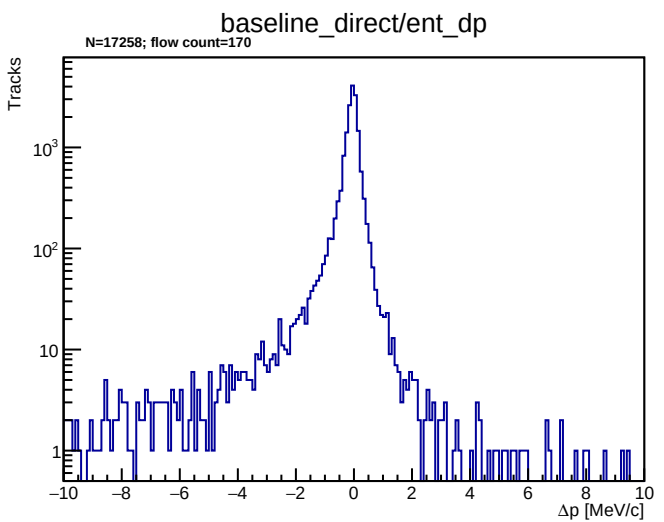

baseline/failure_flags
N=11463; flow count=0
Tracks
Flag
MissingInput
NotElectron
MissingAssociation
AmbiguousAssociation
NotPrimaryTruth
MissingTruth
MissingClock
MultipleMCEntrance
MissingMCEntrance
MultipleRecoEntrances
MissingRecoEntrance
UnsupportedTrajectory
InvalidCovariance
PropagationFailure
MissingProduction
AmbiguousProduction
OutsideTarget
CrossingSeaplane
BoundaryMismatch
ConfigurationDisabled
Exception
Failure

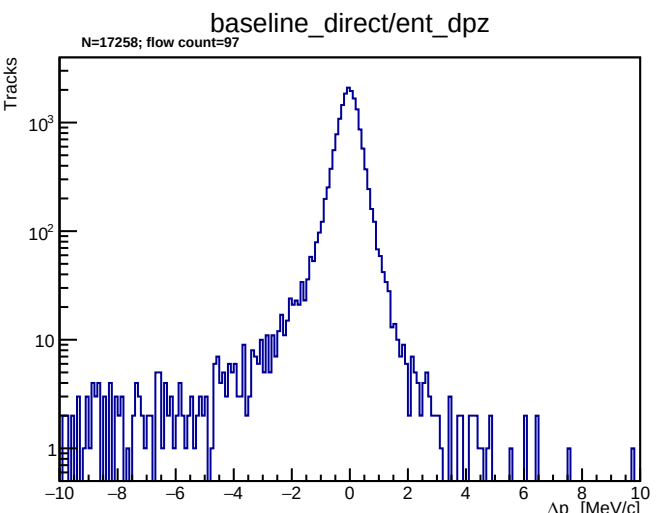

baseline/opening_error_deg
N=9059; flow count=0
Pairs
180-opening angle [degrees]

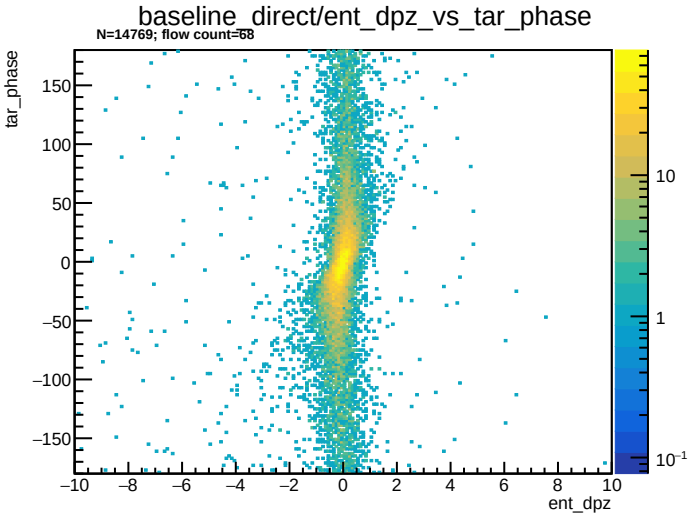

baseline/separation
N=9059; flow count=0
Pairs
Distance [mm]

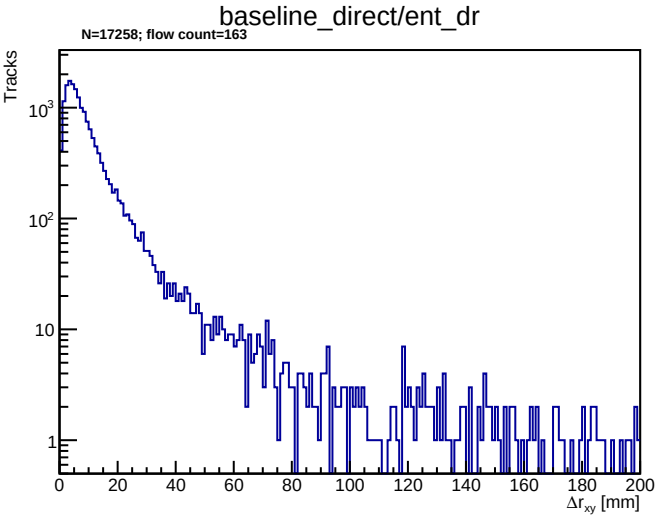

baseline/time_difference
N=9059; flow count=2
Pairs
Δt [ns]

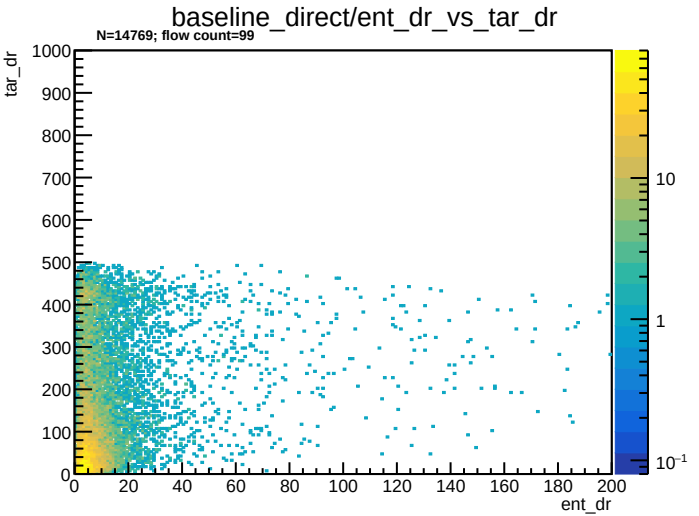

baseline/time_difference_error
N=9059; flow count=2
Pairs
δ(Δt) [ns]

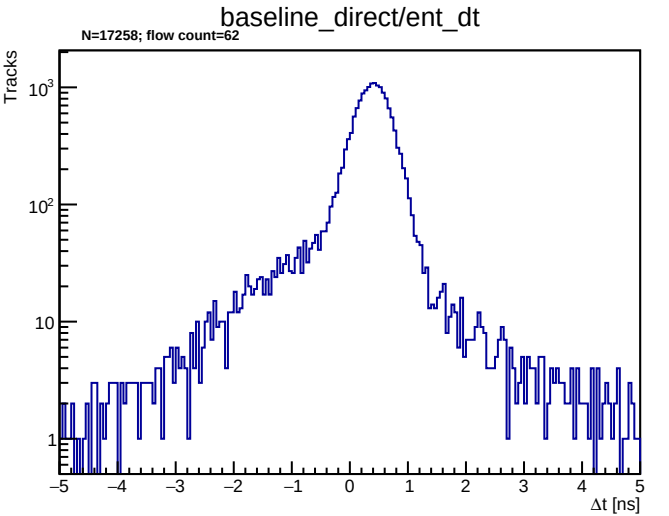

baseline_direct/ent_dp
N=17258; flow count=170
Tracks
Δp [MeV/c]

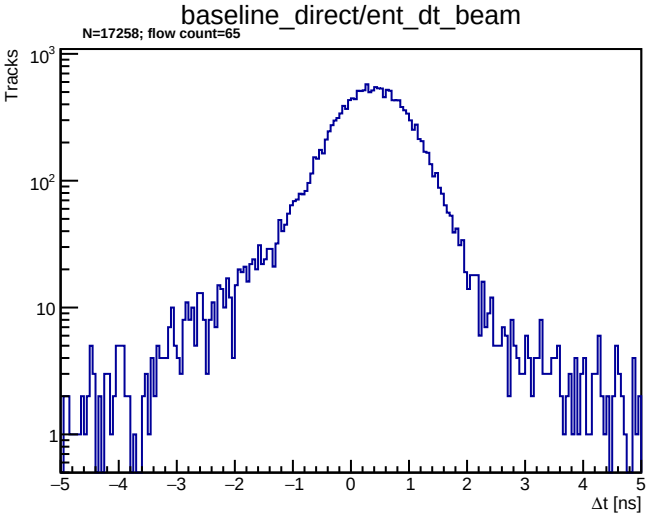

baseline_direct/ent_dpz
N=17258; flow count=97
Tracks
Δp_z [MeV/c]

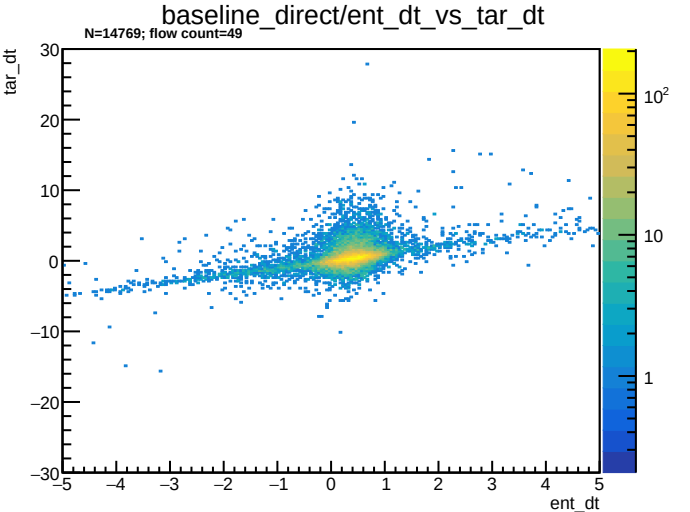

baseline_direct/ent_dpz_vs_tar_phase
N=14769; flow count=68
tar_phase
ent_dpz

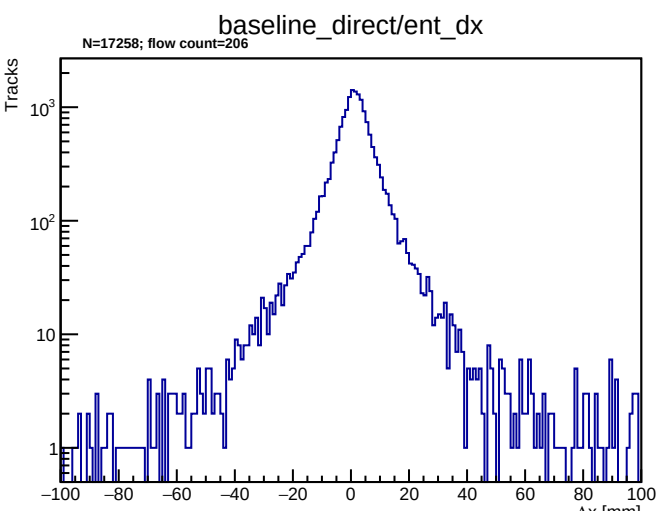

baseline_direct/ent_dr
N=17258; flow count=163
Tracks
Δr_xy [mm]

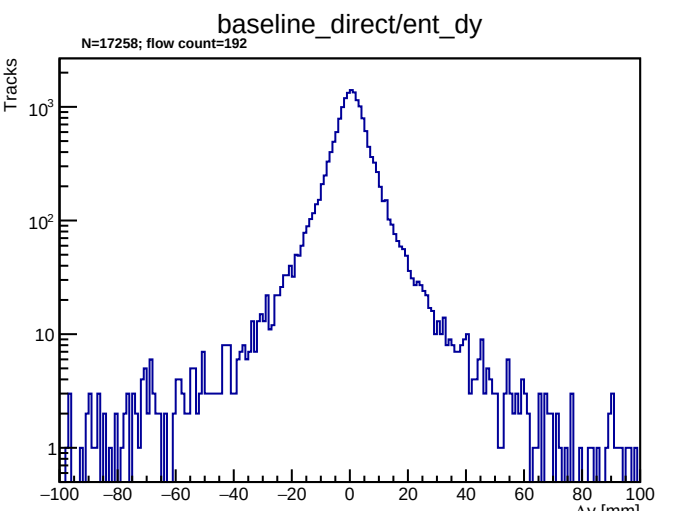

baseline_direct/ent_dr_vs_tar_dr
N=14769; flow count=99
tar_dr
ent_dr

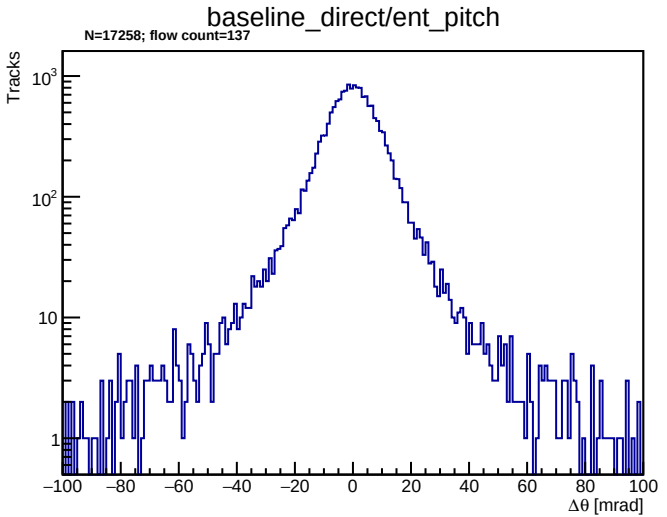

baseline_direct/ent_dt
N=17258; flow count=62
Tracks
Δt [ns]

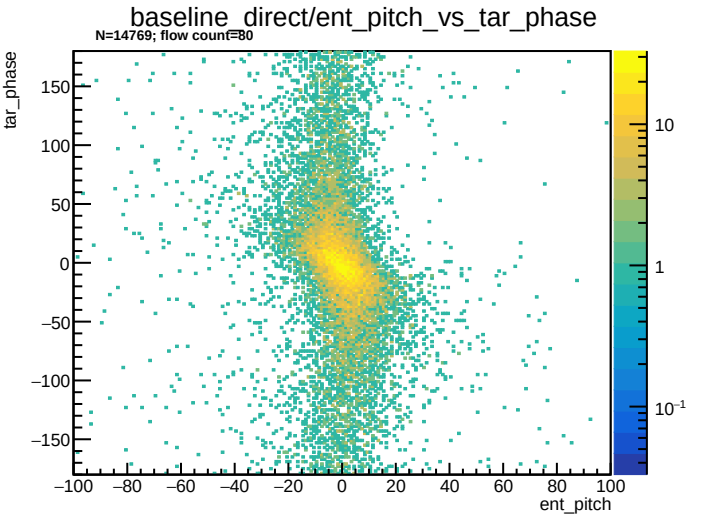

baseline_direct/ent_dt_beam
N=17258; flow count=65
Tracks
Δt [ns]

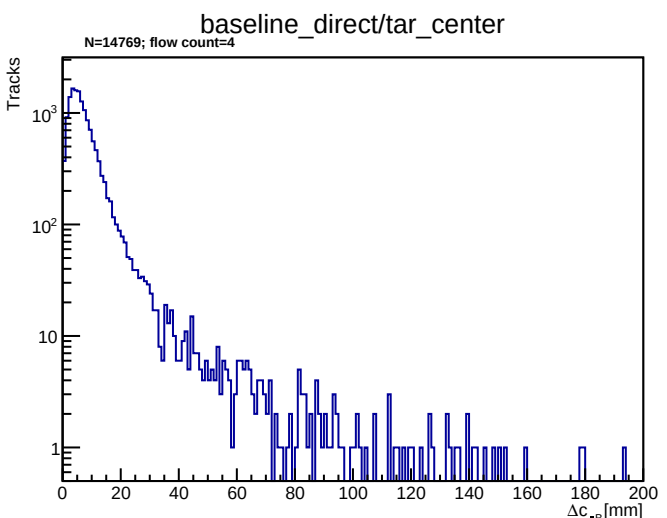

baseline_direct/ent_dt_vs_tar_dt
N=14769; flow count=49
tar_dt
ent_dt

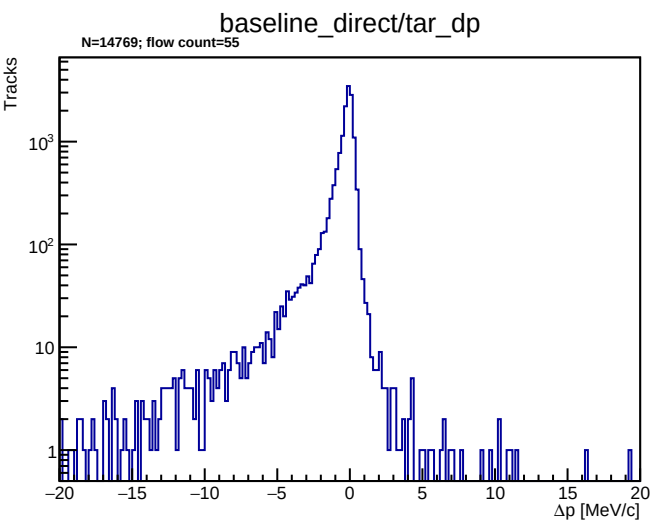

baseline_direct/ent_dx
N=17258; flow count=206
Tracks
Δx [mm]

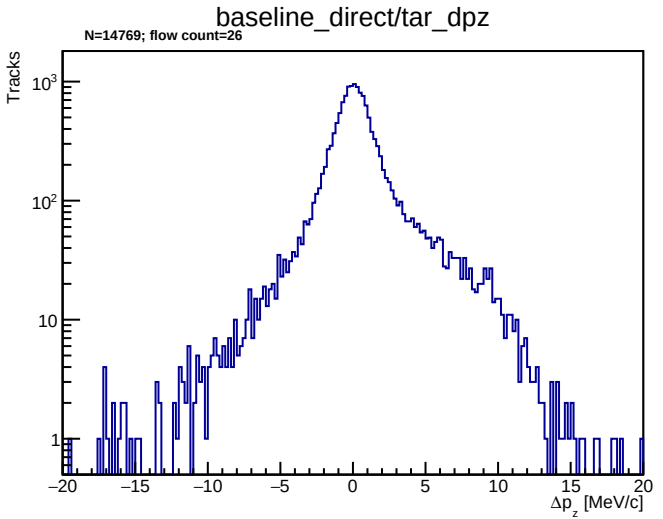

baseline_direct/ent_dy
N=17258; flow count=192
Tracks
Δy [mm]

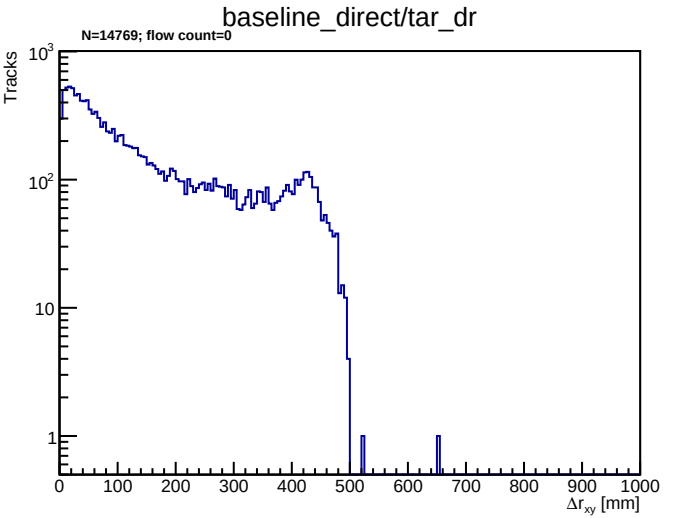

baseline_direct/ent_pitch
N=17258; flow count=137
Tracks
Δθ [mrad]

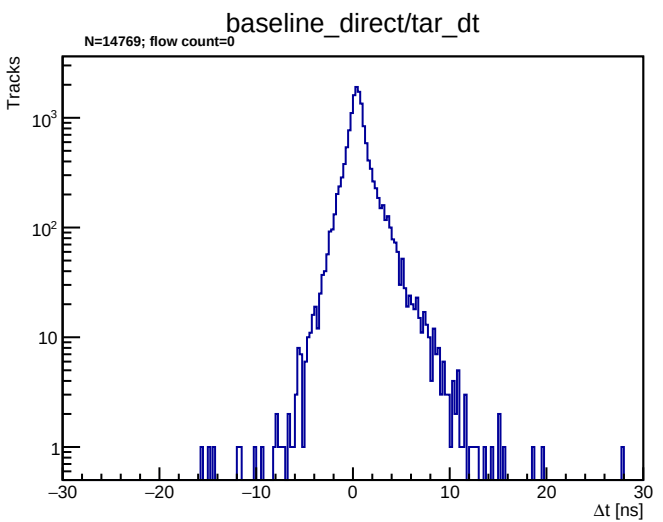

baseline_direct/ent_pitch_vs_tar_phase
N=14769; flow count=80
tar_phase
ent_pitch

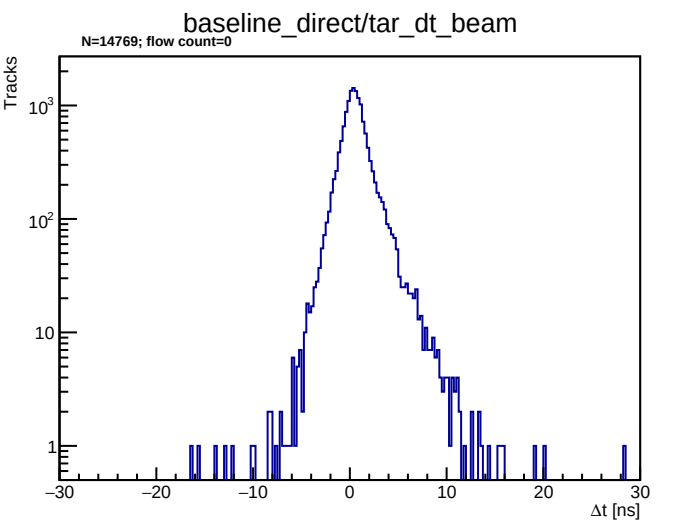

baseline_direct/tar_center
N=14769; flow count=4
Tracks
Δc_B [mm]

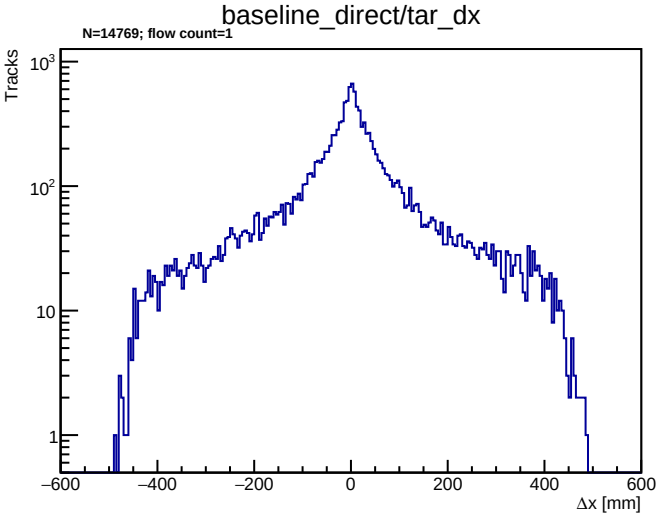

baseline_direct/tar_dp
N=14769; flow count=55
Tracks
Δp [MeV/c]

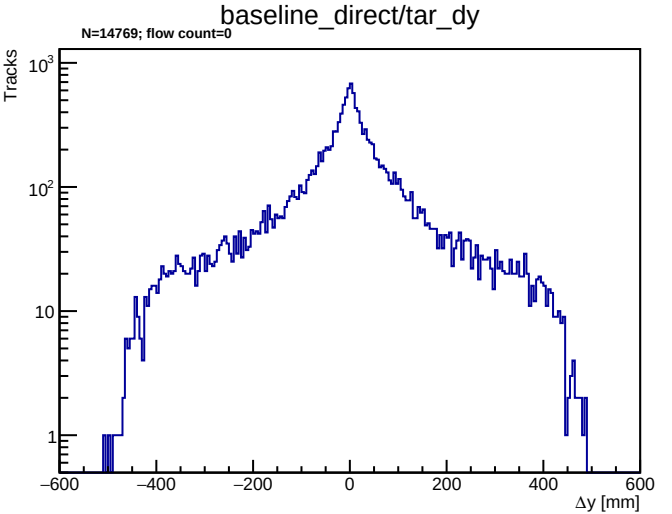

baseline_direct/tar_dpz
N=14769; flow count=26
Tracks
10^3
10^2
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

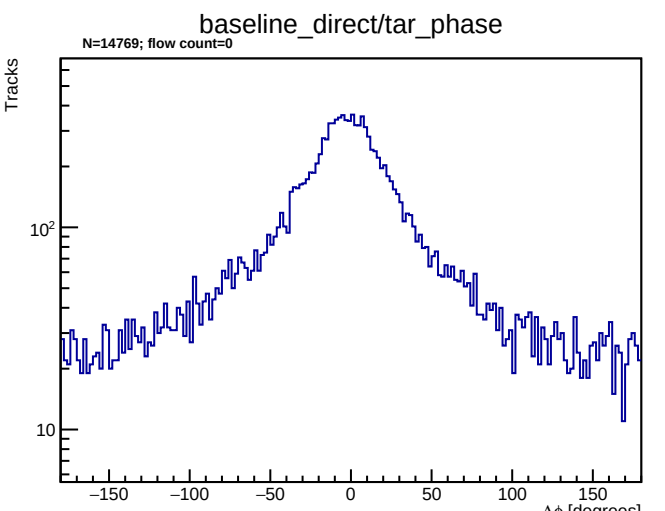

baseline_direct/tar_dr
N=14769; flow count=0
Tracks
10^3
10^2
10
1
0 100 200 300 400 500 600 700 800 900 1000
Δr_{xy} [mm]

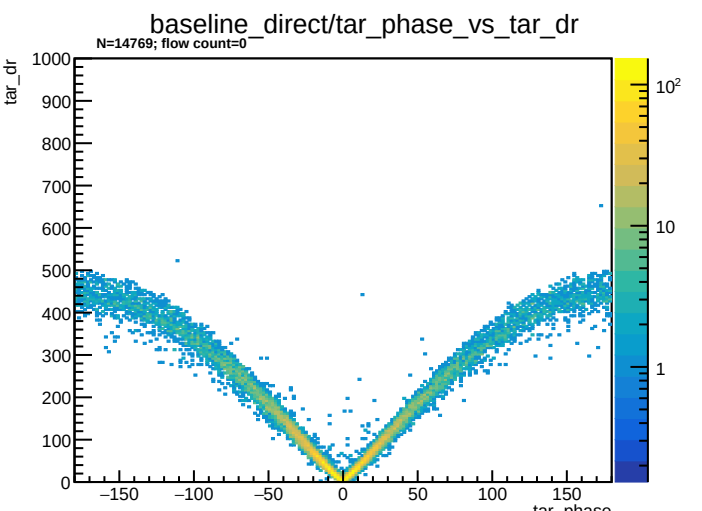

baseline_direct/tar_dt
N=14769; flow count=0
Tracks
10^3
10^2
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

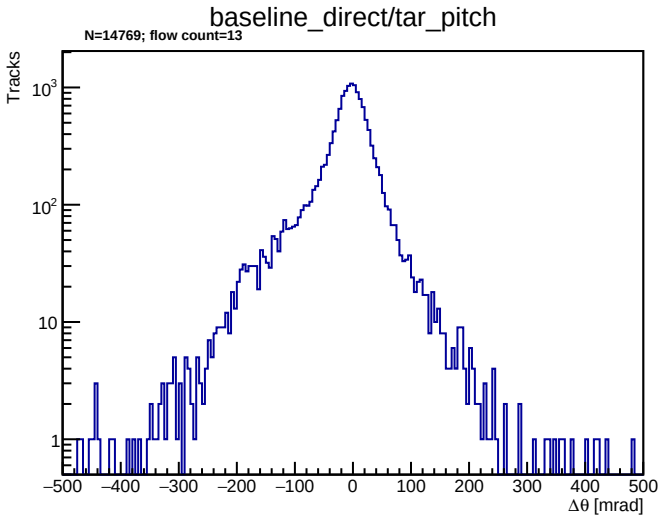

baseline_direct/tar_dt_beam
N=14769; flow count=0
Tracks
10^3
10^2
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

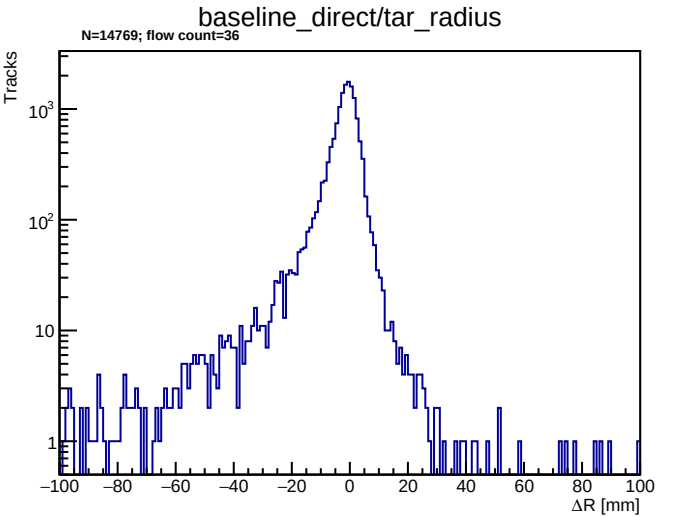

baseline_direct/tar_dx
N=14769; flow count=1
Tracks
Δx [mm]

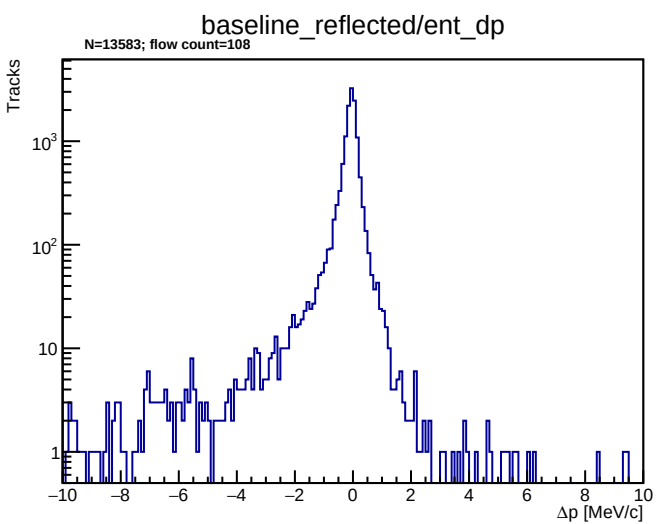

baseline_direct/tar_dy
N=14769; flow count=0
Tracks
Δy [mm]

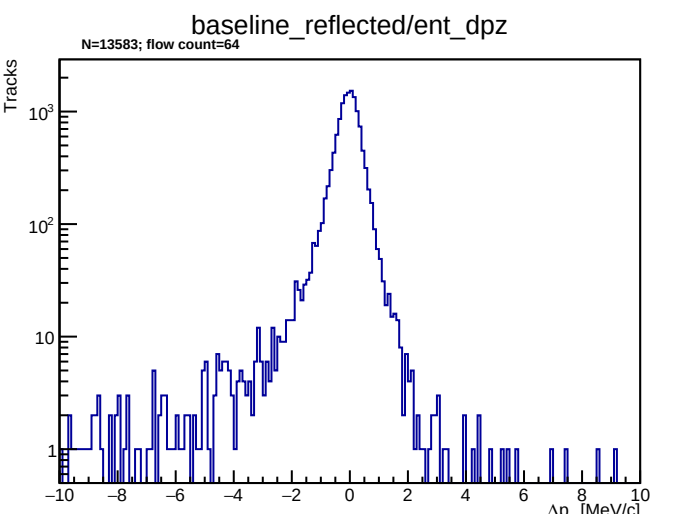

baseline_direct/tar_phase
N=14769; flow count=0
Tracks
Δϕ [degrees]

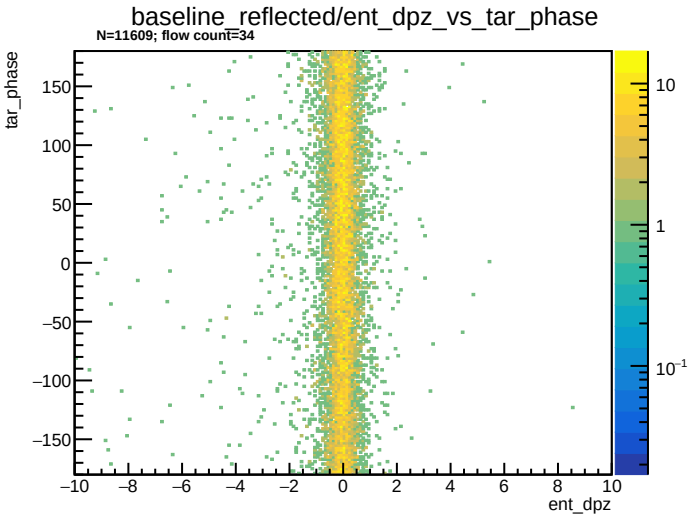

baseline_direct/tar_phase_vs_tar_dr
N=14769; flow count=0
tar_dr
tar_phase

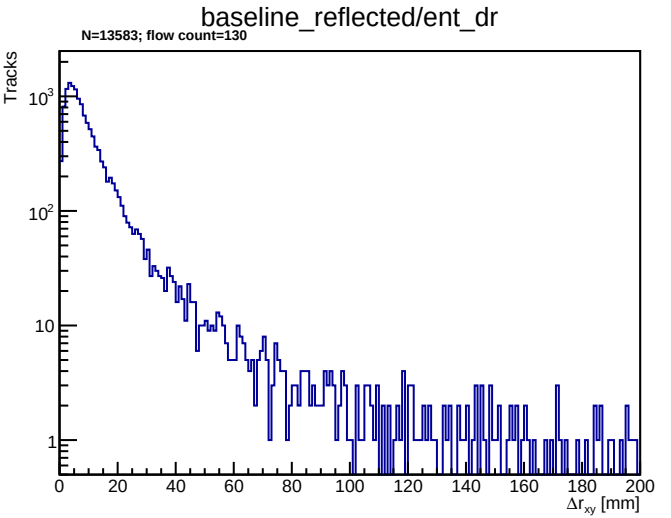

baseline_direct/tar_pitch
N=14769; flow count=13
Tracks
Δθ [mrad]

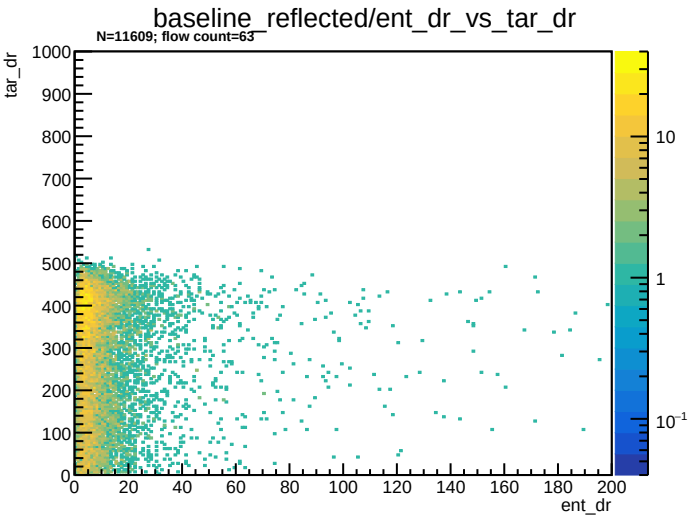

baseline_direct/tar_radius
N=14769; flow count=36
Tracks
ΔR [mm]

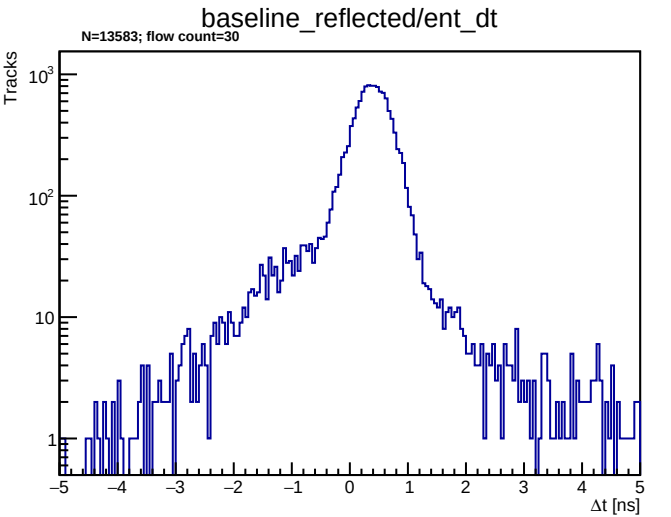

baseline_reflected/ent_dp
N=13583; flow count=108
Tracks
Δp [MeV/c]

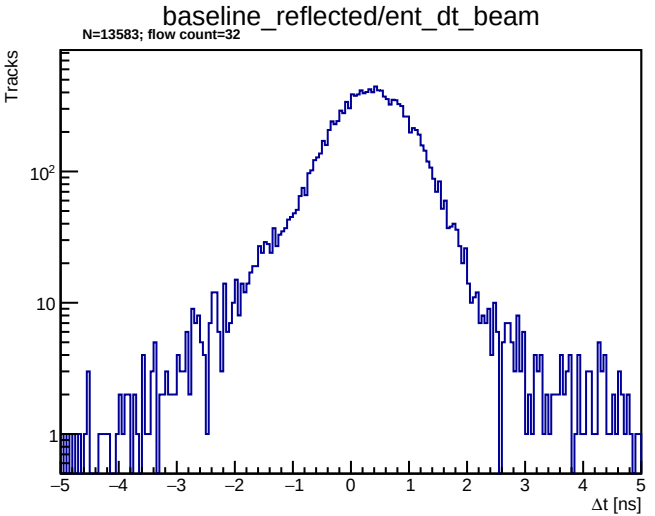

baseline_reflected/ent_dpz
N=13583; flow count=64
Tracks
Δp_z [MeV/c]

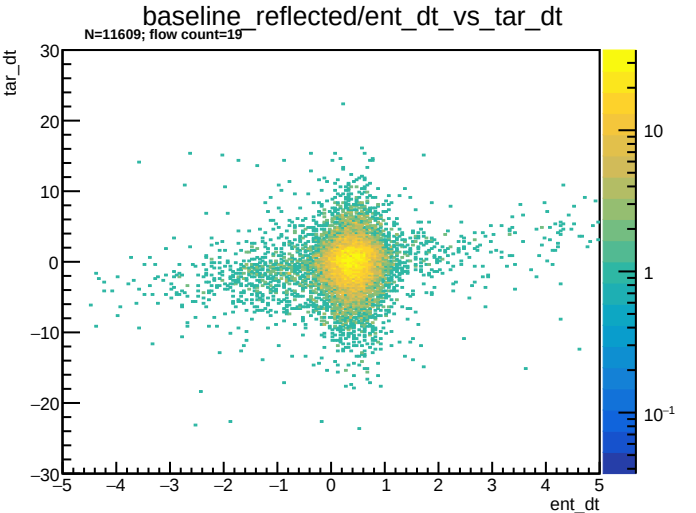

baseline_reflected/ent_dpz_vs_tar_phase
N=11609; flow count=34
tar_phase
ent_dpz

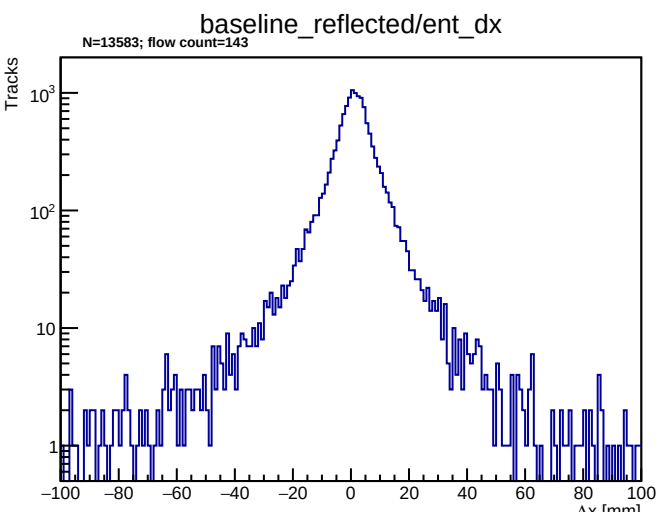

baseline_reflected/ent_dr
N=13583; flow count=130
Tracks
Δr_xy [mm]

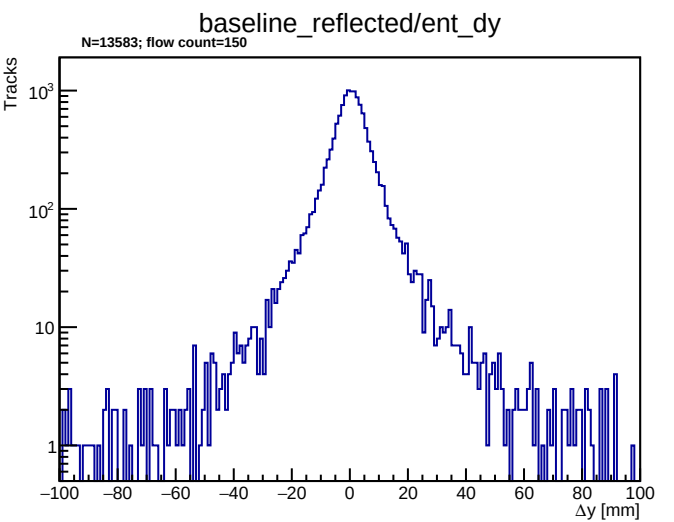

baseline_reflected/ent_dr_vs_tar_dr
N=11609; flow count=63
tar_dr
ent_dr

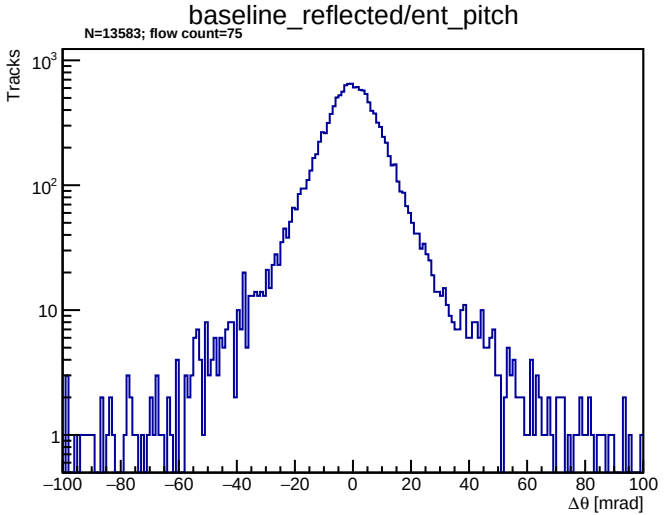

baseline_reflected/ent_dt
N=13583; flow count=30
Tracks
Δt [ns]

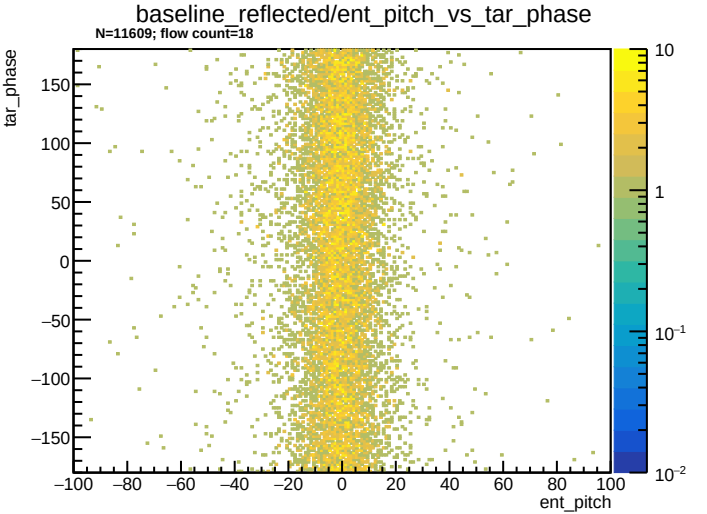

baseline_reflected/ent_dt_beam
N=13583; flow count=32
Tracks
Δt [ns]

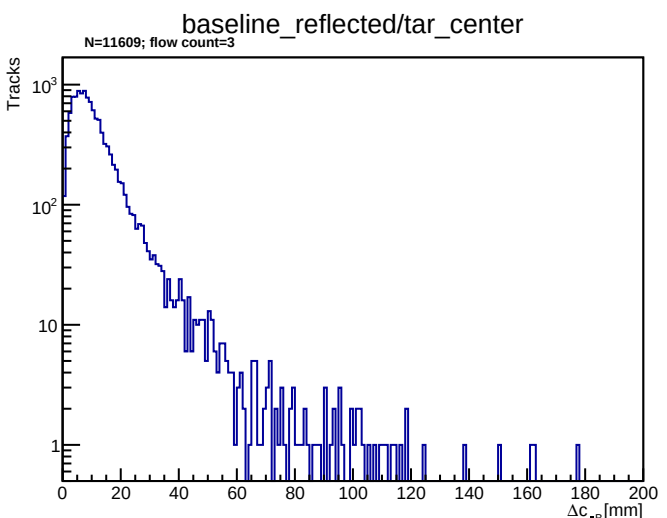

baseline_reflected/ent_dt_vs_tar_dt
N=11609; flow count=19
tar_dt
ent_dt

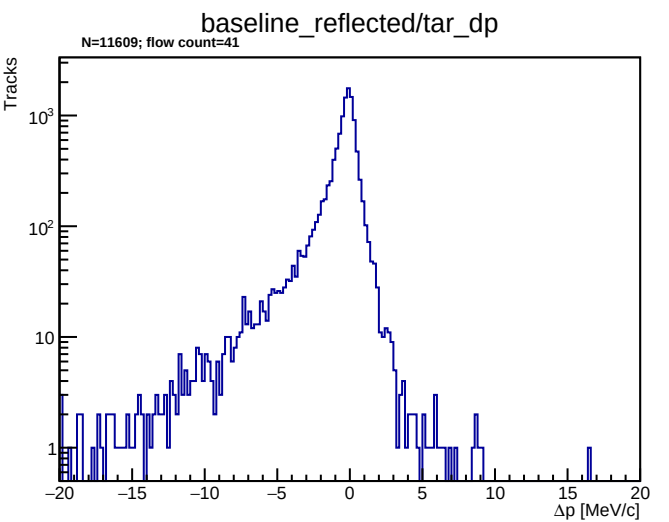

baseline_reflected/ent_dx
N=13583; flow count=143
Tracks
Δx [mm]

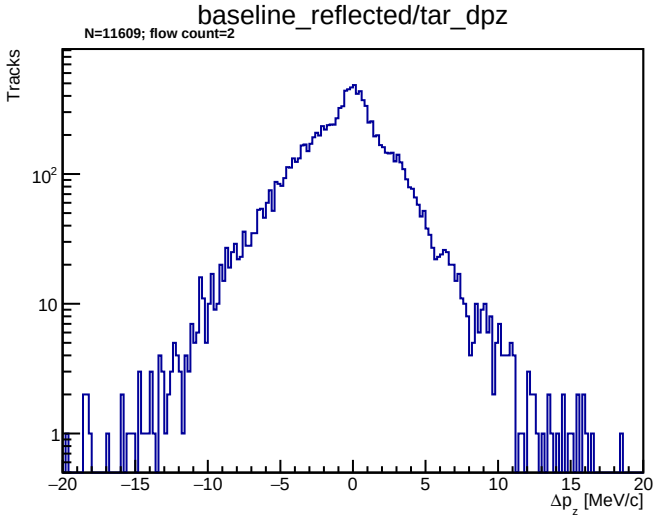

baseline_reflected/ent_dy
N=13583; flow count=150
Tracks
Δy [mm]

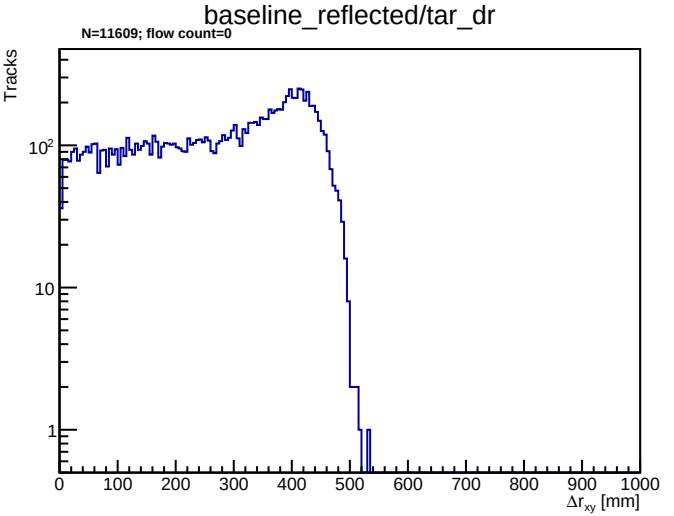

baseline_reflected/ent_pitch
N=13583; flow count=75
Tracks
Δθ [mrad]

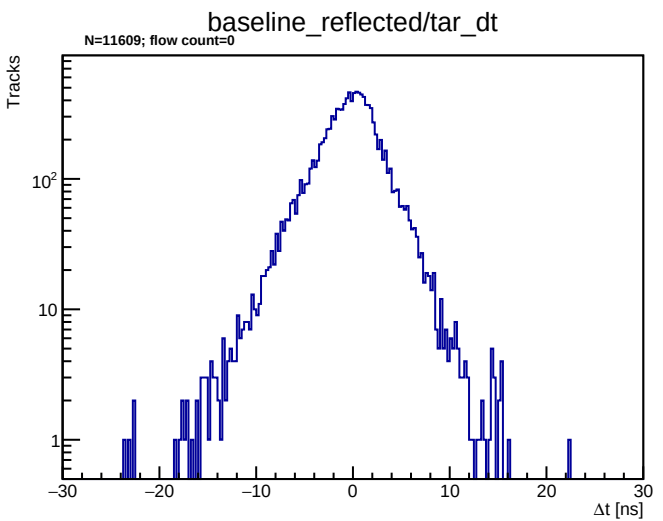

baseline_reflected/ent_pitch_vs_tar_phase
N=11609; flow count=18
tar_phase
ent_pitch

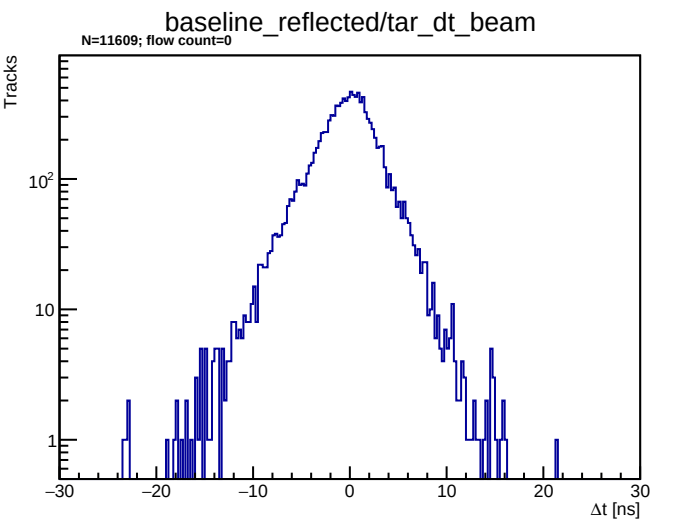

baseline_reflected/tar_center
N=11609; flow count=3
Tracks
Δc_B [mm]

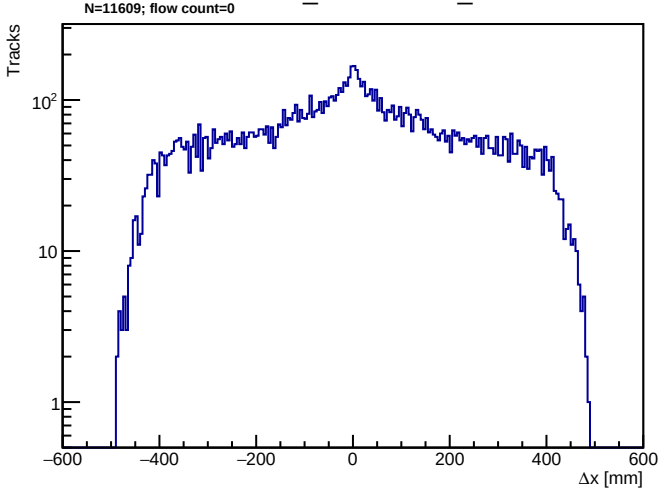

baseline_reflected/tar_dp
N=11609; flow count=41
Tracks
Δp [MeV/c]

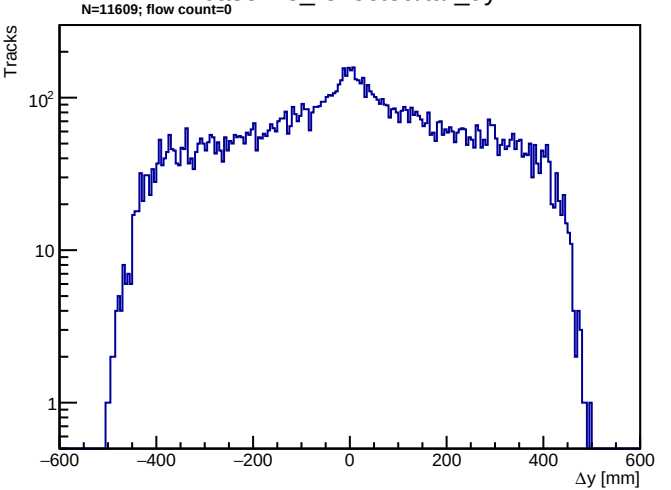

baseline_reflected/tar_dpz
N=11609; flow count=2
Tracks
10^2
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

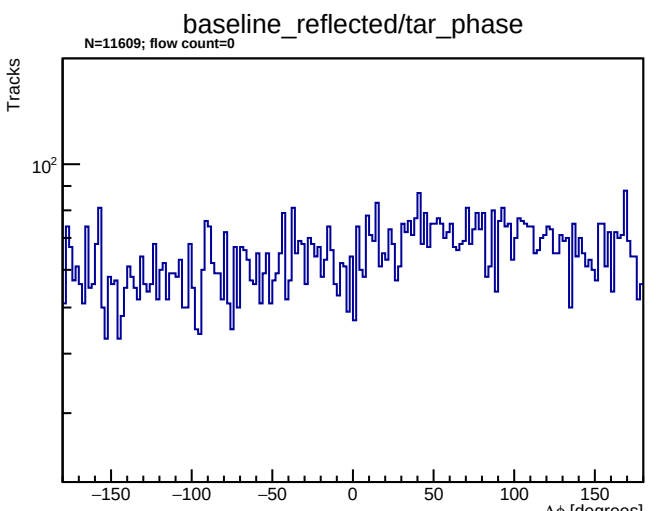

baseline_reflected/tar_dr
N=11609; flow count=0
Tracks
10^2
10
1
0 100 200 300 400 500 600 700 800 900 1000
Δr_{xy} [mm]

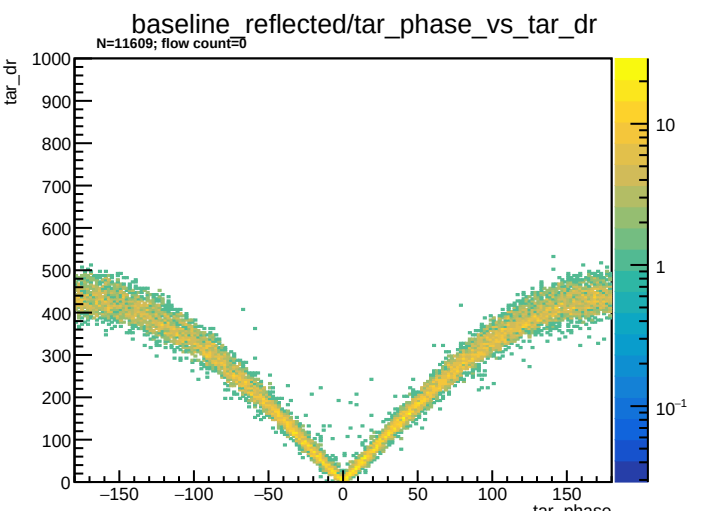

baseline_reflected/tar_dt
N=11609; flow count=0
Tracks
10^2
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

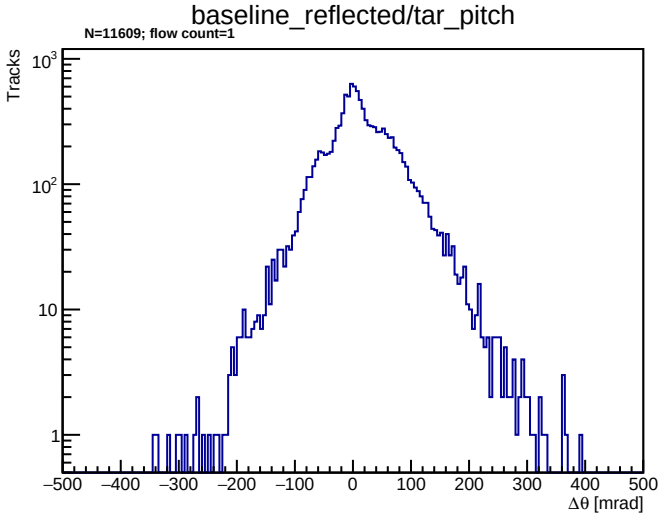

baseline_reflected/tar_dt_beam
N=11609; flow count=0
Tracks
10^2
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

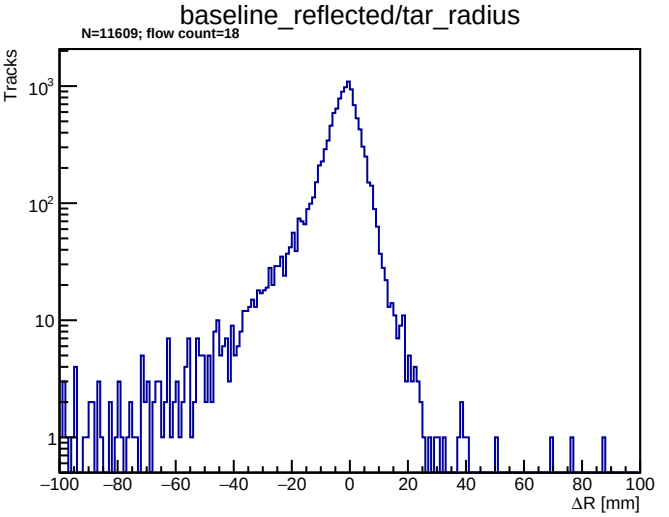

baseline_reflected/tar_dx
N=11609; flow count=0
Tracks
10^2
10
1
−600
−400
−200
0
200
400
600
Δx [mm]

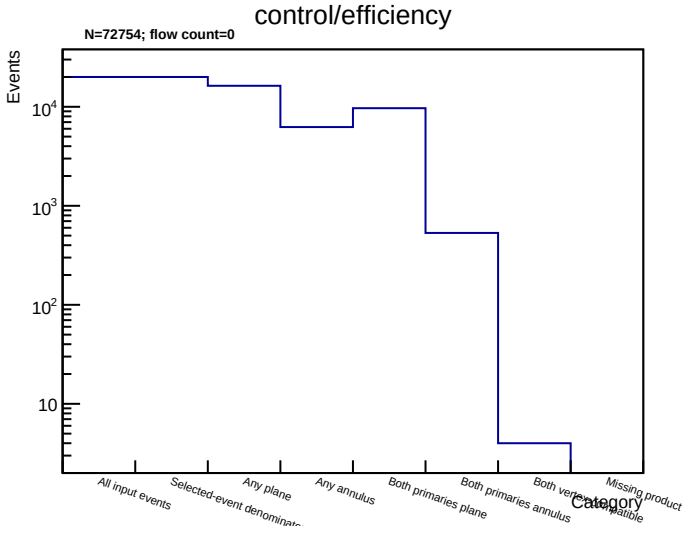

baseline_reflected/tar_dy
N=11609; flow count=0
Tracks
10^2
10
1
−600
−400
−200
0
200
400
600
Δy [mm]

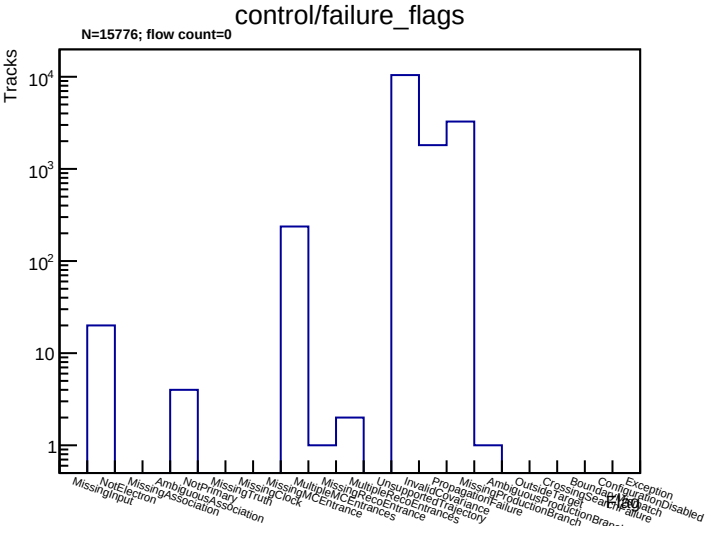

baseline_reflected/tar_phase
N=11609; flow count=0
Tracks
10^2
−150
−100
−50
0
50
100
150
Δϕ [degrees]

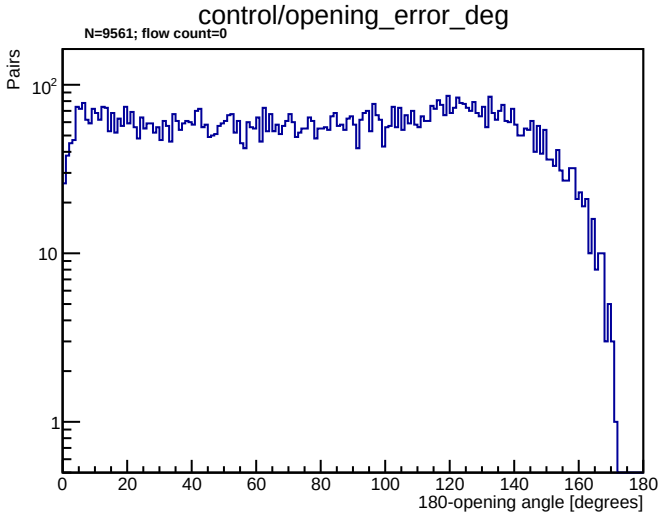

baseline_reflected/tar_phase_vs_tar_dr
N=11609; flow count=0
tar_dr
1000
900
800
700
600
500
400
300
200
100
0
−150
−100
−50
0
50
100
150
tar_phase
10
1
10^{-1}

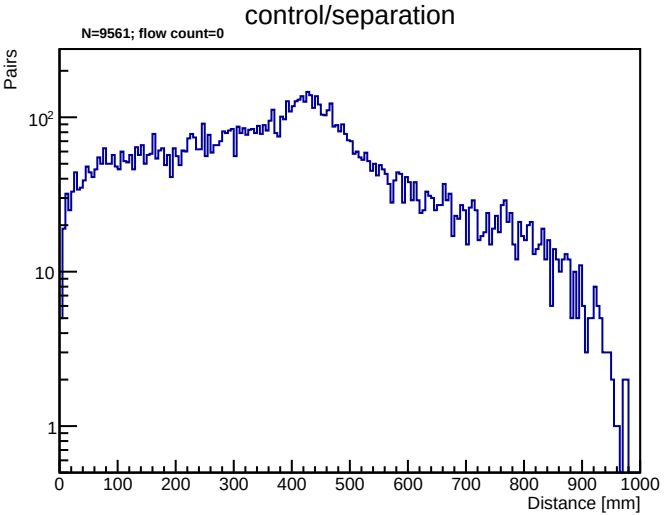

baseline_reflected/tar_pitch
N=11609; flow count=1
Tracks
Δθ [mrad]

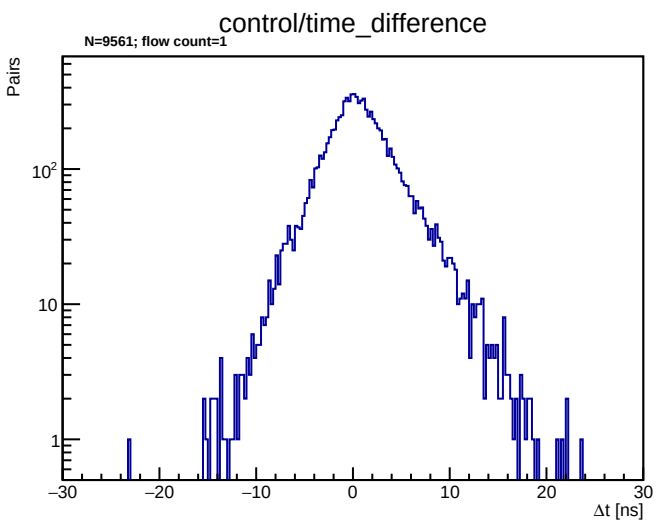

baseline_reflected/tar_radius
N=11609; flow count=18
Tracks
ΔR [mm]

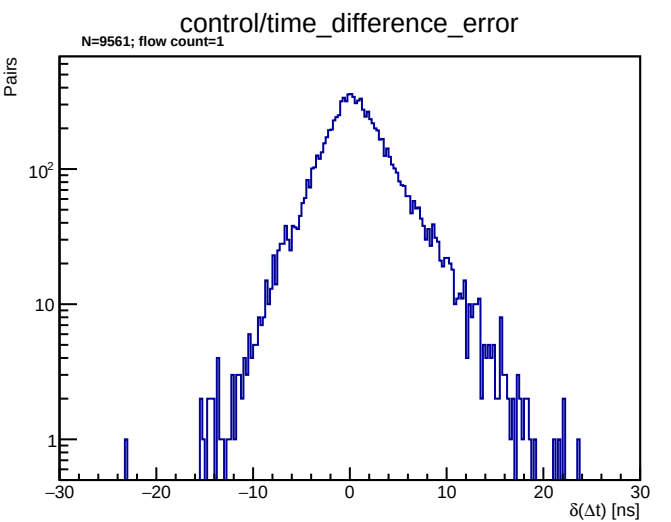

control/efficiency
N=72754; flow count=0
Events
Category
All input events
Any plane
Any annulus
Both primaries plane
Both primaries annulus
Missing product

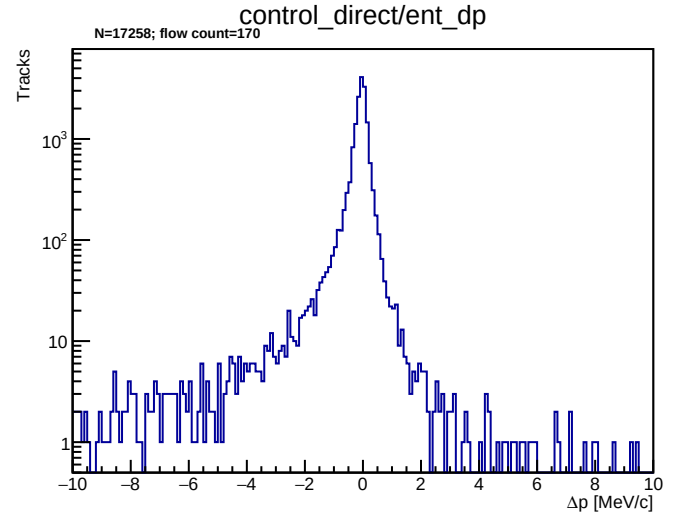

control/failure_flags
N=15776; flow count=0
Tracks
MissingInput
NotElectron
MissingAssociation
AmbiguousAssociation
NotPrimary
MissingTruth
MissingClock
MultipleMCEntrance
MultipleRecoEntrance
MultipleRecoEntrances
MissingRecoEntrance
UnsupportedTrajectory
InvalidCoordinate
PropagationFailure
MissingProductionBranch
AmbiguousProduction
OutsideTarget
CrossingSeam
BoundaryBranch
ConfigurationDisabled
Exception

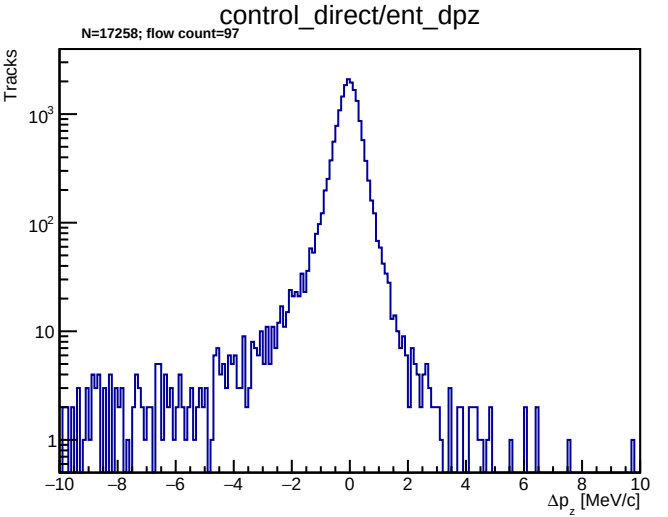

control/opening_error_deg
N=9561; flow count=0
Pairs
10^2
10
1
0
20
40
60
80
100
120
140
160
180
180-opening angle [degrees]

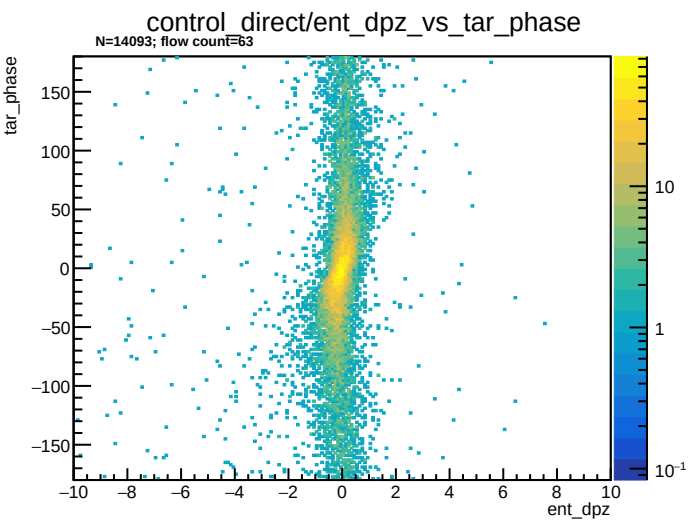

control/separation
N=9561; flow count=0
Pairs
10^2
10
1
0
100
200
300
400
500
600
700
800
900
1000
Distance [mm]

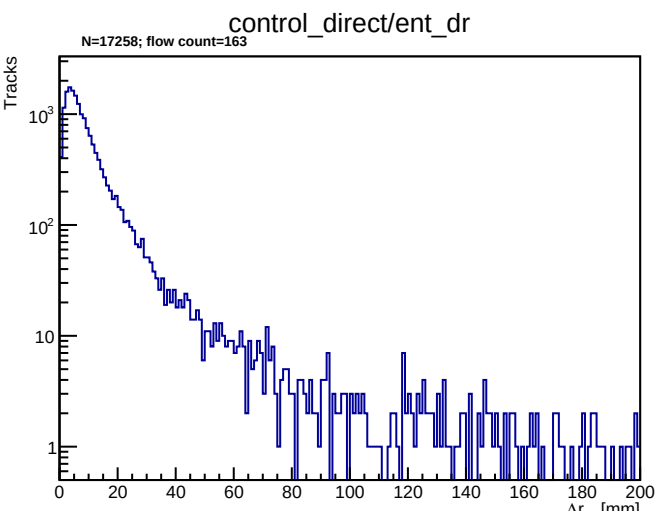

control/time_difference
N=9561; flow count=1
Pairs
10^2
10
1
−30
−20
−10
0
10
20
30
Δt [ns]

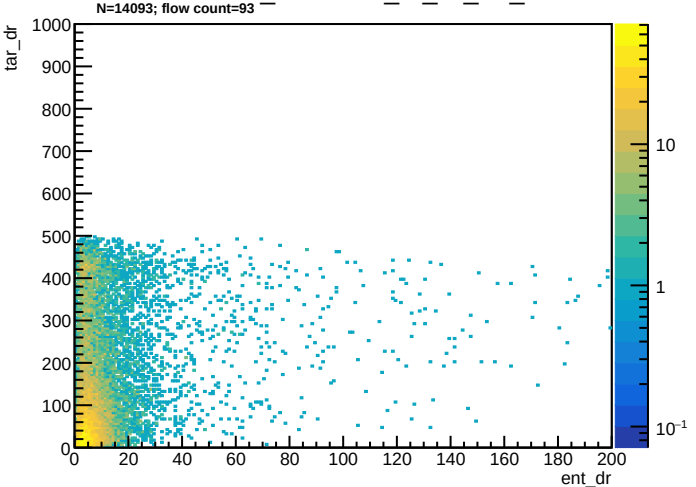

control/time_difference_error
N=9561; flow count=1
Pairs
10^2
10
1
−30
−20
−10
0
10
20
30
δ(Δt) [ns]

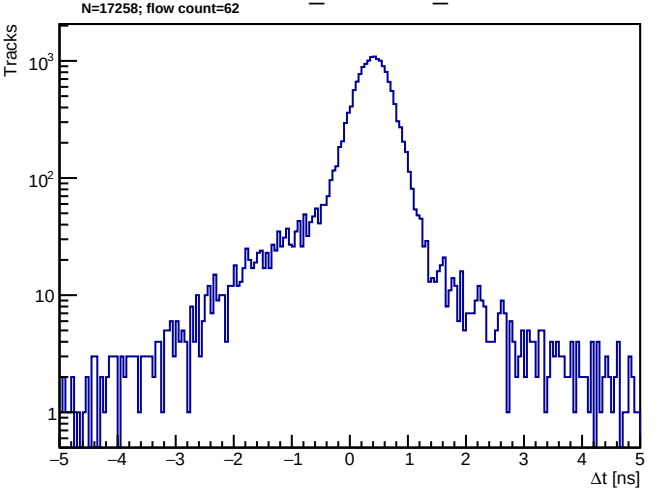

control_direct/ent_dp
N=17258; flow count=170
Tracks
Δp [MeV/c]

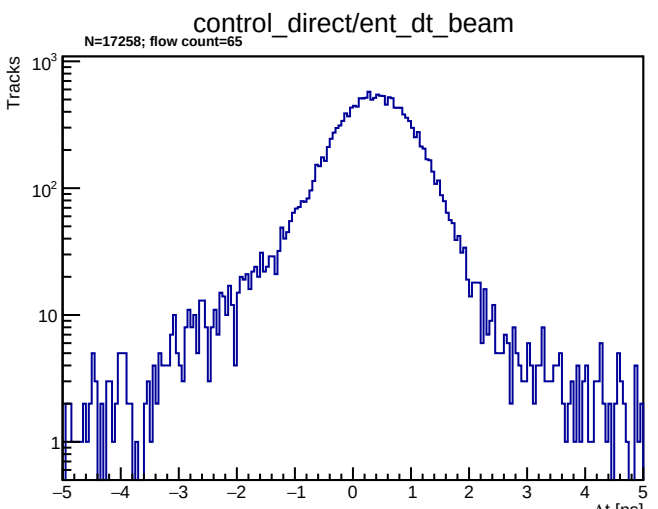

control_direct/ent_dpz
N=17258; flow count=97
Tracks
Δp_z [MeV/c]

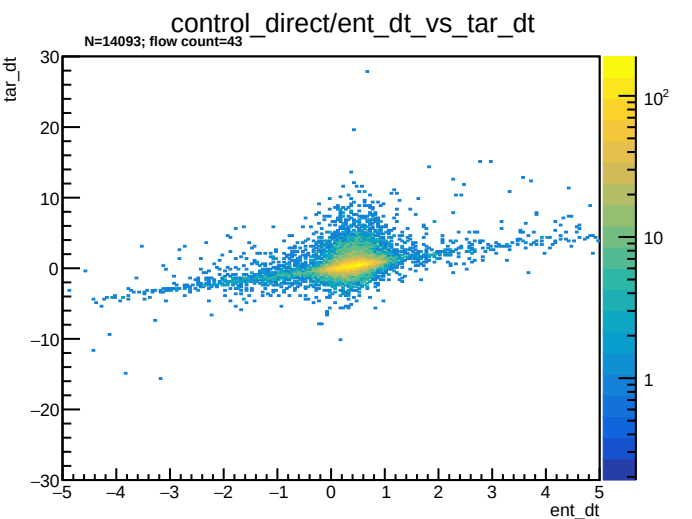

control_direct/ent_dpz_vs_tar_phase
N=14093; flow count=63
tar_phase
ent_dpz

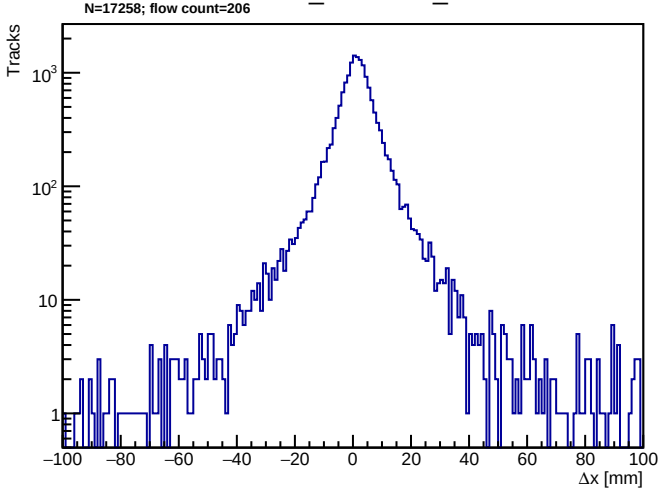

control_direct/ent_dr
N=17258; flow count=163
Tracks
Δr_xy [mm]

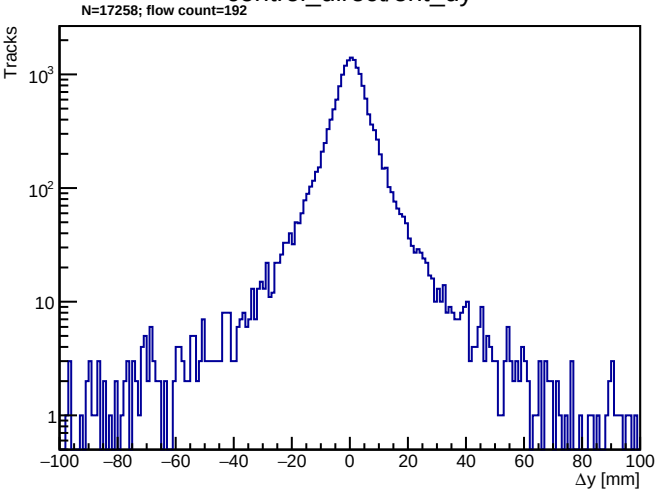

control_direct/ent_dr_vs_tar_dr
N=14093; flow count=93
tar_dr
ent_dr

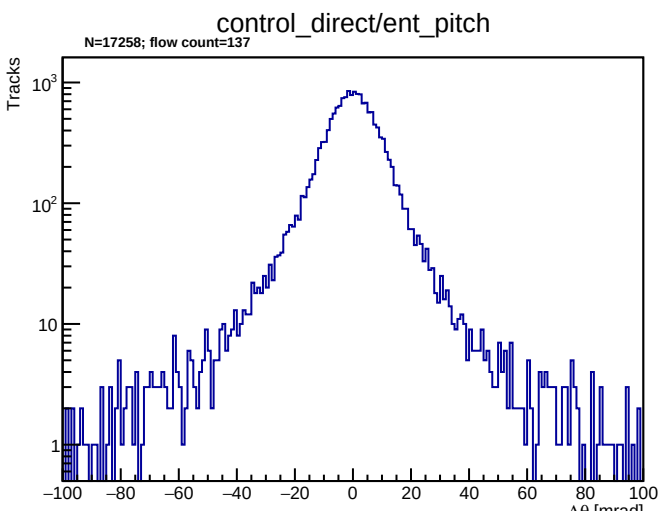

control_direct/ent_dt
N=17258; flow count=62
Tracks
Δt [ns]

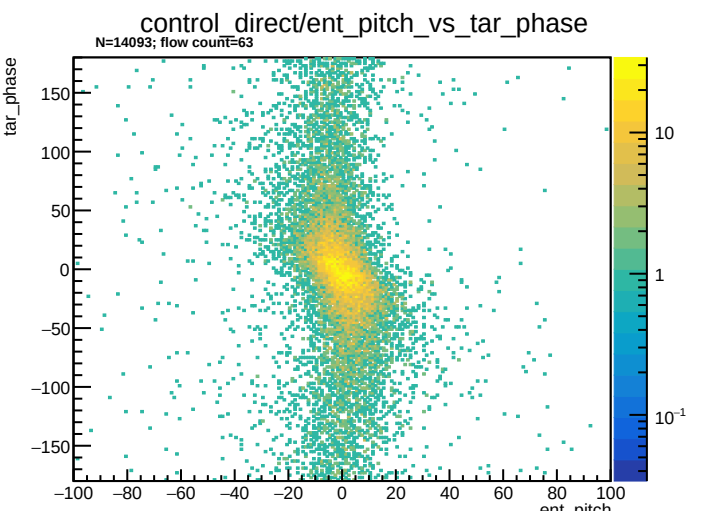

control_direct/ent_dt_beam
N=17258; flow count=65
Tracks
Δt [ns]

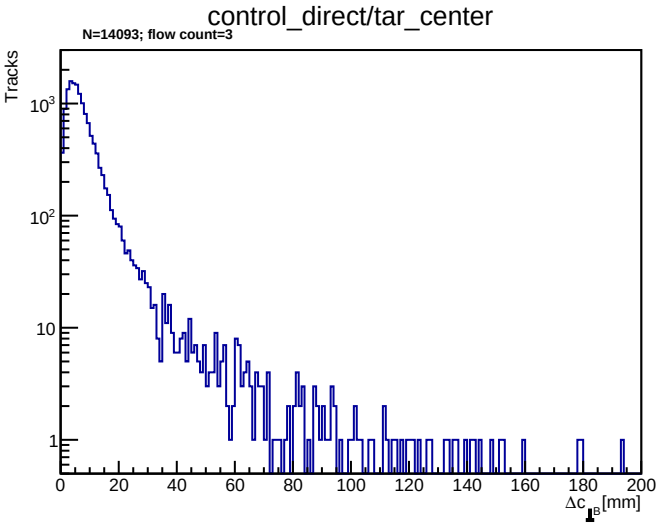

control_direct/ent_dt_vs_tar_dt
N=14093; flow count=43
tar_dt
ent_dt

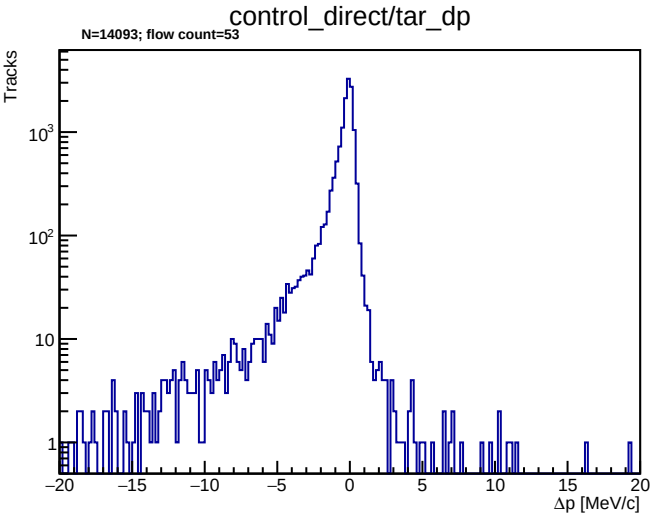

control_direct/ent_dx
N=17258; flow count=206
Tracks
Δx [mm]

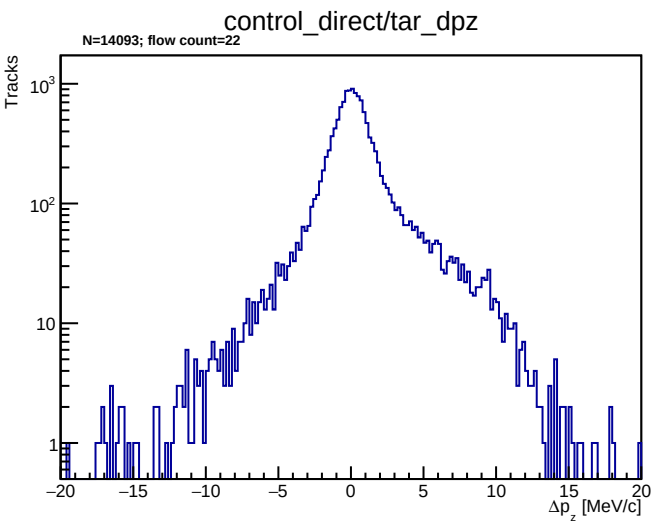

control_direct/ent_dy
N=17258; flow count=192
Tracks
Δy [mm]

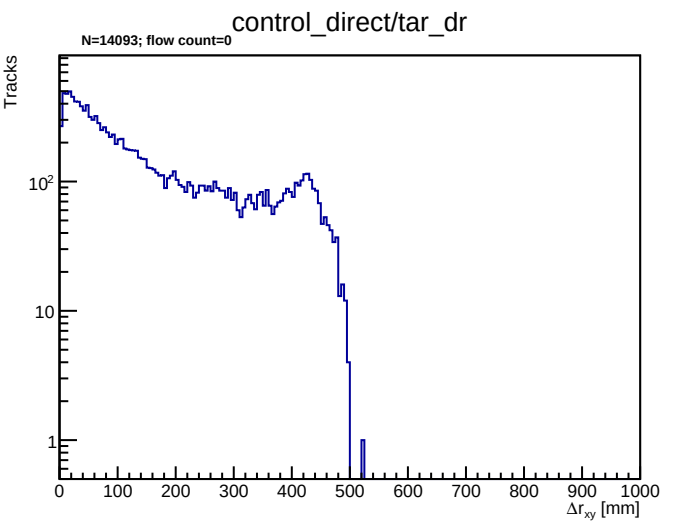

control_direct/ent_pitch
N=17258; flow count=137
Tracks
Δθ [mrad]

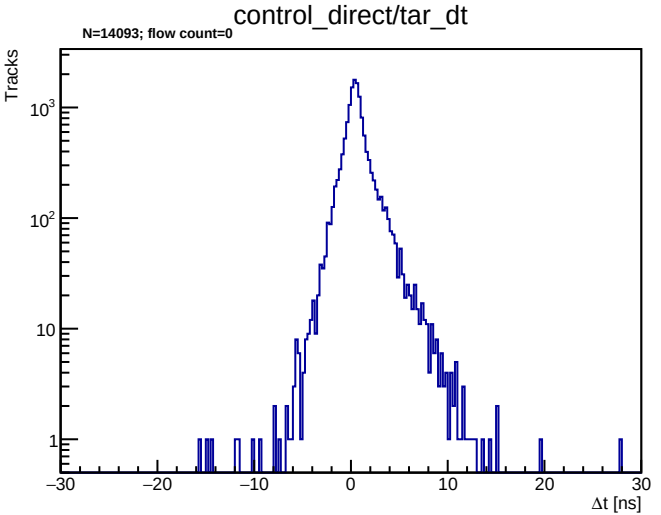

control_direct/ent_pitch_vs_tar_phase
N=14093; flow count=63
tar_phase
ent_pitch

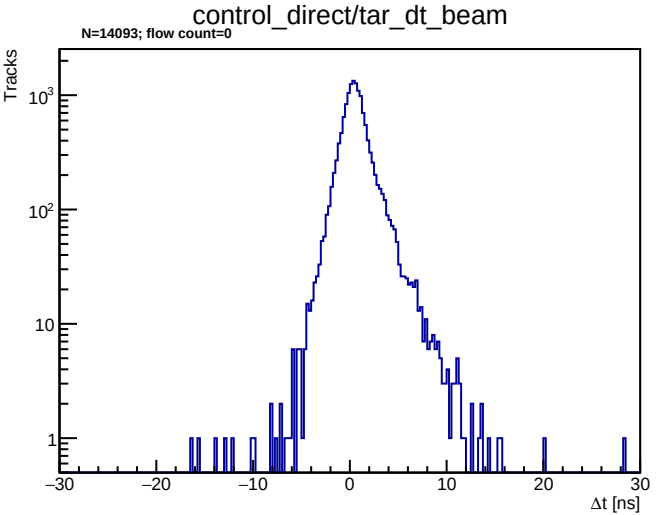

control_direct/tar_center
N=14093; flow count=3
Tracks
10^3
10^2
10
1
0 20 40 60 80 100 120 140 160 180 200
Δc_B [mm]

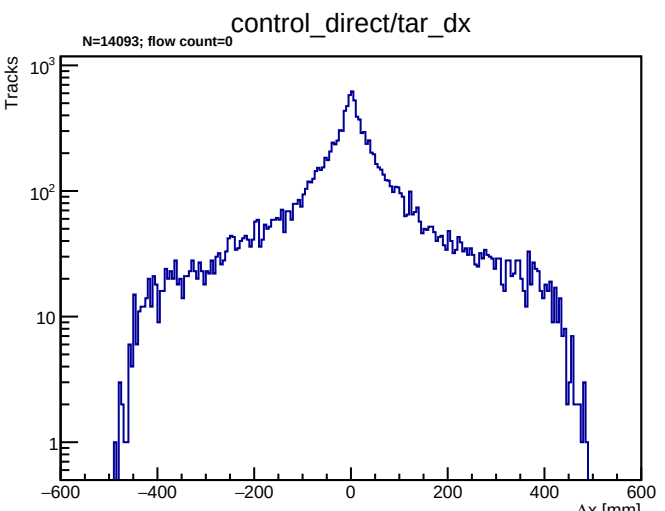

control_direct/tar_dp
N=14093; flow count=53
Tracks
10^3
10^2
10
1
–20 –15 –10 –5 0 5 10 15 20
Δp [MeV/c]

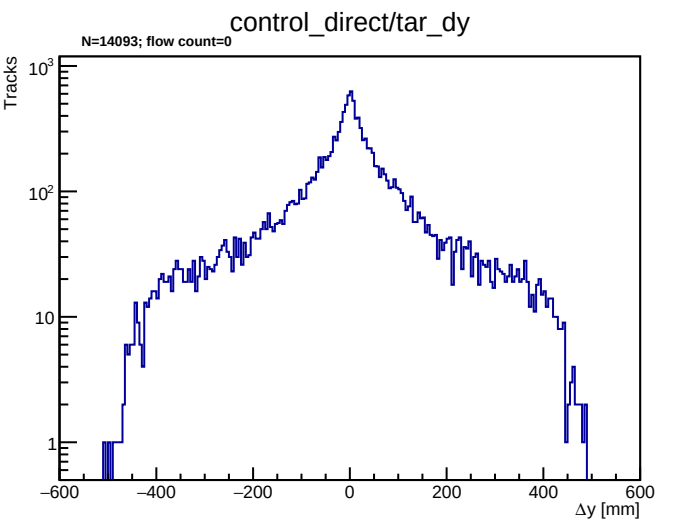

control_direct/tar_dpz
N=14093; flow count=22
Tracks
10^3
10^2
10
1
–20 –15 –10 –5 0 5 10 15 20
Δp_z [MeV/c]

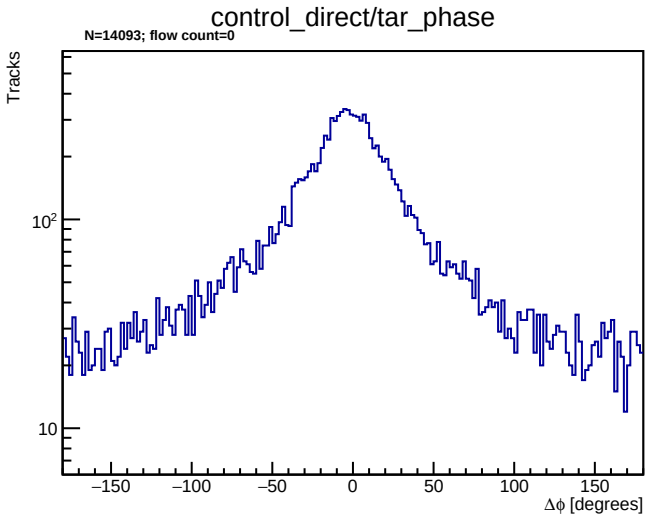

control_direct/tar_dr
N=14093; flow count=0
Tracks
10^2
10
1
0 100 200 300 400 500 600 700 800 900 1000
Δr_{xy} [mm]

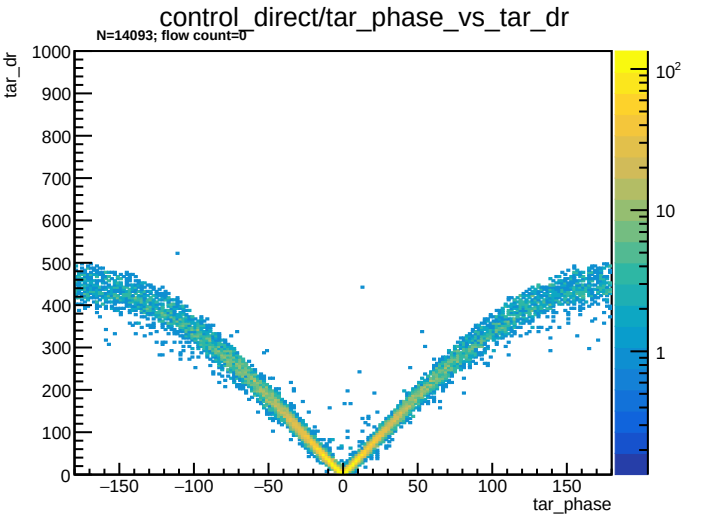

control_direct/tar_dt
N=14093; flow count=0
Tracks
Δt [ns]

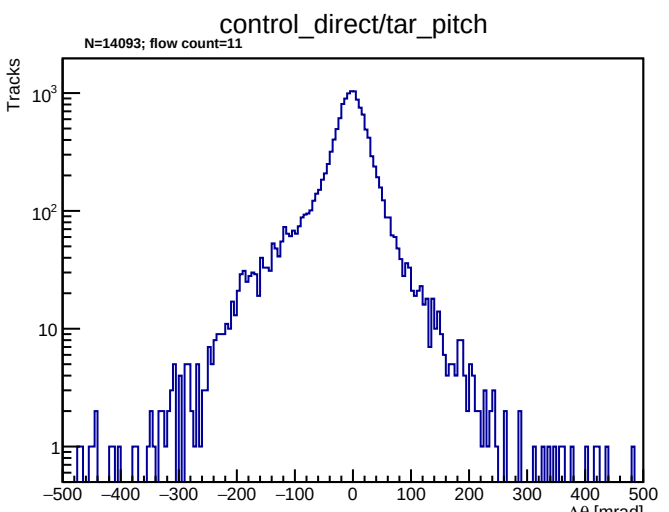

control_direct/tar_dt_beam
N=14093; flow count=0
Tracks
Δt [ns]

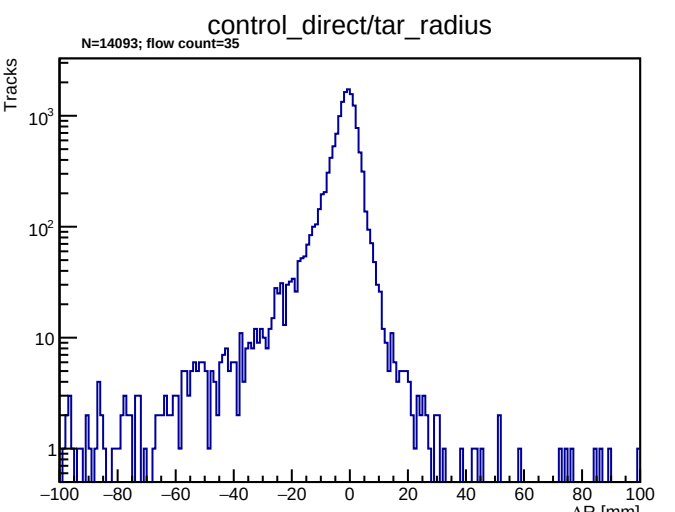

control_direct/tar_dx
N=14093; flow count=0
Tracks
Δx [mm]

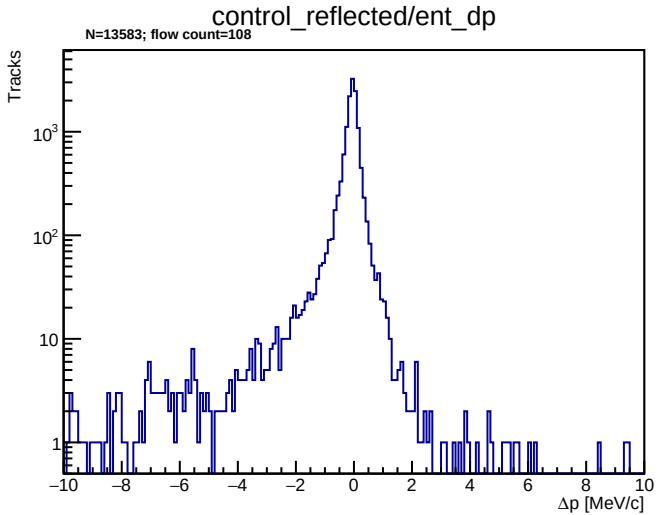

control_direct/tar_dy
N=14093; flow count=0
Tracks
Δy [mm]

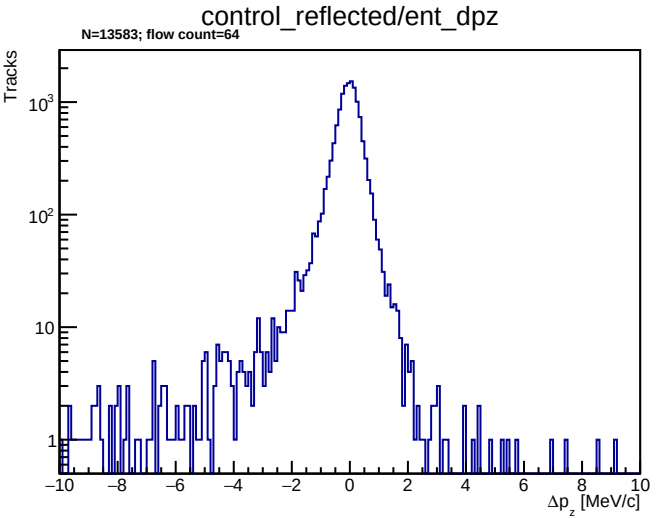

control_direct/tar_phase
N=14093; flow count=0
Tracks
10^2
10
−150 −100 −50 0 50 100 150
Δϕ [degrees]

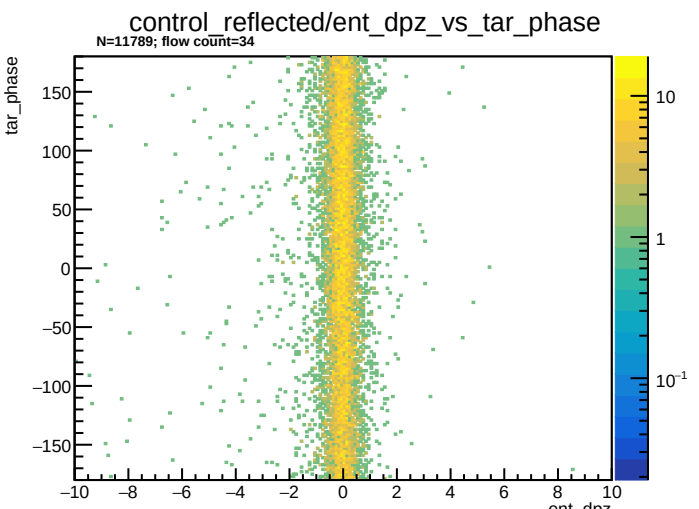

control_direct/tar_phase_vs_tar_dr
N=14093; flow count=0
tar_dr
1000 900 800 700 600 500 400 300 200 100 0
−150 −100 −50 0 50 100 150
tar_phase
10^2
10
1

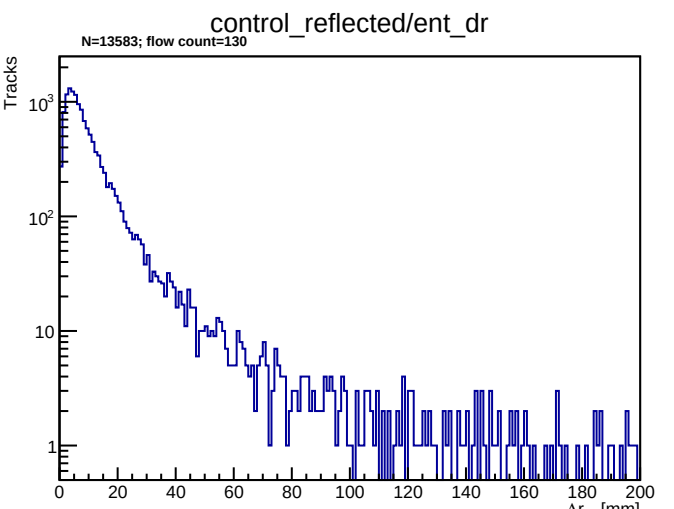

control_direct/tar_pitch
N=14093; flow count=11
Tracks
10^3
10^2
10
1
−500 −400 −300 −200 −100 0 100 200 300 400 500
Δθ [mrad]

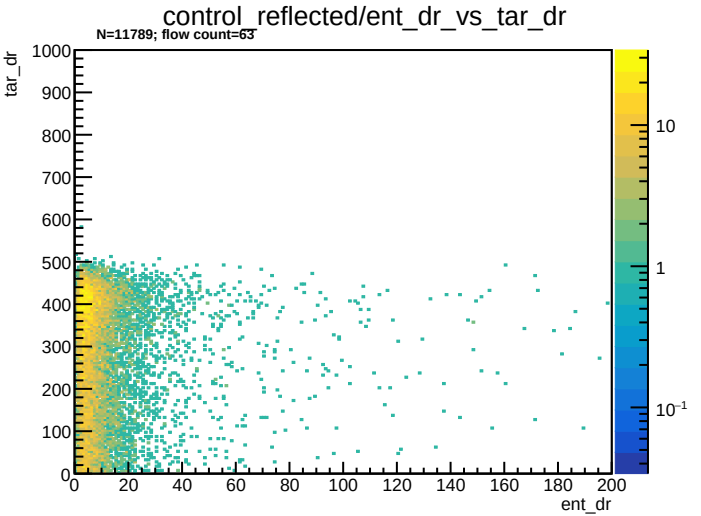

control_direct/tar_radius
N=14093; flow count=35
Tracks
10^3
10^2
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
ΔR [mm]

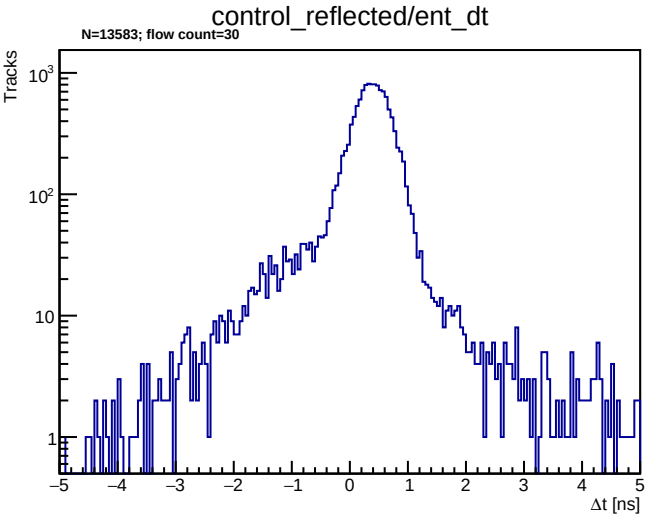

control_reflected/ent_dp
N=13583; flow count=108
Tracks
Δp [MeV/c]

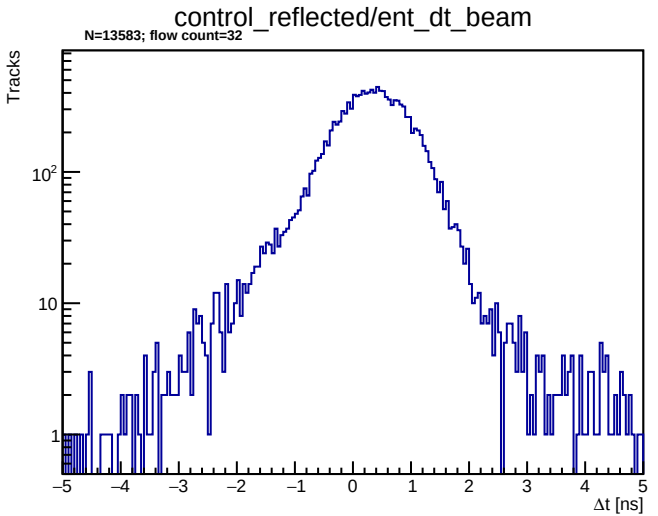

control_reflected/ent_dpz
N=13583; flow count=64
Tracks
Δp_z [MeV/c]

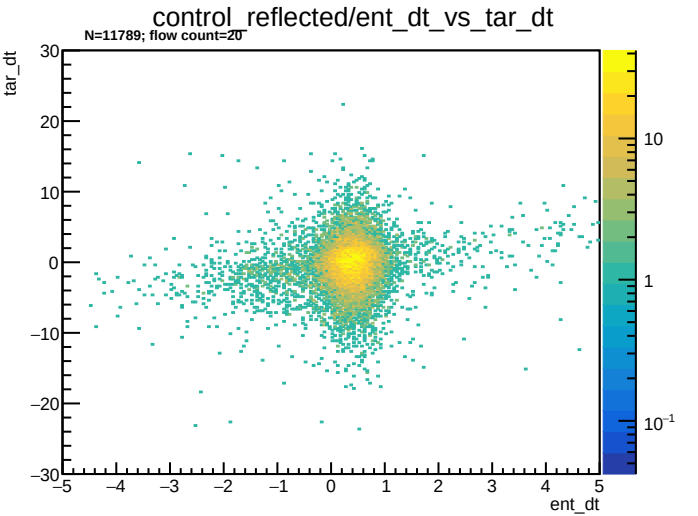

control_reflected/ent_dpz_vs_tar_phase
N=11789; flow count=34
tar_phase
ent_dpz

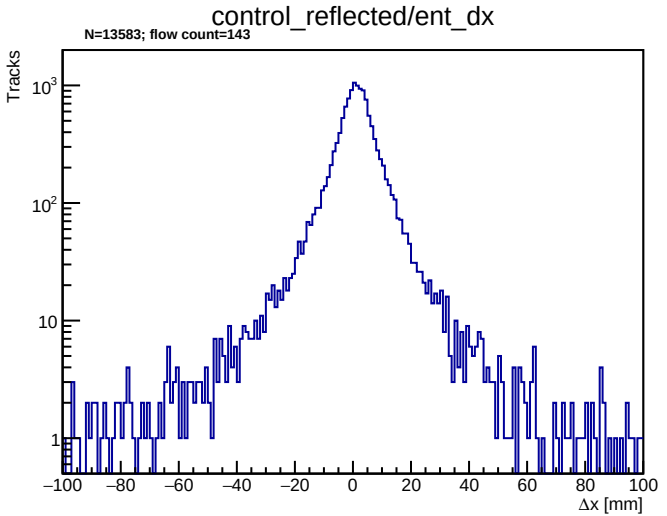

control_reflected/ent_dr
N=13583; flow count=130
Tracks
Δr_xy [mm]

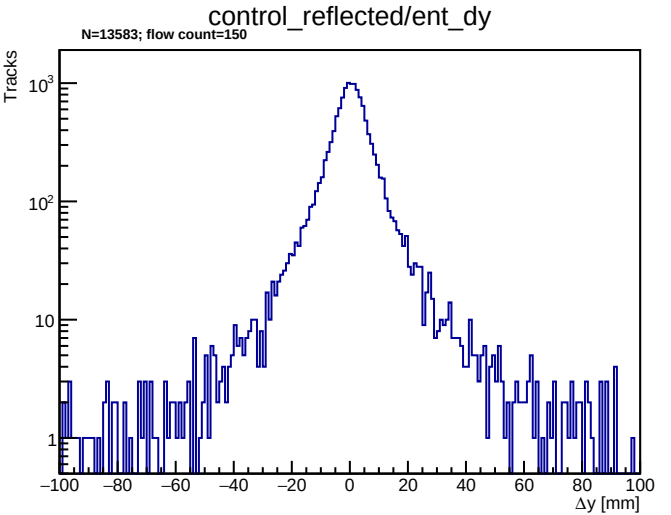

control_reflected/ent_dr_vs_tar_dr
N=11789; flow count=63
tar_dr
ent_dr

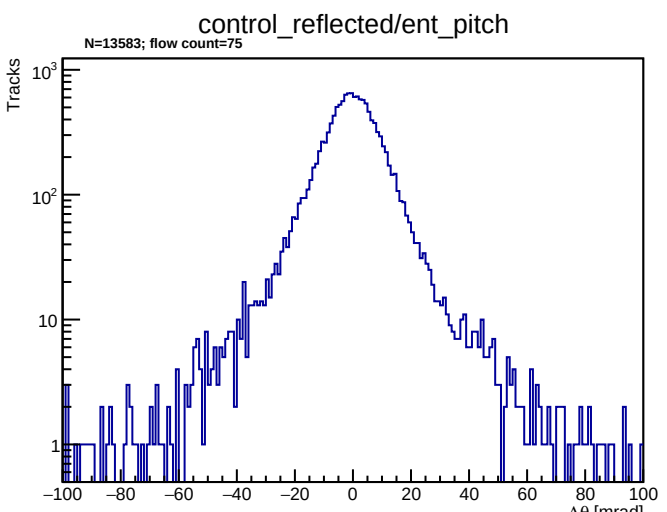

control_reflected/ent_dt
N=13583; flow count=30
Tracks
Δt [ns]

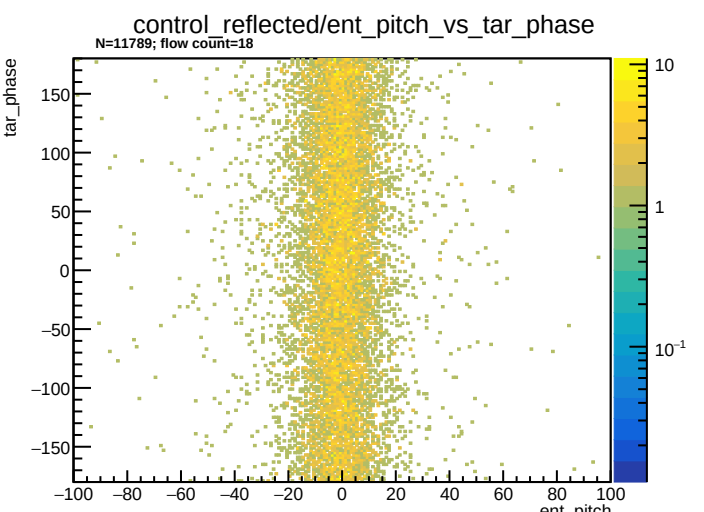

control_reflected/ent_dt_beam
N=13583; flow count=32
Tracks
Δt [ns]

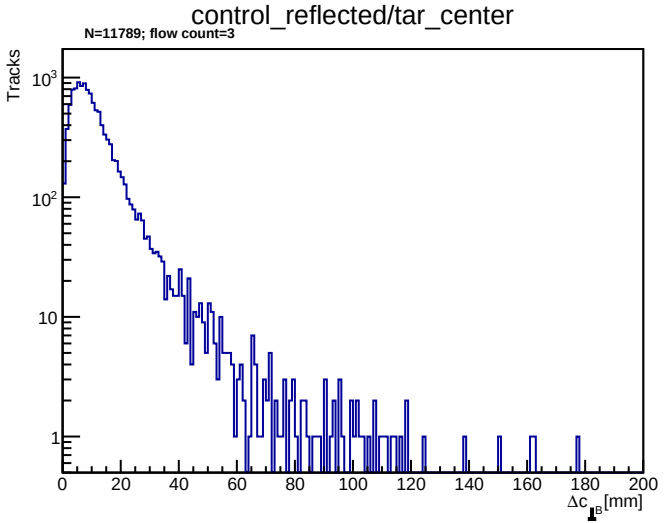

control_reflected/ent_dt_vs_tar_dt
N=11789; flow count=20
tar_dt
ent_dt

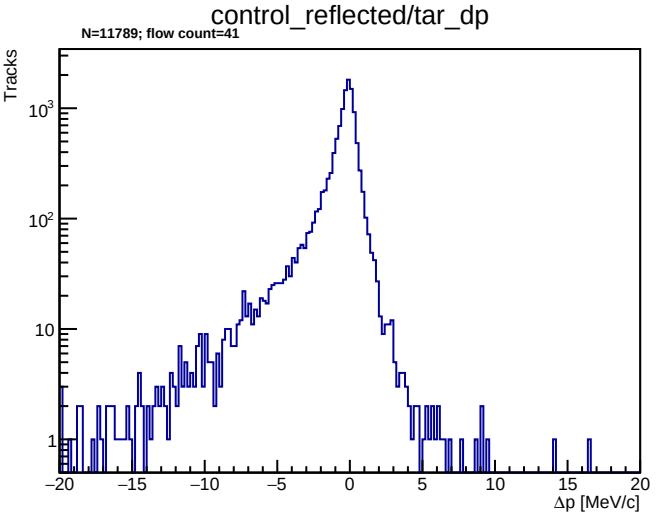

control_reflected/ent_dx
N=13583; flow count=143
Tracks
Δx [mm]

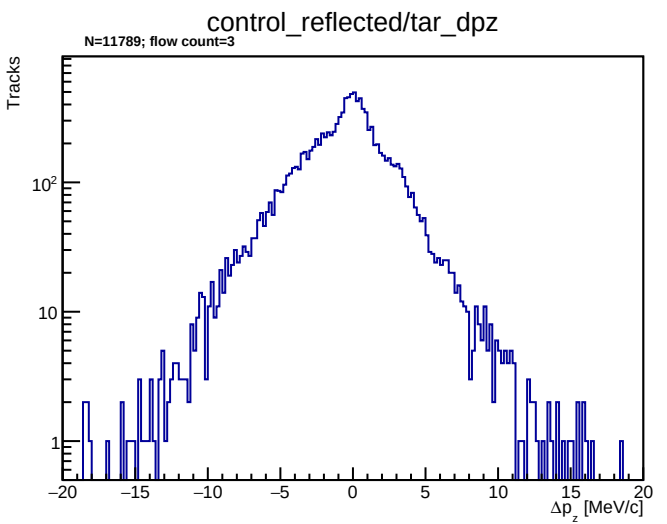

control_reflected/ent_dy
N=13583; flow count=150
Tracks
Δy [mm]

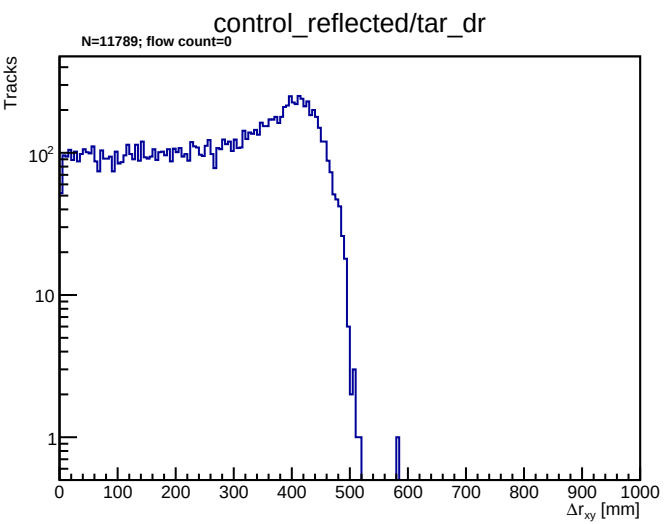

control_reflected/ent_pitch
N=13583; flow count=75
Tracks
Δθ [mrad]

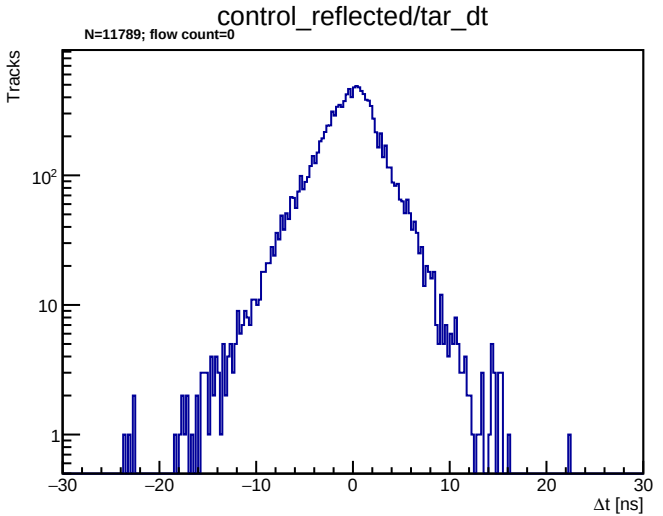

control_reflected/ent_pitch_vs_tar_phase
N=11789; flow count=18
tar_phase
ent_pitch

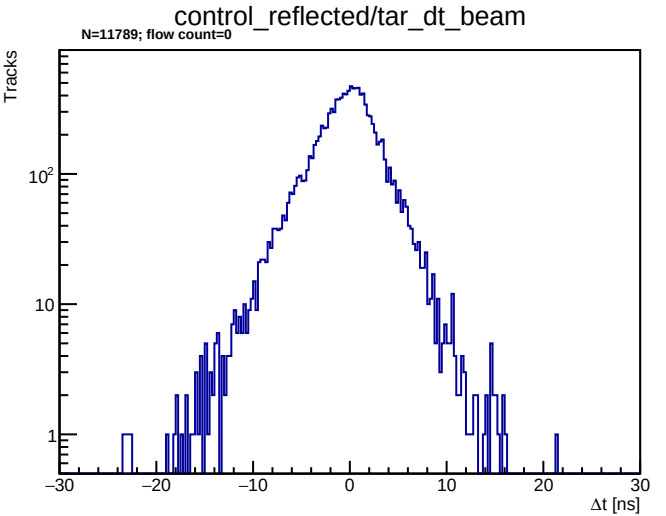

control_reflected/tar_center
N=11789; flow count=3
Tracks
10^3
10^2
10
1
0 20 40 60 80 100 120 140 160 180 200
Δc_B [mm]

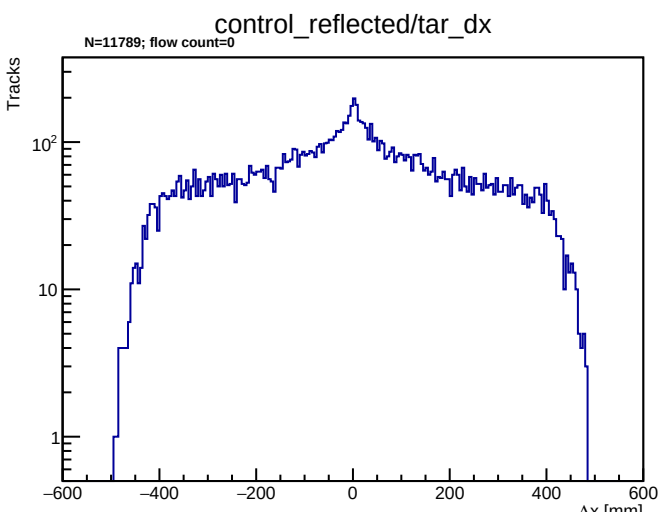

control_reflected/tar_dp
N=11789; flow count=41
Tracks
10^3
10^2
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp [MeV/c]

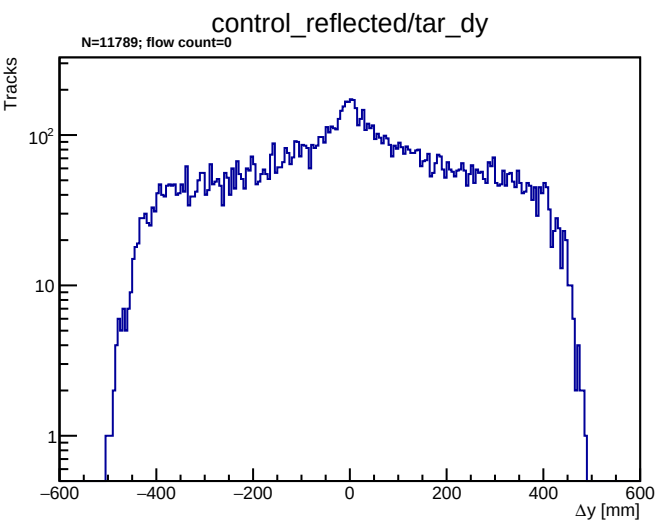

control_reflected/tar_dpz
N=11789; flow count=3
Tracks
10^2
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

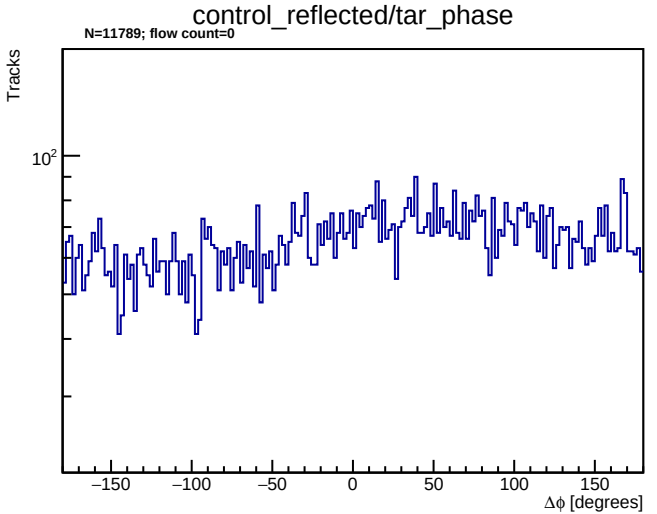

control_reflected/tar_dr
N=11789; flow count=0
Tracks
10^2
10
1
0 100 200 300 400 500 600 700 800 900 1000
Δr_{xy} [mm]

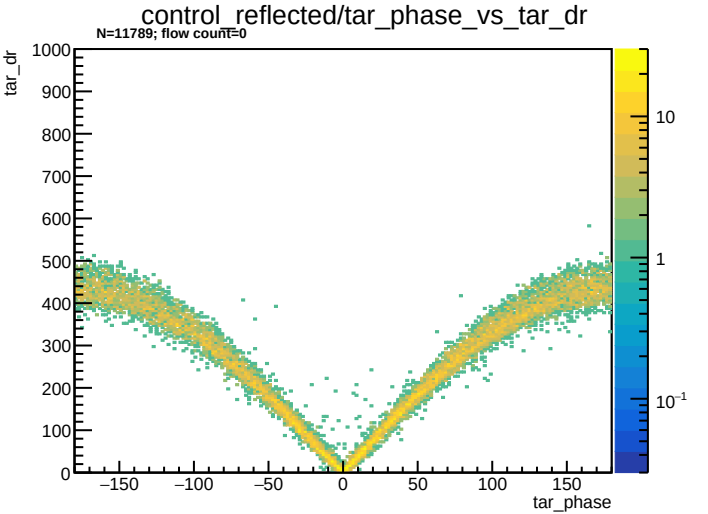

control_reflected/tar_dt
N=11789; flow count=0
Tracks
10^2
10
1
−30
−20
−10
0
10
20
30
Δt [ns]

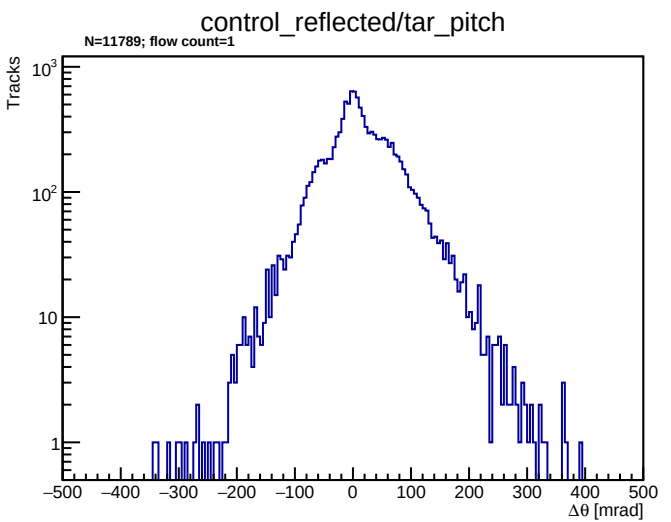

control_reflected/tar_dt_beam
N=11789; flow count=0
Tracks
10^2
10
1
−30
−20
−10
0
10
20
30
Δt [ns]

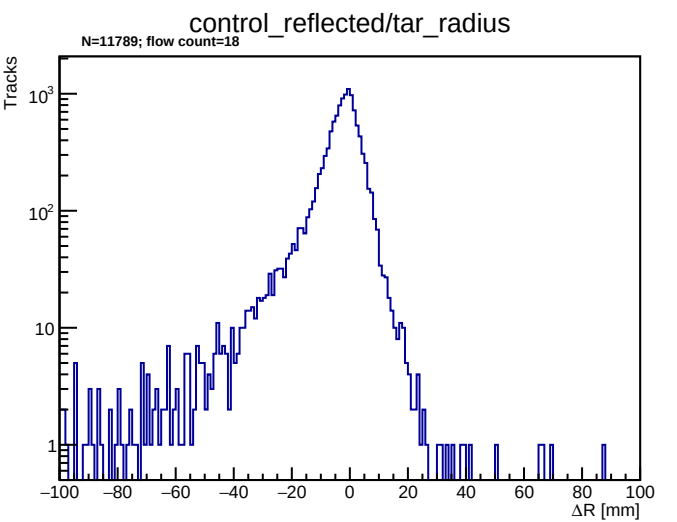

control_reflected/tar_dx
N=11789; flow count=0
Tracks
10^2
10
1
−600
−400
−200
0
200
400
600
Δx [mm]

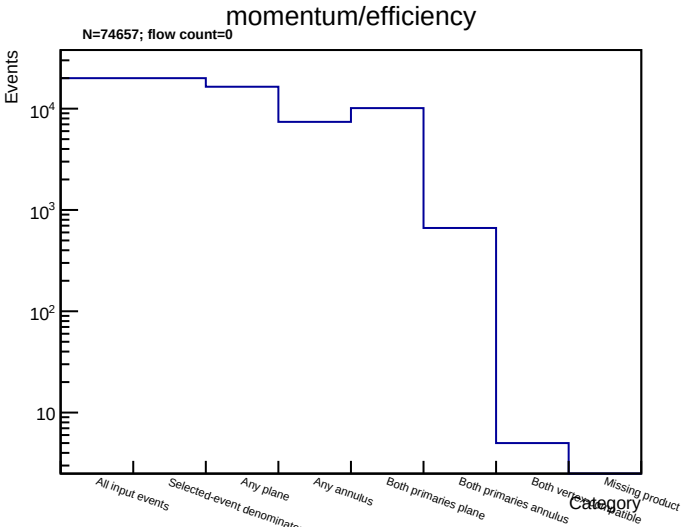

control_reflected/tar_dy
N=11789; flow count=0
Tracks
10^2
10
1
−600
−400
−200
0
200
400
600
Δy [mm]

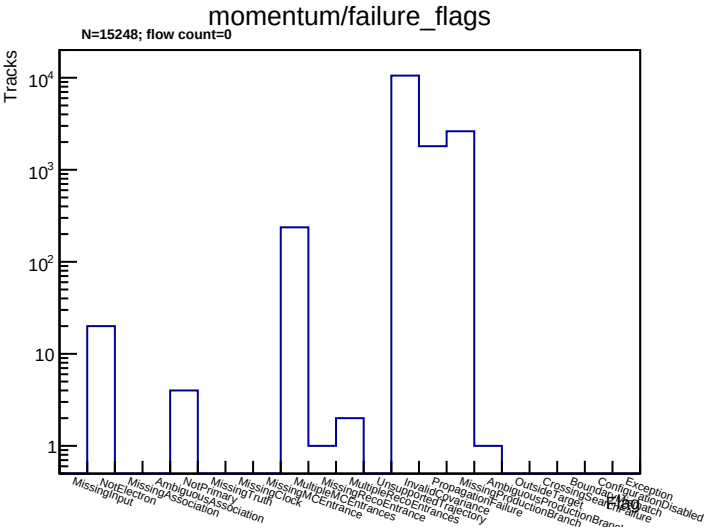

control_reflected/tar_phase
N=11789; flow count=0
Tracks
10^2
−150 −100 −50 0 50 100 150
Δϕ [degrees]

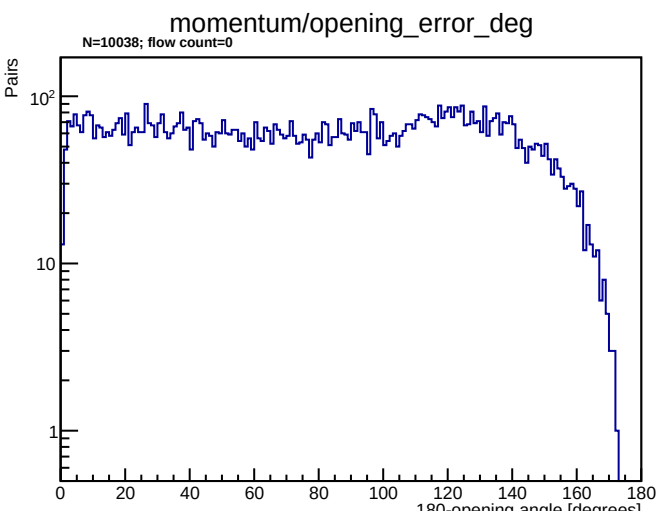

control_reflected/tar_phase_vs_tar_dr
N=11789; flow count=0
tar_dr
1000 900 800 700 600 500 400 300 200 100 0
−150 −100 −50 0 50 100 150
tar_phase
10
1
10^{-1}

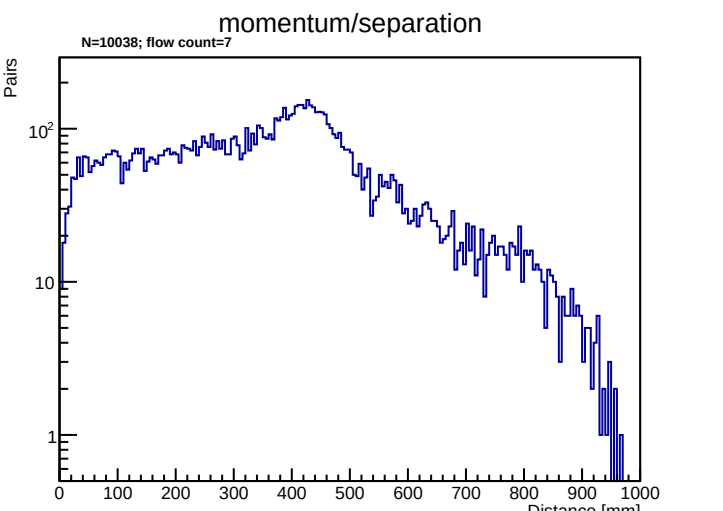

control_reflected/tar_pitch
N=11789; flow count=1
Tracks
10^3
10^2
10
1
−500 −400 −300 −200 −100 0 100 200 300 400 500
Δθ [mrad]

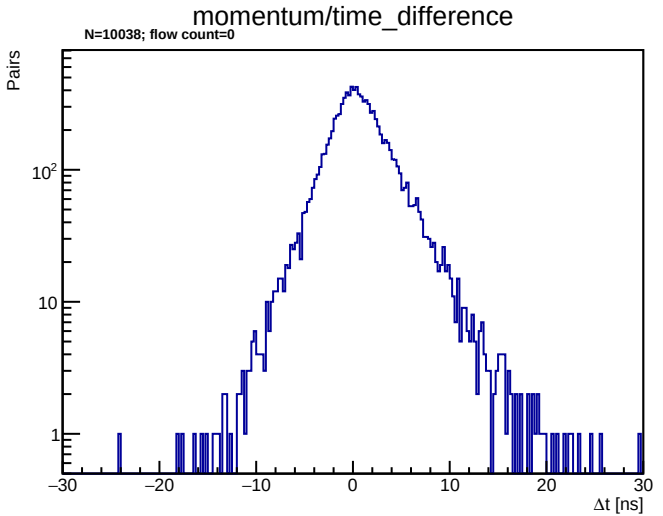

control_reflected/tar_radius
N=11789; flow count=18
Tracks
10^3
10^2
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
ΔR [mm]

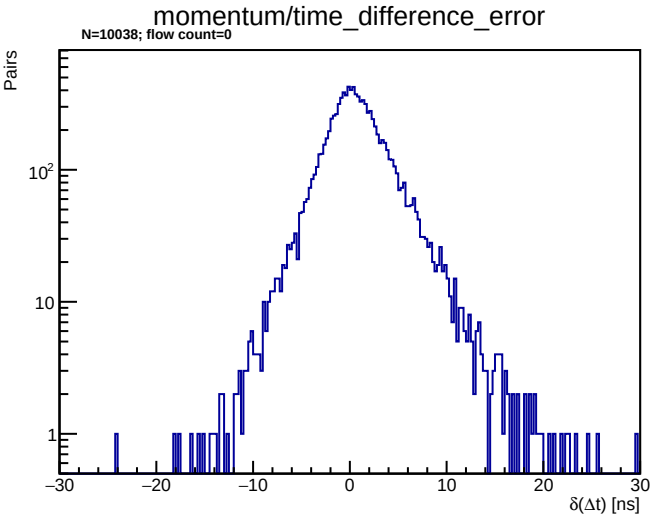

momentum/efficiency
N=74657; flow count=0
Events
10^4
10^3
10^2
10
All input events
Selected-event denomina
Any plane
Any annulus
Both primaries plane
Both primaries annulus
Both ver
Missing product
Category

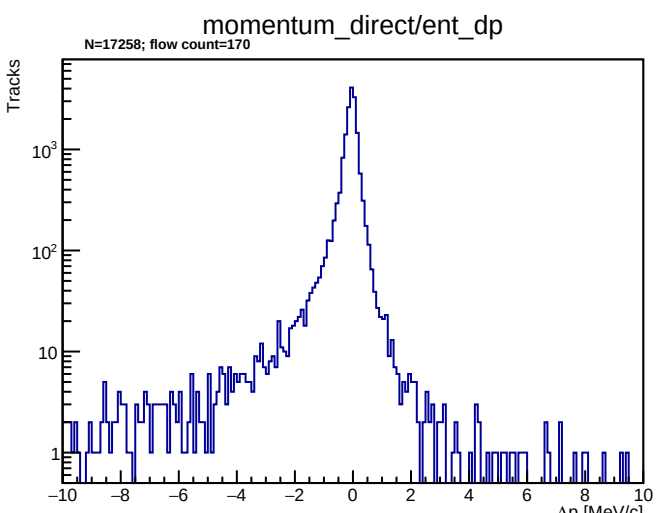

momentum/failure_flags
N=15248; flow count=0
Tracks
10^4
10^3
10^2
10
1
MissingInput
NotElectron
MissingAssociation
AmbiguousAssociation
NotPrimaryTruth
MissingTruth
MissingClock
MultipleMCEntrance
MissingMCEntrance
MultipleRecoEntrance
MissingRecoEntrances
MultipleRecoEntrances
UnsupportedTrajectory
InvalidCovariance
PropagationFailure
MissingProduction
AmbiguousProduction
OutsideTarget
CrossingSeam
BoundaryMismatch
ConfigurationDisabled
Exception
Failure
Cut

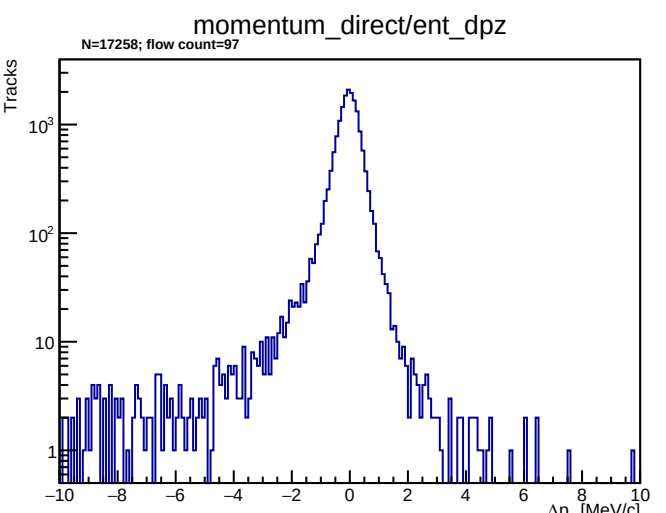

momentum/opening_error_deg
N=10038; flow count=0
Pairs
10^2
10
1
0
20
40
60
80
100
120
140
160
180
180-opening angle [degrees]

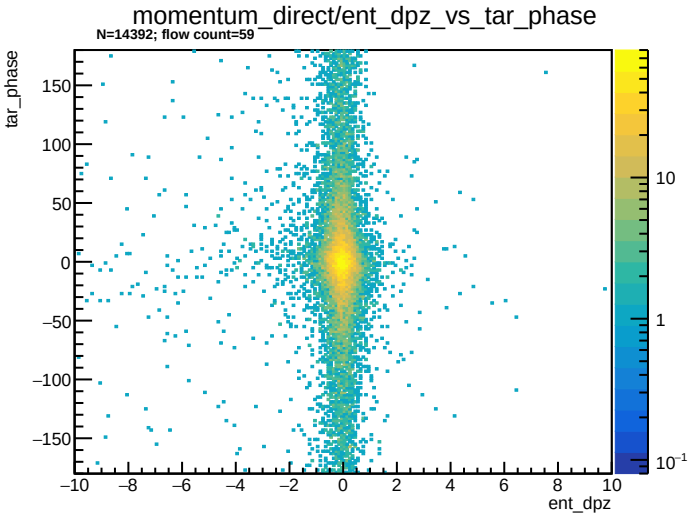

momentum/separation
N=10038; flow count=7
Pairs
10^2
10
1
0
100
200
300
400
500
600
700
800
900
1000
Distance [mm]

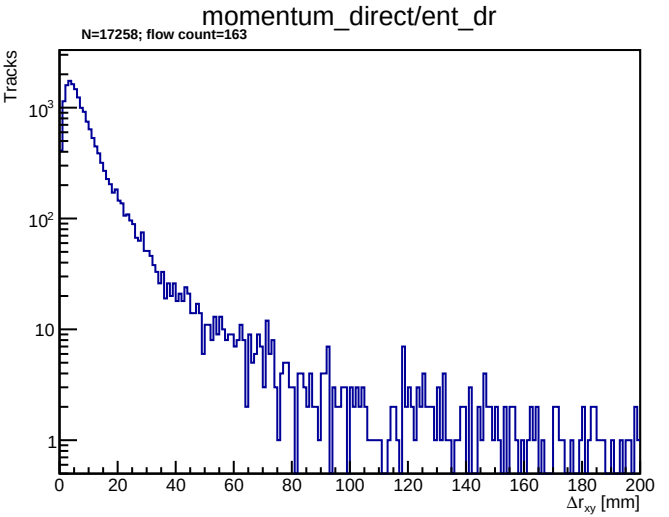

momentum/time_difference
N=10038; flow count=0
Pairs
Δt [ns]

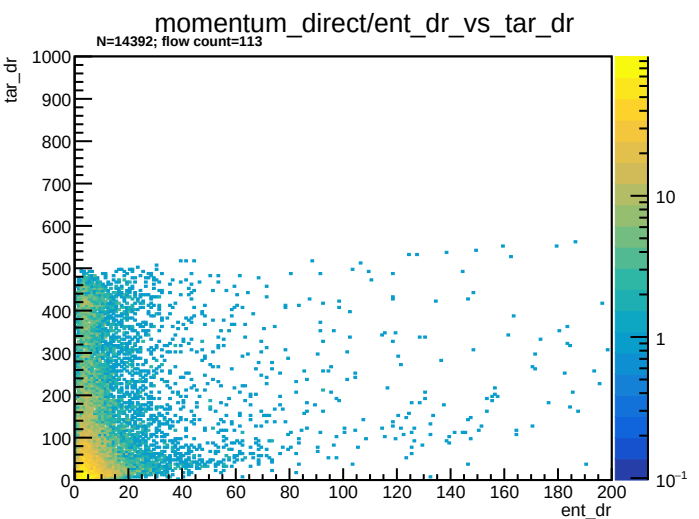

momentum/time_difference_error
N=10038; flow count=0
Pairs
δ(Δt) [ns]

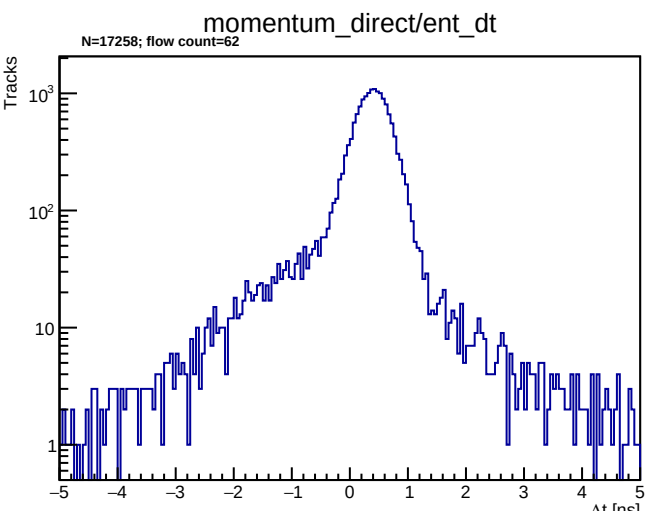

momentum_direct/ent_dp
N=17258; flow count=170
Tracks
Δp [MeV/c]

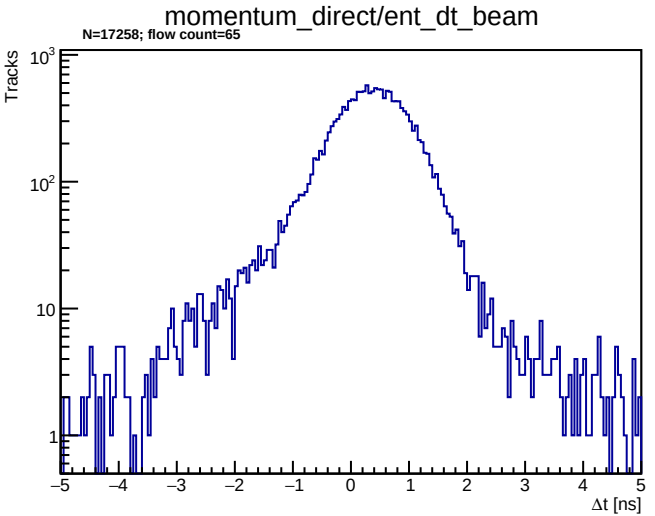

momentum_direct/ent_dpz
N=17258; flow count=97
Tracks
Δp_z [MeV/c]

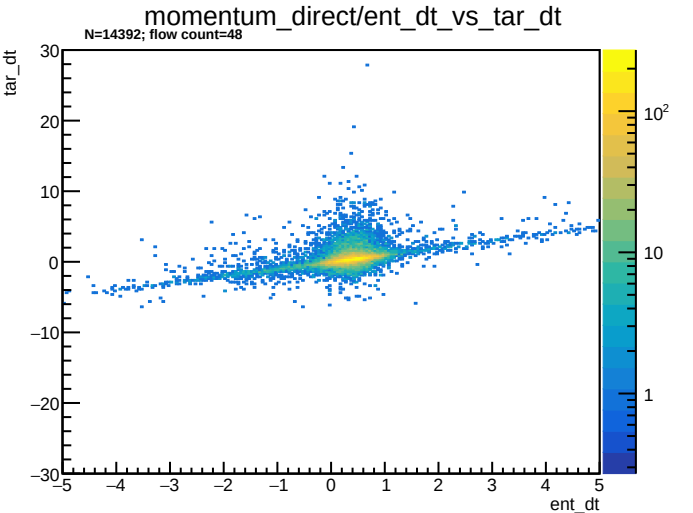

momentum_direct/ent_dpz_vs_tar_phase
N=14392; flow count=59
tar_phase
ent_dpz

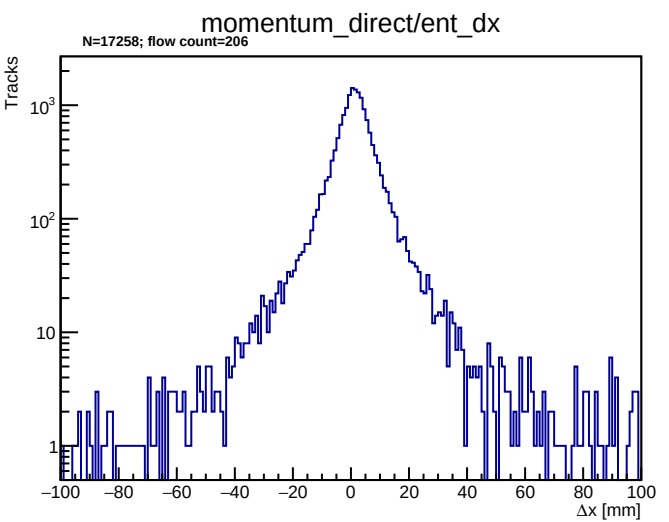

momentum_direct/ent_dr
N=17258; flow count=163
Tracks
Δr_xy [mm]

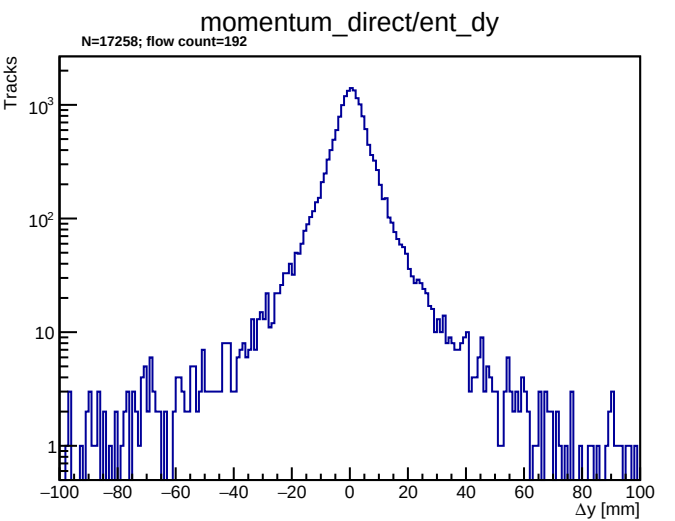

momentum_direct/ent_dr_vs_tar_dr
N=14392; flow count=113
tar_dr
ent_dr

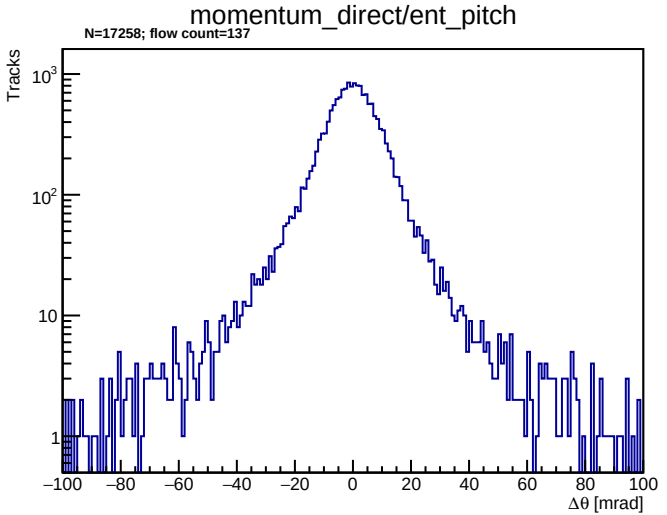

momentum_direct/ent_dt
N=17258; flow count=62
Tracks
Δt [ns]

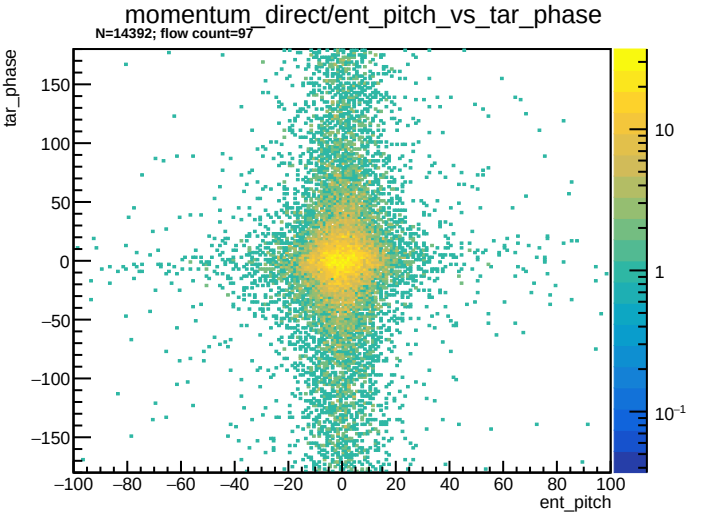

momentum_direct/ent_dt_beam
N=17258; flow count=65
Tracks
Δt [ns]

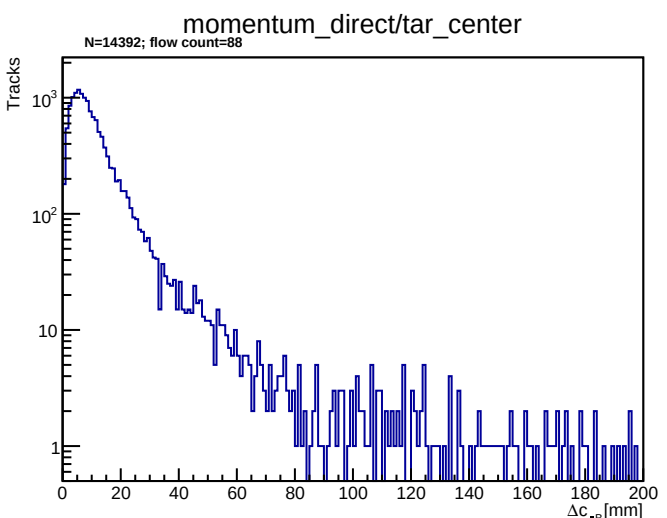

momentum_direct/ent_dt_vs_tar_dt
N=14392; flow count=48
tar_dt
ent_dt

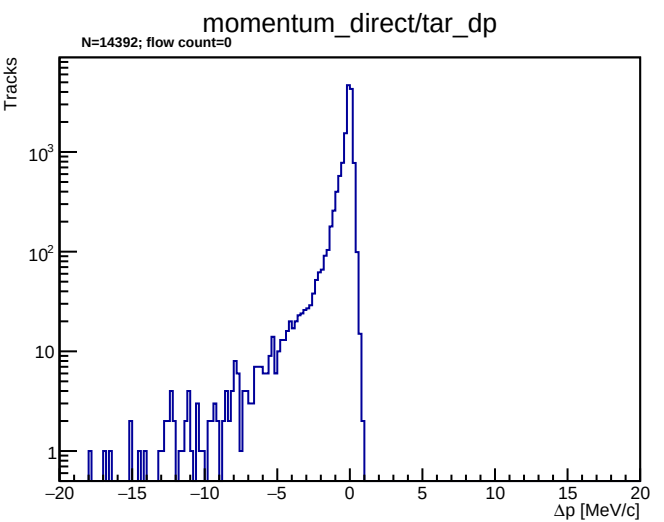

momentum_direct/ent_dx
N=17258; flow count=206
Tracks
Δx [mm]

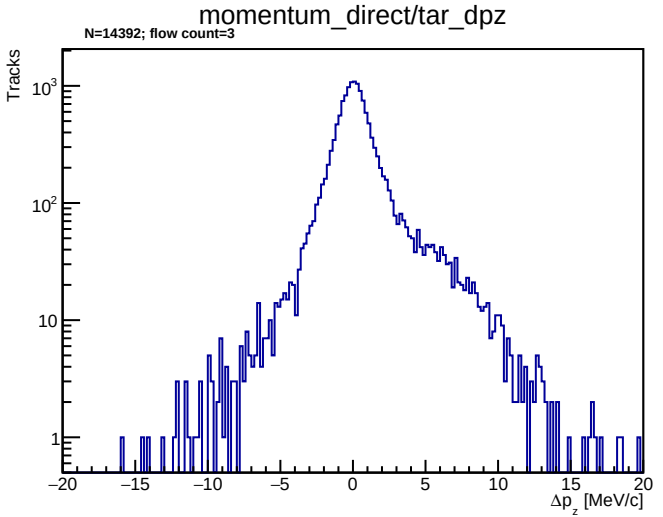

momentum_direct/ent_dy
N=17258; flow count=192
Tracks
Δy [mm]

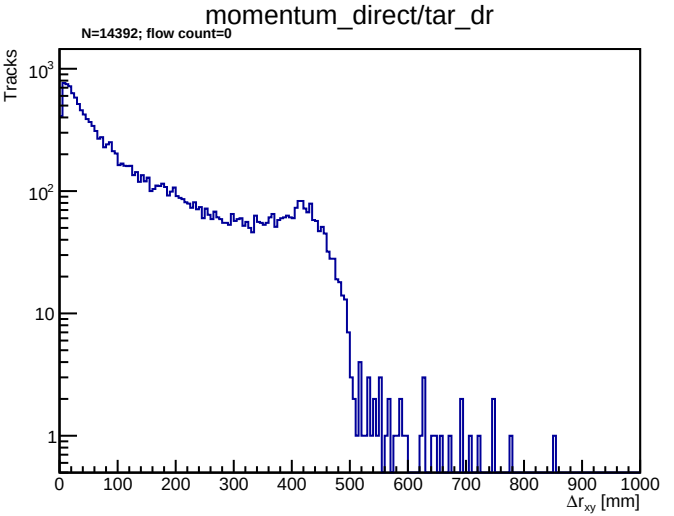

momentum_direct/ent_pitch
N=17258; flow count=137
Tracks
Δθ [mrad]

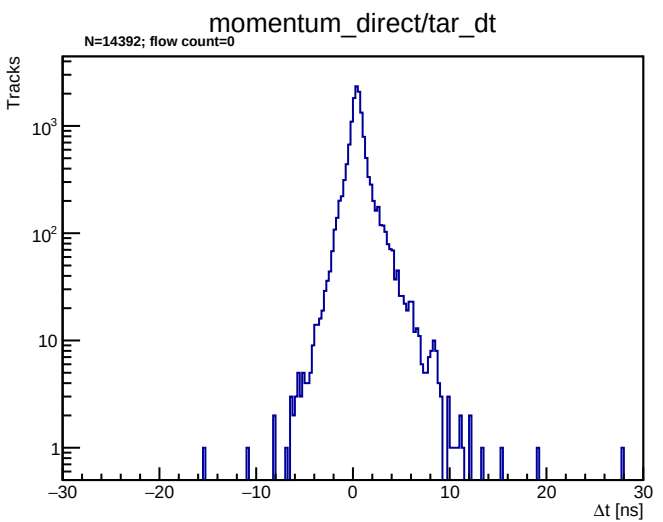

momentum_direct/ent_pitch_vs_tar_phase
N=14392; flow count=97
tar_phase
ent_pitch

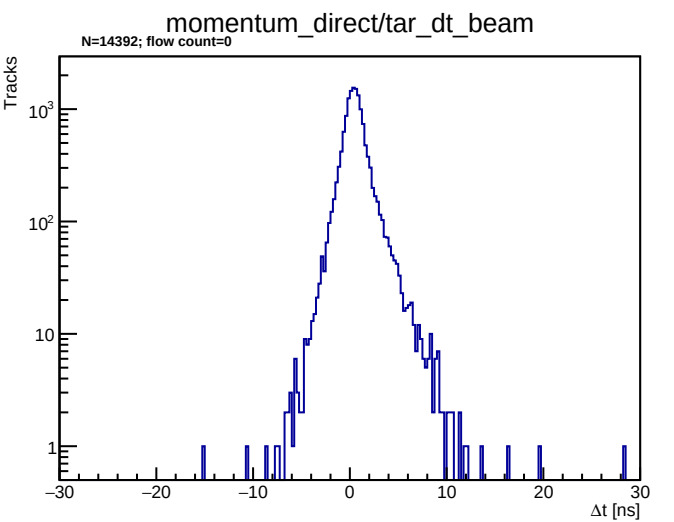

momentum_direct/tar_center
N=14392; flow count=88
Tracks
Δc_tB [mm]

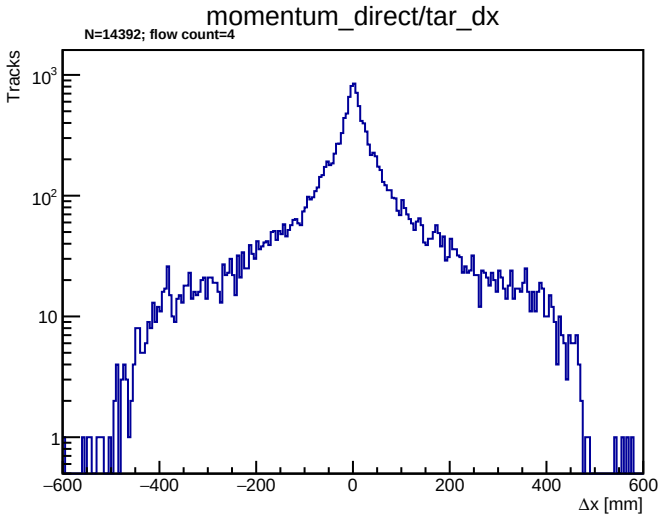

momentum_direct/tar_dp
N=14392; flow count=0
Tracks
Δp [MeV/c]

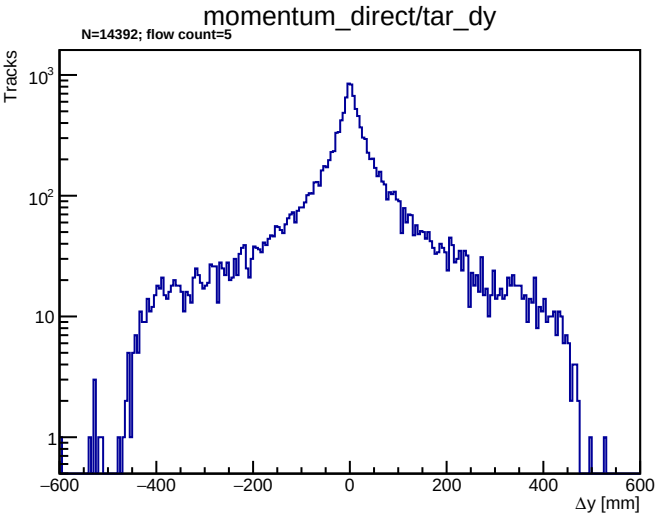

momentum_direct/tar_dpz
N=14392; flow count=3
Tracks
10^3
10^2
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

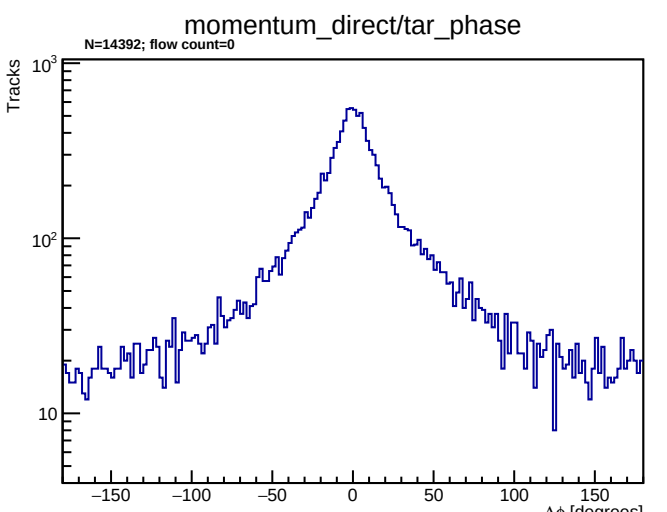

momentum_direct/tar_dr
N=14392; flow count=0
Tracks
10^3
10^2
10
1
0 100 200 300 400 500 600 700 800 900 1000
Δr_{xy} [mm]

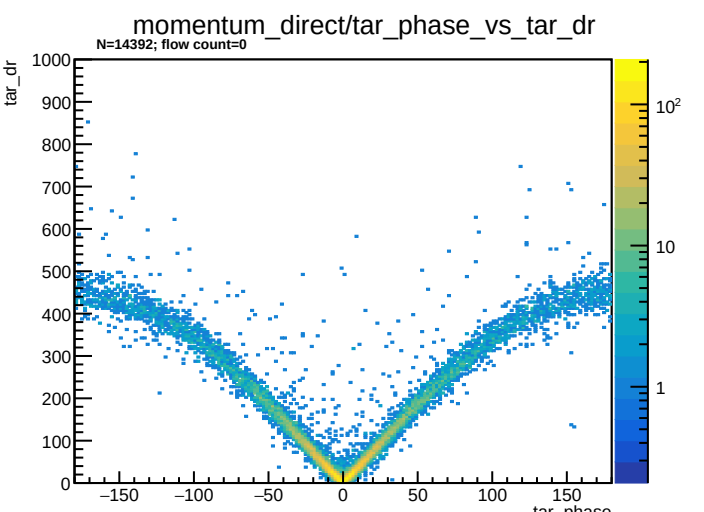

momentum_direct/tar_dt
N=14392; flow count=0
Tracks
10^3
10^2
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

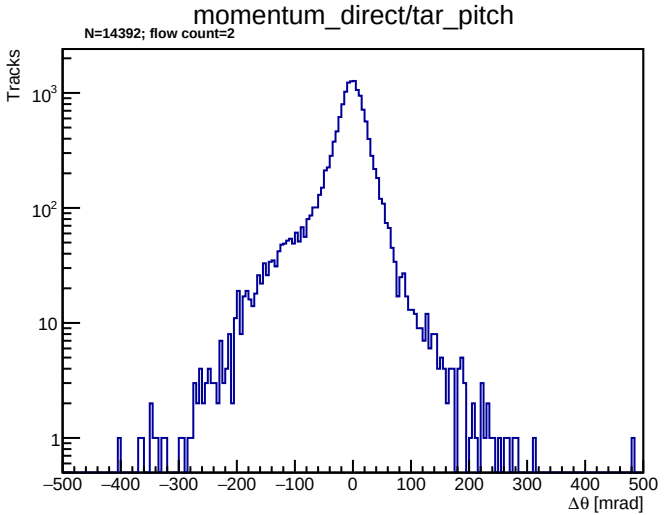

momentum_direct/tar_dt_beam
N=14392; flow count=0
Tracks
10^3
10^2
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

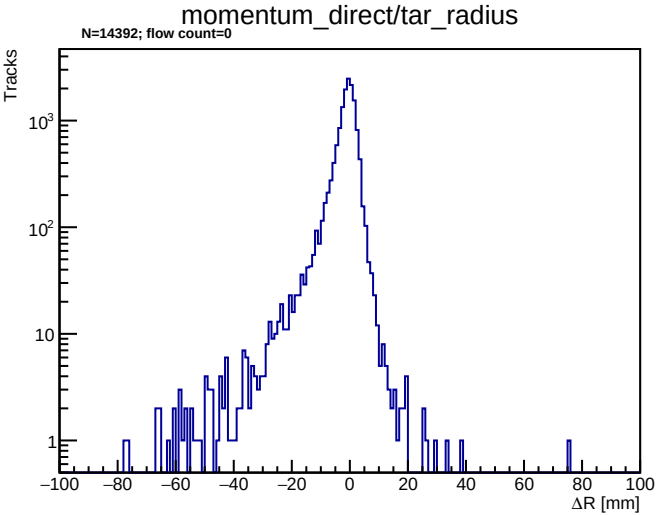

momentum_direct/tar_dx
N=14392; flow count=4
Tracks
Δx [mm]

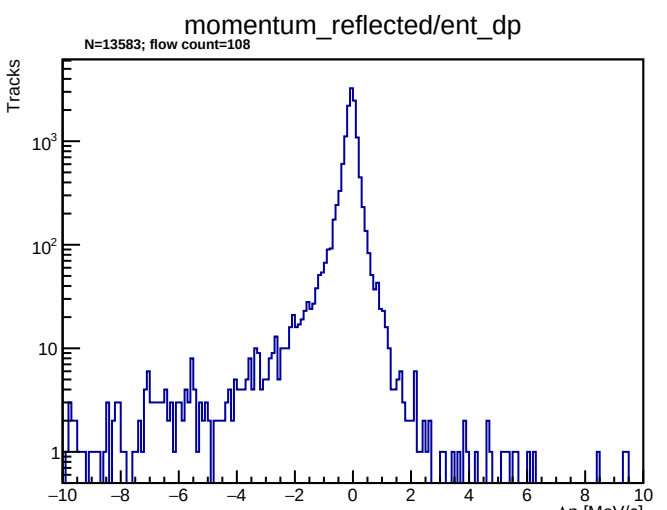

momentum_direct/tar_dy
N=14392; flow count=5
Tracks
Δy [mm]

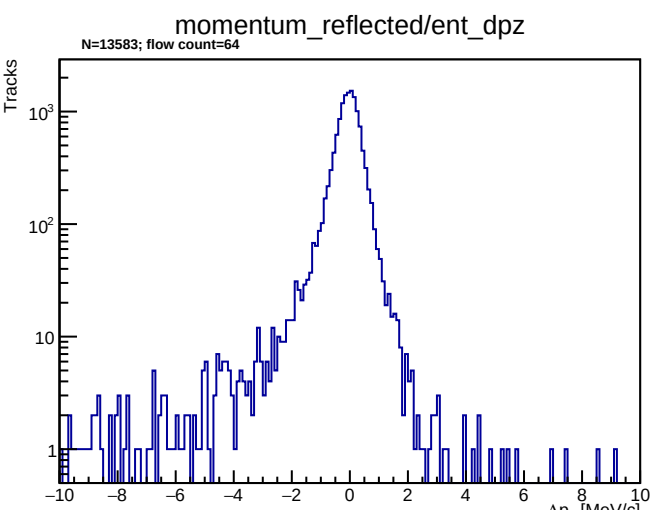

momentum_direct/tar_phase
N=14392; flow count=0
Tracks
Δϕ [degrees]

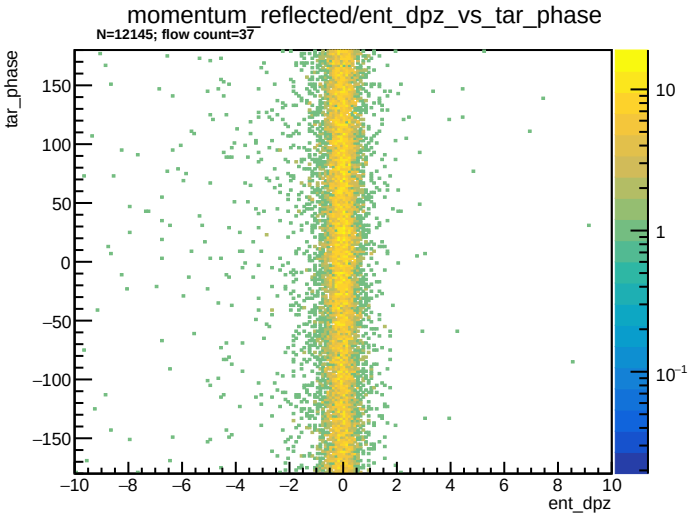

momentum_direct/tar_phase_vs_tar_dr
N=14392; flow count=0
tar_dr
tar_phase

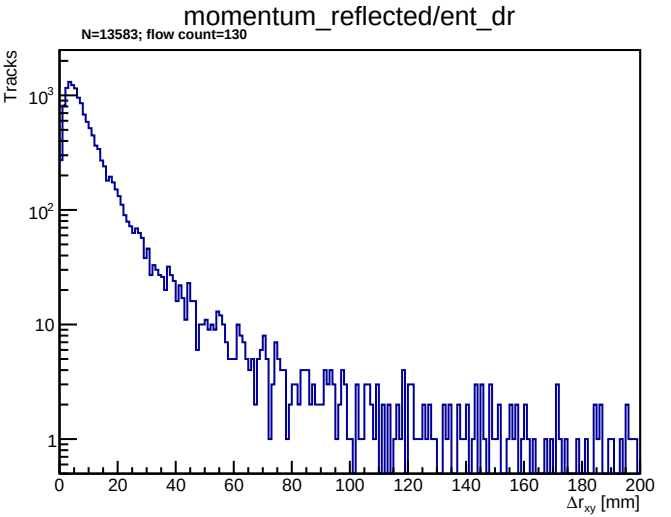

momentum_direct/tar_pitch
N=14392; flow count=2
Tracks
Δθ [mrad]

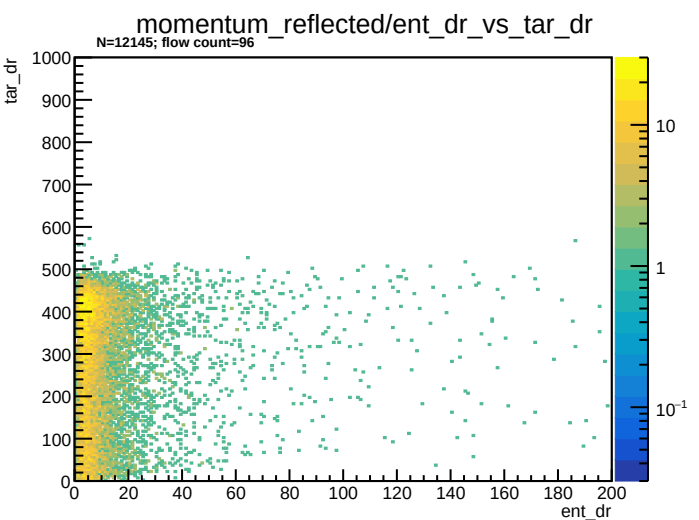

momentum_direct/tar_radius
N=14392; flow count=0
Tracks
ΔR [mm]

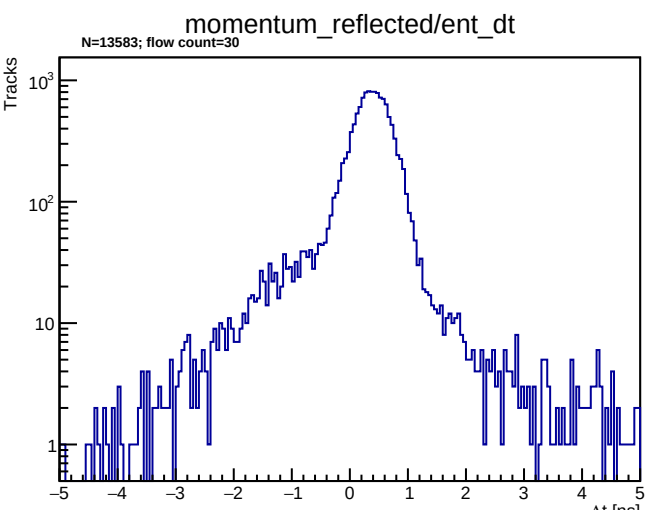

momentum_reflected/ent_dp
N=13583; flow count=108
Tracks
Δp [MeV/c]

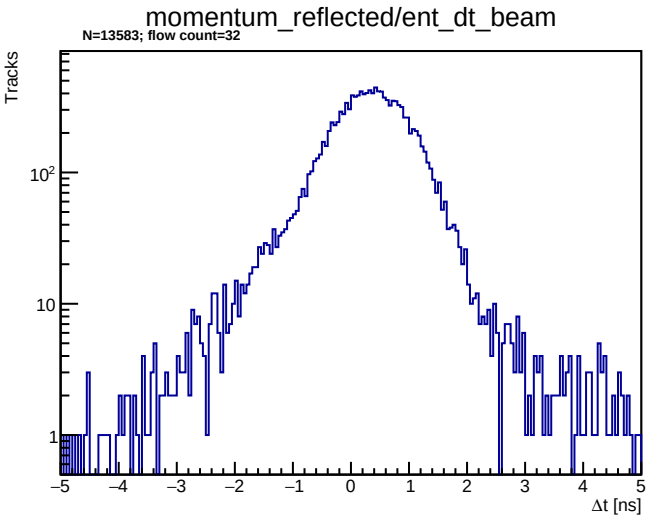

momentum_reflected/ent_dpz
N=13583; flow count=64
Tracks
Δp_z [MeV/c]

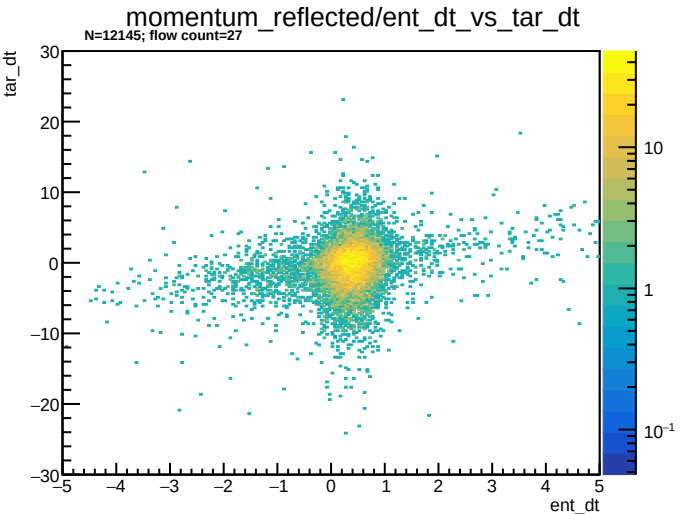

momentum_reflected/ent_dpz_vs_tar_phase
N=12145; flow count=37
tar_phase
ent_dpz

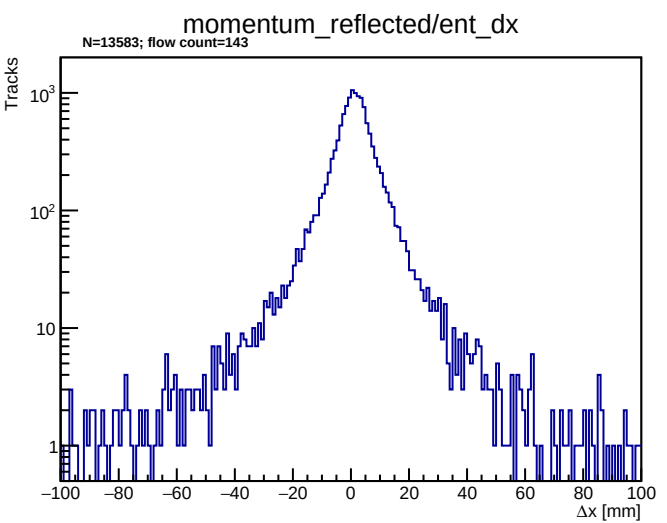

momentum_reflected/ent_dr
N=13583; flow count=130
Tracks
Δr_xy [mm]

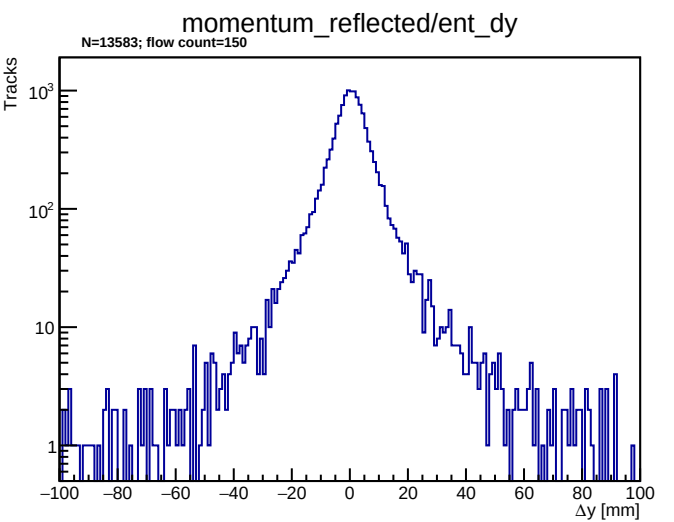

momentum_reflected/ent_dr_vs_tar_dr
N=12145; flow count=96
tar_dr
ent_dr

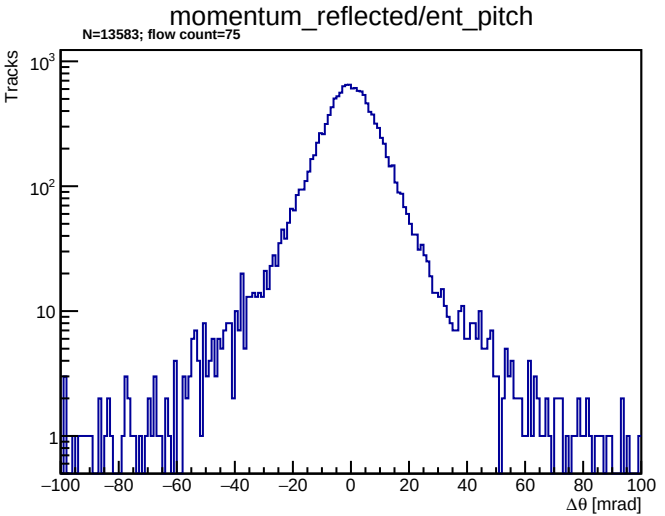

momentum_reflected/ent_dt
N=13583; flow count=30
Tracks
Δt [ns]

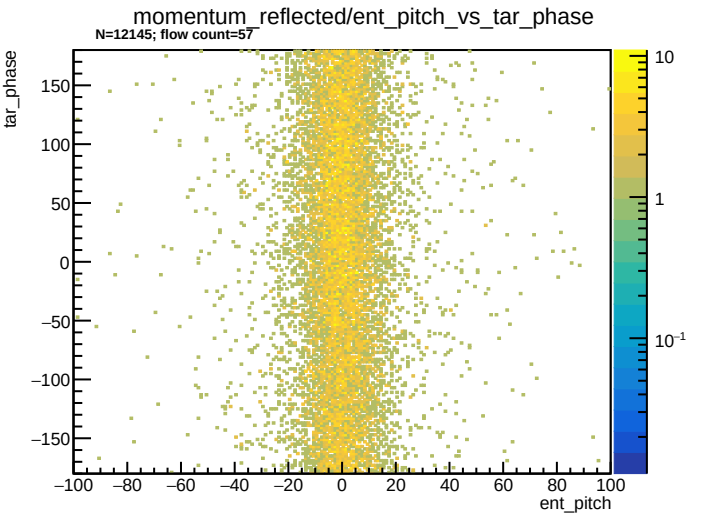

momentum_reflected/ent_dt_beam
N=13583; flow count=32
Tracks
Δt [ns]

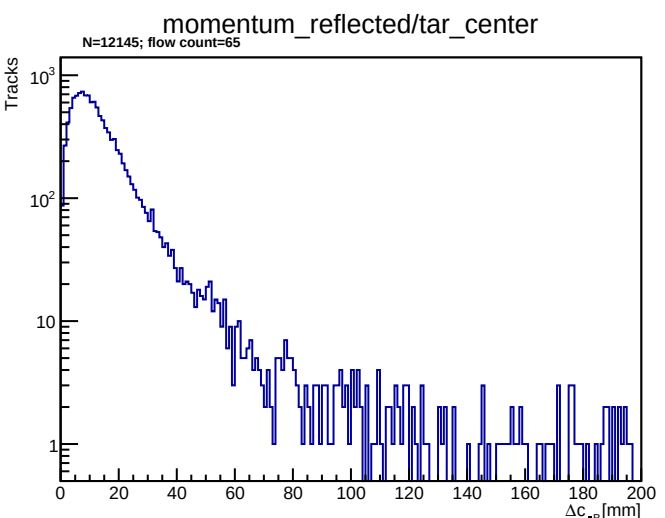

momentum_reflected/ent_dt_vs_tar_dt
N=12145; flow count=27
tar_dt
ent_dt

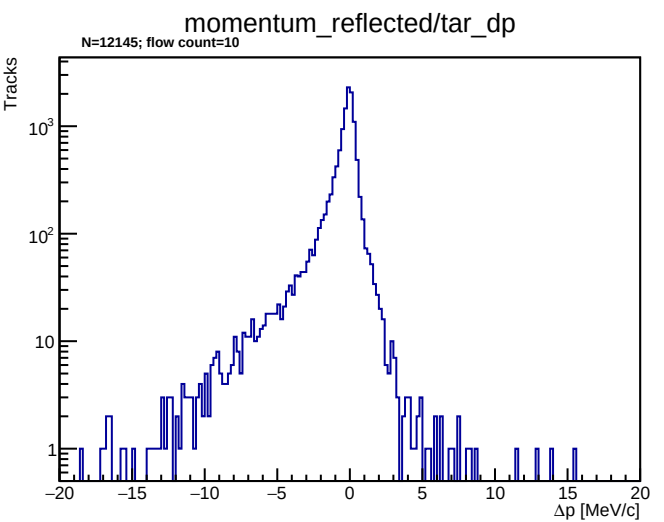

momentum_reflected/ent_dx
N=13583; flow count=143
Tracks
Δx [mm]

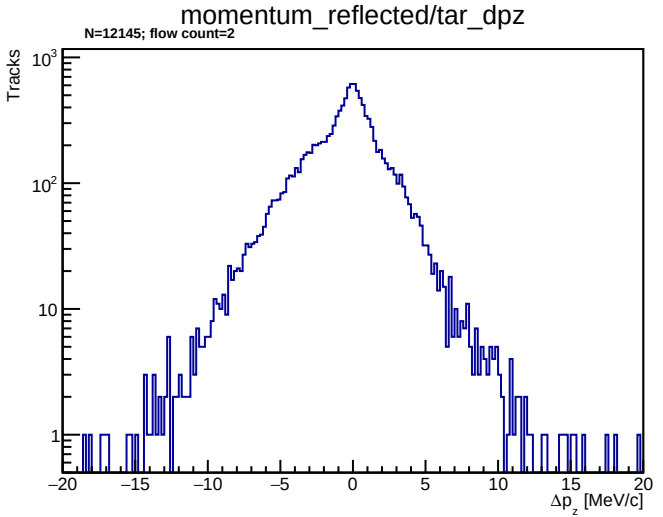

momentum_reflected/ent_dy
N=13583; flow count=150
Tracks
Δy [mm]

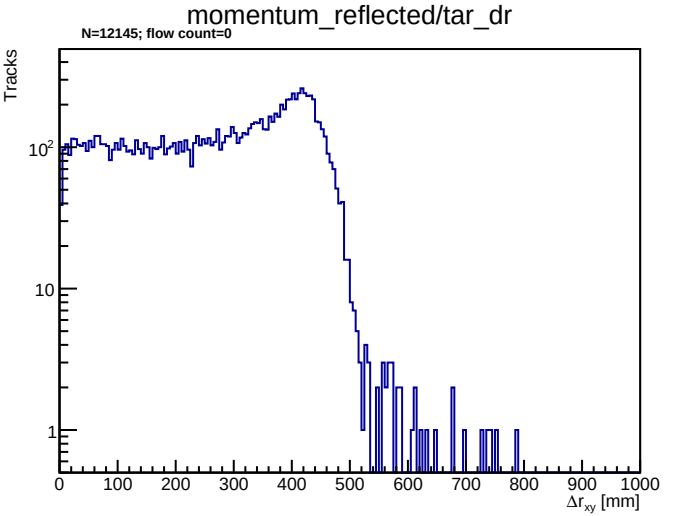

momentum_reflected/ent_pitch
N=13583; flow count=75
Tracks
Δθ [mrad]

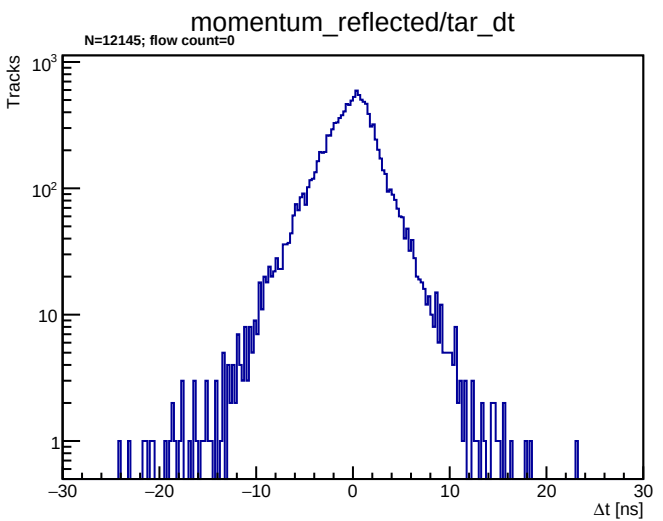

momentum_reflected/ent_pitch_vs_tar_phase
N=12145; flow count=57
tar_phase
ent_pitch

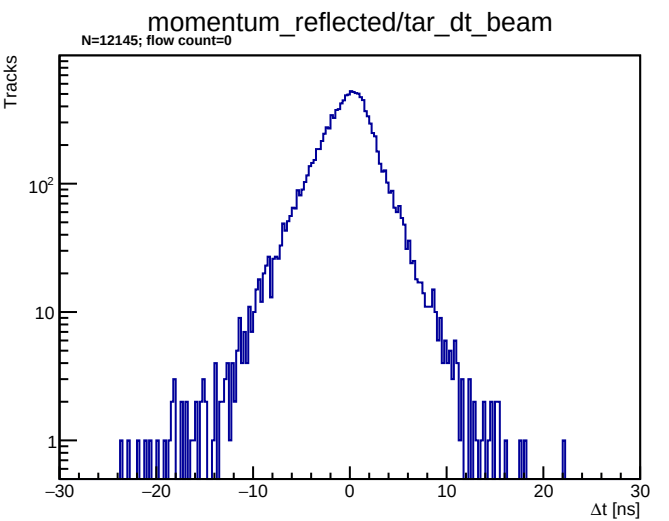

momentum_reflected/tar_center
N=12145; flow count=65
Tracks
Δc_B [mm]

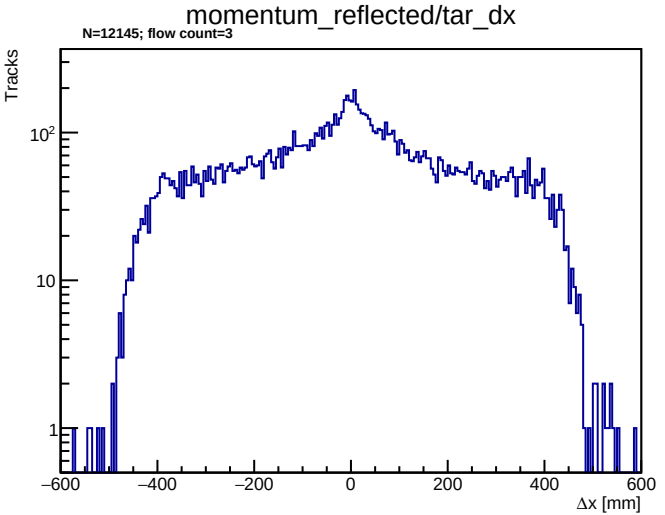

momentum_reflected/tar_dp
N=12145; flow count=10
Tracks
Δp [MeV/c]

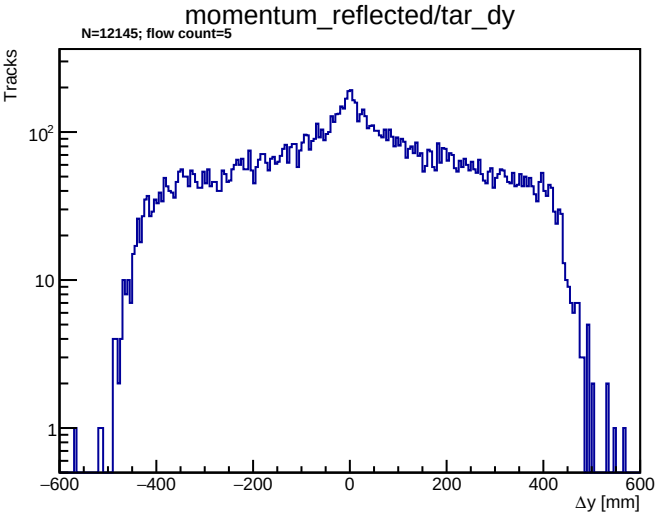

momentum_reflected/tar_dpz
N=12145; flow count=2
Tracks
10^3
10^2
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

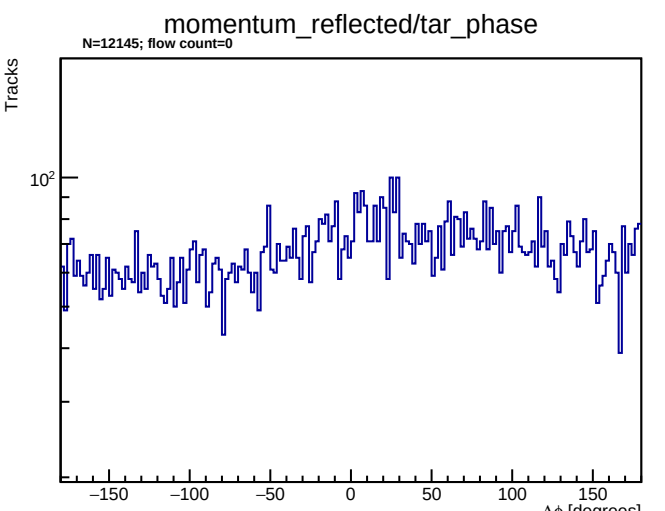

momentum_reflected/tar_dr
N=12145; flow count=0
Tracks
10^2
10
1
0 100 200 300 400 500 600 700 800 900 1000
Δr_{xy} [mm]

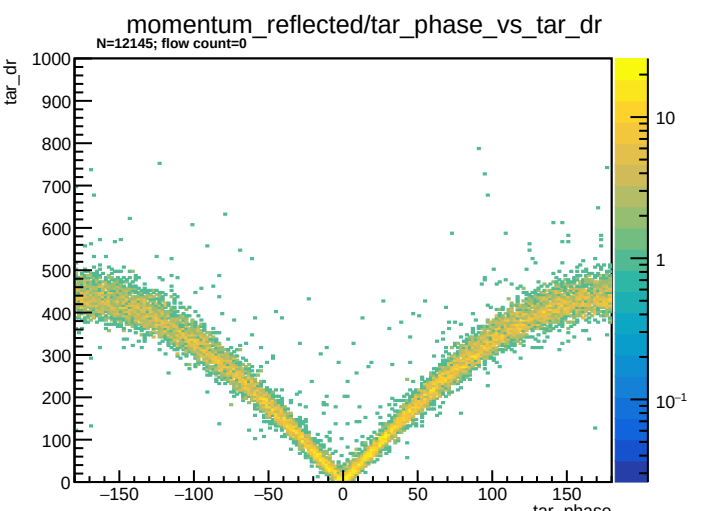

momentum_reflected/tar_dt
N=12145; flow count=0
Tracks
10^3
10^2
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

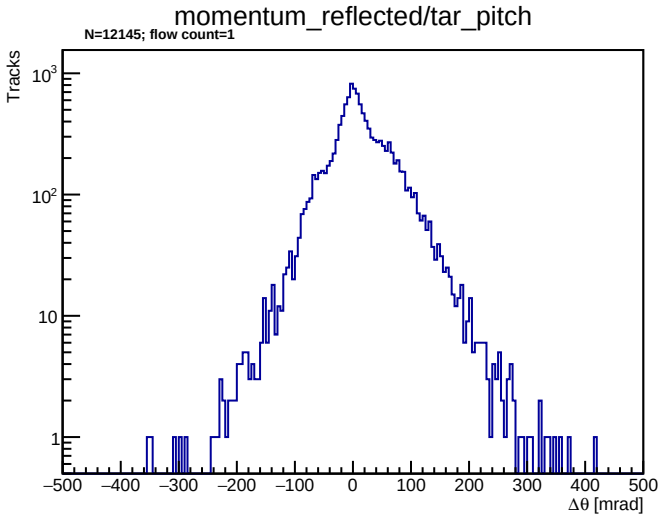

momentum_reflected/tar_dt_beam
N=12145; flow count=0
Tracks
10^2
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

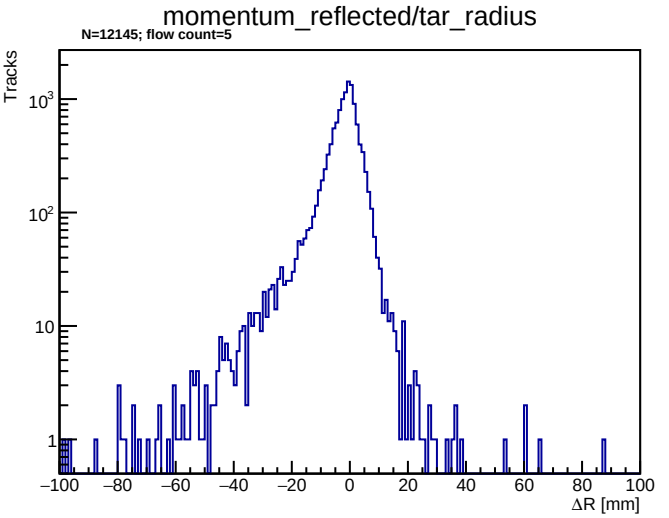

momentum_reflected/tar_dx
N=12145; flow count=3
Tracks
10^2
10
1
−600
−400
−200
0
200
400
600
Δx [mm]

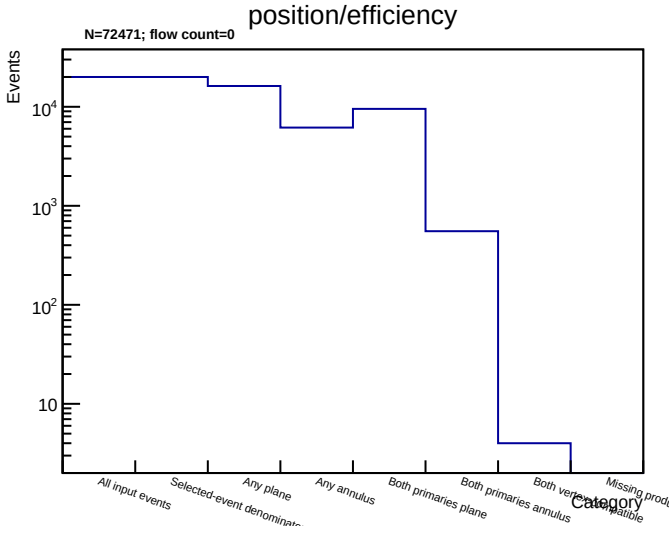

momentum_reflected/tar_dy
N=12145; flow count=5
Tracks
10^2
10
1
−600
−400
−200
0
200
400
600
Δy [mm]

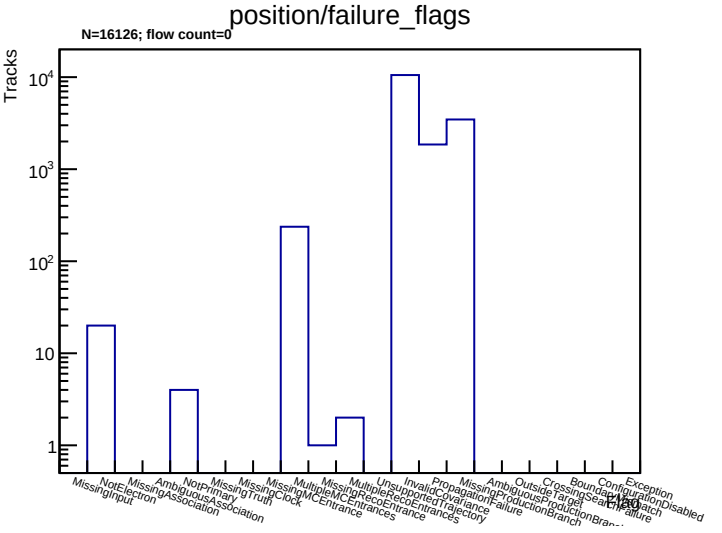

momentum_reflected/tar_phase
N=12145; flow count=0
Tracks
10^2
−150
−100
−50
0
50
100
150
Δϕ [degrees]

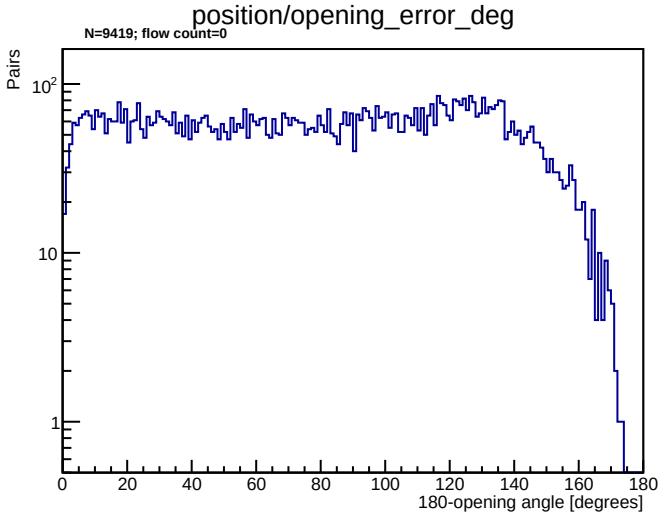

momentum_reflected/tar_phase_vs_tar_dr
N=12145; flow count=0
tar_dr
1000
900
800
700
600
500
400
300
200
100
0
−150
−100
−50
0
50
100
150
tar_phase
10
1
10^{-1}

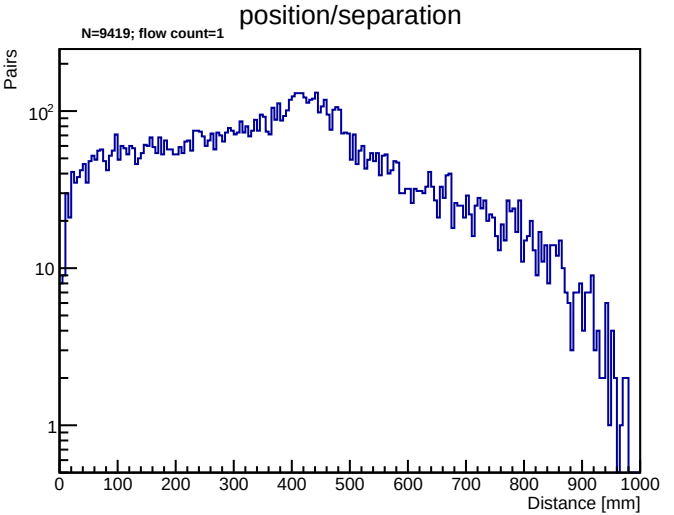

momentum_reflected/tar_pitch
N=12145; flow count=1
Tracks
Δθ [mrad]

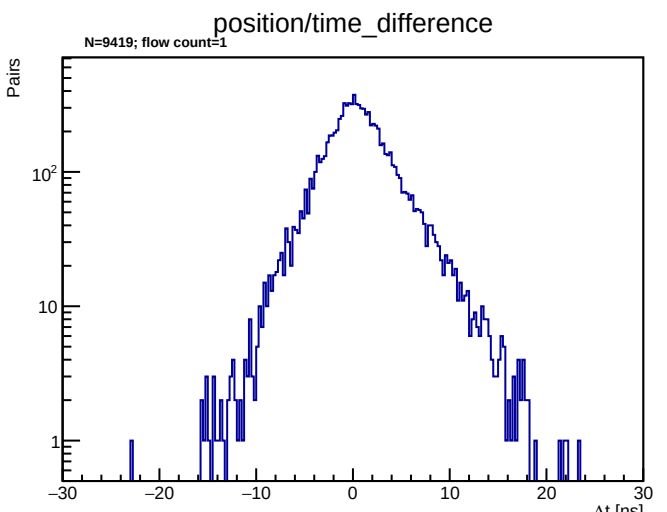

momentum_reflected/tar_radius
N=12145; flow count=5
Tracks
ΔR [mm]

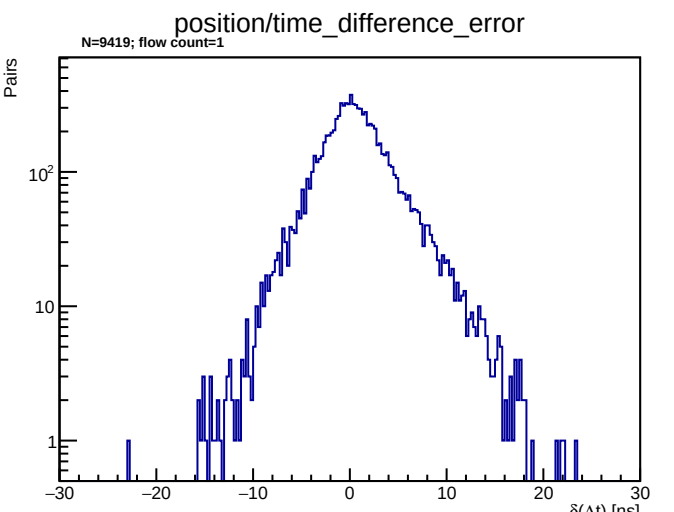

position/efficiency
N=72471; flow count=0
Events
All input events
Any plane
Any annulus
Both primaries plane
Both primaries annulus
Missing product

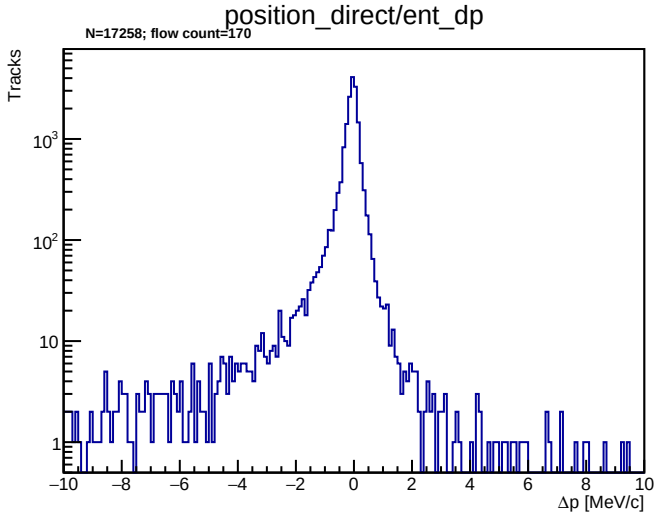

position/failure_flags
N=16126; flow count=0
Tracks
MissingInput
NotElectron
MissingAssociation
AmbiguousAssociation
NotPrimary
MissingTruth
MissingClock
MultipleMCEntrance
MultipleReferenceEntrances
MultipleMCEntrances
UnsupportedTrajectory
InvalidCorrelation
PropagationFailure
MissingProductionBranch
AmbiguousProduction
OutsideTarget
CrossingSeam
BoundaryBranch
ConfigurationDisabled
Exception

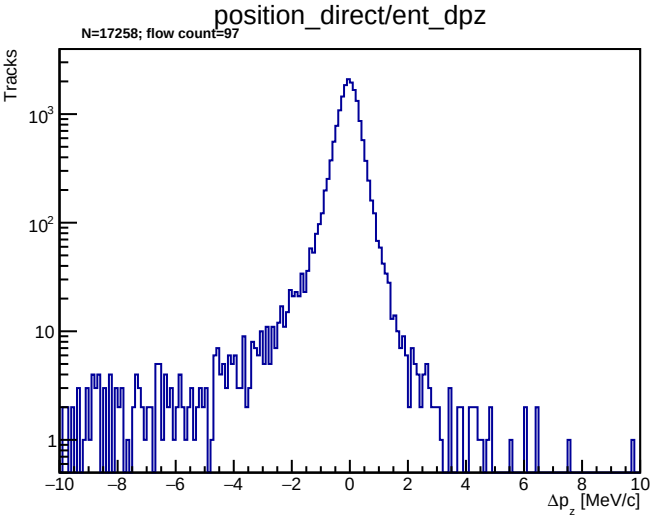

position/opening_error_deg
N=9419; flow count=0
Pairs
180-opening angle [degrees]

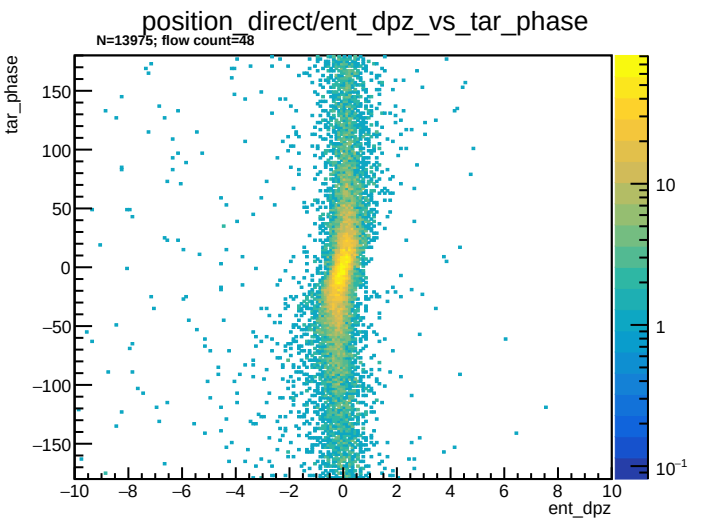

position/separation
N=9419; flow count=1
Pairs
Distance [mm]

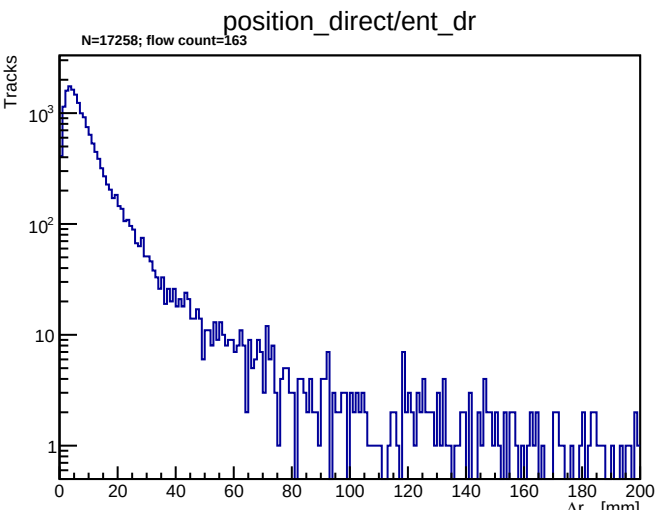

position/time_difference
N=9419; flow count=1
Pairs
Δt [ns]

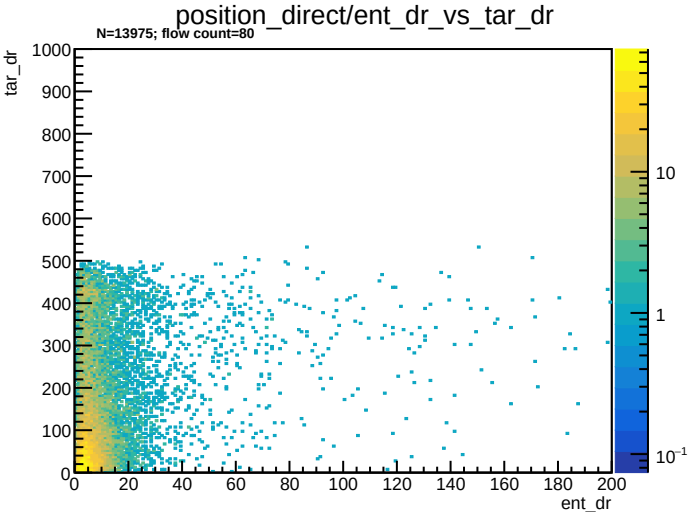

position/time_difference_error
N=9419; flow count=1
Pairs
δ(Δt) [ns]

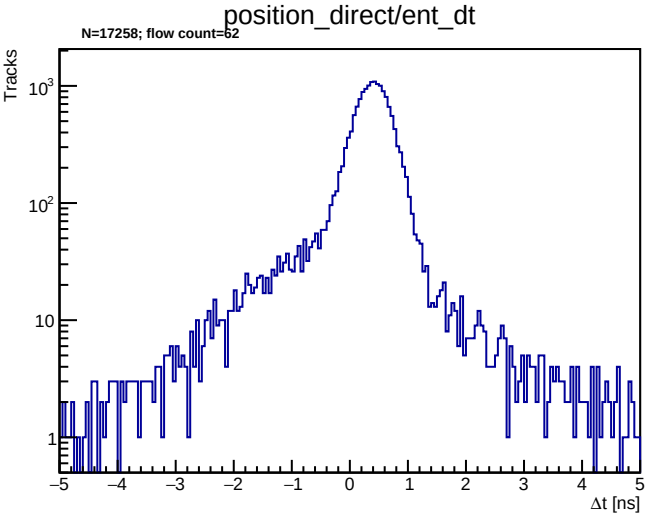

position_direct/ent_dp
N=17258; flow count=170
Tracks
Δp [MeV/c]

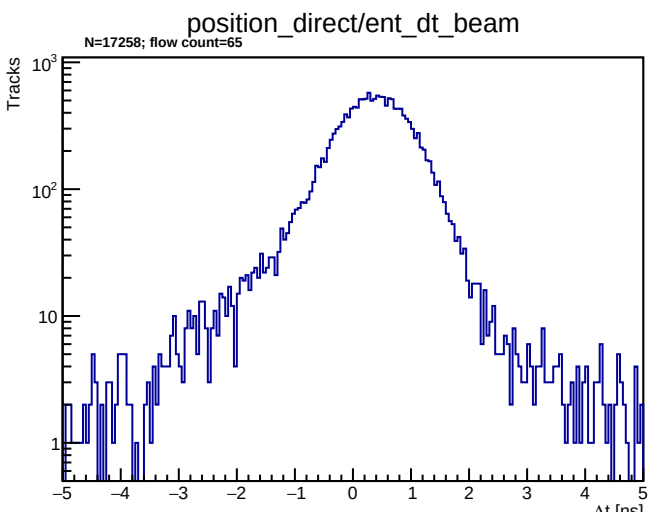

position_direct/ent_dpz
N=17258; flow count=97
Tracks
Δp_z [MeV/c]

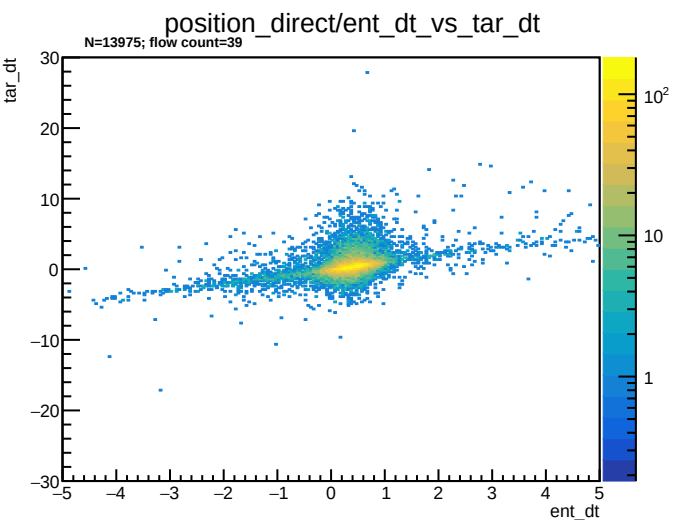

position_direct/ent_dpz_vs_tar_phase
N=13975; flow count=48
tar_phase
ent_dpz

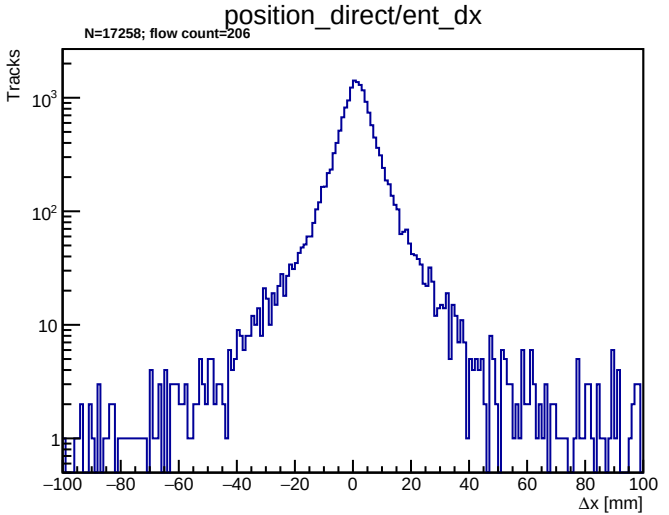

position_direct/ent_dr
N=17258; flow count=163
Tracks
Δr_xy [mm]

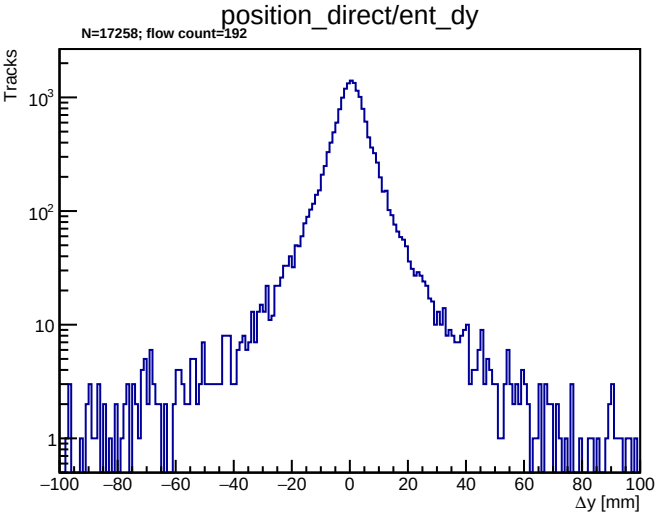

position_direct/ent_dr_vs_tar_dr
N=13975; flow count=80
tar_dr
ent_dr

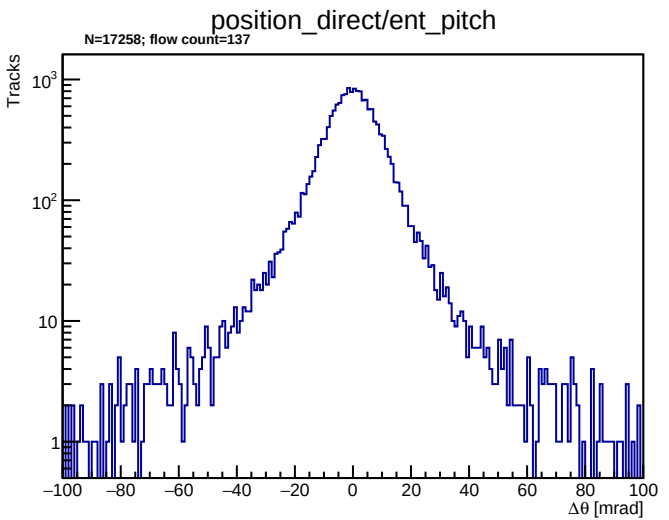

position_direct/ent_dt
N=17258; flow count=62
Tracks
Δt [ns]

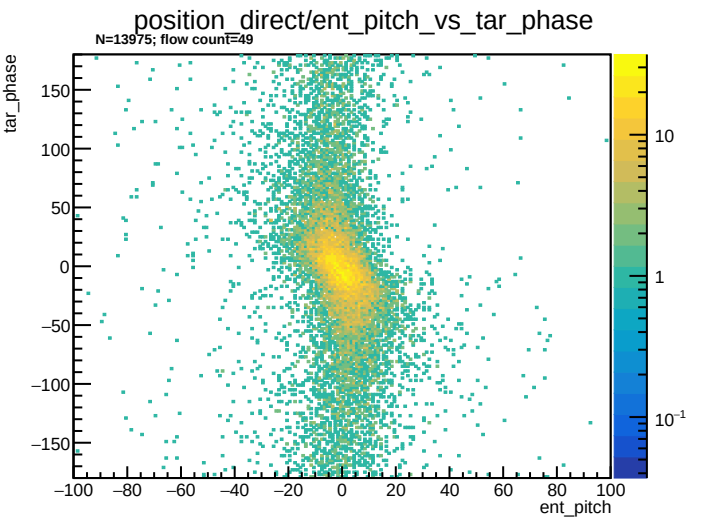

position_direct/ent_dt_beam
N=17258; flow count=65
Tracks
Δt [ns]

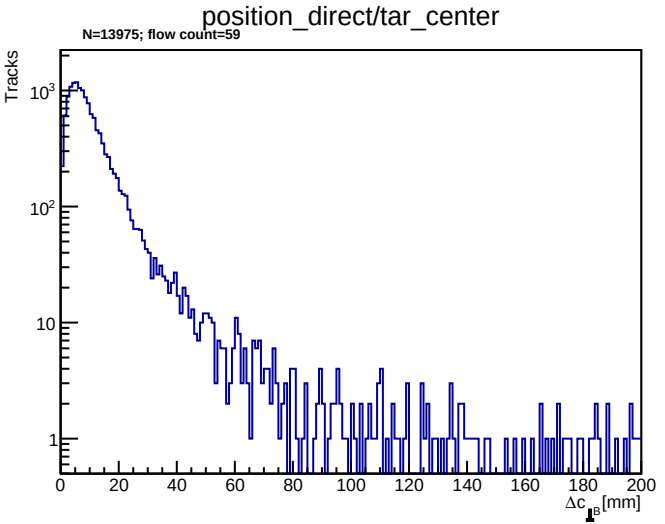

position_direct/ent_dt_vs_tar_dt
N=13975; flow count=39
tar_dt
ent_dt

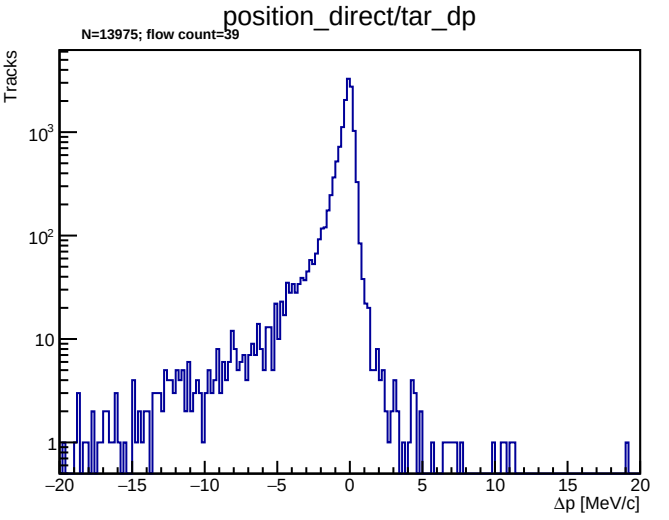

position_direct/ent_dx
N=17258; flow count=206
Tracks
Δx [mm]

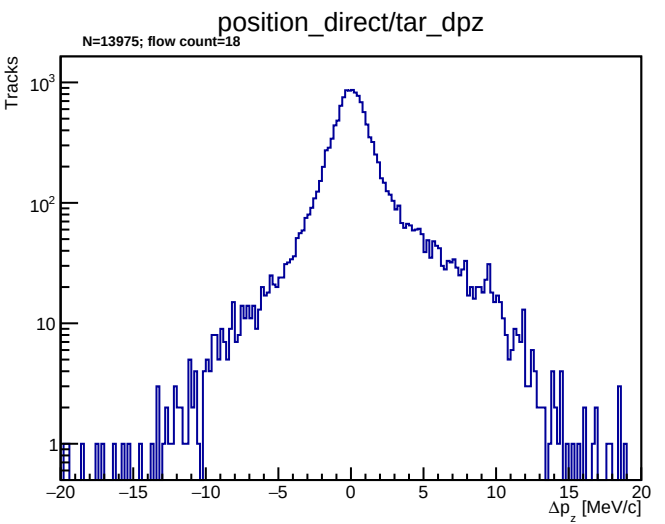

position_direct/ent_dy
N=17258; flow count=192
Tracks
Δy [mm]

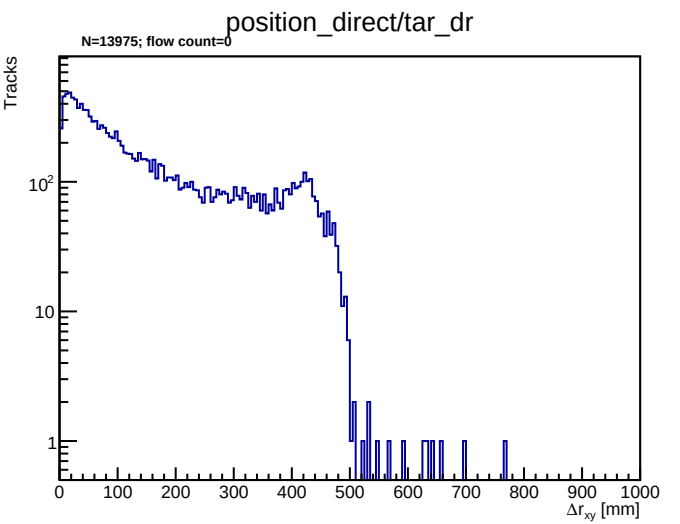

position_direct/ent_pitch
N=17258; flow count=137
Tracks
Δθ [mrad]

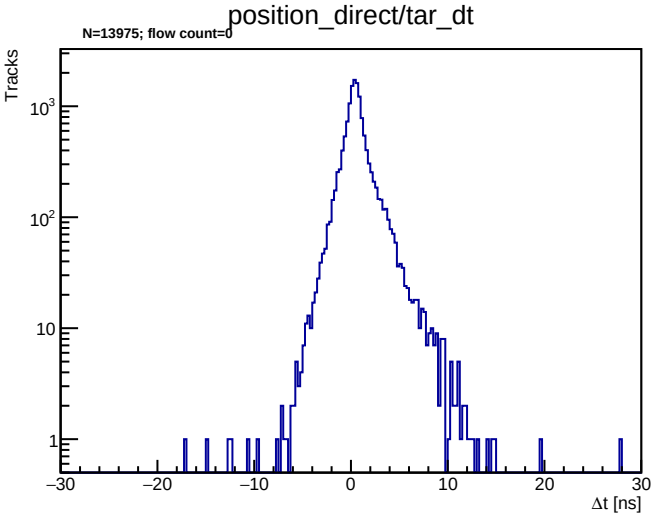

position_direct/ent_pitch_vs_tar_phase
N=13975; flow count=49
tar_phase
ent_pitch

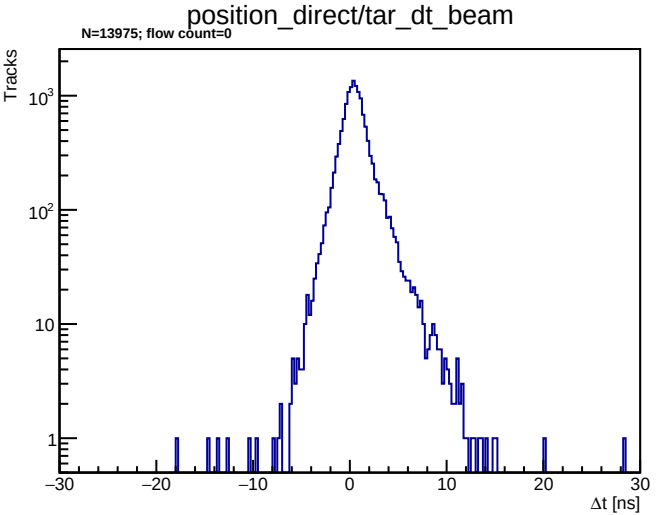

position_direct/tar_center
N=13975; flow count=59
Tracks
10^3
10^2
10
1
0 20 40 60 80 100 120 140 160 180 200
Δc_B [mm]

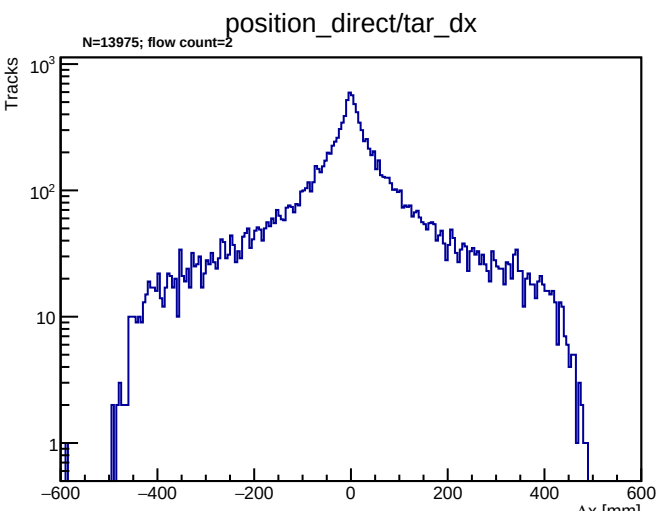

position_direct/tar_dp
N=13975; flow count=39
Tracks
10^3
10^2
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp [MeV/c]

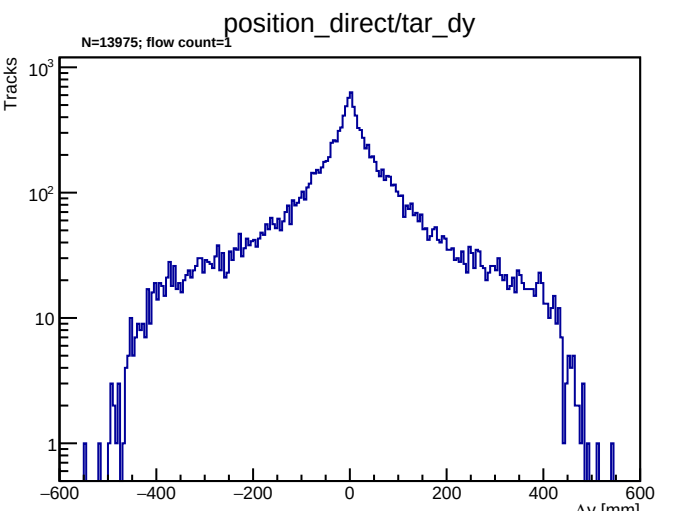

position_direct/tar_dpz
N=13975; flow count=18
Tracks
10^3
10^2
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

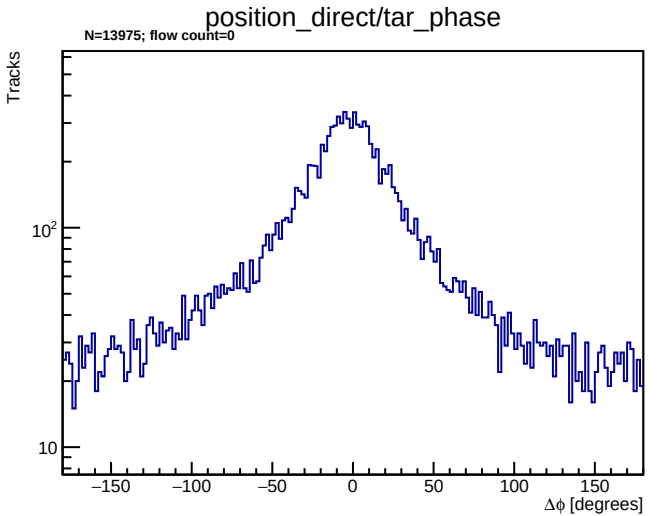

position_direct/tar_dr
N=13975; flow count=0
Tracks
10^2
10
1
0 100 200 300 400 500 600 700 800 900 1000
Δr_{xy} [mm]

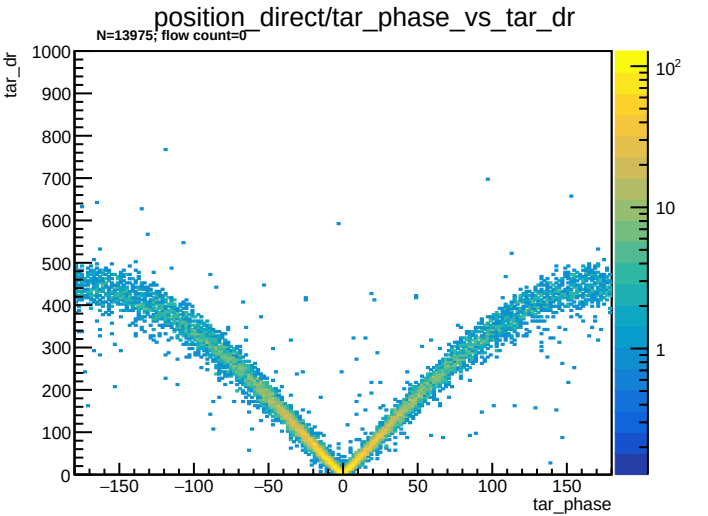

position_direct/tar_dt
N=13975; flow count=0
Tracks
Δt [ns]

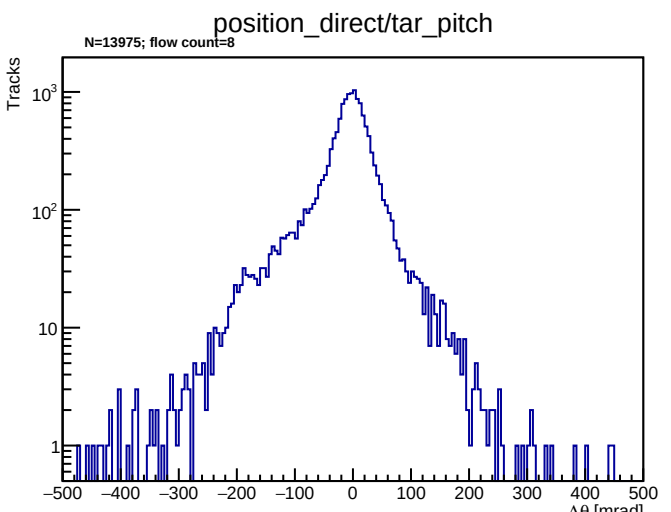

position_direct/tar_dt_beam
N=13975; flow count=0
Tracks
Δt [ns]

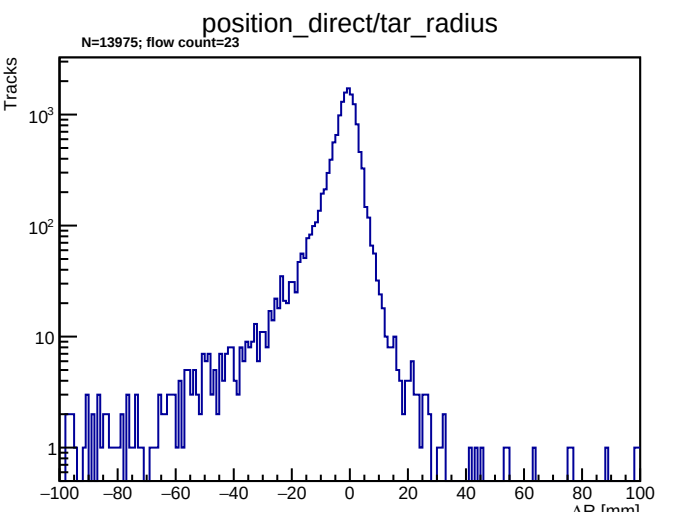

position_direct/tar_dx
N=13975; flow count=2
Tracks
Δx [mm]

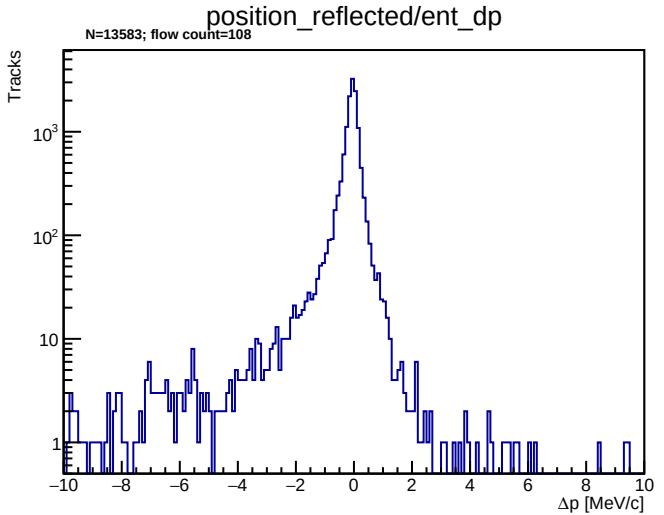

position_direct/tar_dy
N=13975; flow count=1
Tracks
Δy [mm]

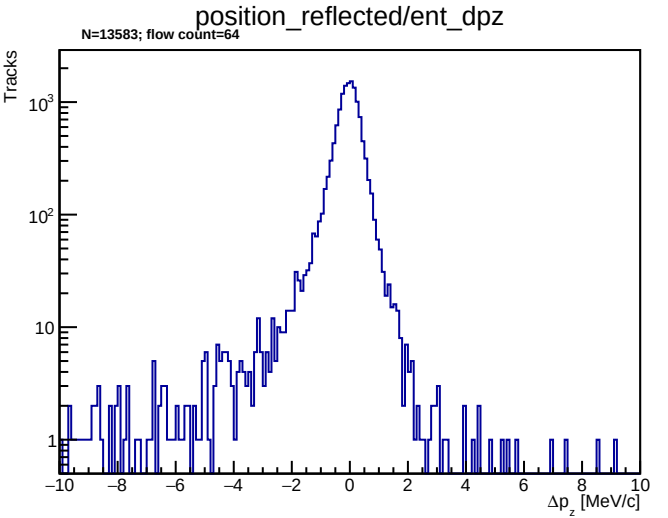

position_direct/tar_phase
N=13975; flow count=0
Tracks
10^2
10
−150 −100 −50 0 50 100 150
Δϕ [degrees]

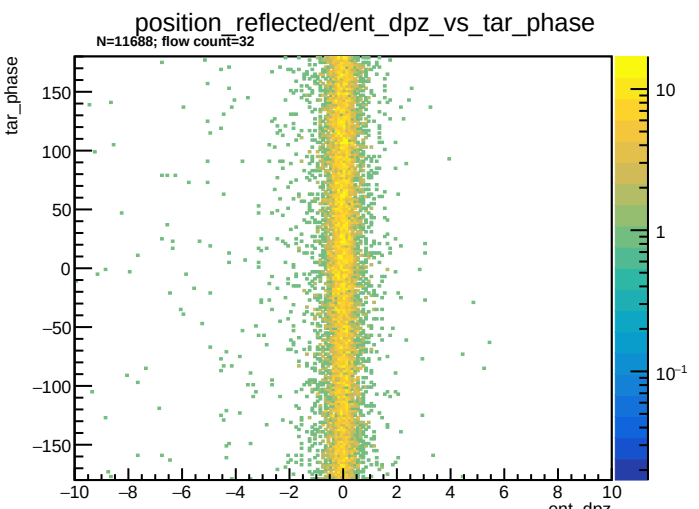

position_direct/tar_phase_vs_tar_dr
N=13975; flow count=0
tar_dr
1000 900 800 700 600 500 400 300 200 100 0
−150 −100 −50 0 50 100 150
tar_phase
10^2
10
1

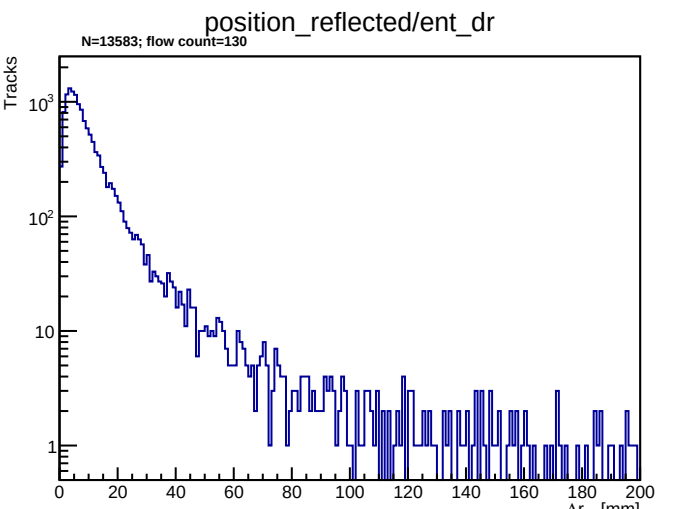

position_direct/tar_pitch
N=13975; flow count=8
Tracks
10^3
10^2
10
1
−500 −400 −300 −200 −100 0 100 200 300 400 500
Δθ [mrad]

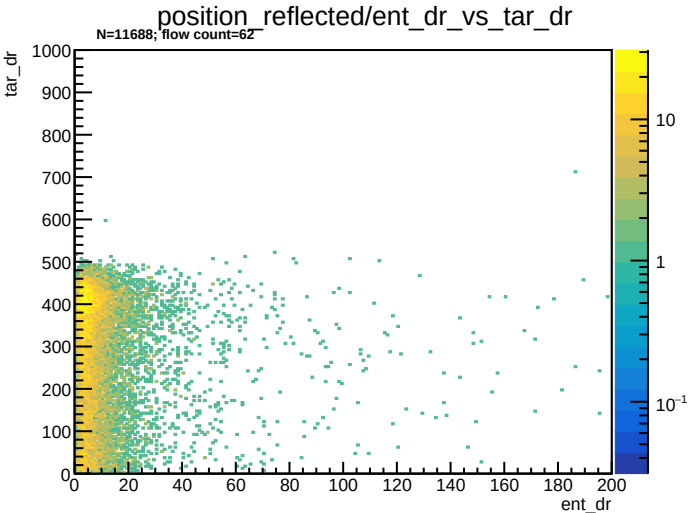

position_direct/tar_radius
N=13975; flow count=23
Tracks
10^3
10^2
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
ΔR [mm]

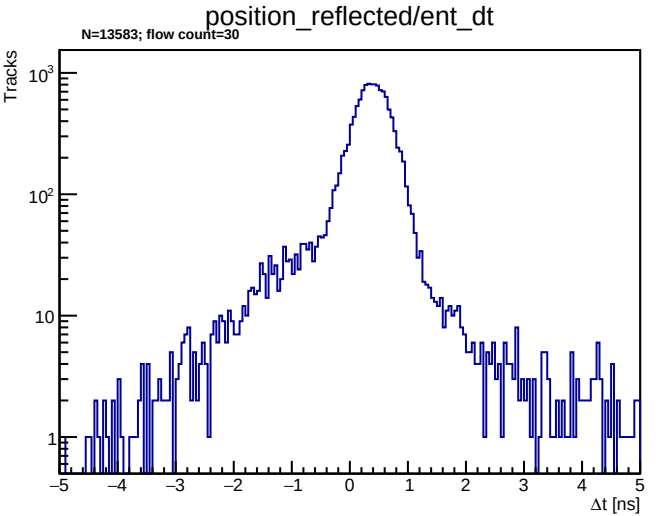

position_reflected/ent_dp
N=13583; flow count=108
Tracks
Δp [MeV/c]

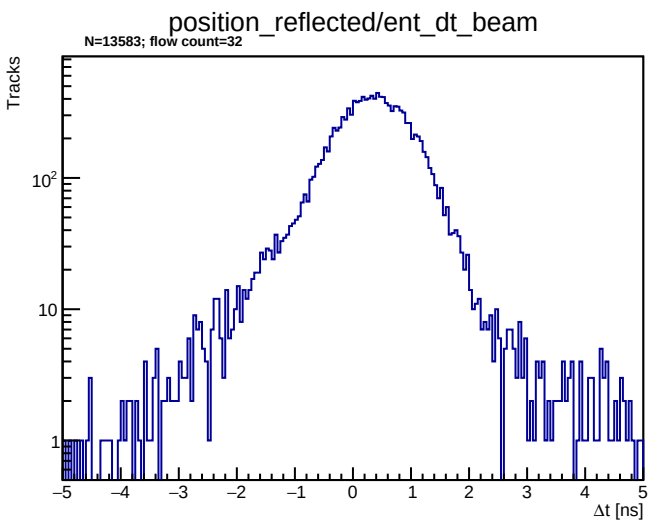

position_reflected/ent_dpz
N=13583; flow count=64
Tracks
Δp_z [MeV/c]

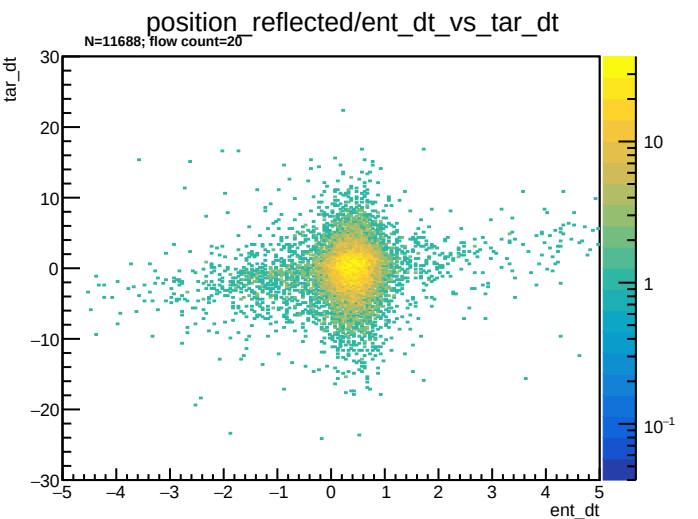

position_reflected/ent_dpz_vs_tar_phase
N=11688; flow count=32
tar_phase
ent_dpz

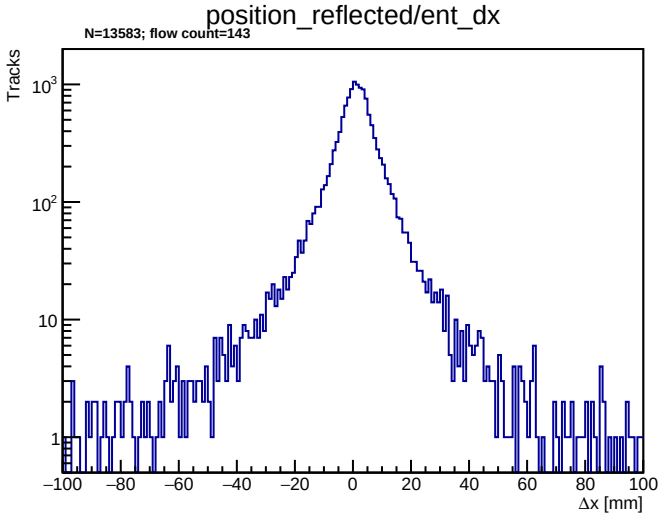

position_reflected/ent_dr
N=13583; flow count=130
Tracks
Δr_xy [mm]

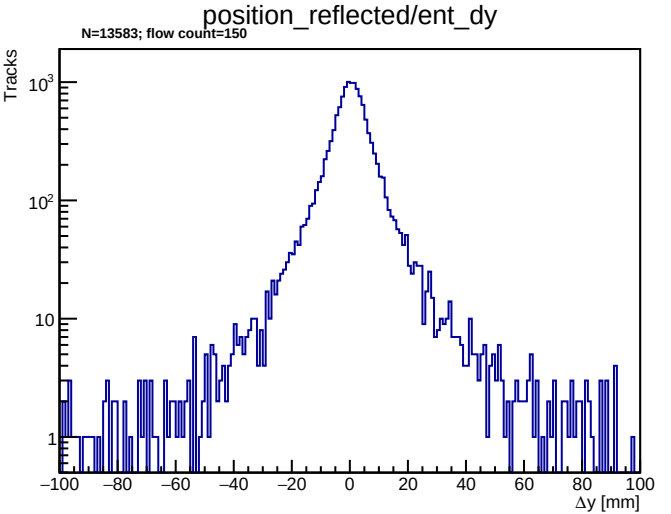

position_reflected/ent_dr_vs_tar_dr
N=11688; flow count=62
tar_dr
ent_dr

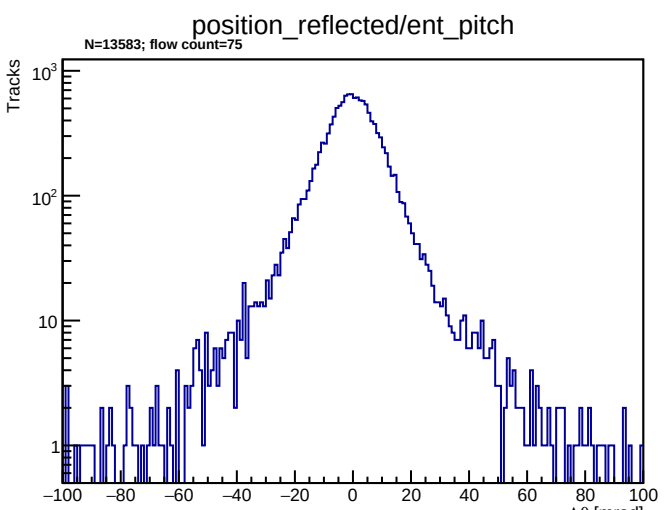

position_reflected/ent_dt
N=13583; flow count=30
Tracks
Δt [ns]

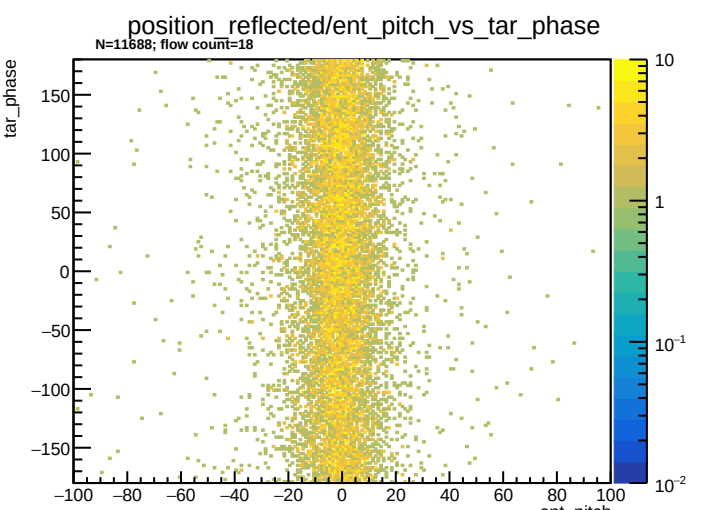

position_reflected/ent_dt_beam
N=13583; flow count=32
Tracks
Δt [ns]

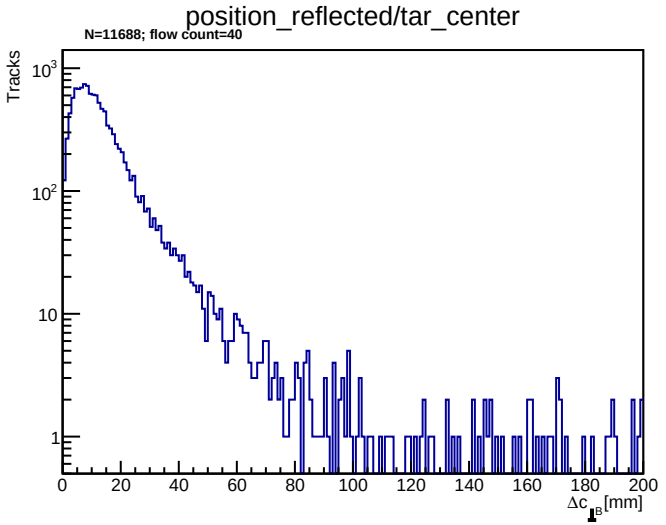

position_reflected/ent_dt_vs_tar_dt
N=11688; flow count=20
tar_dt
ent_dt

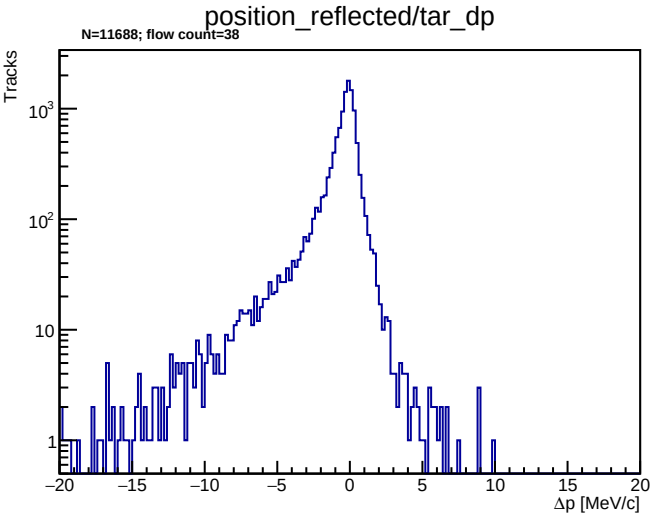

position_reflected/ent_dx
N=13583; flow count=143
Tracks
Δx [mm]

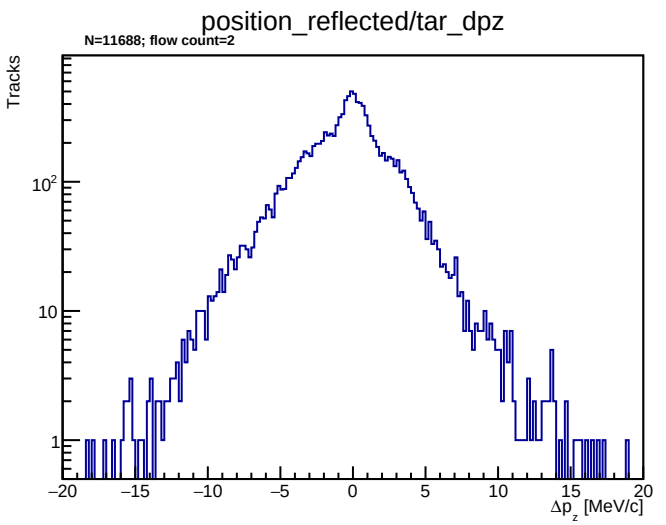

position_reflected/ent_dy
N=13583; flow count=150
Tracks
Δy [mm]

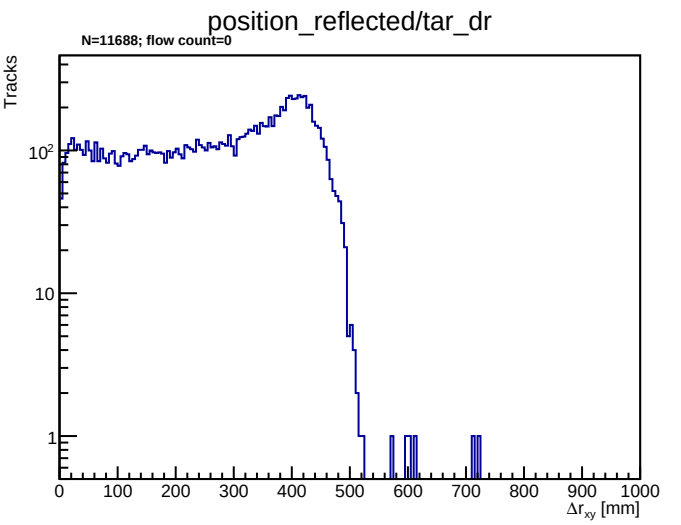

position_reflected/ent_pitch
N=13583; flow count=75
Tracks
Δθ [mrad]

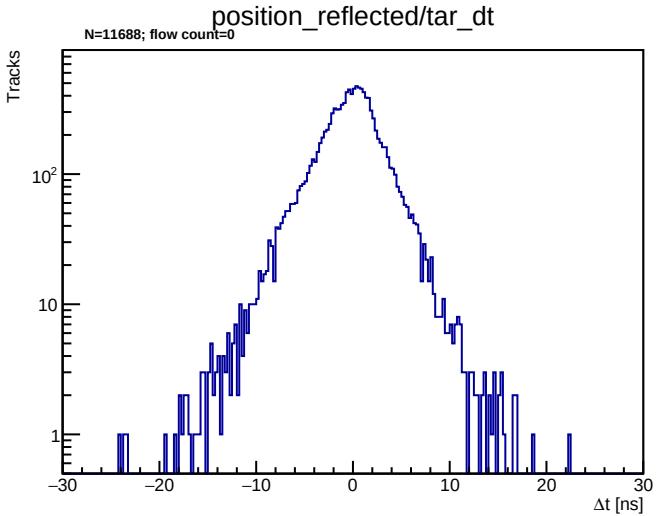

position_reflected/ent_pitch_vs_tar_phase
N=11688; flow count=18
tar_phase
ent_pitch

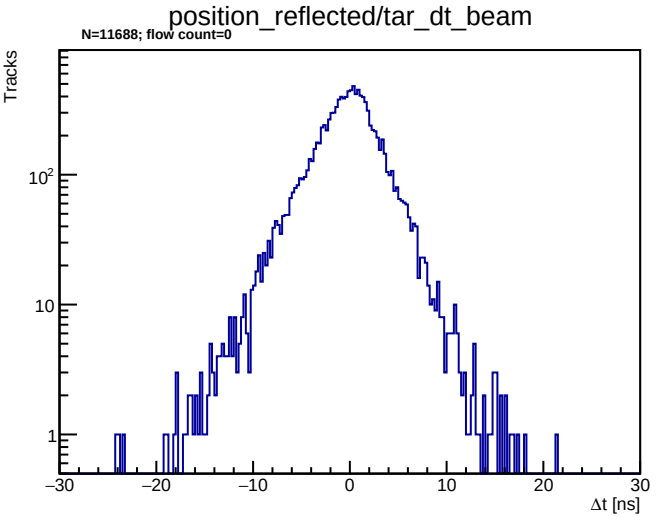

position_reflected/tar_center
N=11688; flow count=40
Tracks
10^3
10^2
10
1
0 20 40 60 80 100 120 140 160 180 200
Δc_B [mm]

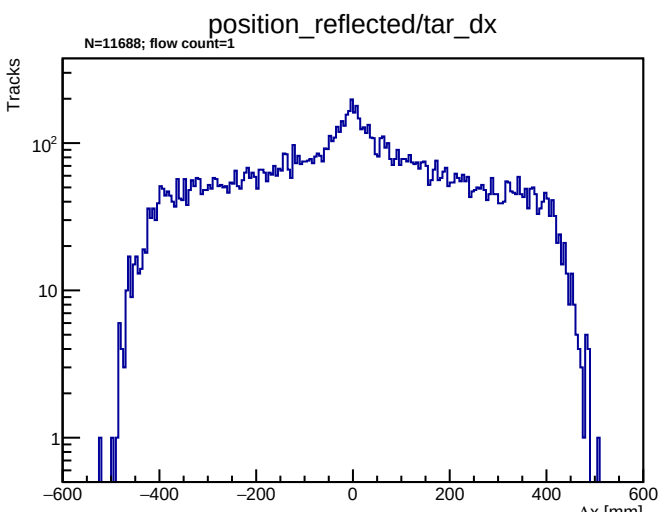

position_reflected/tar_dp
N=11688; flow count=38
Tracks
10^3
10^2
10
1
–20 –15 –10 –5 0 5 10 15 20
Δp [MeV/c]

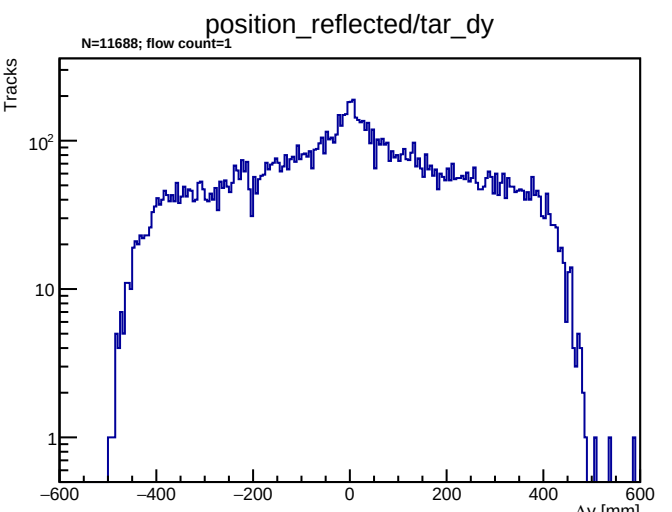

position_reflected/tar_dpz
N=11688; flow count=2
Tracks
10^2
10
1
–20 –15 –10 –5 0 5 10 15 20
Δp_z [MeV/c]

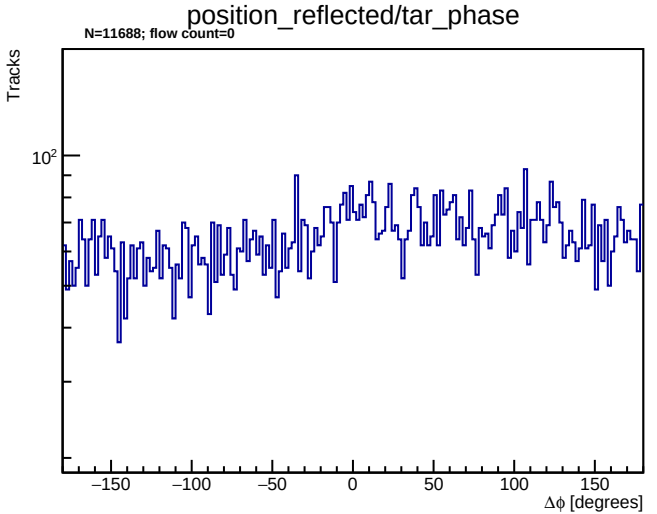

position_reflected/tar_dr
N=11688; flow count=0
Tracks
10^2
10
1
0 100 200 300 400 500 600 700 800 900 1000
Δr_{xy} [mm]

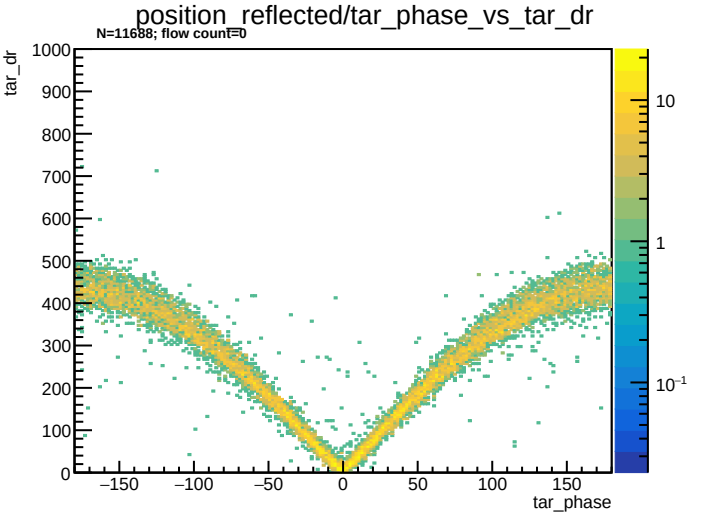

position_reflected/tar_dt
N=11688; flow count=0
Tracks
10^2
10
1
−30
−20
−10
0
10
20
30
Δt [ns]

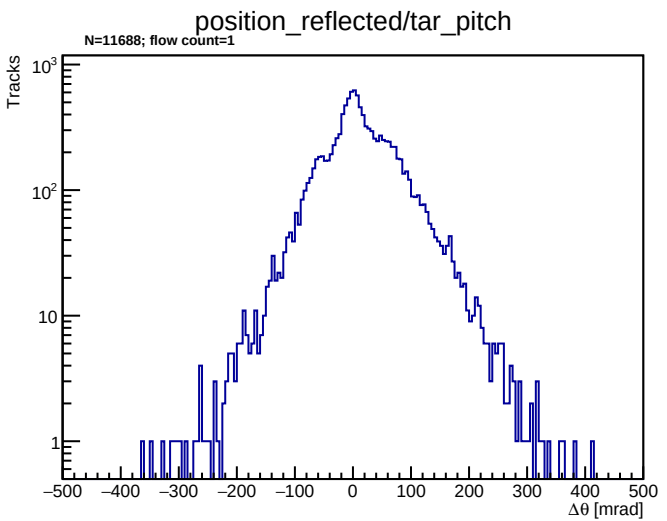

position_reflected/tar_dt_beam
N=11688; flow count=0
Tracks
10^2
10
1
−30
−20
−10
0
10
20
30
Δt [ns]

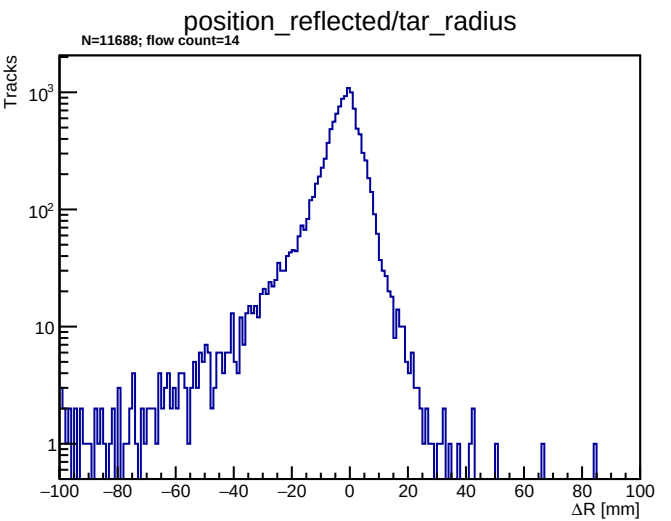

position_reflected/tar_dx
N=11688; flow count=1
Tracks
10^2
10
1
−600
−400
−200
0
200
400
600
Δx [mm]

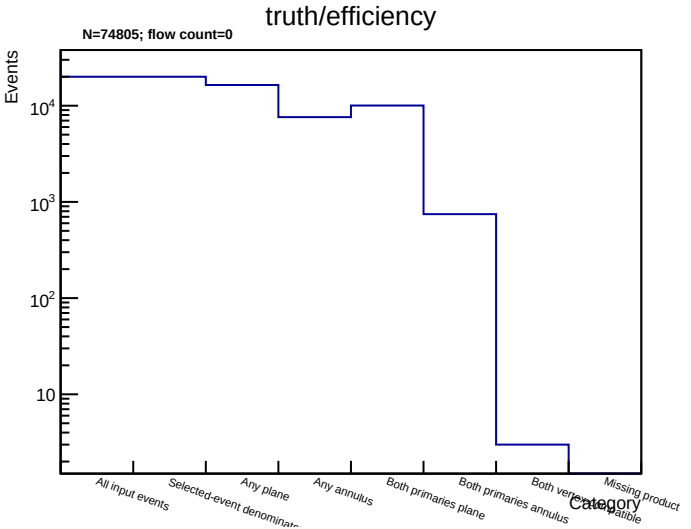

position_reflected/tar_dy
N=11688; flow count=1
Tracks
10^2
10
1
−600
−400
−200
0
200
400
600
Δy [mm]

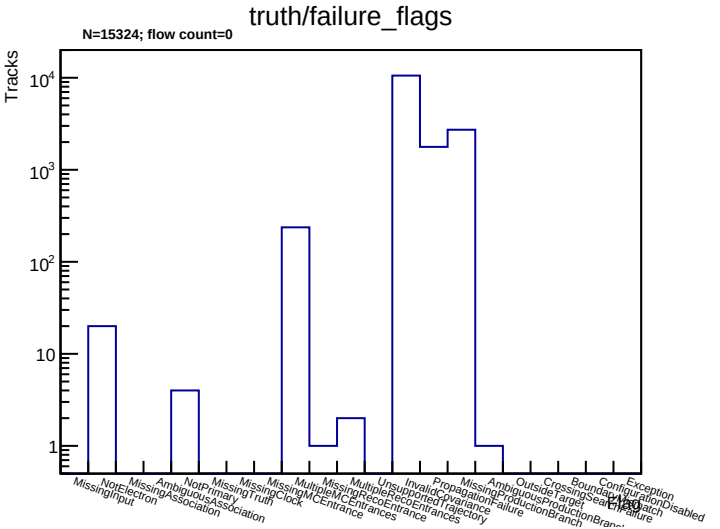

position_reflected/tar_phase
N=11688; flow count=0
Tracks
10^2
−150 −100 −50 0 50 100 150
Δϕ [degrees]

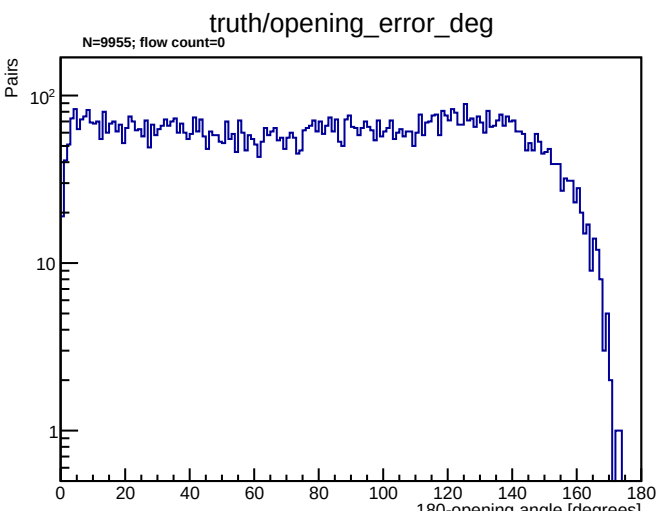

position_reflected/tar_phase_vs_tar_dr
N=11688; flow count=0
tar_dr
1000 900 800 700 600 500 400 300 200 100 0
−150 −100 −50 0 50 100 150
tar_phase
10
1
10^{-1}

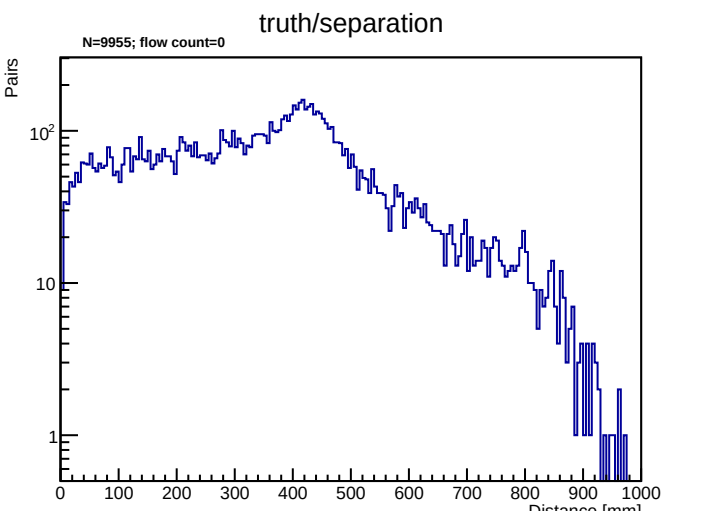

position_reflected/tar_pitch
N=11688; flow count=1
Tracks
10^3
10^2
10
1
−500 −400 −300 −200 −100 0 100 200 300 400 500
Δθ [mrad]

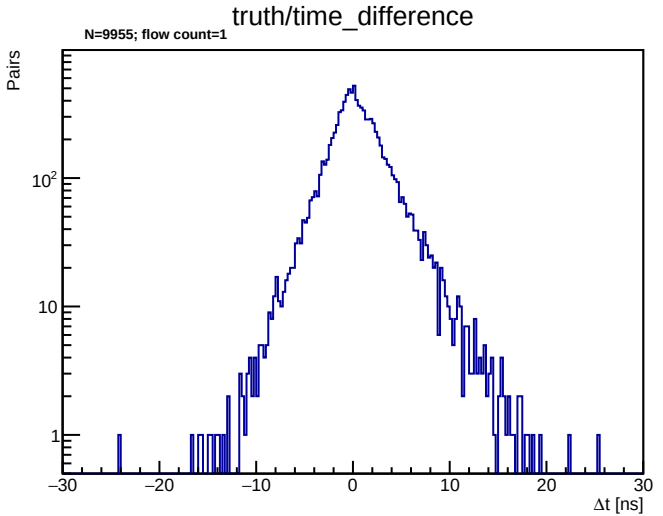

position_reflected/tar_radius
N=11688; flow count=14
Tracks
10^3
10^2
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
ΔR [mm]

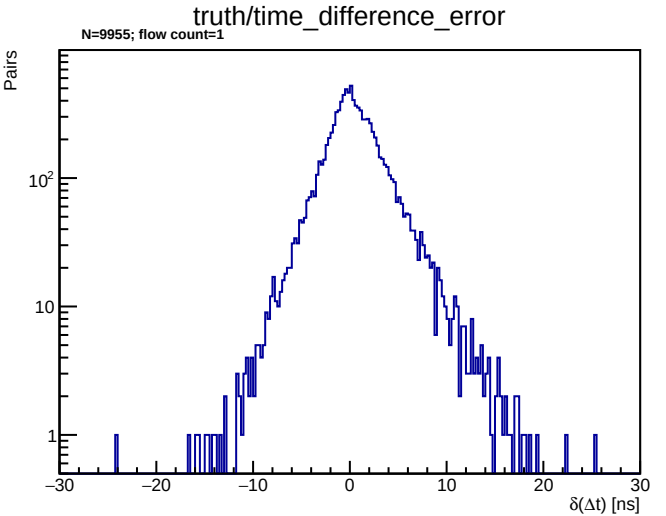

truth/efficiency
N=74805; flow count=0
Events
Category
All input events
Any plane
Any annulus
Both primaries plane
Both primaries annulus
Missing product

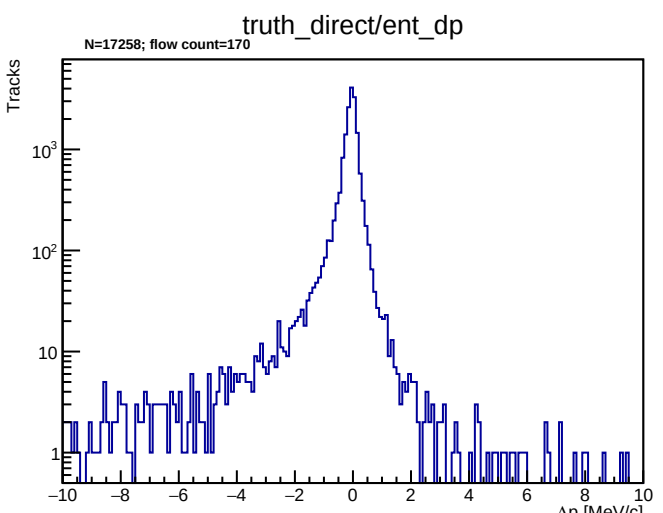

truth/failure_flags
N=15324; flow count=0
Tracks
Category
MissingInput
NotElectron
MissingAssociation
AmbiguousAssociation
NotPrimaryTruth
MissingTruth
MissingClock
MultipleMCEntrance
MissingRecoEntrance
MultipleRecoEntrances
MissingMCEntrances
MultipleRecoEntrances
UnsupportedTrajectory
InvalidCovariance
PropagationFailure
MissingProduction
AmbiguousProduction
OutsideTarget
CrossingSeed
BoundaryMismatch
Configuration
Exception
Disabled
Failure

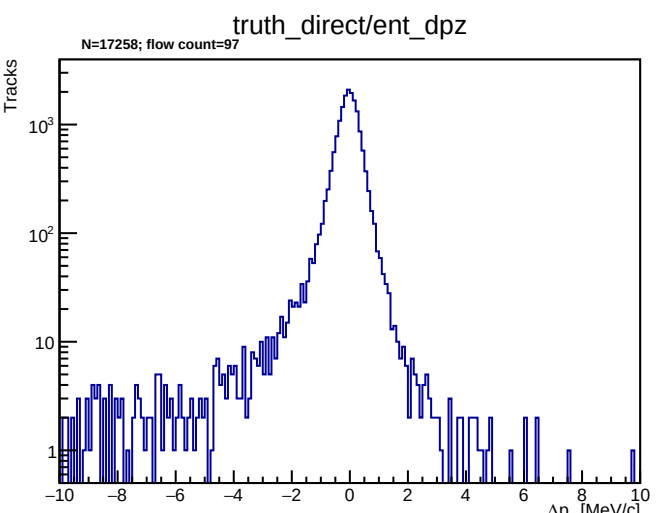

truth/opening_error_deg
N=9955; flow count=0
Pairs
180-opening angle [degrees]

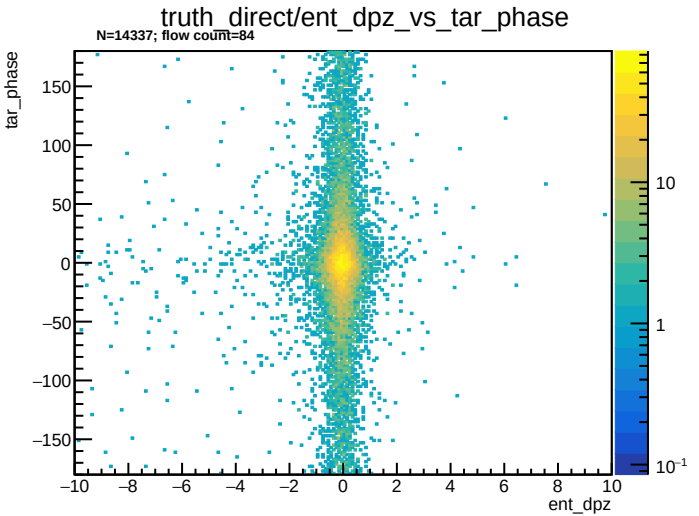

truth/separation
N=9955; flow count=0
Pairs
Distance [mm]

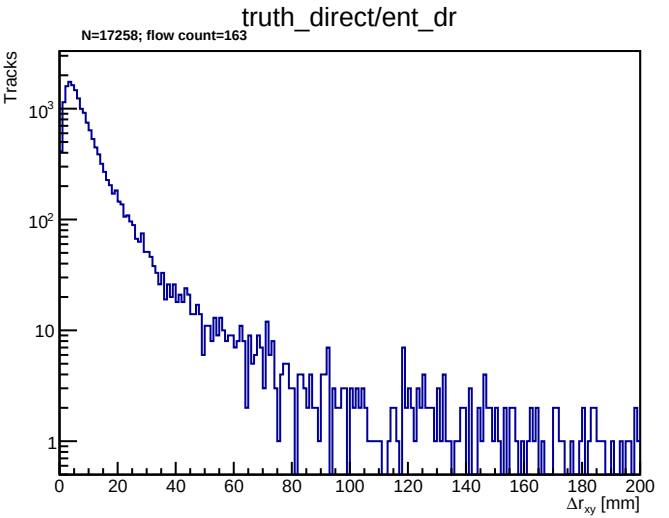

truth/time_difference
N=9955; flow count=1
Pairs
10^2
10
1
−30
−20
−10
0
10
20
30
Δt [ns]

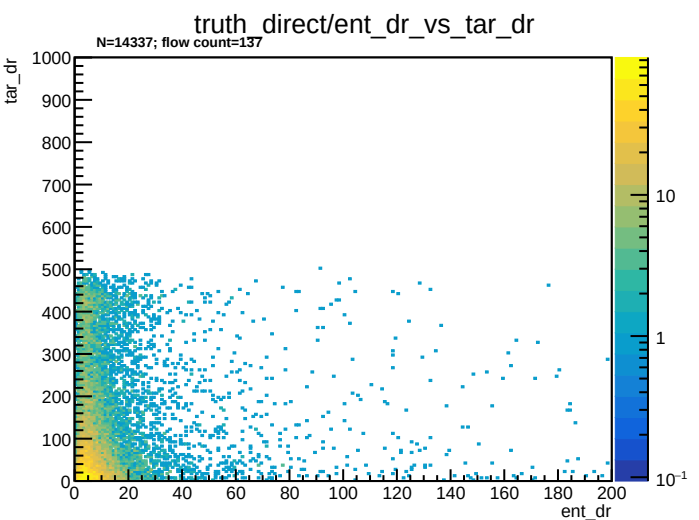

truth/time_difference_error
N=9955; flow count=1
Pairs
10^2
10
1
−30
−20
−10
0
10
20
30
δ(Δt) [ns]

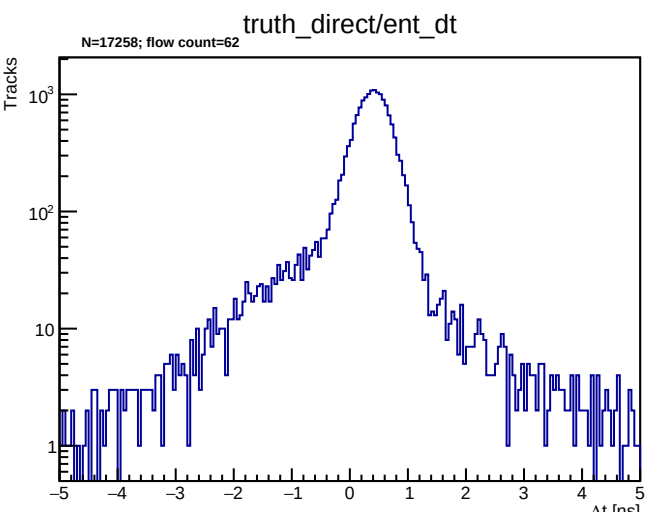

truth_direct/ent_dp
N=17258; flow count=170
Tracks
10^3
10^2
10
1
−10
−8
−6
−4
−2
0
2
4
6
8
10
Δp [MeV/c]

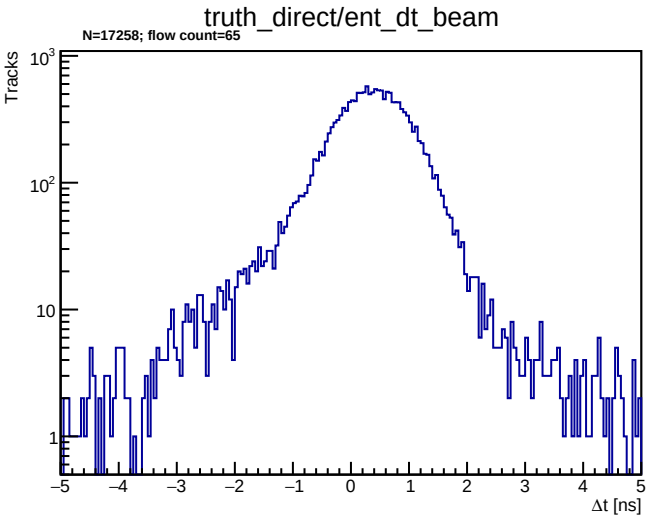

truth_direct/ent_dpz
N=17258; flow count=97
Tracks
10^3
10^2
10
1
−10
−8
−6
−4
−2
0
2
4
6
8
10
Δp_z [MeV/c]

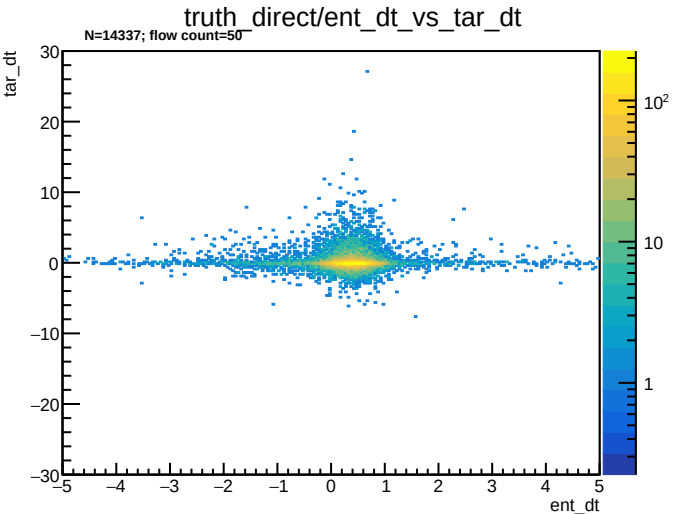

truth_direct/ent_dpz_vs_tar_phase
N=14337; flow count=84
tar_phase
ent_dpz

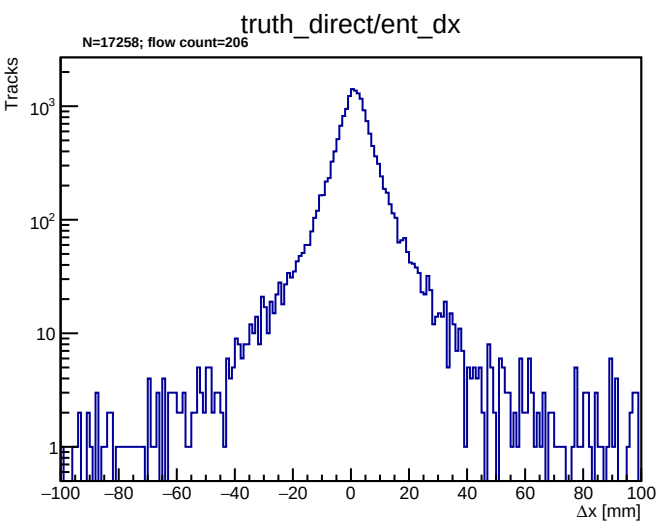

truth_direct/ent_dr
N=17258; flow count=163
Tracks
Δr_xy [mm]

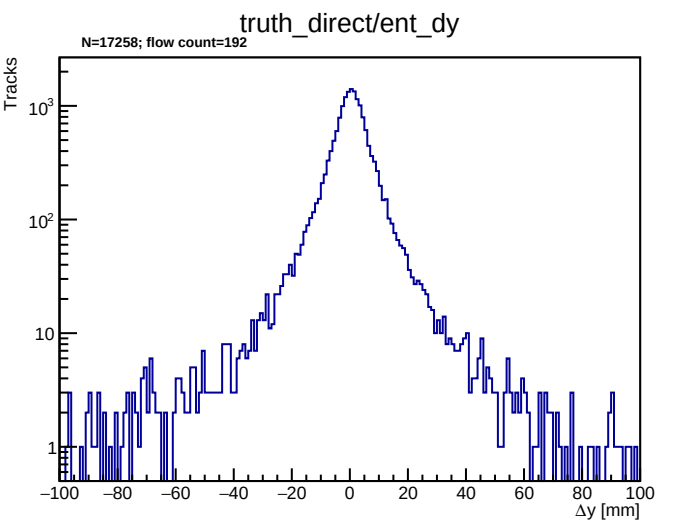

truth_direct/ent_dr_vs_tar_dr
N=14337; flow count=137
tar_dr
ent_dr

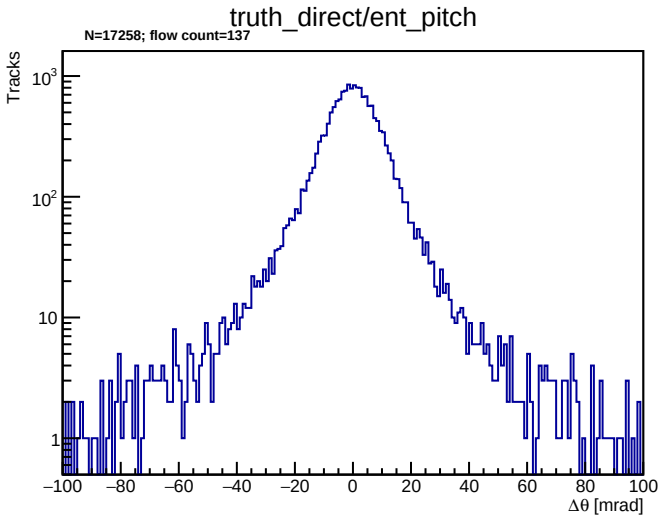

truth_direct/ent_dt
N=17258; flow count=62
Tracks
Δt [ns]

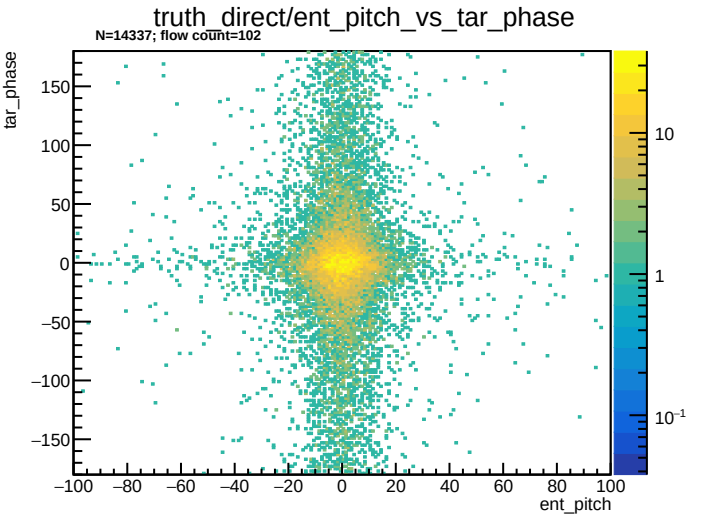

truth_direct/ent_dt_beam
N=17258; flow count=65
Tracks
Δt [ns]

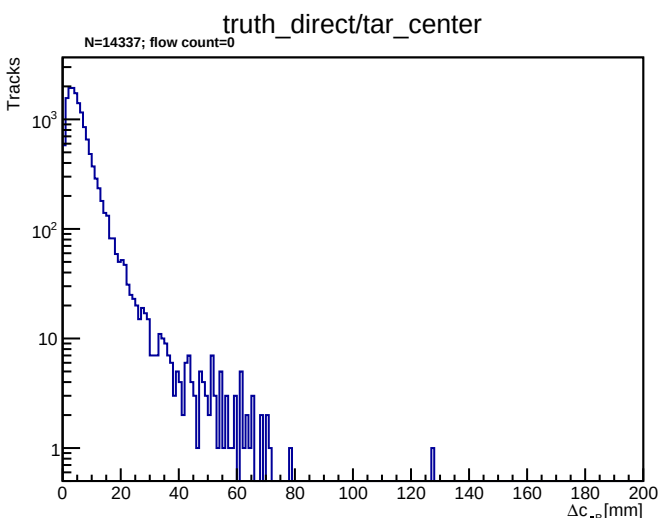

truth_direct/ent_dt_vs_tar_dt
N=14337; flow count=50
tar_dt
ent_dt

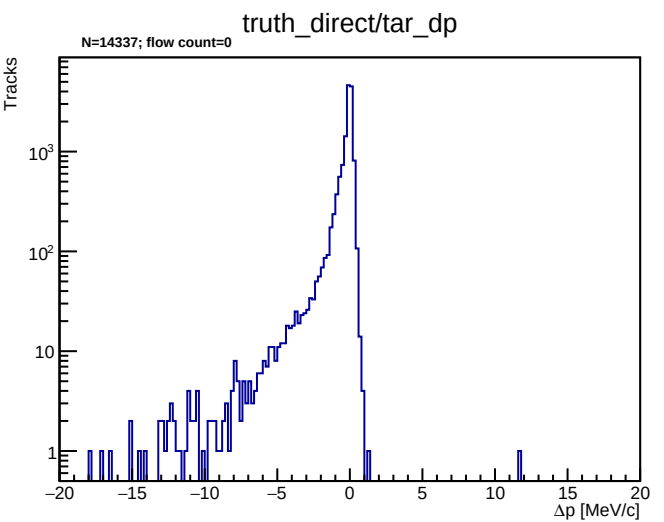

truth_direct/ent_dx
N=17258; flow count=206
Tracks
Δx [mm]

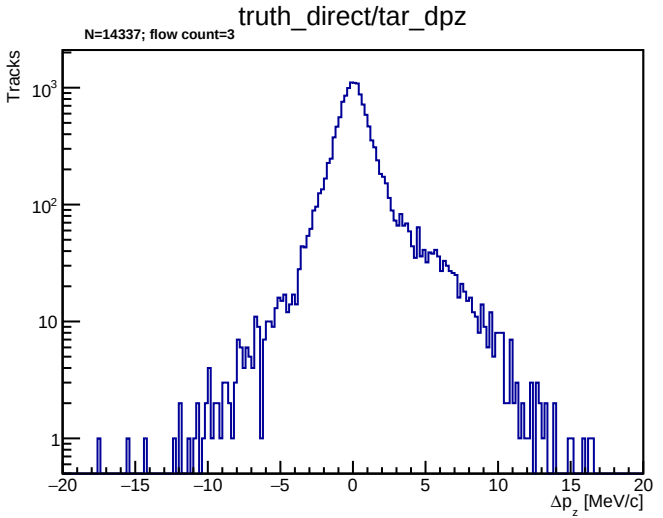

truth_direct/ent_dy
N=17258; flow count=192
Tracks
Δy [mm]

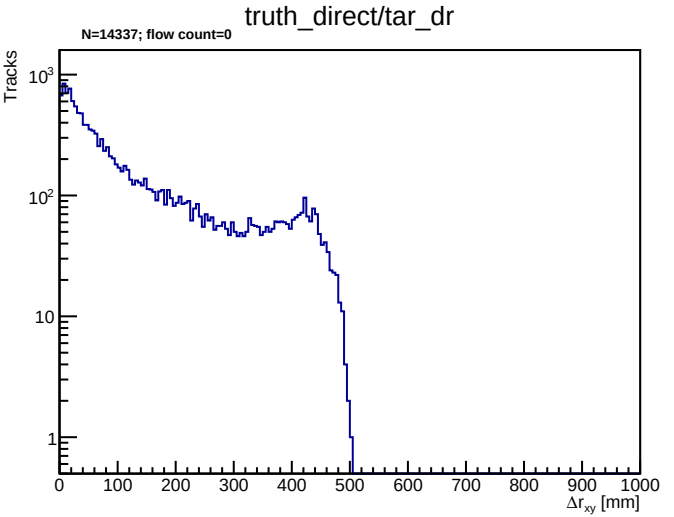

truth_direct/ent_pitch
N=17258; flow count=137
Tracks
Δθ [mrad]

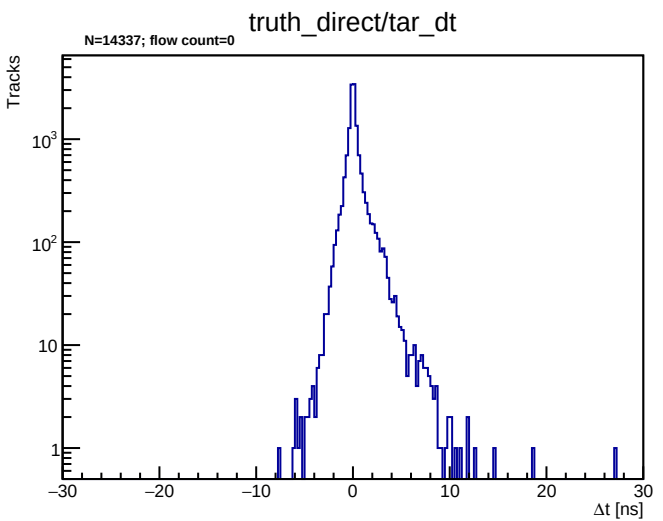

truth_direct/ent_pitch_vs_tar_phase
N=14337; flow count=102
tar_phase
ent_pitch

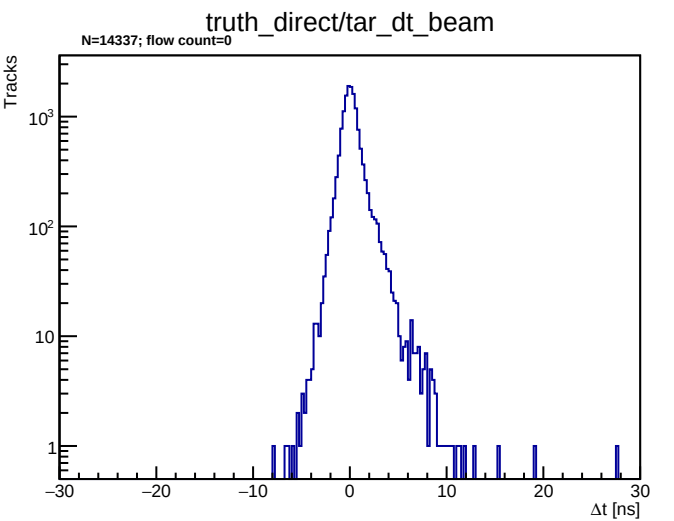

truth_direct/tar_center
N=14337; flow count=0
Tracks
Δc_B [mm]

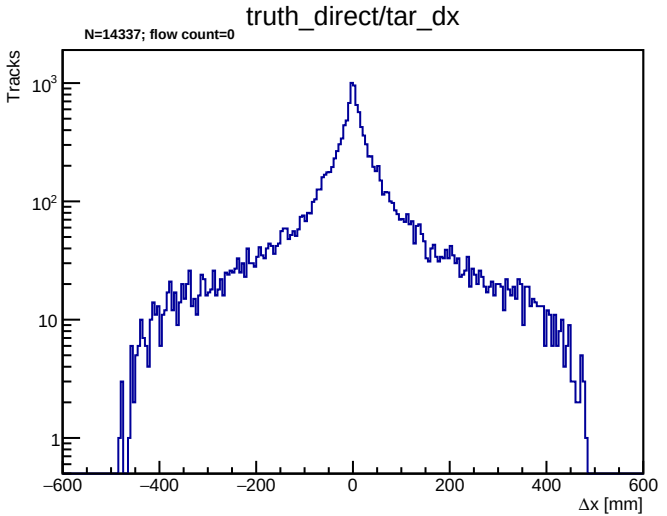

truth_direct/tar_dp
N=14337; flow count=0
Tracks
Δp [MeV/c]

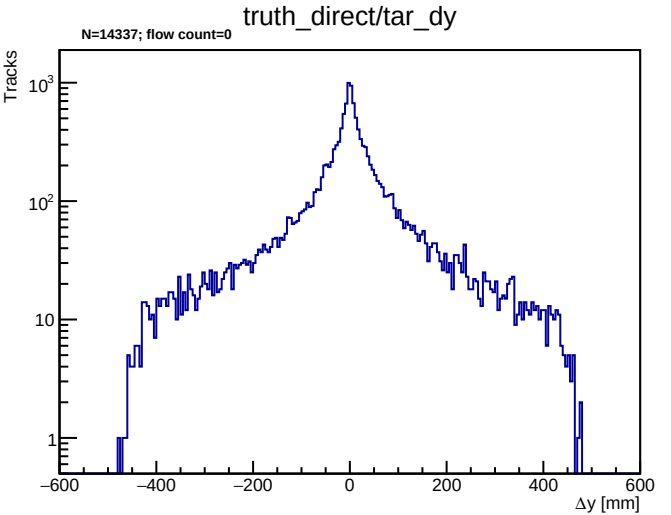

truth_direct/tar_dpz
N=14337; flow count=3
Tracks
10^3
10^2
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

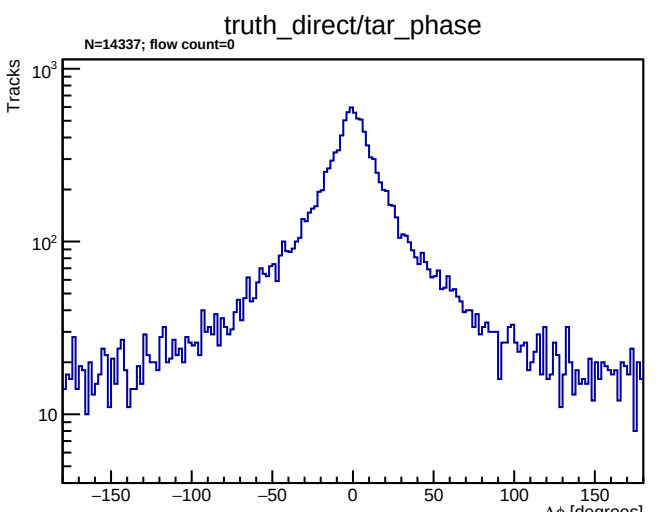

truth_direct/tar_dr
N=14337; flow count=0
Tracks
10^3
10^2
10
1
0 100 200 300 400 500 600 700 800 900 1000
Δr_{xy} [mm]

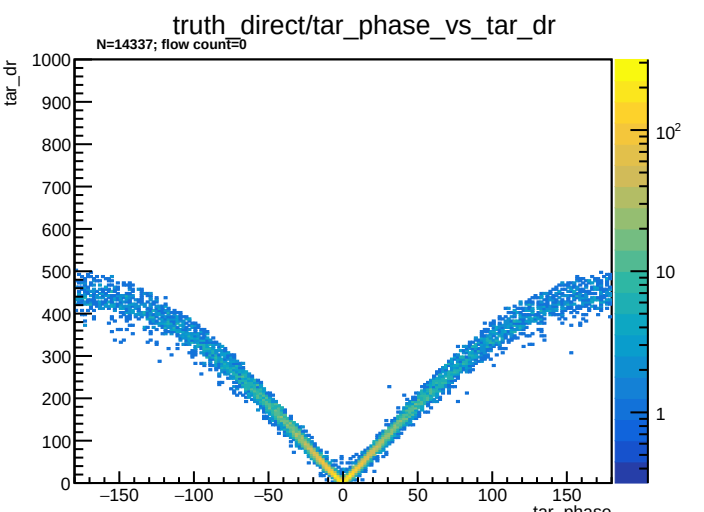

truth_direct/tar_dt
N=14337; flow count=0
Tracks
10^3
10^2
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

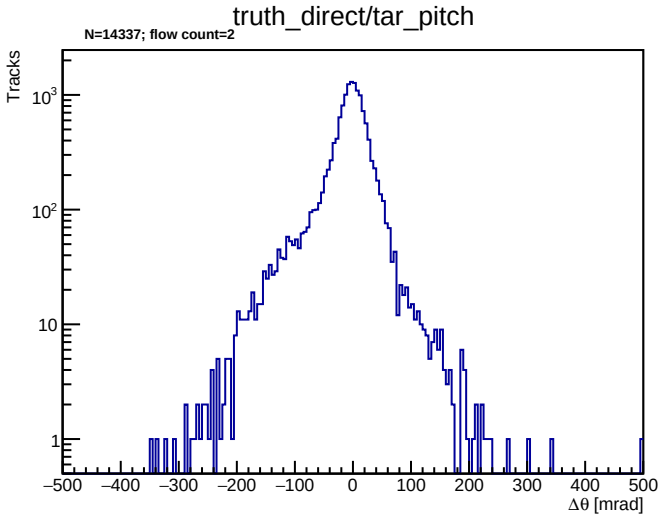

truth_direct/tar_dt_beam
N=14337; flow count=0
Tracks
10^3
10^2
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

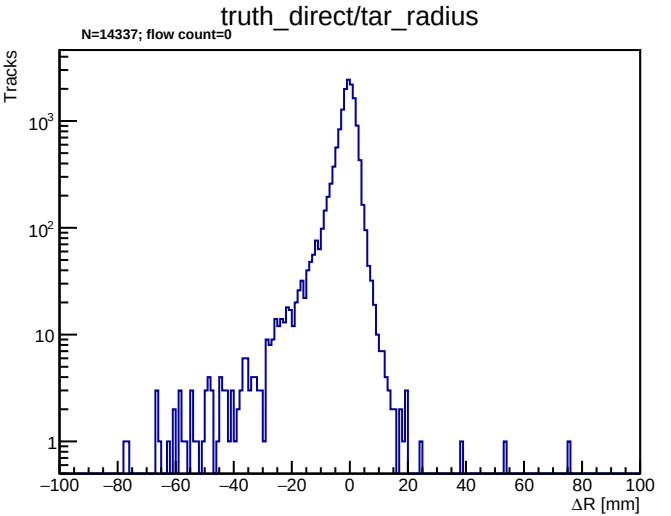

truth_direct/tar_dx
N=14337; flow count=0
Tracks
Δx [mm]

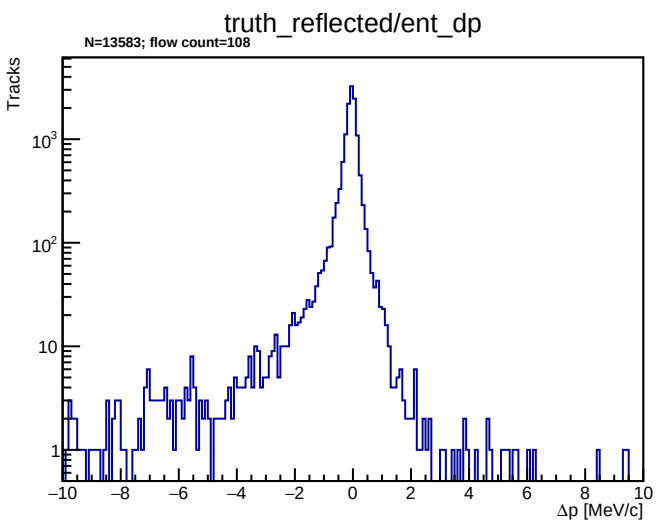

truth_direct/tar_dy
N=14337; flow count=0
Tracks
Δy [mm]

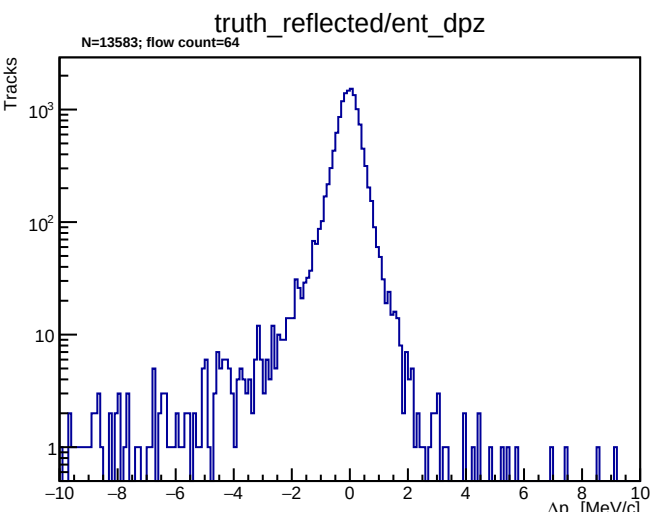

truth_direct/tar_phase
N=14337; flow count=0
Tracks
Δϕ [degrees]

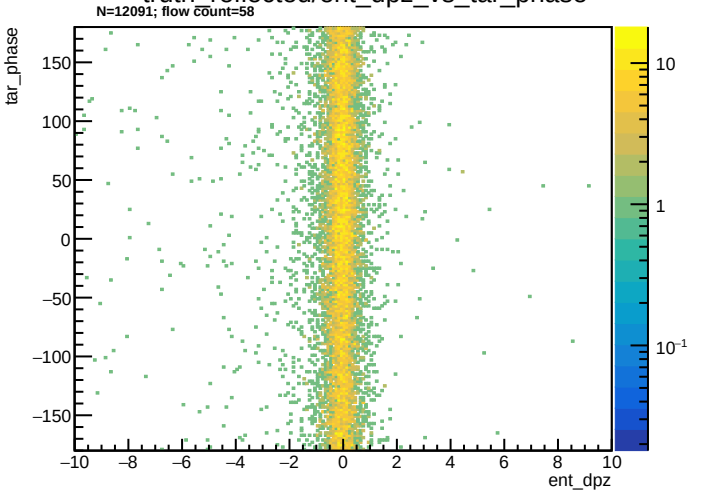

truth_direct/tar_phase_vs_tar_dr
N=14337; flow count=0
tar_dr
tar_phase

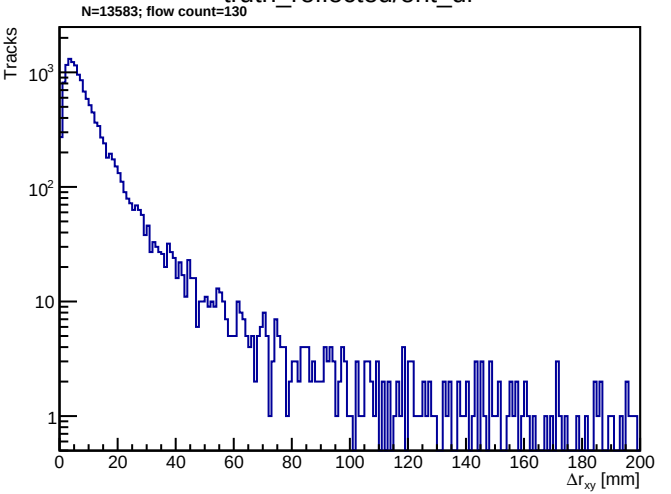

truth_direct/tar_pitch
N=14337; flow count=2
Tracks
10^3
10^2
10
1
−500 −400 −300 −200 −100 0 100 200 300 400 500
Δθ [mrad]

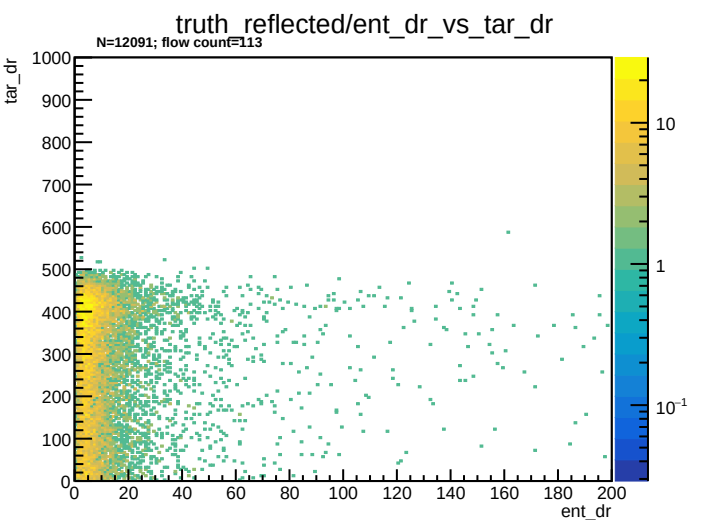

truth_direct/tar_radius
N=14337; flow count=0
Tracks
10^3
10^2
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
ΔR [mm]

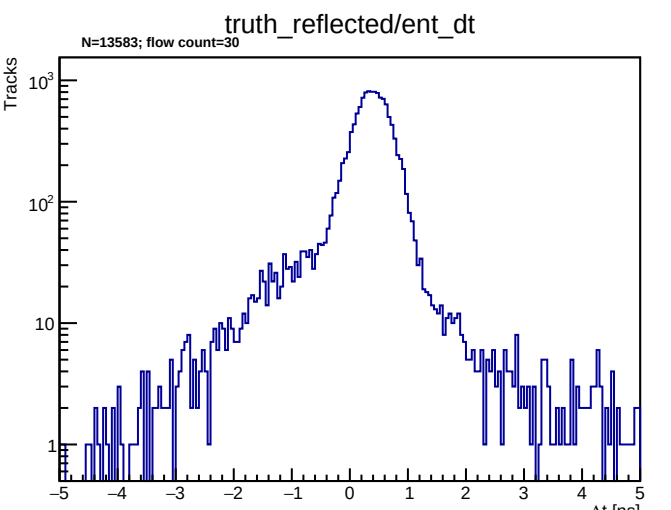

truth_reflected/ent_dp
N=13583; flow count=108
Tracks
10^3
10^2
10
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp [MeV/c]

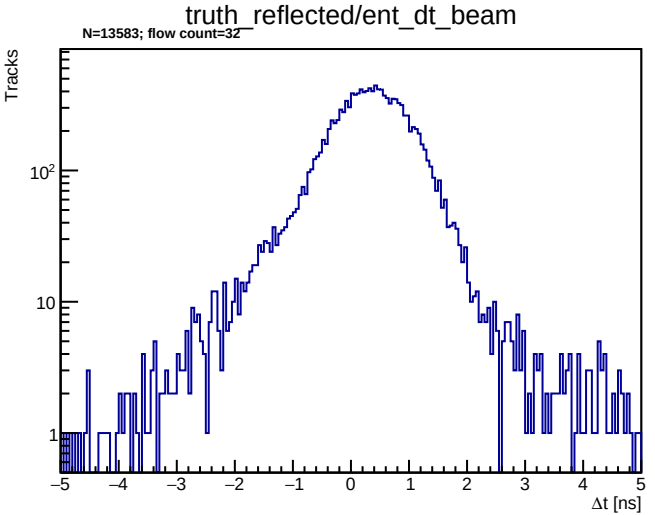

truth_reflected/ent_dpz
N=13583; flow count=64
Tracks
10^3
10^2
10
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp_z [MeV/c]

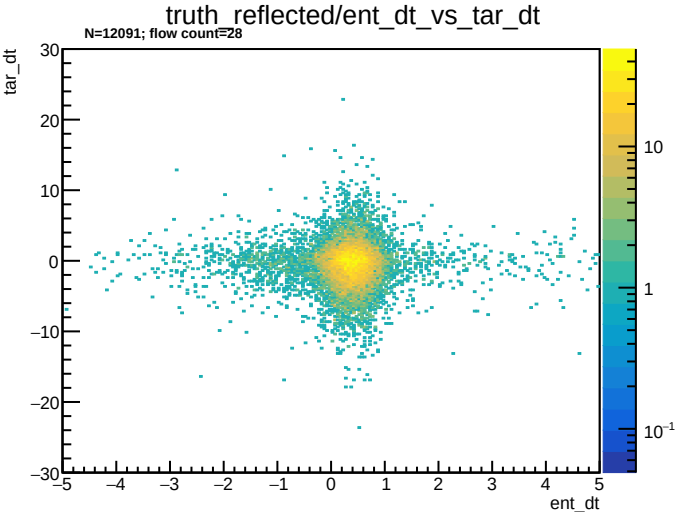

truth_reflected/ent_dpz_vs_tar_phase
N=12091; flow count=58
tar_phase
ent_dpz

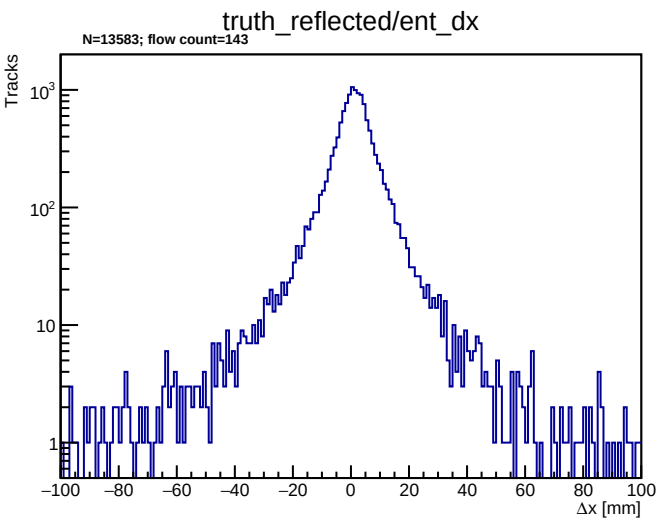

truth_reflected/ent_dr
N=13583; flow count=130
Tracks
Δr_xy [mm]

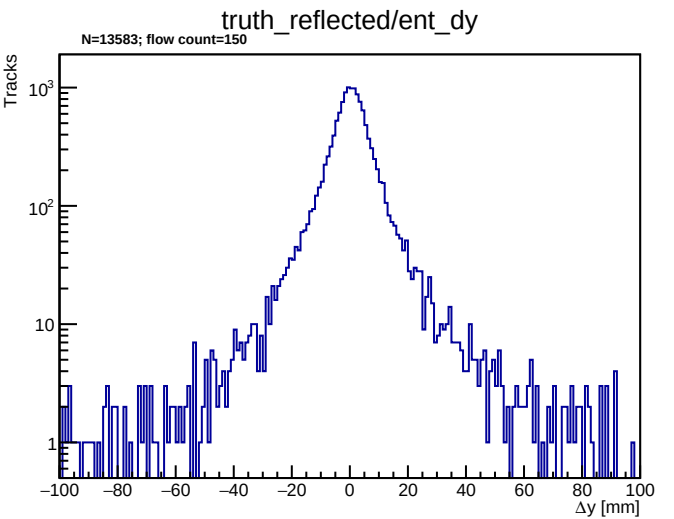

truth_reflected/ent_dr_vs_tar_dr
N=12091; flow count=113
tar_dr
ent_dr

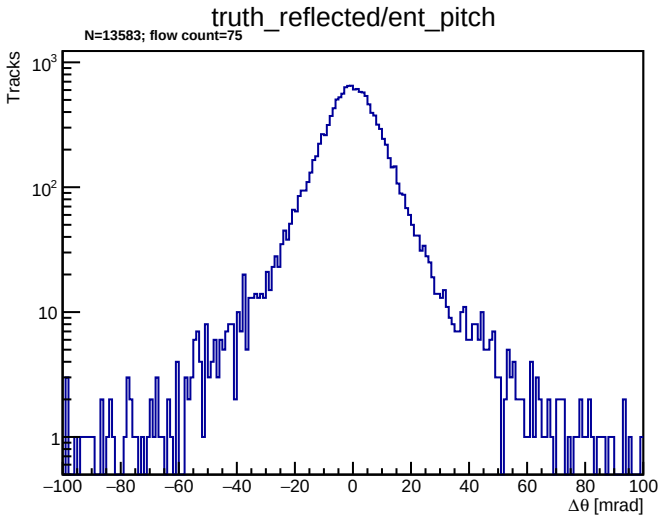

truth_reflected/ent_dt
N=13583; flow count=30
Tracks
Δt [ns]

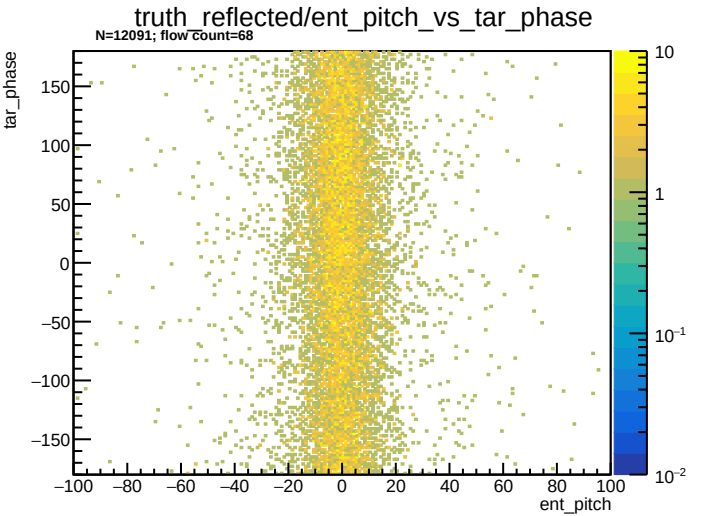

truth_reflected/ent_dt_beam
N=13583; flow count=32
Tracks
Δt [ns]

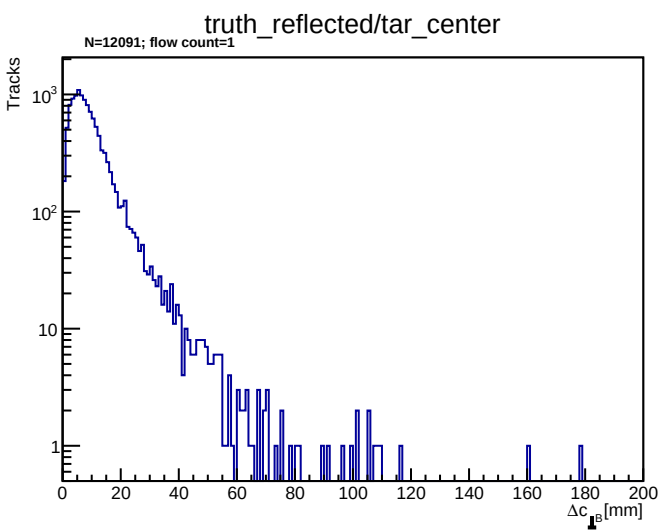

truth_reflected/ent_dt_vs_tar_dt
N=12091; flow count=28
tar_dt
ent_dt

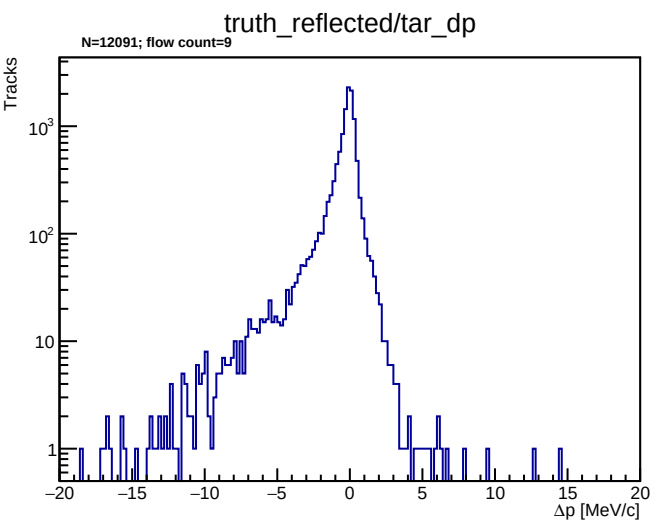

truth_reflected/ent_dx
N=13583; flow count=143
Tracks
Δx [mm]

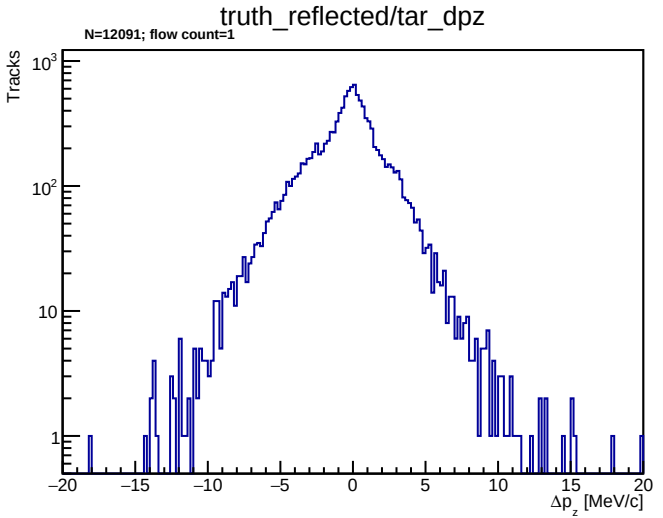

truth_reflected/ent_dy
N=13583; flow count=150
Tracks
Δy [mm]

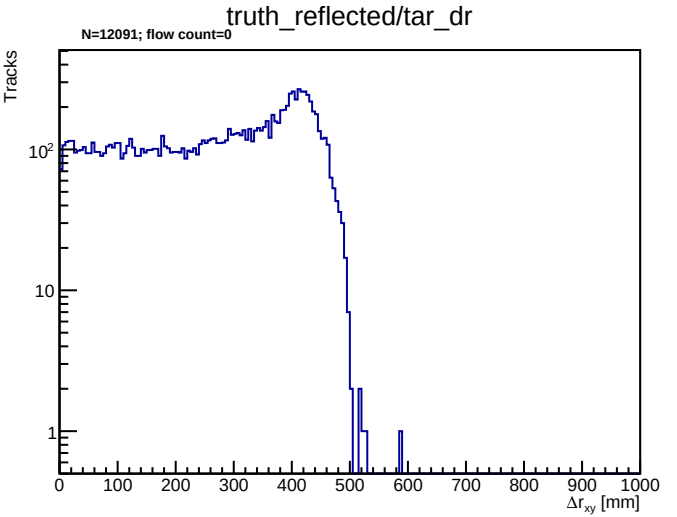

truth_reflected/ent_pitch
N=13583; flow count=75
Tracks
Δθ [mrad]

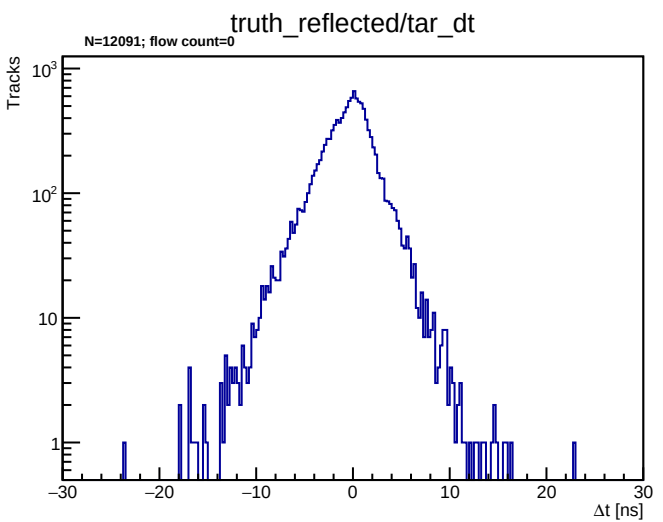

truth_reflected/ent_pitch_vs_tar_phase
N=12091; flow count=68
tar_phase
ent_pitch

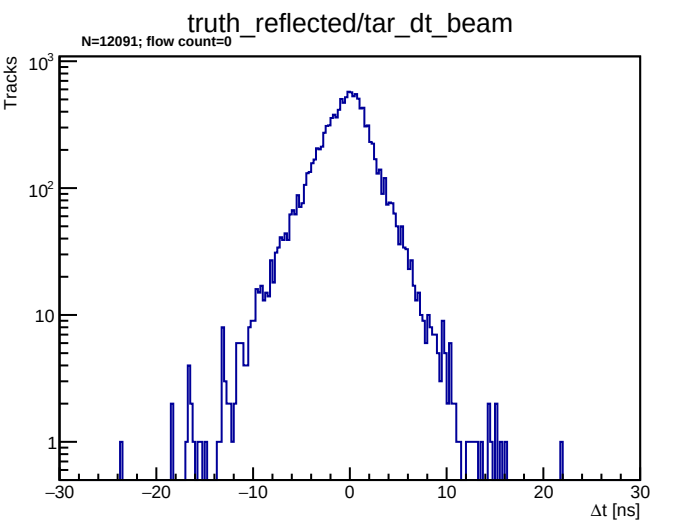

truth_reflected/tar_center
N=12091; flow count=1
Tracks
Δc_B [mm]

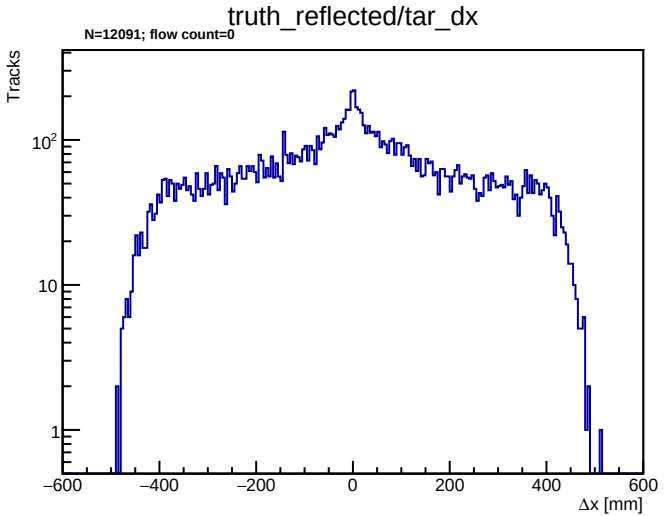

truth_reflected/tar_dp
N=12091; flow count=9
Tracks
Δp [MeV/c]

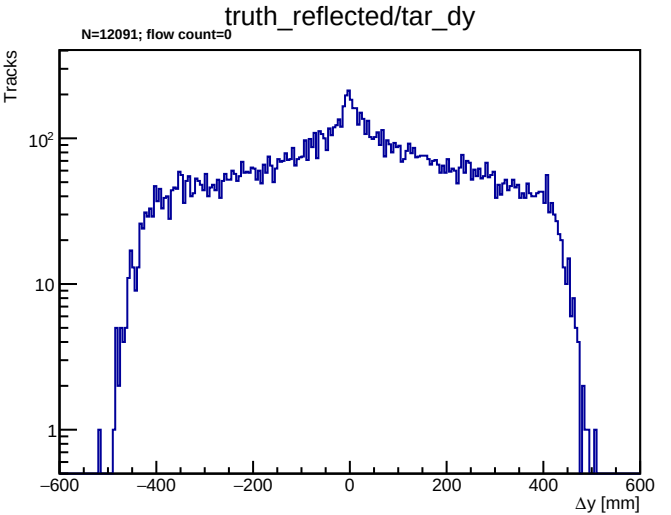

truth_reflected/tar_dpz
N=12091; flow count=1
Tracks
10^3
10^2
10
1
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

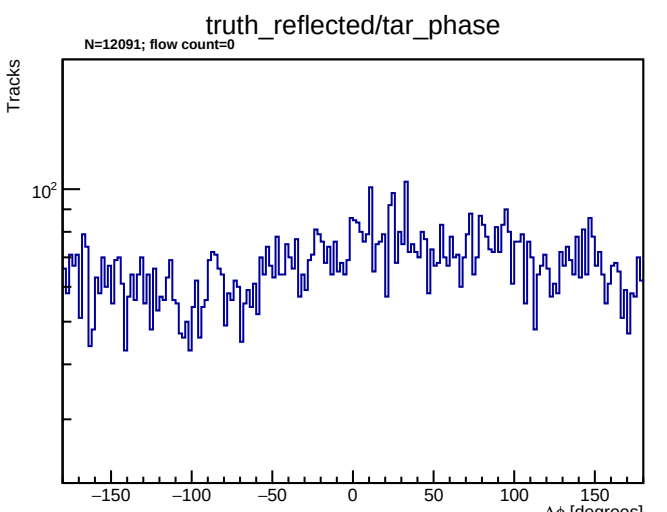

truth_reflected/tar_dr
N=12091; flow count=0
Tracks
10^2
10
1
0 100 200 300 400 500 600 700 800 900 1000
Δr_{xy} [mm]

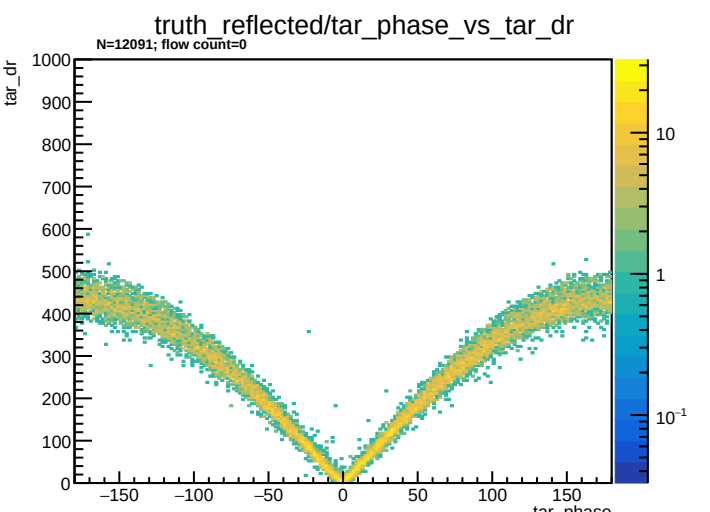

truth_reflected/tar_dt
N=12091; flow count=0
Tracks
10^3
10^2
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

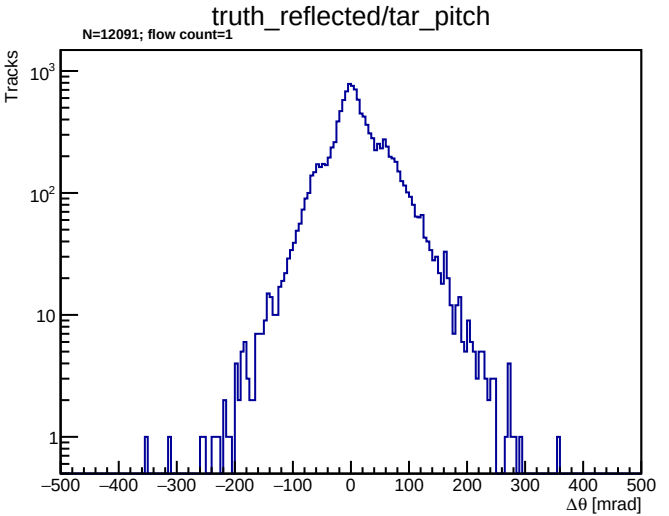

truth_reflected/tar_dt_beam
N=12091; flow count=0
Tracks
10^3
10^2
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

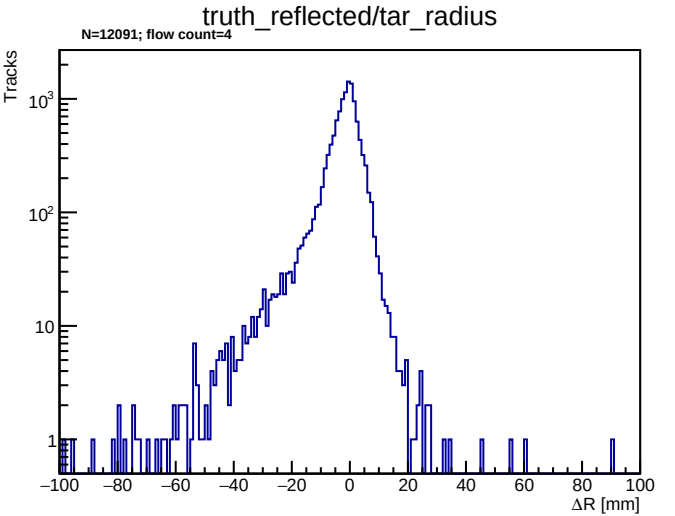

truth_reflected/tar_dx
N=12091; flow count=0
Tracks
10^2
10
1
−600
−400
−200
0
200
400
600
Δx [mm]


truth_reflected/tar_dy
N=12091; flow count=0
Tracks
10^2
10
1
−600
−400
−200
0
200
400
600
Δy [mm]


truth_reflected/tar_phase
N=12091; flow count=0
Tracks
10^2
−150
−100
−50
0
50
100
150
Δϕ [degrees]


truth_reflected/tar_phase_vs_tar_dr
N=12091; flow count=0
tar_dr
1000
900
800
700
600
500
400
300
200
100
0
−150
−100
−50
0
50
100
150
tar_phase
10
1
10^{-1}


truth_reflected/tar_pitch
N=12091; flow count=1
Tracks
10^3
10^2
10
1
−500 −400 −300 −200 −100 0 100 200 300 400 500
Δθ [mrad]


truth_reflected/tar_radius
N=12091; flow count=4
Tracks
10^3
10^2
10
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
ΔR [mm]